

# FALCONAge

## Biological age and aging-clock scoring from multiomic data, in Python and R, on CPU or GPU.

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*Generated from the FALCONAge documentation site, <https://bhagesh-h.github.io/FALCONAge/>*

# Chapter 1. FALCONAge

FALCONAge computes biological age scores from molecular data. It reads what a laboratory actually produces, applies published aging clocks to it, and returns each score with its unit, its provenance, and an estimate of how much of it is measurement noise.

An aging clock is a supervised model mapping molecular features, most often DNA methylation at specific CpG sites, to a single number. Roughly two hundred have been published since 2013, and they were not fitted to the same target. Some were regressed on calendar age, some on a survival-weighted composite and then rescaled into years, some on the rate of change in organ-system biomarkers tracked longitudinally, some on cumulative stem-cell divisions. Those are four different quantities. Reported together as “biological age” they look comparable, and the arithmetic that mixes them has no defined meaning.

This package treats that as the problem to solve rather than a caveat to mention. Every catalogue entry declares the target its clock was trained on, the unit the output carries, and which downstream operations are consequently defined. Asking for age acceleration on a pace-of-aging clock returns an error naming the units, not a number. Scoring a placenta clock on blood is refused. Scoring saliva on a whole-blood clock warns and cites the size of the error measured in people sampled both ways. The refusals are the design, and each one names the measurement it rests on.

The second half of the package is about what a single number is worth. Technical replicates of the same DNA differ by up to nine years on prominent clocks, and only 18% of probes on the 450K array reach an intraclass correlation of 0.5 in whole blood. A score reported without that context invites a reader to believe a difference that the assay alone could have produced. FALCONAge propagates published per-probe reliability through each clock’s own weights to give a standard error per sample, and calibrates distribution-free prediction intervals against known ages.

It is one numerical core in Python, called from R through reticulate, so a result obtained in either language is the same bits rather than two implementations that agree approximately. The catalogue holds 175 clocks; 46 of them score offline, and the rest are documented with the reason they cannot, which is licensing or untraced provenance rather than missing code.

These are instruments for population research, not a clinical judgement about one person. FALCONAge computes what the papers specify and records how it did so; whether a given clock is fit for a given decision is not something a scoring tool can certify, and the limits below say what else it cannot do.

## Input

| Input                      | Reader                            | Notes                                                                                                         |
|----------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------|
| Raw Illumina IDATs         | <code>read_idat_dir()</code>      | 27K, 450K, EPIC v1, EPIC v2. Full chain to betas: pOOBAH detection, noob background correction, optional BMIQ |
| Beta matrix (CSV, Parquet) | <code>read_betas()</code>         |                                                                                                               |
| GEO series matrix          | <code>read_series_matrix()</code> |                                                                                                               |

| Input               | Reader                              | Notes                                                                                             |
|---------------------|-------------------------------------|---------------------------------------------------------------------------------------------------|
| Nanopore bedMethyl  | <code>read_bedmethyl_dir()</code>   | <code>modkit</code> output, with per-site coverage kept alongside; 5mC and 5hmC read separately   |
| RRBS coverage files | <code>read_rrbs_dir()</code>        |                                                                                                   |
| Targeted panels     | <code>read_panel()</code>           | Pyrosequencing, EpiTYPER, ampli-con. The declared CpG list is checked, not intersected            |
| Olink NPX           | <code>read_olink()</code>           | Proteomics. Already log2 and relative; the scale is left alone                                    |
| SomaScan RFU        | <code>read_somascan()</code>        | Proteomics, log2 by default because that is what the models were fitted on                        |
| Bulk RNA-seq counts | <code>read_counts()</code>          | Transcriptomics, with the full RLE $\rightarrow$ log10 $\rightarrow$ z $\rightarrow$ YuGene chain |
| Clinical chemistry  | <code>read_clinical()</code>        | Units declared rather than guessed                                                                |
| ComputAgeBench      | <code>read_computage_bench()</code> | 65 harmonised case/control studies, pinned to a commit                                            |

## Output

- One score per sample per clock, for as many of the 175 catalogued clocks as your features support
- **A standard error on every score**, propagated from published per-probe reliability through each clock's own weights, and a distribution-free prediction interval
- A coverage figure per clock, and what the probes you are missing cost in *years*
- A run manifest holding the package and registry versions, the SHA-256 of every coefficient file, the array manifest that decoded your IDATs, the reliability table behind the intervals, the device and floating-point precision, the imputation policy, and every warning raised

**Execution.** One numerical core in Python, called from R through reticulate, on CPU or GPU.

**Refusals.** A saliva sample scored on a whole-blood clock (measured at 3.83–16.46 years of error in the same people). Cell-free DNA on any array clock. Age acceleration on a pace-of-aging clock. A cohort-centred clock given one sample. Each refusal names the measurement behind it.

## Installation

Docker is the supported path. One image carries Python, R, the CLI and all 43 bundled weight files at pinned versions, so the same input gives the same numbers on your machine and on a reviewer's. Both images are published on [Docker Hub](#) and built from the Dockerfiles in this repository. Two commands from nothing to a proven install:

```
docker pull bhagesh/falconage:1.0.0-cpu
docker run --rm bhagesh/falconage:1.0.0-cpu python -m pytest /opt/falconage/src/python/
tests -q
```

The second line proves it by running the 462-test suite inside the image, against the source it was built from. To build the same image yourself instead of pulling it:

```
git clone https://github.com/bhagesh-h/FALCONAge.git && cd FALCONAge
docker build -f docker/Dockerfile.cpu -t bhagesh/falconage:1.0.0-cpu .
```

On Windows PowerShell write "\${PWD}". Open a session with `docker run --rm -it -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu python`, or ... R, and every example below works unchanged.

**Getting started** walks the same ground one command at a time, including fetching real public data, the GPU image, and a **native pip** / `remotes::install_github install` for anyone who already manages their own environment.

## Quick start

### Python

```
import falconage as fa

data = fa.prepare(fa.read_betas("betas.csv"))
res = fa.score(data, clocks="compatible")

res.summary()
fa.acceleration(res, method="both")
fa.plot.ba_vs_ca(res, "horvath2013")
fa.report.write_report(res, "report.html")
```

### R

```
library(FALCONAge)

d <- prepare(read_betas("betas.csv"))
res <- score(d, clocks = "compatible")

summary(res)
acceleration(res, method = "both")
plot_ba_vs_ca(res, "horvath2013")
report(res, "report.html")
```

Both produce the same numbers, because both run the same code. The R package delegates all arithmetic to the Python core through reticulate, and the R suite asserts equality at tolerance exactly zero rather than approximate agreement.

## The clock catalogue

175 clocks, in three availability groups. Availability is about redistribution rights, not about whether the algorithm works: every architecture is implemented and tested whatever the group.

| Availability | n  | What you do                                             |
|--------------|----|---------------------------------------------------------|
| bundled      | 46 | Nothing. Forty-three ship a weights file; PhenoAge, KDM |

| Availability | n  | What you do                                                                                                                                                      |
|--------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| untraced     | 89 | and homeostatic dysregulation are formulas with none to ship. Catalogued, but no primary source for the numbers has been established yet, so nothing is bundled. |
| licensed     | 40 | Obtain a coefficient file and register it. The architecture is implemented and tested; the numbers are research-use-only.                                        |

```
fa.registry.load().filter(availability="A")
```

## Training targets, and what a score means

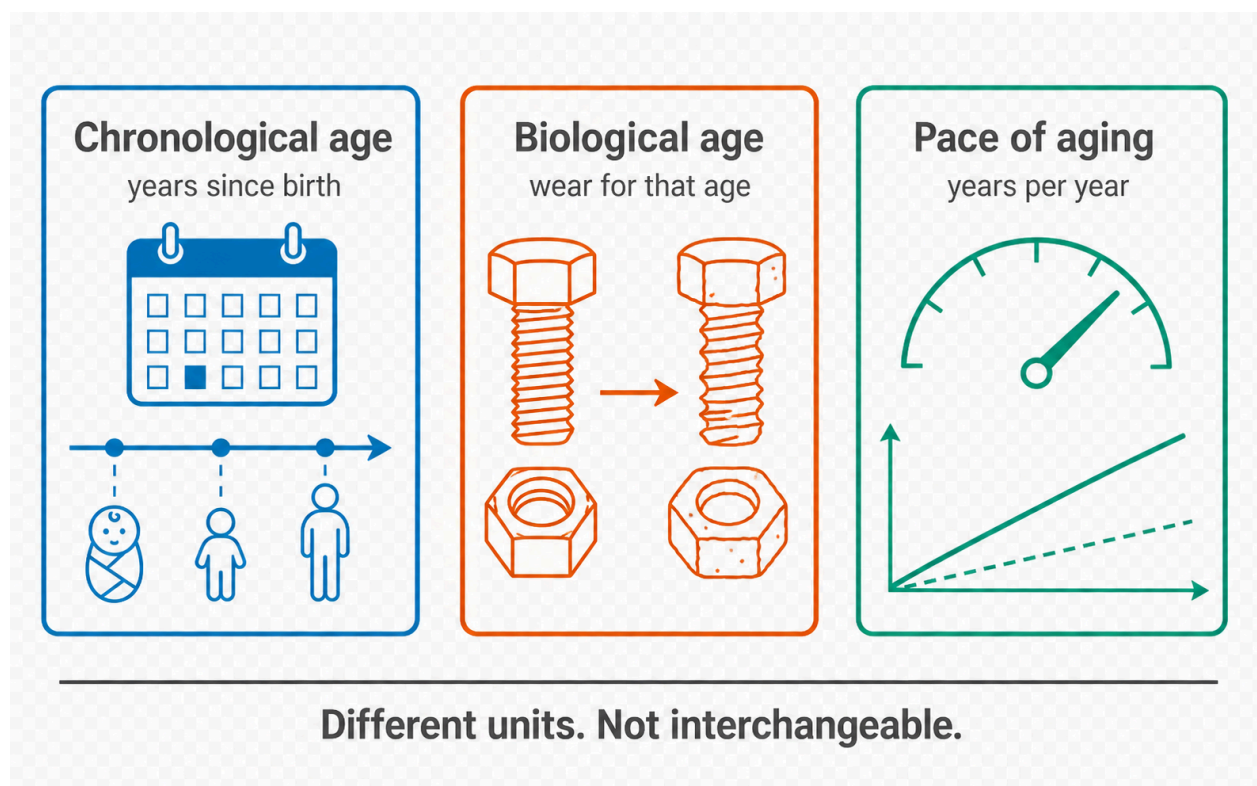


Figure 1: Three quantities that all get called “biological age”, and the units that keep them apart

Chronological age is elapsed time, and it is known without measuring anything. Biological age is an estimate of how worn a body is for that elapsed time. Pace of aging is a rate, years of wear per year of calendar time, and it is the only one of the three that a single sample cannot establish on its own: the Dunedin measure was fitted against two decades of repeated biomarker readings. Because the three carry different units, none of them converts into another, which is why a registry entry declares a `scale_type` and not only a unit.

What a clock’s output *means* is set by the outcome it was regressed on. Six families are in common use, and they are not variants of one quantity:

- **Chronological-age clocks** (Horvath 2013 [5], Hannum 2013 [4]) are fitted to calendar age. Their error bars are prediction error against a known truth.
- **Mortality- and morbidity-trained clocks** (PhenoAge [8], GrimAge [11]) are fitted to a survival-weighted composite and then rescaled into years. The unit is years; the target was never age.
- **Pace-of-aging clocks** (DunedinPACE [15]) are fitted to the *rate of change* in organ-system biomarkers tracked longitudinally. Belsky and colleagues draw the distinction directly:

[Pace-of-aging measures capture] the rate of deterioration in system integrity occurring over a fixed time interval, whereas epigenetic clocks estimate the progress of aging up to a cross-section in time.

Belsky et al., *eLife* 2022

- **Mitotic clocks** (epiTOC2 [20], StemTOC) estimate cumulative stem-cell divisions. The quantity is a count, not elapsed time.
- **Causality-enriched clocks** (CausAge, DamAge, AdaptAge) restrict features to CpGs with Mendelian-randomisation support, separating damaging from adaptive change (Ying et al., *Nature Aging* 2024, [19]).
- **Deconvolution models** return cell-type proportions, which are a composition and must sum to one.

Reporting all six as “biological age” discards the information needed to interpret any of them.

## Scale types and permitted operations

|                      | Correlate | Difference | Residual | Acceleration |
|----------------------|-----------|------------|----------|--------------|
| Years                | ✓         | ✓          | ✓        | ✓            |
| Years, no fixed zero | ✓         | ✓          | ✓        | ✗            |
| Pace ratio           | ✓         | ✓          | ✗        | ✗            |
| Divisions            | ✓         | ✓          | ✗        | ✗            |
| Log hazard           | ✓         | ✗          | ✗        | ✗            |
| Proportion           | ✓         | ✓          | ✗        | ✗            |

Figure 2: Six of the nine scales, and the four operations most often applied to them without checking



Reading down the acceleration column is the shortest statement of the problem. Only an age in years with a fixed origin admits it. A pace is already a rate, a division count is not elapsed time, and a log hazard is not even on an additive scale, so subtracting a chronological age from any of them produces a number with no units behind it. The matrix drawn here is the same one the code enforces, in `LEGAL_OPS`.

Every registry entry declares a `scale_type` alongside its unit, platform, species and the tissue its coefficients were fitted in. That field decides which downstream operations FALCONAge will perform:

| <b>scale_type</b>    | <b>n</b> | <b>What the number is</b>                                                                                                                                                                                     | <b>Units in the registry</b>                                     | <b>Operations permitted</b>                         |
|----------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------|
| age_years            | 77       | An age. The model predicts elapsed time and is reported on the same scale as chronological age.                                                                                                               | years, months, days                                              | acceleration, correlate, difference, mean, residual |
| relative_score       | 54       | A score with no external unit. Comparable within a cohort, meaningless as an absolute value.                                                                                                                  | unitless, picograms per milliliter, probability, and 12 more     | correlate, rank                                     |
| proportion           | 18       | A composition. The estimates are fractions of a whole and are constrained to sum to one.                                                                                                                      | proportion                                                       | compositional, correlate, difference, mean          |
| divisions            | 10       | A count of cell divisions, not elapsed time.                                                                                                                                                                  | beta value, proportion, cell divisions per stem cell, and 1 more | correlate, difference, mean, rank                   |
| gestational_weeks    | 8        | An age, before birth.                                                                                                                                                                                         | weeks, days, units per week                                      | acceleration, correlate, difference, mean, residual |
| telomere_kb          | 2        | An estimated physical length.                                                                                                                                                                                 | kilobases, base pairs                                            | acceleration, correlate, difference, mean           |
| pace_ratio           | 2        | A rate: how much biological ageing per year of calendar time, at the moment of sampling.                                                                                                                      | biological years per chronological year                          | correlate, difference, mean, rank                   |
| mortality_log_hazard | 2        | A risk on the log-hazard scale. Exponentiating a difference gives a hazard ratio.                                                                                                                             | log hazard, unitless                                             | correlate, hazard ratio, mean, rank                 |
| age_years_relative   | 2        | An age in years with a slope near one against chronological age, but no fixed origin: the offset moves between cohorts. The residual and group differences are defined; predicted minus chronological is not. | years                                                            | correlate, difference, mean, residual               |

Age acceleration is the residual of predicted age on chronological age. It is defined when the prediction is in years and undefined when it is a rate, so asking for it on DunedinPACE is an error with a message, not a number:

```
IllegalOperationError: 'acceleration' is not defined for dunedinpace, whose output is
pace_ratio (biological years per chronological year).
Legal here: correlate, difference, mean, rank.
A pace of aging is already a rate; subtracting chronological age from it is a units
error, not a conservative choice.
```

## Scope and limitations

What the tool is for, what it declines to do, and what it cannot do yet. Three different questions with three different answers, in the order a **model card** puts them.

### Intended use

The reason, rather than the disclaimer. Reliability varies sharply between clocks, and technical reproducibility and biological stability are separate properties that do not track together: a clock can be highly repeatable on split samples and still move with a meal or a night's sleep. So a score that survives replication is not thereby measuring a stable trait, and the two failures need different evidence to rule out. **The science of aging clocks** sets out, clock by clock, which is established and which is contested.

### Out-of-scope use

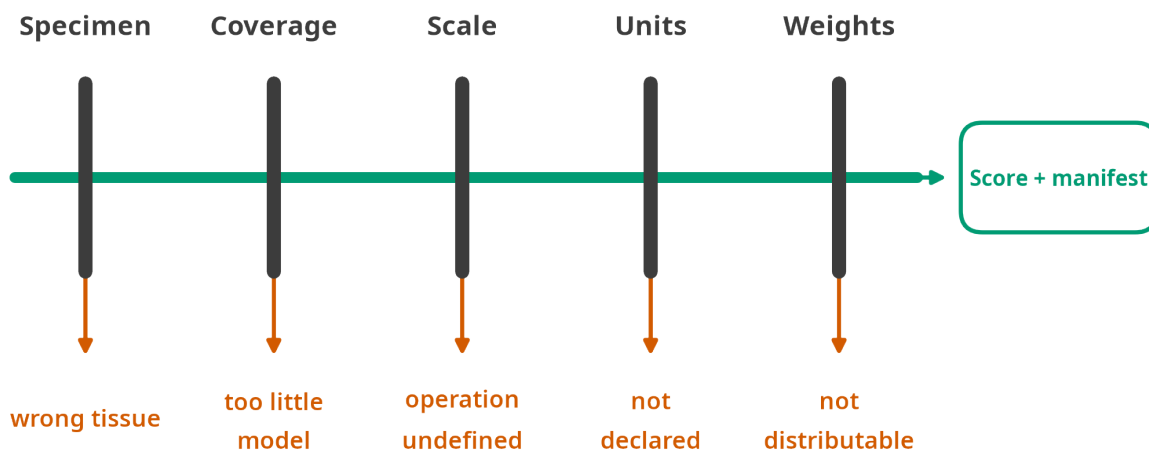


Figure 3: The gates a scoring call passes, and what stops it at each one

Each gate is a place where the wrong answer is available and cheap. A clock scored on the wrong specimen, on too little of its own model, on a scale that does not admit the operation asked for, on clinical values whose units were never stated, or from a weights file that cannot be redistributed, all return a number that looks exactly like a right one. What comes out of the last gate is a score and a manifest recording which gates it passed and on what evidence.

Each of these is a silent wrong answer somewhere else, so each is a hard failure here:

- **Fill a missing probe with zero.** In beta space zero means completely unmethylated, which is a real and extreme measurement. Absent features take the clock's published reference value, or the cohort column mean, and the substitution is counted in the manifest.
- **Guess clinical units.** PhenoAge has SI and conventional variants that disagree by years. `units=` has no default.
- **Score below the coverage floor.** A clock whose features are largely absent is skipped and named, rather than scored on mostly-imputed input.
- **Treat probes as interchangeable.** Coverage is measured twice: by count, and by the share of the model's total |coefficient| the present features carry. A clock can clear a 92% feature floor having lost the probes that do a third of the work, and only the second number sees it. `probe_loss()` reports both before you score anything.
- **Claim coverage means validity.** High coverage says the probes are present. It says nothing about whether coefficients fitted in one tissue, species or population transfer to yours; that judgement stays with you, and [Choosing a clock](#) lays out the grounds for it.
- **Let a paper-versus-implementation disagreement pass silently.** Eleven clocks in the catalogue are scored from coefficients that differ from what their paper describes: a feature count, a changed statistic, a constant. Each carries the note, and it is raised as a warning at score time, not left on a documentation page.

### Known limitations

Said here rather than left to be inferred. [The science page](#) §19 and §20 cover each in full, with the algorithms and the reason each is or is not blocked.

- **It does not repair cross-platform probe loss.** It measures it twice over: by count, by coefficient mass, and now in *years* per clock per platform. It cannot undo it. The leftover mapping tables encode concordance measured on paired samples and are not derivable from the array manifests.
- **It has no proteomic or transcriptomic clocks in the catalogue.** The readers and the full preparation chains ship; no entry declares either `data_type`, because every published model in both families is licence-restricted and would enter as a `licensed` scaffold. Scoring a proteomic matrix today refuses with “no clocks to score”, which is accurate rather than broken.
- **It does not correct dye bias by default.** The step exists and is opt-in. Measured on real IDATs our implementation moves the median beta by +0.10 to +0.12, because a correct correction needs the normalisation control probes and their addresses are not in the manifest we can fetch.
- **It has no foundation-model backend.** The `NeuralClock` architecture ships, safetensors only. CpGPT and MethylGPT are interesting here for zero-shot imputation and array conversion rather than as another age predictor.
- **147 of the 175 catalogued clocks have no traced coefficient source.** A per-clock literature hunt, not a technical problem, and `primary_source_traced: false` is the honest record of it.

Removed from this list, because they now exist: array normalisation (pOOBAH, noob, BMIQ, probe masks), the raw-IDAT path, single-cell scoring, and the transcriptomic per-dataset centring refusal.

## Further reading

- [Getting started](#), install, score, interpret, in both languages.
- [Choosing a clock](#), by the question you asked, by what the scale permits, and by whether the coefficients transfer to your data at all.
- [Clock catalogue](#), all 175, searchable.
- [Python reference](#) · [R reference](#)
- [GPU support](#), what was verified on real hardware, and why `device="auto"` is CPU.

The whole site is also downloadable as PDF or markdown; see the **Download** menu.

## Citation

APA:

```
Hunakunti, B. (2026). FALCONAge: Multiomic biological age and aging clock scoring in Python and R (Version 1.0.0) [Computer software]. https://github.com/bhagesh-h/FALCONAge
```

BibTeX:

```
@software{falconage2026,
  author = {Bhagesh Hunakunti},
  title = {FALCONAge: Multiomic Biological Age and Aging Clock Scoring in Python and R},
  year = {2026},
  version = {1.0.0},
  url = {https://github.com/bhagesh-h/FALCONAge},
  license = {GPL-3.0-or-later}
}
```

In R, `citation("FALCONAge")`. The machine-readable version is [CITATION.cff](#), which is what GitHub's *Cite this repository* button reads.

The clock FALCONAge computed for you is somebody else's work, and citing FALCONAge does not cite it. Every registry entry carries its primary reference, so the list for an analysis comes out of the analysis:

## Python

```
fa.registry.load().get("grimage2").cite("bibtex")
```

## R

```
cite_clock("grimage2", "bibtex")
```

## Chapter 2. Getting started

FALCONAge scores DNA methylation, clinical chemistry, proteomic and transcriptomic data against a catalogue of 161 published aging clocks. The numerical work happens in one Python core; the R package calls it through reticulate, so an R result and a Python result are the same bits rather than two implementations that agree approximately.

### 💡 How to read this page

Blocks with a **copy button** are things to run. Blocks tagged **Output** are what you should see back. Reading them is the point; pasting one into a terminal only produces an error.

```
this is a command: copy it
```

```
this is what it printed
```

## Installation

**Docker is the supported path**, and it is the one the rest of this page assumes. One image carries Python, R, the CLI and all 32 bundled clocks at pinned versions, so the same input gives the same numbers on your machine and on a reviewer's. Nothing else to install.

```
docker pull bhagesh/falconage:1.0.0-cpu
```

Or build the same image from a clone, which takes about ten minutes:

```
git clone https://github.com/bhagesh-h/FALCONAge.git
cd FALCONAge
docker build -f docker/Dockerfile.cpu -t bhagesh/falconage:1.0.0-cpu .
```

Then either open a session in the image:

```
docker run --rm -it -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu python
docker run --rm -it -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu R
```

Or run a single command without opening one:

```
docker run --rm -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu \
  report betas.csv --outdir results/
```

The image's entrypoint is the `falconage` CLI, so the first argument is a verb: `report`, not `falconage report`. The exceptions are `python`, `R`, `Rscript`, `pytest` and `bash`, which are run as given.

If you have never used a terminal, start at the step-by-step walkthrough below instead; it assumes nothing and covers fetching real public data.

### i Which shell you are typing into

"\$PWD" means "the folder I am in now", and the spelling differs:

| Where you are typing                 | Use       |
|--------------------------------------|-----------|
| macOS Terminal, Linux, Git Bash, WSL | "\$PWD"   |
| Windows PowerShell                   | "\${PWD}" |
| Windows cmd.exe                      | "%cd%"    |

Everything else is identical. A path error on Windows is almost always this.

### 💡 Native install, if you manage your own environment

```
pip install "falconage @ git+https://github.com/bhagesh-h/FALCONAge.git#subdirectory=python"
```

```
import falconage as fa
fa.config()
```

```
remotes::install_github("bhagesh-h/FALCONAge", subdir = "r")

library(FALCONAge)
falconage_install() # builds the Python environment R calls into; a few minutes
falconage_config() # what resolved: versions, devices, registry size
```

The R package delegates every number to the Python core, so it needs a Python environment either way. On reticulate  $\geq 1.41$  an ephemeral one is built on first use and `falconage_install()` is optional; use it when you want a pinned, reproducible one.

Every code example on this page runs unchanged in a native install; only the `docker run` prefix goes away.

`falconage_config()` is the first thing to run when anything looks wrong. It reports the interpreter, the registry version, and how many clocks are usable:

```
FALCONAge 1.0.0
registry    1.1.0 (161 clocks)
tiers      A 23 ship - B 110 need a traced source - C 28 scaffold only
devices     cpu
```

## Complete walkthrough, from an empty machine

Everything above, in order, assuming no Python, no R, and no prior experience with either. Each command can be copied one at a time and each says what it does and what you should see. It ends with a scored public dataset and the figures.

### Step 0: install Docker

Install **Docker Desktop** (Windows, macOS) or Docker Engine (Linux), start it, then run:

```
docker --version
```

You should get a version string. If instead you get “command not found” or “cannot connect to the Docker daemon”, Docker is not installed or not running; fix that before going further, because nothing below will work.

If a later command fails with a path error: it is almost certainly the “\$PWD” spelling for your shell; see the table above.

### Step 1: clone the repository

```
git clone https://github.com/bhagesh-h/FALCONAge.git
cd FALCONAge
```

Every later command is run from inside this **FALCONAge** folder.

### Step 2: obtain the image

One image carries Python, R, the CLI and all 32 bundled clocks. Pull the published one:

```
docker pull bhagesh/falconage:1.0.0-cpu
```

Or build the same thing from the clone, which takes a few minutes the first time and is then cached:

```
docker build -f docker/Dockerfile.cpu -t bhagesh/falconage:1.0.0-cpu .
```

Check it either way:

```
docker run --rm bhagesh/falconage:1.0.0-cpu python -c "import falconage;
print(falconage.__version__)"
```

### Step 3: inspect the corpus before downloading it

A second, deliberately tiny image fetches the test corpus. It is separate on purpose: fetching a fixture is the first thing you do when the package itself is broken, so it must not need the package.

```
docker build -f docker/Dockerfile.testdata -t falconage-testdata:1.0.0 .
```

Now the dry run. **This transfers nothing**. It prints every group, file count and byte total so you can decide before committing to half a gigabyte:



```
docker run --rm -v "$PWD/test/data:/data" falconage-testdata:1.0.0 plan
```

You should see:

| group       | files | size     | what it is for                                |
|-------------|-------|----------|-----------------------------------------------|
| bench       | 11    | 368.4 MB | ComputAgeBench case/control methylation       |
| clinical    | 3     | 5.4 MB   | NHANES blood chemistry with mortality linkage |
| gestational | 1     | 50.8 MB  | Umbilical cord blood with gestational age     |
| epicv2      | 1     | 41.2 MB  | EPIC v2 probe-suffix handling                 |
| idat        | 8     | 60.4 MB  | Raw IDATs, EPIC v1 and EPIC v2                |
| mammalian   | 4     | 9.3 MB   | Mammalian methylation array, three species    |
| mouse       | 5     | 50.2 MB  | Mouse blood RRBS across the lifespan          |
| selected    | 33    | 585.7 MB |                                               |

#### Step 4: download and verify

```
docker run --rm -v "$PWD/test/data:/data" falconage-testdata:1.0.0 python
docker run --rm -v "$PWD/test/data:/data" falconage-testdata:1.0.0 verify
```

The first fetches 33 files from GEO and Hugging Face into `test/data/` on your own disk, never inside the image, so rebuilding never re-downloads. The second re-computes every SHA-256 against `checksums.sha256` and fails loudly on a single wrong byte.

Only smaller pieces, if 586 MB is too much:

```
docker run --rm -v "$PWD/test/data:/data" falconage-testdata:1.0.0 \
python test/data/fetch_test_data.py --groups clinical --out /data
```

#### 💡 Fetching the corpus from R

Not decoration: the R fetcher writes a byte-identical `checksums.sha256`, so the two languages are shown to agree on the input before any clock is scored.

```
docker run --rm -v "$PWD/test/data:/data" falconage-testdata:1.0.0 R
```

#### Step 5: score it

##### Python, CPU

```
docker run --rm -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu \
python test/run_all.py
```

Every clock against every dataset. It writes one folder per input under `test/output/`, every figure under `test/output_figures/`, and rewrites the generated tables in `test/README.md`, so the numbers in that document are the numbers the code just produced and cannot drift from them.

To check the install without the corpus, run the unit suite instead, seconds, no data needed:

```
docker run --rm -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu \
  python -m pytest python/tests -q
```

## R

The same image. R calls the same Python core through reticulate, so these are the same numbers, not a second implementation that agrees approximately.

```
docker run --rm -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu \
  Rscript -e 'testthat::test_local("r")'
```

That suite includes the R-versus-Python conformance tests, which assert equality at tolerance exactly zero.

Scoring from R interactively:

```
docker run --rm -it -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu R
```

```
library(FALCONAge)
falconage_config()
d <- read_betas("test/data/bench/GSE40279.csv")
res <- score(prepare(d), clocks = "compatible")
summary(res)
```

## Python, GPU

Only worth it if you have an NVIDIA card. **Read GPU support first:** for the clocks that ship today the GPU is *slower* than the CPU, which is why `device="auto"` resolves to CPU even inside this image.

Pull the CUDA image. It is 14.6 GB against the CPU image's 2.5 GB, because it carries torch built against CUDA 12.4:

```
docker pull bhagesh/falconage:1.0.0-cuda
```

Or build it:

```
docker build -f docker/Dockerfile.cuda -t bhagesh/falconage:1.0.0-cuda .
```

Check the card is visible inside a container before anything else:

```
docker run --rm --gpus all bhagesh/falconage:1.0.0-cuda nvidia-smi
```

No output or an error here means GPU passthrough is not set up, and every command below will fail. On Linux that is usually a missing `nvidia-container-toolkit`.

```
docker run --rm --gpus all -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cuda \
python test/gpu_check.py
```

This reproduces every number on the GPU page, device resolution, CPU-versus-GPU agreement, the speed table and the profile that explains it. On a machine with no CUDA device it stops after the first step with an explanation rather than an error, so it is safe to run anywhere.

Scoring on the GPU needs `--gpus all` **and** `--device cuda`. Without the flag, `--device cuda` raises rather than falling back, so a run that asked for a GPU cannot quietly become a very slow success:

```
docker run --rm --gpus all -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cuda \
score --input test/data/prepared.h5ad --device cuda --outdir results
```

### Step 6: score your own data

Put a beta matrix or a clinical table in a folder, mount it read-only, and mount somewhere for the results to land:

```
docker run --rm \
-v "$PWD/mydata:/data:ro" \
-v "$PWD/results:/results" \
-v falconage-cache:/opt/falconage/cache \
bhagesh/falconage:1.0.0-cpu \
score --input /data/betas.csv --clocks compatible --outdir /results
```

The named `falconage-cache` volume is worth adding from the start. Nothing in tier A needs it, but the PC clocks are several hundred megabytes each and re-fetching them on every container start is the difference between a two-minute run and a twenty-minute one.

## Scoring

Every Python and R snippet from here on runs unchanged in three places: inside the Docker image (`docker run --rm -it -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cpu python`, or ... R), in a native install, or in the CLI-free path below. Only the way you open the session differs.

### Python

```
d = fa.prepare(fa.read_betas("betas.csv"))
res = fa.score(d, clocks="compatible")

res.summary()
res.scores          # samples x clocks
res.long()          # one row per sample per clock, with the scale
res.coverage        # per-clock feature coverage and skip reasons
res.manifest        # versions, device, SHA-256 of every coefficient file
```

**R**

```
d <- prepare(read_betas("betas.csv"))
res <- score(d, clocks = "compatible")

summary(res)
as.data.frame(res)           # samples x clocks
as.data.frame(res, form = "long") # one row per sample per clock, with the scale
coverage(res)                # per-clock feature coverage and skip reasons
manifest(res)                 # versions, device, and every coefficient checksum
```

`clocks = "compatible"` scores what this dataset can support and reports the rest as skipped with a reason. Naming clocks explicitly is different: then every one must work, because an explicit request should never be dropped quietly.

**Raw IDATs**

New in this release. Before this you had to normalise in `sesame` or `minfi` and bring a beta matrix back.

**Python**

```
d = fa.read_idat_dir("idats/")           # every Grn/Red pair in the folder
res = fa.score(fa.prepare(d), clocks="compatible")
```

**R**

```
# The raw-array chain is Python-only for now; call it through reticulate.
fa <- reticulate::import("falconage")
d <- fa$read_idat_dir("idats/")
```

The chain runs detection first (pOOBAH), then background correction (noob), because pOOBAH's null is the uncorrected out-of-band signal. Undetected probes become `NA` rather than a number. The array manifest is downloaded from Illumina's public bucket the first time and cached. It is the one step in FALCONAge that needs a network.

```
FalconData(2 samples x 865918 features, dna_methylation, EPICv1, 2.0% missing)
```

**Measurement uncertainty**

Also new in this release, and the reason the rest of this page is worth running. Technical replicates of the *same* DNA differ by up to nine years on prominent clocks, so a bare number does not say whether a difference between two samples is real.

**Python**

```
se = fa.technical_se(res, d)
se.diagnostics[["median_se", "se_over_sd", "implied_cohort_icc"]]
```

```
fa.conformal_interval(res, level=0.90)    # the other, larger uncertainty
fa.power("horvath2013", effect=0.5, result=res)  # how many samples you need
```

## R

```
u <- technical_se(res, d)
u$diagnostics

conformal_interval(res, level = 0.90)
power_n("horvath2013", effect = 0.5, result = res)
```

|               | method | n_features | median_se | se_over_sd | implied_cohort_icc |
|---------------|--------|------------|-----------|------------|--------------------|
| horvath2013   | probe  | 353        | 1.58      | 0.14       | 0.98               |
| hannum        | probe  | 71         | 2.46      | 0.22       | 0.95               |
| dunedinpoam38 | probe  | 46         | 0.04      | 0.53       | 0.72               |

Two samples whose intervals overlap are not distinguishable by that clock on that assay, however different their point estimates look.

## Platform coverage

Worth running once on an array you have not used before. It reports, per clock, how many features are present **and** how much of the model's weight they carry. Two different numbers, and the second is the one that catches EPIC v2 probe loss:

## Python

```
fa.probe_loss(d, clocks="scoreable")
```

## R

```
probe_loss(d, clocks = "scoreable")
```

|                         | tier | n_features | n_present | coverage | mass_coverage | bias_years | heaviest_absent |
|-------------------------|------|------------|-----------|----------|---------------|------------|-----------------|
| horvath2013             | A    | 353        | 340       | 0.9632   | 0.6117        | NaN        |                 |
| cg16867657 (14.1%), ... |      |            |           |          |               |            |                 |
| hannum                  | A    | 71         | 64        | 0.9014   | 0.8830        | -7.69      |                 |
| cg22454769 (3.1%), ...  |      |            |           |          |               |            |                 |

`bias_years` is filled in when the platform is one the shift has been measured on; see [what probe loss costs](#). NaN there means *not measured*, never *no bias*.

A clock at 96% feature coverage and 61% weight coverage has kept nearly all its probes and lost the ones that matter. The scoring floor applies to both measures, so this run would be refused with a message naming the heaviest absent probes rather than returning a confident number.

## Interpreting the result

The scale, the unit, which arithmetic is permitted, both coverage measures, the published reliability where one exists, and any documented disagreement between the clock's paper and the coefficients in circulation, all on the object rather than only on this website:

### Python

```
res.interpretation()
```

### R

```
interpretation(res)
```

`technical_icc` and `biological_icc` are separate columns because they measure different things and do not track together: the same sample assayed twice, versus the same person sampled again a few days later. A clock can be excellent on the first and poor on the second, and the clocks most used to claim an intervention worked are where that gap is widest.

## Age acceleration

The three conventions disagree by several years on the same data, and papers frequently do not say which they used.

### Python

```
fa.acceleration(res, method="absolute")      # predicted - chronological
fa.acceleration(res, method="residual")      # residual of predicted on chronological
fa.acceleration(res, method="both")          # both, as <clock>_absolute / <clock>_residual
fa.acceleration(res, method="within_group", group="dataset")
```

### R

```
acceleration(res, method = "absolute")
acceleration(res, method = "residual")
acceleration(res, method = "both")
acceleration(res, method = "within_group", group = "dataset")
```

Absolute is interpretable in years and carries the clock's own bias: a clock five years high on everybody gives everybody five years of acceleration. Residual removes that bias, and removes any real cohort-wide effect with it. Within-group is what the AA2 benchmark needs: it asks whether cases accelerate relative to *their own* controls. "both" returns the first two side by side, suffixed, for when you would otherwise have to pick without knowing which the comparison paper used.

### Adjusting for cell composition

Blood cell composition changes with age, and with most other things that happen to a person. In more than 10,000 blood samples, immune composition was significantly associated with age acceleration for **every one** of six widely used clocks, so an unadjusted acceleration is measuring two things and reporting one number.

The proportions needed to separate them are usually already in the result, computed by the deconvolution clocks in the same run:

### Python

```
fa.cell_composition(res)                # the proportions, as covariates
fa.acceleration(res, adjust="cell_composition") # net of them
fa.acceleration(res, adjust=["cd8t", "mono"])  # or measured counts from obs
```

### R

```
cell_composition(res)
acceleration(res, adjust = "cell_composition")
acceleration(res, adjust = c("cd8t", "mono"))
```

An adjusted acceleration is a different quantity from an unadjusted one, so the returned frame records which covariates went in. Comparing the two as though they were the same measurement is the error this is designed to make visible.

Ask for acceleration on a pace-of-aging clock and it refuses:

```
Error: 'acceleration' is not defined for dunedinpoam38, whose output is
  pace_ratio (biological years per chronological year).
  A pace of aging is already a rate; subtracting chronological age from it is
  a units error, not a conservative choice.
```

That refusal is the point. A wrong number in a column of numbers is invisible.

## The clock catalogue

### Python

```
reg = fa.registry.load()
reg.filter(availability="A")
reg.search("mortality")
reg.get("horvath2013")
reg.get("dnamphenoage").cite("bibtex")
```

### R

```
list_clocks(tier = "A")
list_clocks(search = "mortality")
clock_info("horvath2013")
cite_clock("dnamphenoage", "bibtex")
```

Twenty-eight clocks are scaffolds: the model, the feature list, the transform chain and the expected shapes all ship, but the coefficients are research-use-only and are not ours to distribute. Asking about one says why, where to obtain a file, and which open clocks answer the same question.

## Python

```
fa.registry.register_local_weights("grimage2", "~/licensed/grimage2_coefs.csv")
fa.score(d, clocks=["grimage2"])
```

## R

```
register_local_weights("grimage2", "~/licensed/grimage2_coefs.csv")
score(d, clocks = "grimage2")
```

## Clinical chemistry

```
d <- read_clinical("nhanes.csv", units = list(
  albumin = "g/L", creatinine = "umol/L", glucose = "mmol/L", crp = "mg/dL",
  lymphocyte_percent = "%", mean_cell_volume = "fL",
  red_cell_distribution_width = "%", alkaline_phosphatase = "U/L",
  white_blood_cell_count = "10^3/uL", age = "years"))

score(d, clocks = "phenoage")
```

Units are required and never guessed. Albumin at 4.2 is g/dL and albumin at 42 is g/L; both are clinically normal, and PhenoAge fitted on one returns nonsense for the other. No range check can separate them, so the only correct behaviour is to ask.

KDM and homeostatic dysregulation have no fixed coefficients at all: they are defined relative to a reference cohort:

```
ref <- fit_kdm(nhanes3, markers = markers)
score(d, clocks = "kdm", reference = ref)
```

## Survival and association

Two figures that most aging-clock papers carry, and that need the analysis functions above.

### Survival

#### Python

```
fa.cox_hazard(res, time_col="time", event_col="status")      # hazard ratio per SD
fa.plot.kaplan_meier(res, "horvath2013",
                      time_col="time", event_col="status")  # the figure
```

#### R

```
cox_hazard(res, time_col = "time", event_col = "status")
plot_kaplan_meier(res, "horvath2013", time_col = "time", event_col = "status")
```



The curves compare the fastest-ageing tenth with the slowest, not two halves. The middle of the acceleration distribution is where a clock discriminates least, so a median split dilutes the comparison; the tails are the published convention. The log-rank p is in the subtitle.

If the curves cross: the hazards are not proportional and the single hazard ratio beside them is an average over a period during which the ordering changed. That is worth saying in a caption rather than leaving for a reader to notice.

## Association

### Python

```
assoc = fa.associate(res, "crp", covariates=("age", "sex"))
fa.plot.volcano(assoc, fdr=0.05)
```

### R

```
assoc <- associate(res, "crp", covariates = c("age", "sex"))
plot_volcano(assoc, fdr = 0.05)
```

The dashed line is the Benjamini-Hochberg threshold at the stated FDR, read off the `q` column rather than recomputed. This matters more than it looks: across a few hundred tests a raw 0.05 admits roughly one in twenty of the nulls, and drawing that line instead labels them significant.

## Benchmarking

```
res <- combine(list(r1, r2, r3), keys = c("GSE1", "GSE2", "GSE3"))
b <- run_benchmark(res, condition_col = "condition", control = "HC",
                  dataset_col = "dataset")
b$summary
```

Median absolute error against chronological age is reported but never ranked on: a perfect chronological oracle would score best on it and be useless, because it would have no age acceleration left to detect anything with.

## Reporting

```
write_results(res, "results/")
report(res, "report.html", group = "condition")
```

The HTML report is one self-contained file, inlined CSS, base64 figures, the table embedded, so it still works after somebody emails it, which is the only thing anybody does with a report.

## The Claude skill

The repository ships a **Claude** skill at `.claude/skills/falconage/`. It is four markdown files: the workflow above, the Docker command set, the clock catalogue routed by the question each clock answers, and every refusal with the measurement behind it.

It is active already for anyone working inside a clone. To have it available in any directory, copy it into your personal skills folder with `cp -r .claude/skills/falconage ~/.claude/skills/`, or on Windows PowerShell `Copy-Item -Recurse .claude\skills\falconage $HOME\.claude\skills\`.

Then ask in plain language. “Score these IDATs and tell me which clocks you refused and why” is enough; the skill supplies the rest.

#### **i** Why the skill is checked by CI, and checked harder than the prose

`docs/check_api_docs.py` resolves every `fa.*` name and every `falconage <verb>` in the skill against the running package on each push, exactly as it does for these pages.

The reason it matters more here is that these pages are read by a person, who notices when a function does not exist and goes looking. A skill is read by a model that then *runs* what it says. The first draft named five readers at top level that actually live under `fa.preprocess`, and invented a `falconage probe` verb outright, copied from a specification section of the architecture page that described a verb never built. The check caught all six, and then caught two more invented verbs on that page.

## Chapter 3. Choosing a clock

There is no best clock. There is a clock that answers the question you asked, and a much larger number that answer a different one in the same units.

|                |                              |                                      |
|----------------|------------------------------|--------------------------------------|
| <b>First</b>   | Chronological age            | → An age; the residual is the signal |
| <b>Second</b>  | Mortality-weighted composite | → A hazard in year-like units        |
| <b>Third</b>   | Longitudinal rate            | → A ratio: years per year            |
| <b>Causal</b>  | Damaging vs adaptive         | → Two scores, no shared origin       |
| <b>Mitotic</b> | Division count               | → Divisions, not time                |

Figure 4: Five generations of clock, and what the output of each one actually is

The generations are not successive improvements on one measurement. Each was trained against a different target, so each returns a different quantity: a predicted age whose signal is the residual, a mortality composite rescaled into year-like units, a rate, a pair of scores with no shared origin, and a division count. A later generation is not a better answer to the earlier question, and the first generation is still the right choice when the question really is how old a sample looks.

## Choosing by research question

| Question                                       | Ready to score | Needs a coefficient file                                                                                                                                 | Scale                                           |
|------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| How old does this sample look?                 | —              | altumage, corticalclock, hannum, horvath2013, lin, meer, pedbe, skinandblood, and 41 more                                                                | age_years, relative_score                       |
| How old is this newborn, gestationally?        | —              | knight, leecontrol, leerefinerobust, leerobust, bohlin, epicga, mayne                                                                                    | gestational_weeks                               |
| How fast is this person aging?                 | —              | dunedinpoam38, dunedinpace, ensemblea gestatic, ensemblea gestatictop                                                                                    | age_years, pace_ratio                           |
| Which organ system is aging fastest?           | —              | systemsage, systemsageblood, systemsagebrain, systemsageheart, systemsagehormone, systemsageimmune, systemsageinflammation, systemsagekidney, and 5 more | age_years                                       |
| Who is at risk of dying sooner, or is frailer? | —              | dnamphenophage, hd, hrsinchphenophage, kdm, phenophage, weidner, zhangmortality, cpgptgrimage3, and 13 more                                              | age_years, mortality_log_hazard, relative_score |
| Is damage separable from adaptation?           | —              | yingadaptage, yingdamage                                                                                                                                 | age_years_relative                              |
| How much has this tissue divided?              | —              | epicmithyper, epicmithypo, epitoc1, epitoc2, epitoc3, hypoclock, replitali, stemtoc, and 4 more                                                          | divisions, relative_score                       |

| Question                                              | Ready to score | Needs a coefficient file                                                                                                                                                                                                                                                                                             | Scale                                        |
|-------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| What is the blood's cell composition?                 | —              | deconvolutebloodepi<br>cbcell, deconvolu<br>tebloodepiccd4tcell<br>, deconvolutebloodepi<br>ccd8tcell, deconvolu<br>tebloodepicmonocyte<br>, deconvolutebloodepi<br>cneutrophil, deconvolu<br>tebloodepicnkcell,<br>twelvecelldeconvolu<br>tebloodepicbas,<br>twelvecelldeconvolu<br>tebloodepicbmem, and 10<br>more | proportion                                   |
| How long are the telomeres?                           | —              | dnamt1, pcdnamt1                                                                                                                                                                                                                                                                                                     | telomere_<br>kb                              |
| What is this person exposed to, or how do they live?  | —              | mccartneyalcohol,<br>mccartneybmi, mccartney<br>bodyfat, mccartney<br>education, mccartney<br>hdlcholesterol,<br>mccartneyldlcholest<br>erol, mccartneysmoking,<br>mccartneytotalchole<br>sterol, and 11 more                                                                                                        | gestation<br>al_weeks,<br>relative_<br>score |
| What is a specific protein or lab value likely to be? | —              | grimage2adm, grimage2b<br>2m, grimage2cystatinc,<br>grimage2gdf15, grimage2l<br>eptin, grimage2logalc,<br>grimage2logcrp,<br>grimage2pail, and 12 more                                                                                                                                                               | relative_<br>score                           |
| Is a named disease more likely?                       | —              | adbahadosingh,<br>cvdwesterman, depressio<br>nbarbu, hepatoxu,<br>prostatecancerkirby                                                                                                                                                                                                                                | relative_<br>score                           |
| What is the sample's chromosomal sex, or its species? | —              | mammalianfemale, xchrom,<br>ychrom                                                                                                                                                                                                                                                                                   | relative_<br>score                           |

Every one of the 175 catalogued clocks appears above, routed by its declared `predicts` field. The full table with paper links is the [clock catalogue](#).

“Ready to score” means the coefficients ship in the package. The rest are catalogued and need a file; see Availability below.

### Python

```
reg = fa.registry.load()
reg.filter(generation="second", availability="A")
reg.compatible_with(d)
```

### R

```
list_clocks(generation = "second", tier = "A")
compatible_clocks(d)
```

## Scale type and permitted operations

Every clock carries a `scale_type`, and it decides which downstream operations are defined.

| Scale                             | Clocks | Permitted                                           | Refused, and why                                                                                                                                    |
|-----------------------------------|--------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>age_years</code>            | 77     | acceleration, correlate, difference, mean, residual | —                                                                                                                                                   |
| <code>relative_score</code>       | 54     | correlate, rank                                     | everything except correlation and rank: there is no external unit to difference or average                                                          |
| <code>proportion</code>           | 18     | compositional, correlate, difference, mean          | anything ignoring the sum-to-one constraint                                                                                                         |
| <code>divisions</code>            | 10     | correlate, difference, mean, rank                   | acceleration: a division count is not elapsed time                                                                                                  |
| <code>gestational_weeks</code>    | 8      | acceleration, correlate, difference, mean, residual | mixing with an age in years                                                                                                                         |
| <code>telomere_kb</code>          | 2      | acceleration, correlate, difference, mean           | comparison with an age in years; and higher is the <i>younger</i> direction here, unlike every age clock beside it                                  |
| <code>pace_ratio</code>           | 2      | correlate, difference, mean, rank                   | acceleration: it is already a rate                                                                                                                  |
| <code>mortality_log_hazard</code> | 2      | correlate, hazard ratio, mean, rank                 | acceleration: a log-hazard has no zero on the age scale                                                                                             |
| <code>age_years_relative</code>   | 2      | correlate, difference, mean, residual               | acceleration: the clock tracks age with a slope near one but its origin moves between cohorts, so predicted minus chronological measures the cohort |

`acceleration()` enforces this. Naming a clock whose scale does not support it is an error; leaving the clock list empty quietly excludes them.

## Coverage and validity

A clock trained on 450K runs perfectly well on EPIC data that carries its probes, and fails on 450K data filtered to 20,000 probes. Only the feature list can answer whether a clock is usable, which is what `compatible_clocks()` checks.

But high coverage does not make a result meaningful. The mammalian methylation array was designed around CpGs conserved across mammals, and carries 96% of Horvath2013's 353 probes, so a zebra scores at *higher* coverage than many human 450K datasets and returns a confident number from a clock fitted on people. Nothing in the arithmetic can notice. Set `species` and FALCONAge warns:

```
d <- falcon_data(betas, modality = "dna_methylation", species = "Equus grevyi")
res <- score(d, clocks = "horvath2013")
manifest(res)$warnings
#> trained on Homo sapiens, scored on Equus grevyi. Feature coverage says nothing
#> about whether the coefficients transfer across species.
```

## Feature count and coefficient mass

Coverage has two forms and they disagree exactly when it matters: how many of a clock's features are present, and what share of the model's total |coefficient| those present features carry. Check both before committing to a clock on a given array.

```
fa.probe_loss(d, clocks="scoreable")
#> horvath2013 coverage 0.9632 mass_coverage 0.6117 cg16867657 (14.1%), ...
```

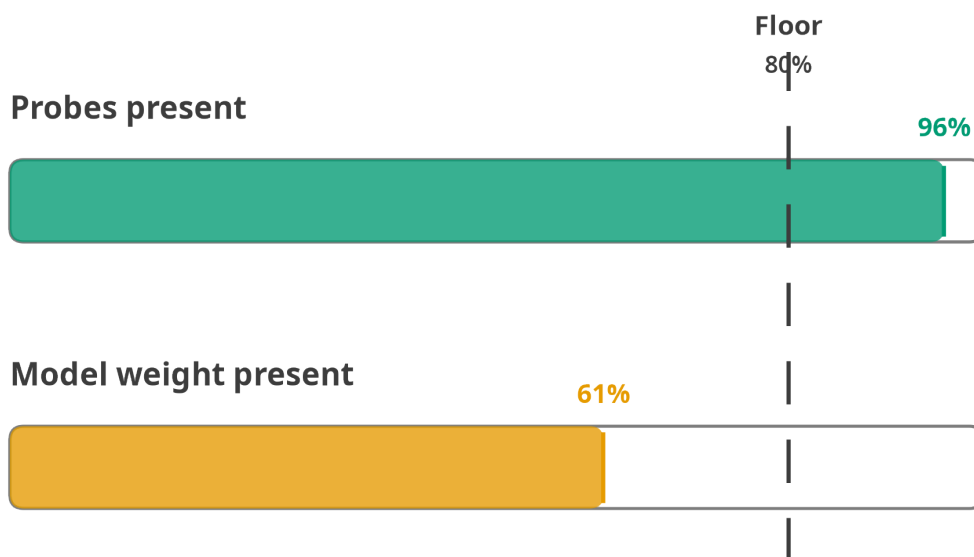


Figure 5: Horvath 2013 on one EPIC v2 array: probes present against model weight present

Ninety-six per cent of probes present, sixty-one per cent of the model. The scoring floor applies to both, so this is refused with the heaviest absent probes named rather than scored into a confident number. The gap opens because elastic-net coefficients are far from uniform: a handful of probes carry a disproportionate

share of the total weight, and losing four per cent of the probes loses thirty-five per cent of the model when the missing four per cent are those.

→ [The science](#), §19.5 prices the same loss in years, per clock per platform, and explains why the PC clocks barely move where first-generation clocks shift by double digits.

## Reliability

“Reliable” means two different things, and they do not track together:

|                       | What is repeated                          | What it measures                                                 |
|-----------------------|-------------------------------------------|------------------------------------------------------------------|
| <b>Technical ICC</b>  | the same DNA, assayed twice               | the assay, and the model’s sensitivity to array noise            |
| <b>Biological ICC</b> | the same person, sampled again days later | whether the number is a property of the person or of the morning |

Nearly every clock is technically excellent; biological ICCs are substantially lower. **GrimAge2 and DunedinPACE are among the most biologically fragile**, and they are the two most often used to claim an intervention worked. `res.interpretation()` carries both columns, with `None` meaning “not established” rather than “fine”.

**The rule:** measuring change within a person, prefer a clock with a published biological ICC, or a principal-component version. → [The science](#), §17.7.

### Measure it on your own cohort

A published ICC is somebody else’s laboratory. Measure yours:

```
se = fa.technical_se(res, data)
se.diagnostics[["median_se", "se_over_sd", "implied_cohort_icc"]]
```

#### `se_over_sd`

measurement error as a fraction of the spread between your own samples. Above 1, the clock cannot distinguish two of your samples however different their point estimates look.

#### `implied_cohort_icc`

the reliability *this* cohort implies. Often below the published figure, because a narrow age range leaves less between-sample variance for the same assay noise.

For scale: on this package’s test corpus Horvath 2013 comes out at  $\pm 1.58$  years, DunedinPoAm38 is least repeatable at 0.72, and the 319,607-probe BLUP clock reaches 1.00. With technical duplicates, `fa.icc_from_replicates(data, "subject")` replaces the published per-probe table with your own plates. → [The science](#), §19.2 for the propagation itself.

### Then ask how many samples you need

```
fa.power("horvath2013", effect=0.5, result=pilot)
```

At ICC 0.9 a tenth of the sample size exists only to average out the instrument, and the function says so. It refuses to default the standard deviation, because `n` scales with its square.



## Specimen type

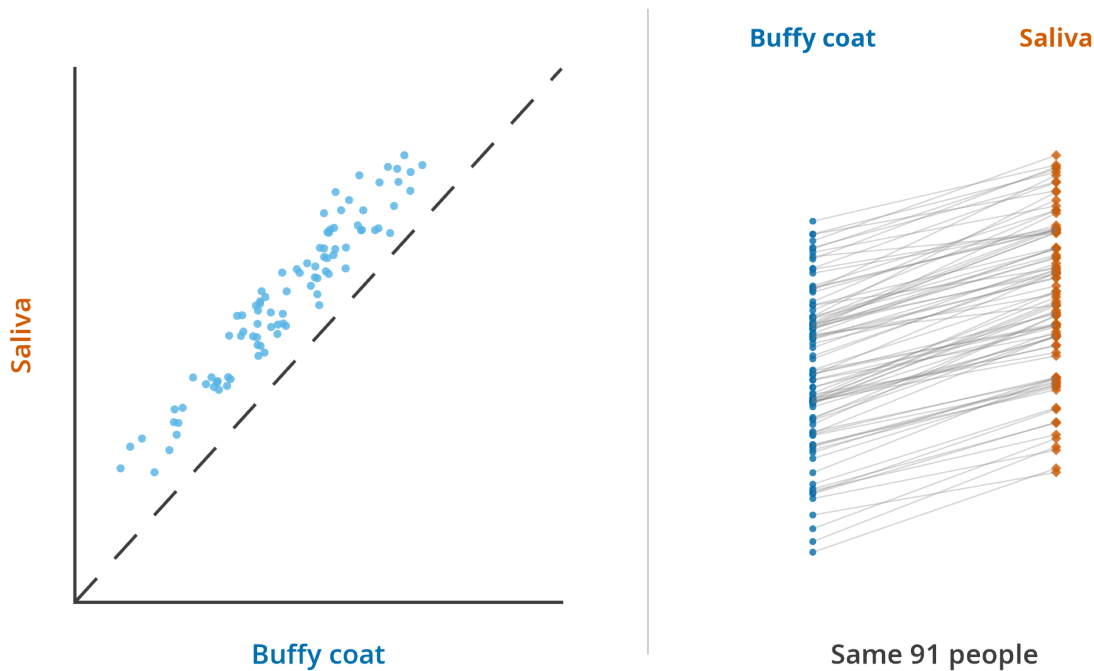


Figure 6: The same 91 donors, scored from saliva and from buffy coat

Saliva clock ages ran **3.83–16.46 years above buffy coat in the same 91 people**, while still correlating with them at Spearman 0.45–0.69. A correlation check passes and the output is a decade out, because correlation is not agreement. The left panel is the whole argument: every point sits above the line of identity, which is a bias, and the scatter around it stays tight, which is what a correlation coefficient reports. The paired lines on the right show the same donors shifting in the same direction, so no amount of averaging removes it.

**The rule:** set `obs["tissue"]` on every dataset. FALCONAge then compares it against what each clock was fitted on. Pan-tissue clocks say nothing, most warn with the measured discordance, and twelve refuse outright. Cell-free DNA is refused by every clock. No `tissue` column gets one warning saying the check did not run.

→ [The science](#), §19.4 for the twelve refusals and the measurements behind each. Reliability, technical against biological, is [39].

## Availability

| Availability          | n  | What you do                                                                            |
|-----------------------|----|----------------------------------------------------------------------------------------|
| <code>bundled</code>  | 46 | Nothing. Coefficients ship inside the package, or the clock is a formula with none.    |
| <code>untraced</code> | 89 | Catalogued, but no primary source has been established, so no coefficients ship.       |
| <code>licensed</code> | 40 | Obtain a coefficient file and register it. The architecture is implemented and tested. |

```
list_clocks(tier = "C")      # the ones needing author permission  
list_clocks(untraced = TRUE)
```

The distinction between B and C matters. C is a licensing decision by the authors. B is an honesty one by us: copying a coefficient set out of another package without knowing where that package got it is how the known paper-versus-implementation discrepancies spread in the first place, and a traced extractor is a welcome contribution.

## Chapter 4. Clock catalogue

This page is written from the registry the package ships, so it cannot describe a catalogue different from the one that scores. Every column below is a field the scoring code reads.

`scale_type` is the load-bearing one. It is not a label: it decides which downstream operations FALCONAge will perform, and asking for age acceleration on a `pace_ratio` clock is refused rather than computed.

**175 clocks**, registry schema 1.1.0.

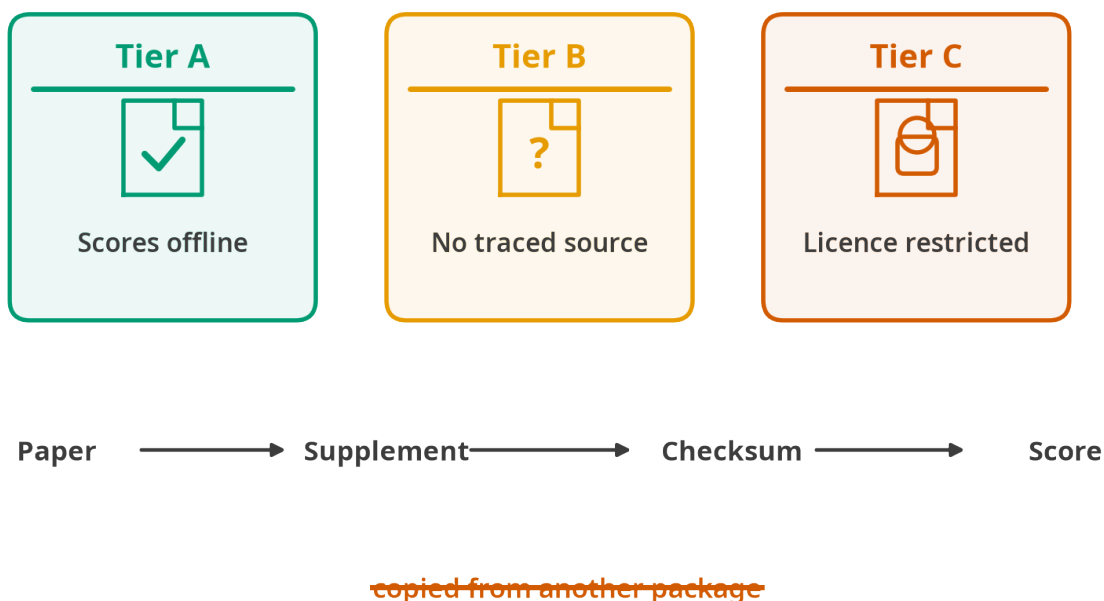


Figure 7: What separates the three availability groups, and what a traced coefficient source means

Availability is about provenance, not about quality. A `bundled` clock has a coefficient file whose origin was followed back to the paper or its supplement and checksummed, so the same numbers come out on any machine with no network. An `untraced` clock has no public source for its coefficients, and a `licensed` one has a source whose terms do not permit redistribution. Copying a coefficient vector out of another package would move clocks into the bundled column and is the one route not taken here, because a number with no source behind it cannot be checked against anything.

## Bundled: 46 clocks

Coefficients ship with the package, or the clock is a formula with none to ship. Works offline, no network, no licence.

| Clock           | Year | Predicts                    | Scale          | Features | Tissue                                                              | Platform                   | Paper                                |
|-----------------|------|-----------------------------|----------------|----------|---------------------------------------------------------------------|----------------------------|--------------------------------------|
| altumage        | 2022 | chronological age           | age_years      | 20318    | multi-tissue                                                        | 27K, 450K                  | <a href="#">de Lima Camillo 2022</a> |
| corticalclock   | 2020 | chronological age           | age_years      | 347      | brain cortex                                                        | 450K                       | <a href="#">Shireby 2020</a>         |
| dnamphenoage    | 2018 | phenotypic age              | age_years      | 513      | whole blood                                                         | 27K, 450K, EPICv1          | <a href="#">Levine 2018</a>          |
| dnaml           | 2019 | leukocyte telomere length   | telomere_kb    | 140      | whole blood                                                         | 450K, EPICv1               | <a href="#">Lu 2019</a>              |
| dunedinpoam38   | 2020 | pace of aging               | pace_ratio     | 46       | whole blood                                                         | 450K                       | <a href="#">Belsky 2020</a>          |
| epicmithyper    | 2020 | mitotic age                 | divisions      | 184      | B cells                                                             | 450K, EPICv1               | <a href="#">Duran-Ferrer 2020</a>    |
| epicmithypo     | 2020 | mitotic age                 | divisions      | 1164     | B cells                                                             | 450K, EPICv1               | <a href="#">Duran-Ferrer 2020</a>    |
| epitoc1         | 2016 | mitotic age                 | divisions      | 385      | multi-tissue, whole blood                                           | 450K                       | <a href="#">Yang 2016</a>            |
| epitoc2         | 2020 | mitotic age                 | divisions      | 163      | whole blood                                                         | 450K                       | <a href="#">Teschendorff 2020</a>    |
| epitoc3         | 2020 | mitotic age                 | divisions      | 170      | cultured primary human cells, whole blood, multi-tissue, cord blood | 450K, EPICv1               | <a href="#">Teschendorff 2020</a>    |
| hannum          | 2013 | chronological age           | age_years      | 71       | whole blood                                                         | 450K                       | <a href="#">Hannum 2013</a>          |
| hd              | 2013 | physiological dysregulation | relative_score | —        | blood                                                               | clinical laboratory assays | <a href="#">Cohen 2013</a>           |
| horvath2013     | 2013 | chronological age           | age_years      | 353      | multi-tissue                                                        | 27K, 450K                  | <a href="#">Horvath 2013</a>         |
| hrsinchphenoage | 2022 | phenotypic age              | age_years      | 959      | whole blood                                                         | 450K, EPICv1               | <a href="#">Higgins-Chen 2022</a>    |
| hypoclock       | 2020 | mitotic age                 | divisions      | 678      | multi-tissue                                                        | 450K                       | <a href="#">Teschendorff 2020</a>    |
| kdm             | 2006 | biological age              | age_years      | —        | blood                                                               | clinical laboratory assays | <a href="#">Klemera 2006</a>         |

| Clock                                                   | Year | Predicts                       | Scale                          | Features | Tissue                           | Platform     | Paper                          |
|---------------------------------------------------------|------|--------------------------------|--------------------------------|----------|----------------------------------|--------------|--------------------------------|
| <code>knight</code>                                     | 2016 | gestational age                | <code>gestational_weeks</code> | 148      | cord blood, neonatal blood spots | 27K, 450K    | <a href="#">Knight 2016</a>    |
| <code>leecontrol</code>                                 | 2019 | gestational age                | <code>gestational_weeks</code> | 546      | placenta                         | 450K, EPICv1 | <a href="#">Lee 2019</a>       |
| <code>leerefine</code><br><code>drobust</code>          | 2019 | gestational age                | <code>gestational_weeks</code> | 395      | placenta                         | 450K, EPICv1 | <a href="#">Lee 2019</a>       |
| <code>leerobust</code>                                  | 2019 | gestational age                | <code>gestational_weeks</code> | 558      | placenta                         | 450K, EPICv1 | <a href="#">Lee 2019</a>       |
| <code>lin</code>                                        | 2016 | chronological age              | <code>age_years</code>         | 99       | whole blood                      | 27K, 450K    | <a href="#">Lin 2016</a>       |
| <code>mccartney</code><br><code>alcohol</code>          | 2018 | alcohol consumption            | <code>gestational_weeks</code> | 450      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartneybmi</code>                               | 2018 | body mass index                | <code>relative_score</code>    | 1109     | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>bodyfat</code>          | 2018 | body fat percentage            | <code>relative_score</code>    | 968      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>education</code>        | 2018 | educational attainment         | <code>relative_score</code>    | 373      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>hdlcholesterol</code>   | 2018 | HDL cholesterol                | <code>relative_score</code>    | 737      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>ldlcholesterol</code>   | 2018 | LDL cholesterol                | <code>relative_score</code>    | 233      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>smoking</code>          | 2018 | smoking exposure               | <code>relative_score</code>    | 233      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>totalcholesterol</code> | 2018 | total cholesterol              | <code>relative_score</code>    | 204      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartney</code><br><code>totalhdlratio</code>    | 2018 | total-to-HDL cholesterol ratio | <code>relative_score</code>    | 412      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |
| <code>mccartneywhr</code>                               | 2018 | waist-to-hip ratio             | <code>relative_score</code>    | 226      | whole blood                      | EPICv1       | <a href="#">McCartney 2018</a> |

| Clock          | Year | Predicts                | Scale                | Features | Tissue                                                                             | Platform                   | Paper            |
|----------------|------|-------------------------|----------------------|----------|------------------------------------------------------------------------------------|----------------------------|------------------|
| meer           | 2018 | chronological age       | age_years            | 435      | multi-tissue                                                                       | RRBS                       | Meer 2018        |
| pedbe          | 2020 | chronological age       | age_years            | 94       | buccal epithelium                                                                  | 450K, EPICv1               | McEwen 2020      |
| phenoage       | 2018 | phenotypic age          | age_years            | 10       | blood                                                                              | clinical laboratory assays | Levine 2018      |
| replitali      | 2022 | replicative history     | divisions            | 87       | cultured primary human cells                                                       | EPICv1                     | Endicott 2022    |
| skinandblood   | 2018 | chronological age       | age_years            | 391      | buccal epithelium, whole blood, epithelium, cultured fibroblasts, skin, cord blood | 450K, EPICv1               | Horvath 2018     |
| stemtoc        | 2024 | mitotic age             | divisions            | 371      | multi-tissue, cultured human cells, whole blood                                    | 450K, EPICv1               | Zhu 2024         |
| stemtocvitro   | 2024 | mitotic age             | divisions            | 629      | multi-tissue, cultured human cells                                                 | 450K, EPICv1               | Zhu 2024         |
| vidalbralo     | 2016 | chronological age       | age_years            | 8        | whole blood                                                                        | 27K                        | Vidal-Bralo 2016 |
| weidner        | 2014 | biological age          | age_years            | 3        | whole blood                                                                        | 27K, bisulfite sequencing  | Weidner 2014     |
| yingadaptage   | 2024 | adaptive epigenetic age | age_years_relative   | 999      | whole blood                                                                        | 450K                       | Ying 2024        |
| yingcausage    | 2024 | chronological age       | age_years            | 585      | whole blood                                                                        | 450K                       | Ying 2024        |
| yingdamage     | 2024 | damaging epigenetic age | age_years_relative   | 1089     | whole blood                                                                        | 450K                       | Ying 2024        |
| zhangblup      | 2019 | chronological age       | age_years            | 319607   | whole blood, saliva                                                                | 450K, EPICv1               | Zhang 2019       |
| zhangen        | 2019 | chronological age       | age_years            | 514      | whole blood, saliva                                                                | 450K, EPICv1               | Zhang 2019       |
| zhangmortality | 2017 | mortality risk          | mortality_log_hazard | 10       | whole blood                                                                        | 450K                       | Zhang 2017       |

## Untraced: 89 clocks

Catalogued, but no primary source for the coefficients has been established, so nothing is bundled. The metadata is real; the numbers are absent.

| Clock                            | Year | Predicts                       | Scale             | Features | Tissue                    | Platform     | Paper                             |
|----------------------------------|------|--------------------------------|-------------------|----------|---------------------------|--------------|-----------------------------------|
| abec                             | 2020 | chronological age              | age_years         | —        | whole blood               | EPICv1       | <a href="#">Lee 2020</a>          |
| adbahadosingh                    | 2021 | late-onset Alzheimer's disease | relative_score    | —        | whole blood               | EPICv1       | <a href="#">Bahado-Singh 2021</a> |
| bocklandt                        | 2011 | EDARADD methylation            | relative_score    | —        | saliva                    | 27K          | <a href="#">Bocklandt 2011</a>    |
| bohlin                           | 2016 | gestational age                | gestational_weeks | —        | cord blood                | 450K         | <a href="#">Bohlin 2016</a>       |
| cabec                            | 2020 | chronological age              | age_years         | —        | whole blood               | 450K, EPICv1 | <a href="#">Lee 2020</a>          |
| cellpopage                       | 2024 | cell-population passage age    | relative_score    | —        | cultured fibroblasts      | EPICv1       | <a href="#">Lujan 2024</a>        |
| compil6                          | 2021 | interleukin-6                  | relative_score    | —        | whole blood               | 450K         | <a href="#">Stevenson 2021</a>    |
| ctsliver                         | 2024 | chronological age              | age_years         | —        | liver                     | EPICv1       | <a href="#">Tong 2024</a>         |
| cvdwesterman                     | 2020 | cardiovascular disease risk    | relative_score    | —        | whole blood               | 450K         | <a href="#">Westerman 2020</a>    |
| deconvolutedblooddepicbcell      | 2018 | B cell proportion              | proportion        | —        | purified blood leukocytes | EPICv1       | <a href="#">Salas 2018</a>        |
| deconvolutedblooddepiccd4tcell   | 2018 | CD4+ T cell proportion         | proportion        | —        | purified blood leukocytes | EPICv1       | <a href="#">Salas 2018</a>        |
| deconvolutedblooddepiccd8tcell   | 2018 | CD8+ T cell proportion         | proportion        | —        | purified blood leukocytes | EPICv1       | <a href="#">Salas 2018</a>        |
| deconvolutedblooddepicmonocyte   | 2018 | monocyte proportion            | proportion        | —        | purified blood leukocytes | EPICv1       | <a href="#">Salas 2018</a>        |
| deconvolutedblooddepicneutrophil | 2018 | neutrophil proportion          | proportion        | —        | purified blood leukocytes | EPICv1       | <a href="#">Salas 2018</a>        |

| Clock                              | Year | Predicts                           | Scale              | Features | Tissue                                         | Platform     | Paper           |
|------------------------------------|------|------------------------------------|--------------------|----------|------------------------------------------------|--------------|-----------------|
| deconvolu<br>tebloodepi<br>cnkcell | 2018 | natural killer cell<br>proportion  | proportion         | —        | purified blood<br>leukocytes                   | EPICv1       | Salas 2018      |
| depressionbarbu                    | 2021 | major depressive<br>disorder       | relative_<br>score | —        | whole blood                                    | EPICv1       | Barbu 2021      |
| dnamfili                           | 2022 | frailty risk                       | relative_<br>score | —        | whole blood                                    | EPICv1, 450K | Li 2022         |
| dnamfitage                         | 2023 | physical-fitness<br>biological age | age_years          | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamfitagegaitf                    | 2023 | gait speed                         | relative_<br>score | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamfitagegaitm                    | 2023 | gait speed                         | relative_<br>score | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamfitagegripf                    | 2023 | grip strength                      | relative_<br>score | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamfitagegripm                    | 2023 | grip strength                      | relative_<br>score | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamfitage<br>evo2max              | 2023 | VO2max                             | relative_<br>score | —        | whole blood                                    | 450K         | McGreevy 2023   |
| dnamic                             | 2025 | intrinsic capacity                 | relative_<br>score | —        | whole blood                                    | EPICv1       | Fuentealba 2025 |
| dnamstress                         | 2023 | stress exposure                    | relative_<br>score | —        | whole blood                                    | EPICv1       | Jung 2023       |
| downsyndrome                       | 2021 | Down syndrome<br>methylation score | relative_<br>score | —        | neonatal blood<br>spots                        | EPICv1       | Muskens 2021    |
| eabec                              | 2020 | chronological age                  | age_years          | —        | whole blood                                    | EPICv1       | Lee 2020        |
| efrs                               | 2022 | frailty index                      | relative_<br>score | 20       | whole blood                                    | 450K, EPICv1 | Gao 2022        |
| encen100                           | 2023 | chronological age                  | age_years          | —        | whole blood,<br>saliva, buccal ep-<br>ithelium | 450K, EPICv1 | Dec 2023        |
| encen40                            | 2023 | chronological age                  | age_years          | —        | whole blood,<br>saliva, buccal ep-<br>ithelium | 450K, EPICv1 | Dec 2023        |



| Clock                     | Year | Predicts                               | Scale             | Features | Tissue                       | Platform                                                | Paper                                   |
|---------------------------|------|----------------------------------------|-------------------|----------|------------------------------|---------------------------------------------------------|-----------------------------------------|
| ensemblea<br>gehumanmouse | 2025 | relative age                           | relative_score    | —        | multi-tissue,<br>whole blood | mammalian<br>methylation array                          | Haghani 2025                            |
| ensemblea<br>gestatic     | 2025 | intervention-responsive epigenetic age | age_years         | —        | multi-tissue                 | Horvath MammalMethylChip40, Horvath MammalMethylChip320 | Haghani 2025                            |
| ensemblea<br>gestatictop  | 2025 | intervention-responsive epigenetic age | age_years         | —        | multi-tissue                 | Horvath MammalMethylChip40, Horvath MammalMethylChip320 | Haghani 2025                            |
| epicga                    | 2021 | gestational age                        | gestational_weeks | —        | cord blood                   | EPICv1                                                  | Haftorn 2021                            |
| fiage                     | 2025 | frailty index                          | relative_score    | 53       | whole blood                  | EPICv1                                                  | no primary source traced, catalogued at |
| garagnani                 | 2012 | ELOVL2 methylation                     | relative_score    | —        | whole blood                  | 450K                                                    | Garagnani 2012                          |
| gliasin                   | 2024 | chronological age                      | age_years         | —        | brain cortex                 | 450K                                                    | Tong 2024                               |
| han                       | 2020 | chronological age                      | age_years         | —        | whole blood                  | 450K                                                    | Han 2020                                |
| hep                       | 2024 | chronological age                      | age_years         | —        | liver                        | EPICv1                                                  | Tong 2024                               |
| hepatoxu                  | 2017 | hepatocellular carcinoma               | relative_score    | —        | plasma cell-free DNA         | targeted bisulfite sequencing                           | Xu 2017                                 |
| intrinclock               | 2024 | chronological age                      | age_years         | —        | multi-tissue                 | 450K, EPICv1                                            | Tomusiak 2024                           |
| mammalian1                | 2023 | chronological age                      | age_years         | —        | multi-tissue                 | Horvath MammalMethylChip40                              | Lu 2023                                 |
| mammalian2                | 2023 | chronological age                      | age_years         | —        | multi-tissue                 | Horvath MammalMethylChip40                              | Lu 2023                                 |
| mammalian3                | 2023 | chronological age                      | age_years         | —        | multi-tissue                 | Horvath MammalMethylChip40                              | Lu 2023                                 |
| mammalian<br>blood2       | 2023 | chronological age                      | age_years         | —        | blood                        | Horvath MammalMethylChip40                              | Lu 2023                                 |
| mammalian<br>blood3       | 2023 | chronological age                      | age_years         | —        | blood                        | Horvath MammalMethylChip40                              | Lu 2023                                 |

| Clock               | Year | Predicts                  | Scale             | Features | Tissue                                  | Platform                    | Paper                             |
|---------------------|------|---------------------------|-------------------|----------|-----------------------------------------|-----------------------------|-----------------------------------|
| mammalianfemale     | 2023 | sex                       | relative_score    | —        | multi-tissue                            | mammalian methylation array | <a href="#">Li 2023</a>           |
| mammalian lifespan  | 2024 | species maximum lifespan  | age_years         | —        | multi-tissue                            | Horvath MammalMethylChip40  | <a href="#">Li 2024</a>           |
| mammalian skin2     | 2023 | chronological age         | age_years         | —        | skin                                    | Horvath MammalMethylChip40  | <a href="#">Lu 2023</a>           |
| mammalian skin3     | 2023 | chronological age         | age_years         | —        | skin                                    | Horvath MammalMethylChip40  | <a href="#">Lu 2023</a>           |
| mayne               | 2017 | gestational age           | gestational_weeks | —        | placenta                                | 27K, 450K                   | <a href="#">Mayne 2017</a>        |
| neusin              | 2024 | chronological age         | age_years         | —        | brain cortex                            | 450K                        | <a href="#">Tong 2024</a>         |
| pcdnamt1            | 2022 | leukocyte telomere length | telomere_kb       | —        | whole blood                             | 450K                        | <a href="#">Higgins-Chen 2022</a> |
| pchannum            | 2022 | chronological age         | age_years         | —        | whole blood                             | 450K                        | <a href="#">Higgins-Chen 2022</a> |
| pchorvath 2013      | 2022 | chronological age         | age_years         | —        | multi-tissue                            | 450K, EPICv1                | <a href="#">Higgins-Chen 2022</a> |
| pcphenoage          | 2022 | phenotypic age            | age_years         | —        | whole blood                             | 450K, EPICv1                | <a href="#">Higgins-Chen 2022</a> |
| pcskinandblood      | 2022 | chronological age         | age_years         | —        | skin, whole blood, cultured fibroblasts | 450K, EPICv1                | <a href="#">Higgins-Chen 2022</a> |
| petkovich           | 2017 | chronological age         | age_years         | —        | whole blood                             | bisulfite sequencing        | <a href="#">Petkovich 2017</a>    |
| pipekelasticnet     | 2022 | chronological age         | age_years         | —        | multi-tissue                            | 27K, 450K, EPICv1           | <a href="#">Pipek 2022</a>        |
| pipekfilteredh      | 2022 | chronological age         | age_years         | —        | multi-tissue                            | 27K, 450K, EPICv1           | <a href="#">Pipek 2022</a>        |
| pipekretrainedh     | 2022 | chronological age         | age_years         | —        | multi-tissue                            | 27K, 450K, EPICv1           | <a href="#">Pipek 2022</a>        |
| prostatecancerkirby | 2017 | prostate cancer           | relative_score    | —        | prostate                                | 450K                        | <a href="#">Kirby 2017</a>        |
| reedbmi             | 2020 | BMI methylation score     | relative_score    | —        | whole blood, cord blood                 | 450K                        | <a href="#">Reed 2020</a>         |

| Clock                                                             | Year | Predicts                 | Scale                                | Features | Tissue                                                                                               | Platform     | Paper                         |
|-------------------------------------------------------------------|------|--------------------------|--------------------------------------|----------|------------------------------------------------------------------------------------------------------|--------------|-------------------------------|
| <a href="#">replitalinorm</a>                                     | 2022 | replicative history      | <a href="#">divisions</a>            | —        | cultured fibroblasts                                                                                 | EPICv1       | <a href="#">Endicott 2022</a> |
| <a href="#">retroelem<br/>entagev1</a>                            | 2024 | chronological age        | <a href="#">age_years</a>            | —        | whole blood                                                                                          | EPICv1       | <a href="#">Ndhlovu 2024</a>  |
| <a href="#">retroelem<br/>entagev2</a>                            | 2024 | chronological age        | <a href="#">age_years</a>            | —        | whole blood                                                                                          | EPICv1       | <a href="#">Ndhlovu 2024</a>  |
| <a href="#">senchronage</a>                                       | 2026 | chronological age        | <a href="#">age_years</a>            | —        | whole blood                                                                                          | 450K, EPICv1 | <a href="#">Kasamoto 2026</a> |
| <a href="#">sencultureage</a>                                     | 2026 | cellular senescence      | <a href="#">relative_score</a>       | —        | cultured fibroblasts, cultured mesenchymal stromal cells                                             | 450K, EPICv1 | <a href="#">Kasamoto 2026</a> |
| <a href="#">senmortalityage</a>                                   | 2026 | mortality risk           | <a href="#">mortality_log_hazard</a> | —        | whole blood                                                                                          | 450K, EPICv1 | <a href="#">Kasamoto 2026</a> |
| <a href="#">stoch</a>                                             | 2024 | chronological age        | <a href="#">age_years</a>            | —        | sorted monocytes                                                                                     | 450K         | <a href="#">Tong 2024</a>     |
| <a href="#">stocp</a>                                             | 2024 | chronological age        | <a href="#">age_years</a>            | —        | sorted monocytes                                                                                     | 450K         | <a href="#">Tong 2024</a>     |
| <a href="#">stocz</a>                                             | 2024 | chronological age        | <a href="#">age_years</a>            | —        | sorted monocytes                                                                                     | 450K         | <a href="#">Tong 2024</a>     |
| <a href="#">stubbs</a>                                            | 2017 | chronological age        | <a href="#">age_years</a>            | —        | multi-tissue, liver, lung, heart, brain cortex, skeletal muscle, cerebellum, spleen                  | RRBS         | <a href="#">Stubbs 2017</a>   |
| <a href="#">thompson</a>                                          | 2018 | chronological age        | <a href="#">age_years</a>            | —        | adipose tissue, blood, cerebellum, brain cortex, heart, kidney, liver, lung, skeletal muscle, spleen | RRBS         | <a href="#">Thompson 2018</a> |
| <a href="#">twelvecel<br/>ldeconvolu<br/>tebloodepi<br/>cbas</a>  | 2022 | basophil proportion      | <a href="#">proportion</a>           | —        | purified blood leukocytes                                                                            | EPICv1       | <a href="#">Salas 2022</a>    |
| <a href="#">twelvecel<br/>ldeconvolu<br/>tebloodepi<br/>cbmem</a> | 2022 | memory B cell proportion | <a href="#">proportion</a>           | —        | purified blood leukocytes                                                                            | EPICv1       | <a href="#">Salas 2022</a>    |
| <a href="#">twelvecel<br/>ldeconvolu</a>                          | 2022 | naive B cell proportion  | <a href="#">proportion</a>           | —        | purified blood leukocytes                                                                            | EPICv1       | <a href="#">Salas 2022</a>    |

| Clock                                            | Year | Predicts                          | Scale      | Features | Tissue                       | Platform | Paper      |
|--------------------------------------------------|------|-----------------------------------|------------|----------|------------------------------|----------|------------|
| tebloodepi<br>cbnv                               |      |                                   |            |          |                              |          |            |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>ccd4mem | 2022 | memory CD4+ T<br>cell proportion  | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>ccd4nv  | 2022 | naive CD4+ T cell<br>proportion   | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>ccd8mem | 2022 | memory CD8+ T<br>cell proportion  | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>ccd8nv  | 2022 | naive CD8+ T cell<br>proportion   | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>ceos    | 2022 | eosinophil pro-<br>portion        | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>cmono   | 2022 | monocyte pro-<br>portion          | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>cneu    | 2022 | neutrophil pro-<br>portion        | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu<br>tebloodepi<br>cnk     | 2022 | natural killer cell<br>proportion | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |
| twelvecel<br>ldeconvolu                          | 2022 | regulatory T-cell<br>proportion   | proportion | —        | purified blood<br>leukocytes | EPICv1   | Salas 2022 |

| Clock               | Year | Predicts              | Scale          | Features | Tissue      | Platform     | Paper     |
|---------------------|------|-----------------------|----------------|----------|-------------|--------------|-----------|
| tebloodepi<br>ctreg |      |                       |                |          |             |              |           |
| wu                  | 2019 | chronological age     | age_years      | —        | whole blood | 27K, 450K    | Wu 2019   |
| xchrom              | 2021 | X-chromosome dosage   | relative_score | —        | whole blood | 450K, EPICv1 | Wang 2021 |
| ychrom              | 2021 | Y-chromosome presence | relative_score | —        | whole blood | 450K, EPICv1 | Wang 2021 |

## Licensed: 40 clocks

The architecture is implemented and tested; the coefficients are research-use-only and are not ours to distribute. Obtain a file and register it.

| Clock                 | Year | Predicts                      | Scale          | Features | Tissue      | Platform     | Paper                                |
|-----------------------|------|-------------------------------|----------------|----------|-------------|--------------|--------------------------------------|
| cpgptgrim<br>age3     | 2024 | biological age                | age_years      | —        | whole blood | 450K         | <a href="#">de Lima Camillo 2024</a> |
| cpgptpcgr<br>image3   | 2024 | biological age                | age_years      | —        | whole blood | 450K         | <a href="#">de Lima Camillo 2024</a> |
| dnamemrage            | 2026 | electronic medical record age | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Chen 2026</a>            |
| dunedinpace           | 2022 | pace of aging                 | pace_ratio     | —        | whole blood | EPICv1       | <a href="#">Belsky 2022</a>          |
| grimage               | 2019 | mortality risk                | age_years      | —        | whole blood | 450K         | <a href="#">Lu 2019</a>              |
| grimage2              | 2022 | mortality risk                | age_years      | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2adm           | 2022 | adrenomedullin                | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2b2m           | 2022 | beta-2-microglobulin          | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2c<br>ystatinc | 2022 | cystatin C                    | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2g<br>df15     | 2022 | GDF-15                        | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2l<br>eptin    | 2022 | leptin                        | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2l<br>ogalc    | 2022 | hemoglobin A1c                | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2l<br>ogcrp    | 2022 | C-reactive protein            | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2p<br>ackyrs   | 2022 | smoking exposure              | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2p<br>ail      | 2022 | PAI-1                         | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |
| grimage2t<br>impl     | 2022 | TIMP-1                        | relative_score | —        | whole blood | 450K         | <a href="#">Lu 2022</a>              |

| Clock                   | Year | Predicts                             | Scale          | Features | Tissue      | Platform     | Paper                             |
|-------------------------|------|--------------------------------------|----------------|----------|-------------|--------------|-----------------------------------|
| omicmage                | 2026 | biological age                       | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Chen 2026</a>         |
| pcgrimage               | 2022 | mortality risk                       | age_years      | —        | whole blood | 450K         | <a href="#">Higgins-Chen 2022</a> |
| pcgrimageadm            | 2022 | adrenomedullin                       | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimageb2m            | 2022 | beta-2 microglobulin                 | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimagecystatinc      | 2022 | cystatin C                           | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimagegdf15          | 2022 | growth differentiation factor 15     | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimageleptin         | 2022 | leptin                               | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimagepackyrs        | 2022 | smoking pack-years                   | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimagepail           | 2022 | plasminogen activator inhibitor 1    | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| pcgrimagetimpl          | 2022 | TIMP metalloproteinase inhibitor 1   | relative_score | —        | whole blood | 450K, EPICv1 | <a href="#">Higgins-Chen 2022</a> |
| retroclock              | 2025 | retroelement methylation age         | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |
| systemsage              | 2025 | multisystem biological age           | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |
| systemsageageprediction | 2025 | chronological age                    | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |
| systemsageblood         | 2025 | blood-system biological age          | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |
| systemsagebrain         | 2025 | brain-system biological age          | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |
| systemsageheart         | 2025 | cardiovascular-system biological age | age_years      | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a>       |

| Clock                                  | Year | Predicts                              | Scale                  | Features | Tissue      | Platform     | Paper                       |
|----------------------------------------|------|---------------------------------------|------------------------|----------|-------------|--------------|-----------------------------|
| <code>systemsagehormone</code>         | 2025 | endocrine-system biological age       | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsageimmune</code>          | 2025 | immune-system biological age          | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsageinflammation</code>    | 2025 | inflammatory-system biological age    | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsagekidney</code>          | 2025 | renal-system biological age           | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsageliver</code>           | 2025 | hepatic-system biological age         | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsagelung</code>            | 2025 | pulmonary-system biological age       | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsagemetabolic</code>       | 2025 | metabolic-system biological age       | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |
| <code>systemsagemusculoskeletal</code> | 2025 | musculoskeletal-system biological age | <code>age_years</code> | —        | whole blood | 450K, EPICv1 | <a href="#">Sehgal 2025</a> |



## Querying it from code

### Python

```
reg = fa.registry.load()
reg.filter(availability="bundled", scale_type="age_years")
reg.search("mortality")
reg.compatible_with(data)
reg.get("horvath2013")
```

### R

```
list_clocks(tier = "bundled")
list_clocks(search = "mortality")
compatible_clocks(d)
clock_info("horvath2013")
```

## Chapter 5. GPU support: what was tested, on what, and what it is worth

Verified on real hardware, re-measured in full on 2026-08-10. The short version: the GPU path works, it is numerically sound, and for the clocks that ship today it is **slower than the CPU**, so `device="auto"` resolves to CPU and the GPU is opt-in.

Every figure below comes from `bhagesh/falconage:1.0.0-cuda`, built from the shipping `docker/Dockerfile.cuda`. The 2026-08-08 run of this page reported the same shape on an earlier build; where a number moved, the current one is here and the old one is named next to it.

### The machine

| GPU               | NVIDIA GeForce RTX 4060 Laptop, 8 GB, compute capability sm_89 (Ada) |
|-------------------|----------------------------------------------------------------------|
| Driver            | 610.88, CUDA UMD 13.3                                                |
| Container runtime | Docker 29.6.2, GPU passthrough working via <code>--gpus all</code>   |
| Second adapter    | AMD Radeon 780M (integrated; not used - FALCONAge has no ROCm path)  |

### What was already present, and what had to be installed

Nothing was installed on the host. Everything went into a throwaway container.

| Component                             | State                                                      | Action                                                                                                                                |
|---------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| NVIDIA driver 610.88                  | already installed                                          | none                                                                                                                                  |
| CUDA user-mode driver 13.3            | already installed                                          | none                                                                                                                                  |
| Docker GPU passthrough                | already working                                            | none, <code>docker run --gpus all nvidia/cuda:12.4.1-base nvidia-smi</code> returned the card first try                               |
| CUDA toolkit on the host              | <b>not installed, and not needed</b>                       | none, the torch wheel bundles its own runtime; only the driver has to be on the host                                                  |
| <code>torch</code> with CUDA          | <b>missing</b> - the CPU image ships without it on purpose | installed <code>torch 2.6.0+cu124</code> inside a container, from <code>docker/Dockerfile.cuda</code> . Nothing was added to the host |
| <code>nvidia-container-toolkit</code> | already configured                                         | none                                                                                                                                  |

The driver reports CUDA 13.3 and the wheel is built against 12.4. That is the supported direction: a newer driver runs an older runtime. The reverse does not work, which is why the shipping `docker/Dockerfile.cuda` pins the runtime rather than tracking the driver.

## 1. Device and dtype resolution

```
resolve('auto',    None,    fp64=False) -> numpy:cpu/float64
resolve('cuda',    None,    fp64=False) -> torch:cuda/float64
resolve('cuda',    float32, fp64=False) -> torch:cuda/float32
resolve('cuda',    float32, fp64=True)  -> torch:cuda/float64  + warning
resolve('mps')      -> refused: no MPS device
```

The `requires_fp64` override fires as designed: a clock the registry flags as ill-conditioned gets double precision back with a warning, rather than silently honouring a `float32` request that would move its answer.

## 2. Which clocks the device actually reaches

`device="cuda"` is a request, and until 2026-08-10 it was granted more often in the manifest than in the arithmetic. All six model classes take a `DeviceSpec`. Of the five that can produce a score, three computed in numpy regardless, and the scoring loop then recorded the request rather than the fact.

| Model class                   | Reaches the device        | Registry entries it serves                       |
|-------------------------------|---------------------------|--------------------------------------------------|
| <code>LinearClock</code>      | yes                       | 20 of the 23 that score offline                  |
| <code>PCLinearClock</code>    | yes                       | 19, none shipping weights                        |
| <code>AggregationClock</code> | yes, since 2026-08-10     | 6, none shipping weights                         |
| <code>NeuralClock</code>      | yes, since 2026-08-10     | AltumAge                                         |
| <code>ClinicalClock</code>    | <b>no, by declaration</b> | PhenoAge, KDM, homeostatic dysregulation         |
| <code>ScaffoldClock</code>    | not applicable            | 28 tier C entries; it raises rather than scoring |

**Why the clinical three decline.** A methylation clock reduces thousands of probes. A clinical clock reduces nine markers: PhenoAge sums ten terms, KDM fits one univariate regression per marker, HD inverts a  $9 \times 9$  covariance. That is less arithmetic than a CUDA kernel launch costs to dispatch, so a device implementation would be slower, and it would pull torch into the one modality that otherwise needs nothing beyond numpy. They set `CPU_ONLY = True`, which `falconage.models.effective_spec()` reads, so the refusal is a declared property rather than an argument quietly dropped.

**Why the other two were routed anyway.** `NeuralClock` is the one architecture here where a device should pay, for the reason section 4 makes measurable: a linear clock is a single dot product over a few thousand features and loses to the CPU because the transfer costs more than the multiply, while AltumAge is dense layers over 20,318 inputs with depth to parallelise. `AggregationClock` takes a mean or a 95th percentile over a probe set and will not repay a device on its own; it was routed so that what the manifest records is what happened.

### What the manifest says now

Per clock, not per run. This is a real run on the card: twenty methylation clocks and one clinical clock, both scored with `device="cuda"` and combined.

```

device / backend : mixed / mixed
device_requested : cuda
compute_summary  : torch:cuda/float64 (20 clocks), numpy:cpu/float64 (1 clock)
compute["METH:horvath2013"] : {device: cuda, dtype: float64, backend: torch}
compute["CLIN:phenoage"]    : {device: cpu, dtype: float64, backend: numpy}

```

Before this, that run reported `device="cuda"` flat, for a manifest in which one of the twenty-one clocks had never touched the card.

The three scalar fields are derived from `compute`: the shared value when every clock agreed, and `"mixed"` when they did not. Before this they were assigned once per clock inside the scoring loop, so a run reported whichever clock ran last. The same overwrite made `dtype` wrong for any run combining a `requires_fp64` clock with a `float32` request, which is all six PC clocks.

Two things make a run legitimately mixed. A `CPU_ONLY` model where the others reached the device, as above; and `combine()` across datasets scored on different machines, which is ordinary in a benchmark. The combined manifest merges the per-clock records rather than copying the first run's device onto all of them.

**This changes no score.** It changes what the run says about itself, which is the whole of the reproducibility claim the next section rests on.

### 3. CPU and GPU do not agree bit for bit

This is the finding that matters most, and it qualifies a claim made elsewhere in the documentation.

| clock        | max abs difference | ulps |
|--------------|--------------------|------|
| dnampheoage  | 9.9e-14            | 14   |
| yingcausage  | 8.5e-14            | 3    |
| horvath2013  | 5.7e-14            | 4    |
| hannum       | 4.3e-14            | 3    |
| lin          | 4.3e-14            | 6    |
| skinandblood | 4.3e-14            | 3    |
| zhangen      | 1.4e-14            | 1    |
| dnamtl       | 1.8e-15            | 2    |

Worst case across eight clocks: **9.9e-14 years**, about three femtoseconds of biological age.

The cause is ordinary: numpy's BLAS and cuBLAS sum a dot product in different orders, and floating-point addition is not associative. It is not a defect and it is not fixable without giving up the vendor kernels.

The per-clock figures moved between the two runs, worst case 1.3e-13 on 2026-08-08 against 9.9e-14 now, horvath2013 the largest mover at 1.3e-13 to 5.7e-14. No code changed for these eight, which are all `LinearClock`; the rebuilt image carries a different numpy and therefore a different BLAS, which sums in a different order again. Numbers that move when the linear algebra library moves are what this section is about, not an exception to it.

**What this means for the conformance guarantee.** FALCONAge claims R and Python return the same bits. That claim is intact, both go through the same Python core, and the R suite asserts equality at tolerance

exactly zero. It is a statement about the two *languages*, not about two *devices*. A CPU result and a GPU result agree to about 1e-13 and are not bit-identical. The run manifest records `device` and `dtype` for exactly this reason: a number is reproducible against the manifest that produced it, not against every possible manifest.

Single precision costs far more: **7e-5 years** between `float32` and `float64` on the same device. Still negligible against effects measured in single-digit years, but four orders of magnitude worse than the device difference, which is the right way round.

## 4. Speed: the GPU makes the shipping clocks slower

Eight clocks, 2,340 distinct features, RTX 4060, best of three timings inside a run and the best of three runs, measured in `bhagesh/falconage:1.0.0-cuda` - the image the command at the foot of this page builds. These are numbers you can reproduce, not numbers from a throwaway environment.

| samples | CPU float64    | CUDA float64 | CUDA float32 | verdict          |
|---------|----------------|--------------|--------------|------------------|
| 128     | <b>0.007 s</b> | 0.013 s      | 0.011 s      | CPU wins by 1.9x |
| 1,024   | <b>0.026 s</b> | 0.048 s      | 0.048 s      | CPU wins by 1.8x |
| 4,096   | <b>0.109 s</b> | 0.333 s      | 0.216 s      | CPU wins by 3.1x |
| 16,384  | <b>0.445 s</b> | 3.275 s      | 2.397 s      | CPU wins by 7.4x |

The gap *widens* with size, which is the opposite of the usual shape. Profiling says why:

```
4096 samples x 2340 features, 8 clocks
feature alignment (pandas, CPU only)    0.132 s
the dot products, on the GPU             0.010 s
the dot products, on the CPU             0.005 s
```

Alignment is 28x the arithmetic it feeds, and at this size that arithmetic is *faster on the CPU*: the GPU's 10 ms is mostly transfer, not multiply. At 16,384 samples it is 2.4 GB moved across PCIe to save nothing. A linear clock over a few thousand features is simply too small a matrix multiplication to be worth a device.

**Read the CUDA columns as a range, not a figure.** Three consecutive runs of the same script on the same idle machine gave 3.275, 3.482 and 3.664 s at 16,384 samples, a 12% spread, while the CPU column moved by under 7% (0.445 to 0.477). This is a laptop card that clocks down as it warms. Each column above is the best of the three, so the columns are not necessarily from one run; computed within a run instead, the CPU margin at 16,384 was 6.9x, 7.7x and 7.8x. A desktop card with a power budget would be steadier and would still lose, because what it is losing to is the transfer.

Two earlier revisions of this page reported 6.5x and then 4.6x at 16,384. The gap has widened again to 7.4x, and the reason is not that the GPU got worse: the CPU column improved from 0.506 s to 0.445 s on a rebuilt image with a newer numpy, while the CUDA column got slower on the same hardware. The shape of the answer has never changed across three measurements on two image builds, which is worth more than any one of the figures.

So `device="auto"` resolves to **CPU**, even when a GPU is present. Picking CUDA because a card exists would make the common case several times slower on every machine that has one, silently. The GPU is opt-in: `device="cuda"` or `FALCONAGE_DEVICE=cuda`.

## When the GPU will be worth it

Not for the 35 tier A clocks. It should pay for itself on:

- **the PC clocks** - 78,464 features and coefficient tensors of 78 MB to 1.2 GB. Thirty times the feature count of the current largest, so the matmul stops being a rounding error.
- **neural architectures** - AltumAge is a multilayer perceptron, not a dot product, and has real depth to parallelise.
- **CpGPT** - a transformer, where the GPU is not optional.

None of those are tier A yet, which is exactly why this document can only report that the path is correct rather than that it is fast.

Both of the first two now route through the device (section 2). That is a correctness statement, not a speed one: no weights ship for either, so there is nothing here to time. The point of routing them before the weights arrive is that the alternative is discovering on the day somebody supplies a `.safetensors` file that `device="cuda"` had been doing nothing all along.

## A performance bug this found

Profiling for the GPU comparison turned up something unrelated to GPUs. Feature alignment was looping `X.iloc[:, i]` once per feature - 2,666 pandas indexing calls for one eight-clock run, 76% of total runtime, against 0.5% for the arithmetic they fed. Replacing the loop with a single `reindex` made the **CPU path 2.1× faster** at 4,096 samples (0.358 s to 0.171 s), which benefits every user whether or not they own a GPU.

It also means the earlier GPU figures were flattering: they were measured against a slow CPU baseline. The table above is after the fix.

## Reproducing this

Every number on this page comes from one script, and it is in the repository:

```
docker pull bhagesh/falconage:1.0.0-cuda
docker run --rm --gpus all -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cuda \
python test/gpu_check.py
```

Or build the image instead of pulling it, which is what produced the numbers above:

```
docker build -f docker/Dockerfile.cuda -t bhagesh/falconage:1.0.0-cuda .
```

`test/gpu_check.py` runs the five steps in the order above, what torch sees, device and dtype resolution, CPU-versus-GPU agreement, the speed and precision table, and the profile that explains it. Each step is meaningless if the one before it failed, so it stops rather than continuing. On a machine with no CUDA device it stops after step 1 with an explanation, which makes it safe to run anywhere and safe to put in a bug report.

The coverage table in section 2 is not from that script, because it needs no hardware. It is asserted by `python/tests/unit/test_device_contract.py`, which hands every model class a spec that counts how often it is reached and requires either a non-empty count or a `CPU_ONLY` declaration. The torch arithmetic is checked against numpy in the same file, on the CPU torch backend, so a runner with no card still verifies everything except the transfer:

```
docker run --rm -v "$PWD:/work" -w /work bhagesh/falconage:1.0.0-cuda \  
python -m pytest python/tests/unit/test_device_contract.py -q
```

**If the build fails at the dependency layer**, the pinned uv is older than the `revision` field in `python/uv.lock` and refuses to parse it. That is what broke both images between the 2026-08-08 and 2026-08-10 runs of this page. `test/check_docker_lock.py` catches it in a second, and CI runs it, but the symptom is worth recognising: `error: Failed to parse `uv.lock``, six minutes into a build.

`--max-samples` caps the speed table for a card with less memory than the 8 GB this was measured on; `--skip-profile` drops the last step.

If your card disagrees with the tables above: the tables are what should change.

## Chapter 6. The science of aging clocks

---

Every number FALCONAge returns comes from somebody’s paper. This is the record of which paper, what it actually says, and where the published description and the circulating implementation disagree, which they do in eleven documented cases.

It covers what an aging score *is*, the chemistry and biology behind each input layer, every algorithm family with its equations, the exact transformations and constants published clocks use, the output types and their units, the figures the field reads, and the benchmarking machinery used to judge a clock. It runs to the state of the field in 2026.

Read it as the substrate rather than as the manual. [Architecture](#) says which file computes what; this says why that is the right computation, and it is the document to check before trusting a number. Where a section describes something FALCONAge does not implement, it says so, the catalogue here is deliberately wider than v1.0, because the point of writing the field down is to know what is missing.

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2. Biology of aging that the scores encode
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### • References

### Appendices

- A - Full pyaging clock registry (173 clocks)



- B - tAge transcriptomic clock registry (60 models)
- C - ComputAgeBench benchmark studies (65 datasets, blood/saliva/buccal)
- D - organAging organ protein panels
- E - Preprocess and postprocess operation inventory
- F - Reproducing the four reference computations by hand

## 1. Definitions and framing

### 1.1 The three quantities

- **Chronological age (CA)** - time since birth. Years for humans, months or days for rodents, hours for *C. elegans*.
- **Biological age (BA)** - the output of a model that maps molecular or clinical features to a number on an age-like scale. In practice BA is whatever a fitted model predicts; its meaning comes entirely from what the model was trained on.
- **Age acceleration (AA)** - the residual. Two conventions, and they are not interchangeable:
  - **Absolute age acceleration (AAA)**, also called delta:  $\Delta = BA - CA$ . This is what ComputAgeBench tests ([ComputAge benchmarking.py:355](#)).
  - **Relative age acceleration (RAA)**: residual of  $\text{lm}(BA \sim CA)$  in the analysis sample. This removes the systematic regression-to-the-mean bias that all penalised regressions carry. Used by the TwinGene/UKB pipeline as `BA_res = residuals(lm(BA ~ age))` ([Biological-Aging-and-PRS README](#)).
  - A third convention, used by spatial single-cell clocks, replaces CA with the per-animal per-cell-type mean prediction, so acceleration is measured within an animal rather than against the calendar ([SpatialAgingClock deploy.py get\\_age\\_acceleration](#)).
- **Pace of aging** - a rate, not a level. Units are biological years per chronological year. DunedinPACE and DunedinPoAm are the two published examples; both are centred near 1.0.

## 1.2 Clock generations

This taxonomy governs everything downstream, what the score means, whether it responds to interventions, and how it must be validated.

| Generation                 | Training target                                                                 | Examples                                                                                                                              | Behaviour                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First                      | Chronological age                                                               | Horvath 2013, Hannum, Lin, Zhang EN/BLUP, AltumAge, PedBE, SkinAnd-Blood                                                              | High age correlation, weak or absent mortality signal. Zhang BLUP has RMSE $\approx 2.0$ y yet its acceleration no longer predicts mortality, the “oracle paradox”. |
| Second                     | Mortality or a composite phenotype                                              | PhenoAge, DNAmPhenoAge, GrimAge, GrimAge2, CpGPTGrimAge3, SystemsAge, OMICmAge                                                        | Lower CA accuracy, much stronger mortality/morbidity association.                                                                                                   |
| Third (“pace”)             | Longitudinal rate of change across many organ systems                           | DunedinPoAm38, DunedinPACE                                                                                                            | Rate, responds fastest to interventions.                                                                                                                            |
| Causal                     | CA, but with features filtered by Mendelian-randomisation evidence of causality | YingCausAge, YingDamAge, YingAdaptAge                                                                                                 | Separates damaging from adaptive/protective methylation.                                                                                                            |
| Mitotic                    | Stem-cell division count / replicative history                                  | epiTOC1, epiTOC2, epiTOC3, stem-TOC, HypoClock, epiCMIT-hyper/hypo, RepliTali, MiAge                                                  | Cancer risk, proliferative history.                                                                                                                                 |
| System- and organ-specific | Mortality or CA, restricted to organ-enriched features                          | SystemsAge (11 systems), organAging (24 organ protein panels), MEAT (muscle), CorticalClock (brain), sclImmuAging (5 PBMC cell types) | Reveals within-person heterogeneity.                                                                                                                                |
| Foundation-model           | Self-supervised pretraining, then fine-tuned                                    | CpGPT, methylGPT                                                                                                                      | Cross-platform imputation, SOTA mortality prediction.                                                                                                               |

### 1.3 Biomarker validation taxonomy (Biomarkers of Aging Consortium)

Moqri et al. classify aging biomarkers as **predictive** (associates with future outcomes), **responsive** (moves under an intervention), and **biological** (has a mechanistic, causal claim). A clock that is only predictive is a risk score; only responsive is a surrogate endpoint; a biological biomarker needs causal evidence. The consortium's Biolearn library and the 2024–25 open competition (chronological age, mortality, multimorbidity; >\$200k prizes) operationalise this. Any new tool should record which of the three claims each clock supports.

### 1.4 Why this matters commercially and scientifically

The geroscience hypothesis holds that aging is the dominant risk factor common to cancer, cardiovascular disease, T2D, Alzheimer's and Parkinson's, so slowing aging delays all of them. A validated surrogate endpoint shortens trials that would otherwise need decades of mortality follow-up. TAME (metformin) has raised ~\$11M of \$40M over seven years with a six-year observation window; a responsive clock is the alternative ([computational\\_aging\\_course/ml/aging\\_clocks.md](https://computational_aging_course/ml/aging_clocks.md)).

Butler's criteria for a usable aging biomarker: changes with age; predicts mortality better than CA alone; predicts early-stage age-related disease; minimally invasive. Moskalev adds: sensitive early, predictable in a practical time window, robust.

## 2. Biology of aging that the scores encode

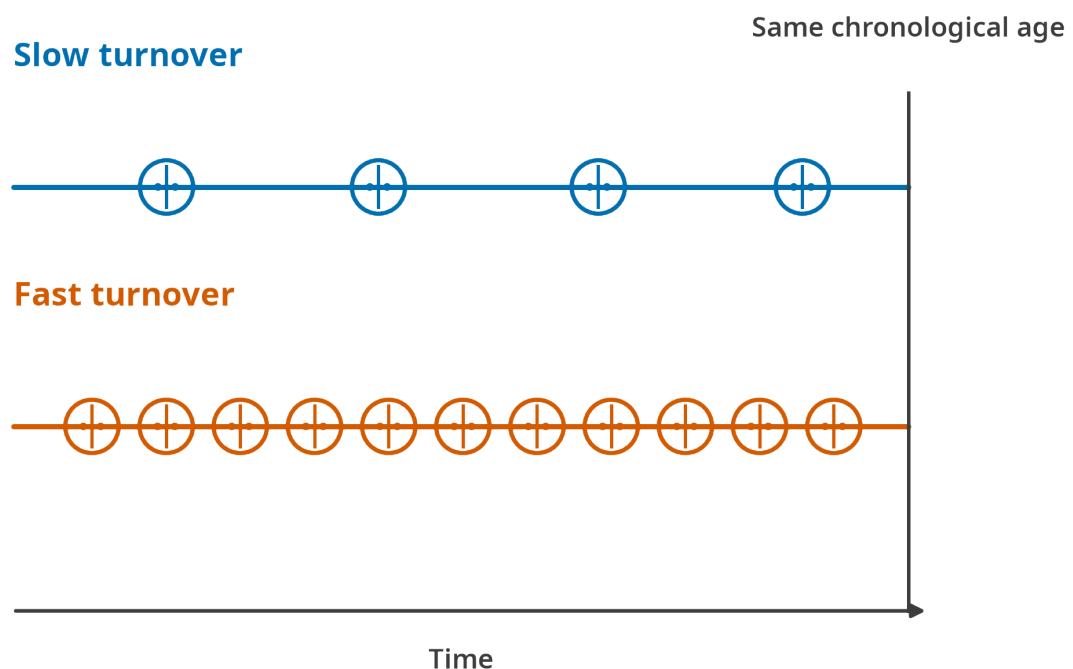


Figure 8: Two tissues of the same chronological age that have divided a different number of times

Elapsed time and accumulated cell divisions are separate axes, and a mitotic clock reads the second one. Colon epithelium turns over in days and brain tissue barely at all, so two samples taken from the same donor on the same day carry the same chronological age and very different division counts. This is why

a division count is not reported in years and why subtracting a chronological age from one is refused: the quantity was never elapsed time.

The 12 hallmarks (López-Otín 2013 [55], updated 2023 [56]) are the biological vocabulary the scores are trying to summarise. Grouped as primary (damage), antagonistic (compensatory responses that turn harmful), integrative (system-level consequence). A hallmark must show age-associated manifestation, experimental acceleration of aging when worsened, and therapeutic reversibility.

### Primary

1. **Genomic instability** - point mutations, indels, single/double-strand breaks (local); translocations, chromosomal rearrangement, nuclear-architecture defects, transposon and viral integration (global). Species somatic mutation rate is inversely correlated with lifespan (Cagan 2022). Pathogenic mechanisms: clonal expansion, somatic evolution, disruption of gene regulatory networks producing transcriptional noise. Mitochondrial DNA is separately unstable
  - many copies, high replicative index, oxidative microenvironment, no histones; most aged-cell mtDNA mutations come from POLG replication error rather than oxidative damage. DNA damage feeds inflammation via cytoplasmic chromatin fragments (CCF), escaped mtDNA, and transposon cDNA. Progeroid syndromes from repair defects: Werner, Bloom, xeroderma pigmentosum, trichothiodystrophy, Cockayne, Seckel.
2. **Telomere attrition** - TTAGGG tandem repeats, several to tens of kb, with a 75–300 nt G-rich 3' overhang forming the protective cap. Telomere dysfunction signals to mitochondria through p53 → PGC-1 $\alpha$  suppression. Measured molecularly (qPCR T/S ratio, Southern TRF, flow-FISH) or estimated from methylation (DNAmTL, PCDNAmTL).
3. **Epigenetic alterations** - histone PTMs, DNA methylation, ncRNA, chromatin remodelling. The direction of travel in aging is de-repression: global hypomethylation, heterochromatin relaxation, transposon (LINE-1/SINE) reactivation, active histone acetylation. Age-related methylation change concentrates in introns, intergenic regions, tumour-suppressor and Polycomb-target promoters. Cross-talk with known longevity pathways: histone demethylases act on IGF-1 signalling; sirtuins are NAD<sup>+</sup>-dependent deacetylases.
4. **Loss of proteostasis** - mistranslation, misfolding, incomplete products; slowed elongation; cumulative oxidative protein damage saturating chaperones. Drives ALS, Alzheimer's.
5. **Disabled macroautophagy** - autophagosome capture then lysosomal digestion. Selective subtypes: glycophagy, lipophagy, mitophagy, ER-phagy, nucleophagy, xenophagy, lysophagy. Autophagy-gene expression falls with age; genetic inhibition accelerates aging in models; CD4<sup>+</sup> T cells from exceptionally long-lived people show enhanced autophagic flux. Reduced flux permits aggregate accumulation, dysfunctional organelles, uncleared pathogens, and inflammasome activation. NAD<sup>+</sup> precursors have clinical-trial evidence in non-melanoma skin cancer chemoprevention, prediabetic insulin resistance, and Parkinson's neuroinflammation.

### Antagonistic

6. **Cellular senescence** - stable proliferative arrest via TP53→CDKN1A/p21 and CDKN2A/p16→RB1. SASP effects: chemokine/cytokine recruitment (CCL2, CXCL2, CXCL3, IL-18, IL-2, IL-6, IL-8); immune suppression (TGF- $\beta$ ); fibroblast activation and collagen deposition (TGF- $\beta$ , IL-11, PAI-1); ECM remodelling (MMPs); progenitor activation (EGF, PDGF); paracrine senescence (TGF- $\beta$ , TNF- $\alpha$ , IL-8). Lamin B1 depletion → SAHF (H3K9me3, HP1 $\alpha$  foci) → LINE-1 de-repression → cytosolic dsDNA → cGAS/STING and TLR activation.

7. **Deregulated nutrient sensing** - insulin/IGF-1 → PI3K-AKT and Ras-MEK-ERK; FOXO and E26 transcription factors; MTORC1 responding to glucose, amino acids (leucine), hypoxia, low energy, acting on SREBP and TFEB. Overnutrition activates MTORC1 and EP300 and inhibits AMPK, SIRT1, SIRT3; fasting and dietary restriction do the reverse and extend lifespan across models including primates.
8. **Mitochondrial dysfunction** - falling ATP output, rising ROS. POLG-mutator mice show fragmented cristae, disrupted outer membrane, and premature aging (alopecia, kyphosis, low body weight and subcutaneous fat, low BMD, anaemia, cardiomyopathy) mirroring telomerase-deficient, p53-hyperactive, and PGC1 $\alpha$ / $\beta$ -null mice.

### Integrative

9. **Stem cell exhaustion** - reduced steady-state renewal and injury repair. OSKM (OCT4, SOX2, KLF4, MYC) full reprogramming resets p16, telomere length, and the DNA methylation clock; partial/transient reprogramming rejuvenates clock, DNA damage, epigenetic patterns and transcriptome without identity loss.
10. **Altered intercellular communication** - pro-aging circulating factors: CCL11/eotaxin and  $\beta$ 2-microglobulin (reduce neurogenesis), IL-6 and TGF- $\beta$  (impair haematopoiesis), complement C1q (compromises muscle repair). A single old-blood transfusion ages young mice within days; dilution of old mouse blood with saline + 5% albumin rejuvenates multiple tissues.
11. **Chronic inflammation (inflammaging)** - rising CRP, IL-6; plasma IL-6 predicts all-cause mortality. Fed by CHIP, cytosolic DNA, SASP, uncleared debris, circadian disruption, gut barrier failure.
12. **Dysbiosis** - gut microbiota diversity declines; SCFA, vitamin, secondary bile acid and amino-acid-derivative production shifts; gut-brain and gut-organ signalling degrades.

**PIP framing** - each hallmark is Part of a complex process, Interdependent with the others, and a Point-of-entry for intervention. No single hallmark accounts for aging.

### What the clocks actually pick up

- Most age-related DNAm changes are small: ~1.5–5% absolute beta shift over decades. Clock signal is an aggregate of many tiny shifts, not a few large ones.
- ~20% of 450K loci can support *some* chronological clock (Porter 2021), so CpG identity across published clocks overlaps very little. Liu 2020 argues the clocks capture genuinely different aging axes rather than being noisy versions of one thing.
- CTCF-binding sites are enriched among top-ranked CpGs in AltumAge SHAP/DeepPINK analyses.
- Epigenetic drift: interindividual methylation variance rises with age (Hannum 2013).
- Men age ~4% faster than women on the Hannum clock.
- Epigenetic age is ~40% heritable (Horvath & Raj 2018).
- Clocks work in immortalised non-senescent B cells and embryonic cells, so they are not simply senescence detectors; they work in non-proliferating brain cells, so they are not simply division counters. Multi-tissue clocks trained on one tissue transfer to others; single-tissue and second-generation clocks may partly reflect cell composition.

### 3. Measurement chemistry, per omics layer

#### 3.1 DNA methylation

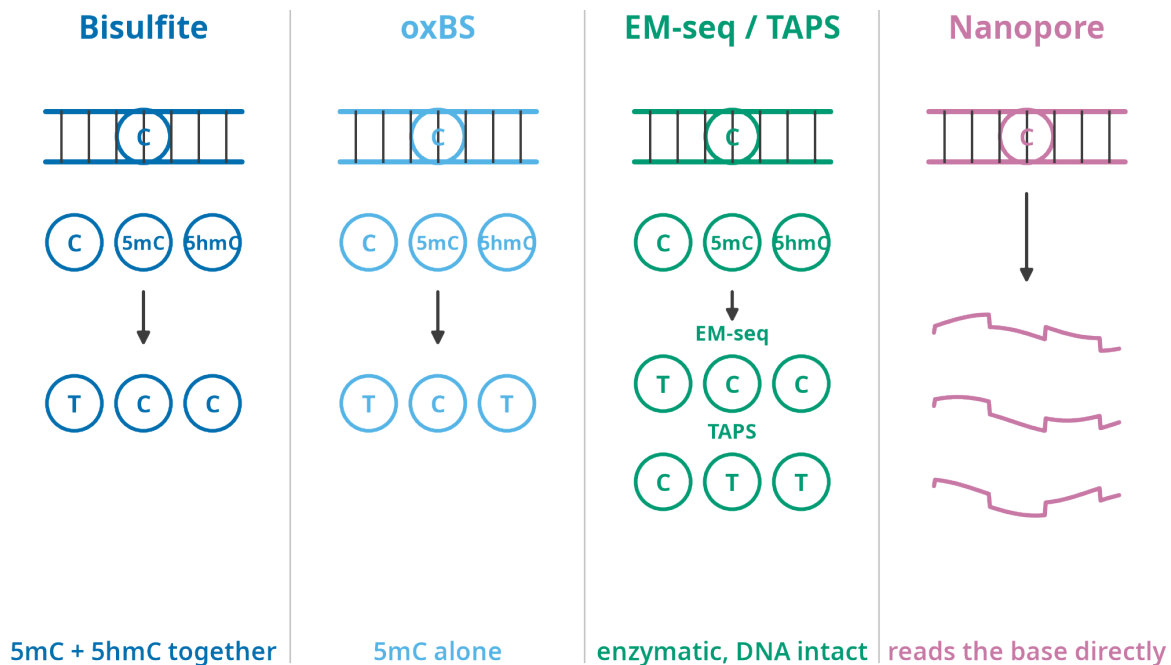


Figure 9: What each conversion chemistry does to an unmodified C, a 5mC and a 5hmC

The four columns answer the same question differently, and the differences are what a clock's coefficients were fitted through. Bisulfite protects 5mC and 5hmC alike, so a bisulfite beta is their sum and every array clock is trained on that sum. Oxidative bisulfite converts 5hmC to a form that then reads as T, which isolates 5mC. EM-seq and TAPS are both enzymatic and both leave the DNA intact, but they are inverses of each other at the readout: EM-seq reads a methylated cytosine as C, TAPS reads it as T. Nanopore skips conversion and calls the base from the current trace. A coefficient fitted on a bisulfite array does not transfer to a TAPS library without accounting for that.

**Chemistry.** 5-methylcytosine at CpG dinucleotides. In mammals 70–80% of CpG cytosines are methylated. Bisulfite treatment deaminates unmethylated cytosine to uracil (read as thymine) and leaves 5mC intact, creating a C/T polymorphism that every downstream assay queries. Enzymatic conversion (EM-seq, TAPS) is the low-damage alternative and is displacing bisulfite for sequencing.

**Beta value.** For a CpG,  $\beta = M / (M + U + \alpha)$  where M and U are methylated and unmethylated intensities and  $\alpha$  is a stabilising offset (Illumina default 100).  $\beta \in [0,1]$ . Nearly every clock consumes  $\beta$ . The M-value  $M = \log_2(\beta / (1 - \beta))$  is preferred for differential testing because it is homoscedastic, but clock coefficients are almost always in  $\beta$  space, do not silently swap.

**Array platforms.**

| Platform                                   | Probes                                | Notes                                                                                                                       |
|--------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Illumina 27K (HumanMethylation27)          | 27,578                                | Horvath 2013 and Bocklandt-era clocks                                                                                       |
| Illumina 450K                              | 485,512                               | The workhorse; most published coefficient sets                                                                              |
| EPIC v1 (850K)                             | 866,554                               | Drops ~40% of 450K probes; adds enhancers                                                                                   |
| EPIC v2                                    | 937,690 probes / 923,452 unique sites | Probe IDs carry suffixes ( <a href="#">cg00000029_TC21</a> ); replicate probes must be aggregated before applying any clock |
| Illumina MSA (Methylation Screening Array) | ~270K                                 | Targeted, cheaper; emerging                                                                                                 |
| HorvathMammalMethylChip40                  | ~37K conserved loci                   | 128+ mammalian species, 59 tissues; basis of the universal mammalian clocks                                                 |
| HorvathMammalMethylChip320                 | expanded                              | Used by EnsembleAge                                                                                                         |

**Probe chemistry inside the array.** Infinium Type I uses two beads per CpG (one for the methylated allele, one for unmethylated) in a single colour channel; Type II uses one bead with two-colour single-base extension. Type II is ~85% of probes on EPIC and has a compressed dynamic range and different beta distribution. This design bias is what BMIQ corrects.

**Sequencing assays.** - **RRBS** - MspI digestion enriches CpG-dense regions; sparse, integer-count coverage. Petkovich (90 CpGs, mouse blood), Stubbs (multi-tissue mouse), Meer, Thompson all use RRBS. - **WGBS** - full coverage, expensive; the substrate for scAge/scEpiAge single-cell work. - **Targeted bisulfite sequencing** - HepatoXu (10 loci, plasma cfDNA HCC detection). - **scM&T-seq** / **snmC-seq** - single-cell methylation, binary and extremely sparse (often <10% of CpGs covered per cell), forcing entirely different algorithms.

**Normalisation pipeline (array).** IDAT → background correction and dye bias ([noob](#)) → between-array ([funnorm](#), quantile, or [SeSAMe](#)) → probe-type bias ([BMIQ](#) or [SWAN](#)) → detection-p filtering, cross-reactive/SNP probe masking, sex-chromosome handling. Consensus from Illumina EPIC comparisons: [noob](#) + [BMIQ](#) and [SeSAMe](#) perform well; [funnorm](#) is preferred when global methylation differences are expected between groups. [sesame\\_450k\\_median.csv](#) in [ComputAge](#) is the field's standard NA-fill vector ([ComputAge clocks dir](#)).

**Cell composition.** Whole blood is a mixture; a clock trained on bulk blood partly reads cell proportion. Two standard reference libraries: Reinus/Houseman 6-cell (450K, [450K\\_reinius\\_12\\_reference.csv](#)) and Salas/IDOL 12-cell EPIC ([EPIC\\_salas\\_18\\_reference.csv](#)). Deconvolution is a constrained regression onto a reference signature matrix; [pyaging](#) exposes 6-cell and 12-cell EPIC deconvolution as clocks in their own right, returning proportions.

**Sex estimation from methylation.** X and Y chromosome probe z-scores against the autosomal mean and SD, projected on PCA loadings, implemented as [XChrom](#) (4,047 probes) and [YChrom](#) (284 probes) in [pyaging\\_models.py](#) lines 1338–1399. Needed because GrimAge, DNAmFitAge and the mammalian clocks take sex as an input.

### 3.2 Transcriptomics

- **Bulk RNA-seq.** Raw counts → library-size normalisation. Three conventions appear in published clocks:

- ▶ **TPM then log1p** - divide by transcript length, rescale to 1e6, `log1p`. Used by OcampoATAC1/2 in `pyaging`.
- ▶ **CPM then binarise** - BiT age (*C. elegans*).
- ▶ **RLE (edgeR `calcNormFactors(method="RLE")`) scaled to 1e7, then `log10(x+1)` - tAge.**
- **Microarray expression** - treat as already normalised; rank or quantile align to the RNA-seq reference distribution.
- **Gene identifier space**. tAge maps everything into **mouse Entrez** space via ortholog tables for human, rat, macaque (`tAge inst/extdata/metadata/`). Pasta/REG use human Ensembl and derive **PastaMouse** by selecting **ENSMUSG\*** features from an 8,113-gene joint space and filling human-only genes with reference values.
- **Single cell**. Sparsity and dropout break bulk assumptions. Three strategies in the literature:
  - ▶ **Pseudobulk** - average n cells per (donor, cell type), repeated B times. `scImmuAging` uses size 15, n = 100 repeats (`scImmuAging/R/main.R`); `SpatialAgingClock` uses n = 20, B = 100.
  - ▶ **Binarisation** - CellBiAge: after log-normalisation, CCA batch integration and scaling, set entry = 1 if scaled value > 0 else 0, on the top 2,000 HVGs. This is the single largest performance lever in that paper (see §4.11).
  - ▶ **Spatial smoothing** - propagate expression over a spatial neighbour graph before prediction (see §4.13).

### 3.3 Chromatin

- **Histone-mark ChIP-seq**. Signal is read from bigWig files, averaged over each Ensembl gene body (chr 1–22, X), then `arcsinh`-transformed (`pyaging bigwig_to_df`). Eight marks have published clocks: H3K4me1, H3K4me3, H3K9ac, H3K9me3, H3K27ac, H3K27me3, H3K36me3, plus a pan-histone model. All are PCA + elastic net + ARD.
- **ATAC-seq**. Counts per open chromatin region (OCR) → length-normalised TPM → `log1p` → subset to the model's OCRs. OcampoATAC1 (228 OCRs) and OcampoATAC2 (380 OCRs), PBMC.

### 3.4 Proteomics

- **Olink Explore** (proximity extension assay). Output is **NPX**, a log2-scaled relative quantification. Explore 3072 vs Explore 1536 have different panels; `organAging` ships separate full and feature-reduced models for each. Application is a plain dot product of NPX with coefficients, plus intercept for first-generation models (`organAging README`).
- **SomaScan** (aptamer). Used by Oh et al. 2023 organ clocks (~5,000 proteins, ~5,700 people).
- **Mass spectrometry**. For any non-Olink platform, `organAging`'s recommendation is: transform to a roughly symmetric distribution (log intensities), z-score each protein, then multiply by the Olink NPX standard deviations published in their Table S3 before applying coefficients.
- **Organ-enrichment definition**. A protein is assigned to an organ if its GTEx expression in that tissue is  $\geq 4\times$  the mean across all other tissues. `organAging`'s feature sets from `organAging GTEx_4x_FC_genes.json`: Adipose 5, Adrenal 4, Artery 14, Brain 118, Esophagus 5, Female 6, Heart 7, Immune 124, Intestine 31, Kidney 9, Liver 79, Lung 11, Male 55, Muscle 20, Pancreas 29, Pituitary 12, Salivary 10, Skin 17, Stomach 12, Thyroid 2, Organismal 2,347, Multi-organ 569, Conventional 2,916.

### 3.5 Clinical chemistry and haematology

The PhenoAge / KDM / HD family. Canonical nine PhenoAge markers plus age:

| Marker  | Unit (Levine) | Alternative unit |
|---------|---------------|------------------|
| Albumin | g/L           | g/dL             |



| Marker                      | Unit (Levine)            | Alternative unit |
|-----------------------------|--------------------------|------------------|
| Creatinine                  | μmol/L                   | mg/dL            |
| Glucose                     | mmol/L                   | mg/dL            |
| C-reactive protein          | ln(mg/dL)                | ln(mg/L)         |
| Lymphocyte percent          | %                        |                  |
| Mean cell volume            | fL                       |                  |
| Red cell distribution width | %                        |                  |
| Alkaline phosphatase        | U/L                      |                  |
| White blood cell count      | 10 <sup>3</sup> cells/μL |                  |

Unit mismatch is the number-one source of wrong PhenoAge values in the wild, see §16.

The BioAge R package extends this to a larger marker menu with clinically-healthy windows used to build the HD reference cohort ([BioAge hd\\_nhanes.R](#)): albumin, ALP, ln(CRP) with CRP < 2, total cholesterol < 200, ln(creatinine), HbA1c 4–5.6, SBP < 120, DBP < 80, mean BP < 93.33, BUN, uric acid (sex-specific 2.7–6.3 F / 3.7–8.0 M), lymphocyte % 20–40, MCV, WBC 4.5–11, RBC (sex-specific), RDW 11.5–14.5, monocytes 3–10, neutrophils 45–74, cystatin C 0.51–0.98, GGT (sex-specific), insulin 2.52–24.1, HDL (sex-specific), LDL 80–130, triglycerides 54–110, pulse 60–100, FEV1 ≥ 0.8 × mean, cadmium 2.7–10.7, vitamins A/E/B12/C.

### 3.6 Other layers

- **Glycomics.** IgG N-glycan composition. GlycanAge from 5,117 European individuals; three glycans (GP6 non-galactosylated, GP14 and GP15 bis-galactosylated) explain ~60% of CA variance; prediction error ~9.7 years. Notable as the first clock class with repeated demonstrations of reversal by lifestyle intervention.
- **Metabolomics.** MetaboAge and the metabolite arm of GOLD BioAge / StackAge / OMICmAge. Nightingale NMR and LC-MS panels dominate; features are usually log-transformed and z-scored.
- **Microbiome.** Deep-learning microbiome clocks exist (Galkin 2018) but predictive accuracy is poor relative to methylation; treat as exploratory.
- **Imaging.** MRI-based multi-organ clocks (Nature Medicine 2025) and retinal/face age. Out of scope for a molecular tool but worth a plug-in slot.
- **EHR / clinical composite.** EMRAge (the target OMICmAge is trained on), frailty index (FI), intrinsic capacity (dynamic clock predicts it from methylation).
- **Telomere length.** Direct assays (qPCR T/S, TRF Southern, flow-FISH, TelSeq from WGS) and DNAm estimators (DNAmTL 140 CpGs → kb; PCDNAmTL → base pairs).

## 4. Algorithm families with full math

### 4.1 Penalised linear regression, the field's default

Objective for elastic net over samples  $i$  and features  $j$ :

$$\min_{\beta} \quad (1/2n) \sum_i (y_i - \beta_0 - \sum_j x_{ij} \beta_j)^2 + \lambda \left[ (1-\alpha)/2 \cdot \sum_j \beta_j^2 + \alpha \cdot \sum_j |\beta_j| \right]$$

$\alpha = 1$  is LASSO,  $\alpha = 0$  is ridge. Horvath used  $\alpha = 0.5$ . Application at inference is a dot product:

$$\text{score} = \beta_0 + \sum_j \beta_j x_j$$

then a clock-specific output transform (§5). Of the 173 clocks in pyaging's registry, 64 are elastic net, 20 LASSO, 5 plain linear regression, 1 ridge, 1 BLUP.

**BLUP** (Zhang 2019) treats all 319,607 probes as random effects with a common variance,  $\beta = (X'X + \lambda I)^{-1}X'y$  in the mixed-model sense. Requires the full probe set, so it degrades less than elastic net when many probes are missing, and Zhang's own script says so explicitly ([DNAm-based-age-predictor pred.R](#)). Crucially, both Zhang models **z-score within each sample across probes** before the dot product:

```
dataNona.norm <- apply(dataNona, 1, scale) # per-individual scaling, probes x individuals
```

## 4.2 PC clocks

Higgins-Chen 2022. Instead of regressing age on individual CpGs, regress on the top principal components of a large CpG block (78,464 probes for most PC clocks, 125,175 for SystemsAge). This averages away probe-level technical noise and raises test-retest ICC dramatically.

$$\text{score} = ((x - \mu) W) \beta + \beta_0$$

where  $\mu$  is the training centre vector,  $W$  the  $p \times k$  rotation (1,933 PCs for PCGrimAge),  $\beta$  the regression weights on PCs. Implemented as `PCLinearModel` ([pyaging\\_base\\_models.py:72](#)). The storage cost is the rotation matrix - 78,464  $\times$   $k$  floats, which is why PC clocks are the memory-heavy members of any registry.

## 4.3 Survival models

**Cox proportional hazards.**  $h(t|x) = h_0(t) \exp(x'\beta)$ . The linear predictor  $x'\beta$  is a log-hazard, unbounded, not in years. GrimAge, GrimAge2, CpGPTGrimAge3, SenMortalityAge and the organAging mortality models all output this.

**Cox  $\rightarrow$  years.** GrimAge standardises the Cox score against training-set moments then rescales to the training age distribution:

$$\text{years} = ((\text{cox} - \text{cox\_mean}) / \text{cox\_std}) \times \text{age\_std} + \text{age\_mean}$$

GrimAge v1: cox\_mean 13.20127, cox\_std 1.086805, age\_mean 59.63951, age\_std 9.049608.  
GrimAge2: cox\_mean 15.370829484122, cox\_std 1.09534876966487, age\_mean 66.0943807965085, age\_std 9.05974444998421 ([pyaging\\_models.py:1683,1769](#)).

**Gompertz mortality  $\rightarrow$  years.** The PhenoAge route (§5.1) and the organAging route. organAging's published human Gompertz fit on the log scale: intercept  $-9.94613787413831$ , slope  $0.0897860500778604$ , mean cohort age 57.29426 y. Convert a relative log-hazard  $r$  to years:

$$\begin{aligned} \text{avg\_r} &= \text{intercept} + \text{slope} \times \text{mean\_age} & \# &= -4.801912 \\ \text{years} &= (-\text{avg\_r} + r) / \text{slope} - \text{intercept} \end{aligned}$$

**Accelerated failure time.** organAging also fits `WeibullAFTFitter(penalizer=0.01)` (lifelines) for time-to-death models ([organAging\\_clock\\_utils.py](#)).

**Cox elastic net.** `CoxnetSurvivalAnalysis(alpha_min_ratio="auto", n_alphas=50)` from scikit-survival, grid-searched over `l1_ratio`  $\in \{0.5, 1.0\}$  with 3-fold CV - organAging's mortality models.

#### 4.4 Kleméra–Doubal method (KDM)

The most statistically principled of the clinical-chemistry methods. Fit one univariate regression of each biomarker on age, then invert them all jointly with precision weights.

For biomarker  $j$ :  $x_j = q_j + k_j \cdot CA + \varepsilon_j$ , with residual SD  $s_j$  and  $R^2$   $r_j$ .

$$BA\_E = \frac{\sum_j (x_j - q_j)(k_j / s_j^2)}{\sum_j (k_j / s_j)^2}$$

KDM corrects  $BA\_E$  toward CA using the residual variance:

$$BA = \frac{[\sum_j (x_j - q_j)(k_j/s_j^2) + CA / s_{BA}^2]}{[\sum_j (k_j/s_j)^2 + 1/s_{BA}^2]}$$

$s_{BA}^2$  is estimated as  $s^2 - s_r$  where  $s^2$  is the observed variance of  $(BA\_E - CA)$  and

$$\begin{aligned} r\_char &= \frac{\sum_j |(k_j/s_j)\sqrt{r_j}|}{\sum_j |k_j/s_j|} \\ s_r &= ((1 - r\_char^2)/r\_char^2) \times ((age\_max - age\_min)^2 / (12\ m)) \end{aligned}$$

with  $m$  the number of biomarkers. Verbatim in [BioAge kdm\\_calc.R](#). BioAge trains separately by sex on NHANES III aged 30–75, non-pregnant, and projects into NHANES IV. Samples with more than 2 missing biomarkers are set to NA. Output columns: `kdm`, `kdm_advance = kdm - age`.

ComputAge also ships a *de novo* KDM estimator ([ComputAge deage/kdm.py](#)) with forward feature selection, LASSO preselection, 10-fold CV, RSE weighting, and stability testing, useful as a template for letting users fit their own KDM on arbitrary biomarker panels.

#### 4.5 Homeostatic dysregulation (HD)

Mahalanobis distance from a young reference cohort. Standardise both the reference and target matrices using the reference means and SDs, then

$$HD\_i = \sqrt{(x_i - \mu_{ref})' \Sigma_{ref}^{-1} (x_i - \mu_{ref})}$$

Reported both raw (`hd`, scaled by its own SD) and logged (`hd_log = log(HD)/sd(log HD)`). The reference must be young (typically 20–30), non-pregnant, and within clinically normal biomarker windows; train separately by sex. [BioAge hd\\_calc.R](#). Unlike KDM and PhenoAge, HD has no age units, it is a dimensionless distance, and it does *not* need to be residualised against CA the same way.

#### 4.6 Mitotic / replicative clocks

- **epiTOC1** - plain mean  $\beta$  over 385 “ground-state hypomethylated” Polycomb-target CpGs. Missing sites encoded as  $-1$  and excluded from the mean ([pyaging \\_models.py:1988](#)).
- **epiTOC2** - an explicit stem-cell division model. Given per-CpG methylation-transmission rate  $\delta_c$  and ground-state  $\beta_{0c}$ :

$$tnsc = (2/k) \sum_c (\beta_c - \beta_{0c}) / (\delta_c (1 - \beta_{0c}))$$

$k$  CpGs (163 for epiTOC2, 170 for epiTOC3). Units: estimated cell divisions per stem cell. [pyaging \\_models.py:2008](#).

- **stemTOC** - 0.95 quantile of  $\beta$  over 371 (or 629 in-vitro) CpGs rather than the mean, which makes it robust to cell-type dilution.
- **HypoClock** - mean  $\beta$  over 678 solo-WCGW hypomethylating sites, returned as `1 - mean`.
- **epiCMIT-hyper** / **epiCMIT-hypo** - mean over 184 hypermethylating and 1,164 hypomethylating B-cell CpGs.
- **RepliTali** - elastic net on 87 EPIC CpGs predicting population doublings in cultured cells; **RepliTaliNorm** is its 218-CpG upstream normaliser exposed separately.

#### 4.7 GrimAge architecture

Two-stage. Stage 1: eight (v1) or ten (v2) independent elastic nets predict plasma protein concentrations and smoking pack-years from methylation. Stage 2: a Cox model on those DNAm surrogates plus age and sex.

v1 components: PACKYRS, ADM (adrenomedullin), B2M, CystatinC, GDF15, Leptin, PAI-1, TIMP-1. v2 adds logCRP and logA1C. Feature counts (pyaging registry): ADM 187, B2M 92, CystatinC 88, GDF15 138, Leptin 187, logA1C 87, logCRP 132, PACKYRS 173, PAI-1 211, TIMP-1 43; the composite model touches 1,032 CpGs plus age and female indicator.

Forward pass ([pyaging \\_models.py:1656](#)), concat order matters and is fixed: [GDF15, B2M, CystatinC, TIMP1, ADM, PAI1, Leptin, PACKYRS, (LogCRP, A1C), Age, Female].

**PCGrimAge** inserts a PCA step: centre 78,464 CpGs, rotate to 1,933 PCs, append Female and Age, then run eight PC-based sub-clocks and the Cox layer.

**CpGPTGrimAge3** / **CpGPTPCGrimAge3** (2024) replace most sub-clocks with CpGPT-derived protein proxies. Required inputs: age, seven GrimAge2 proxies ([grimage2timp1](#), [grimage2packyrs](#), [grimage2logcrp](#), [grimage2adm](#), [grimage2leptin](#), [grimage2gdf15](#), [grimage2pai1](#)) and 16 CpGPT protein proxies ([cpgpt\\_s100a9](#), [tnfrsf13c](#), [tgfb1](#), [tek](#), [ccl14](#), [tnfsf15](#), [lilrb2](#), [tnf](#), [chit1](#), [postn](#), [il34](#), [pdcd1](#), [cst3](#), [cxcl2](#), [gzma](#), [il5](#)). 24 features total for the non-PC variant, 31 for the PC variant (which rotates 30 → 29 PCs). Trained on Framingham Heart Study whole blood, Illumina 450K, mortality endpoint (author clarification recorded in [pyaging audit\\_report.md](#)).

#### 4.8 DunedinPACE and DunedinPoAm

**DunedinPoAm38** (2020) - 46 CpGs, elastic net on the 12-year, 3-timepoint Pace of Aging from the Dunedin cohort. In-sample  $r = 0.6$ .

**DunedinPACE** (2022) - 173 scoring CpGs, elastic net on the 20-year, 4-timepoint Pace of Aging; CpGs restricted to probes with test-retest ICC > 0.4 ( $n = 81,239$  candidates). In-sample  $r = 0.78$ . The published algorithm quantile-normalises the sample against a **gold-standard mean vector** before scoring. pyaging implements the official 20,000-probe panel: 173 scoring CpGs plus 19,827 background probes used only to define the sample's rank distribution ([pyaging \\_models.py:238](#)):

```
sorted_gold = sort(reference_values)                # length 20000
q_idx       = linspace(0, len(sorted_gold)-1, x.shape[1]) # rank → gold quantile map
for each sample row:
    normalized[argsort(row)] = sorted_gold[q_idx]
return normalized[:, scoring_indices]
```

This is a per-sample rank transform, so **the background probes must be present** - dropping them silently changes the answer.

#### 4.9 Mammalian universal clocks

Three variants, all elastic net on the HorvathMammalMethylChip40, each with a different age parameterisation and a species lookup table (AnAge: gestation time, average age at sexual maturity, maximum lifespan).

- **Clock 1** - predicts  $\log(\text{age} + 2)$ ; inverse age =  $\exp(\hat{y}) - 2$ .
- **Clock 2** (relative age), predicts a log-log-transformed relative age. Inverse:

```
rel    = exp(-exp(-ŷ))
age    = rel × (max_age + gestation) - gestation
```

- **Clock 3** (log-linear age relative to maturity), with  $\hat{m} = 5$  (gestation / maturity)<sup>0.38</sup>:

```
ŷ ≥ 0 : age = m̂ (maturity + gestation)(ŷ + 1) - gestation
ŷ < 0 : age = m̂ (maturity + gestation) exp(ŷ) - gestation
```

Species is passed as a one-hot block appended to the CpG vector (1,756 species for clock-2 variants, 1,707 for clock-3 variants); the model splits the input tensor by these widths ([pyaging\\_models.py:513-771](#)). Also in this family: MammalianLifespan (predicts species maximum lifespan, [exp](#) link) and MammalianFemale (sex classifier, sigmoid).

#### 4.10 Deconvolution as a model

Reference-based, constrained. Given a signature matrix  $S$  (cell types  $\times$  CpGs) and a sample  $\beta$  vector, solve for proportions  $p$  with a non-negativity and sum-to-one constraint. `pyaging`'s implementation uses the pseudo-inverse followed by a **Euclidean projection onto the probability simplex** ([pyaging\\_models.py:1269](#)):

```
p_raw = x @ pinv(S).T
sorted_v = sort(p_raw, descending)
cssv    = cumsum(sorted_v)
rho      = max{ i : sorted_v[i] - (cssv[i] - 1)/(i+1) > 0 }
theta    = (cssv[rho] - 1) / (rho + 1)
p        = clamp(p_raw - theta, min=0)
```

This is exact, differentiable almost everywhere, and vectorises across a batch, a good pattern for a GPU tool. 6-cell (600 CpGs, Salas/Reinius) and 12-cell (240 CpGs, Salas EPIC) libraries.

#### 4.11 Binarisation clocks

Two independent lines converged on the same idea: with sparse count data, the *identity* of expressed genes carries more robust age signal than their levels.

- **BiT age** (*C. elegans*, Meyer & Schumacher 2021). Per sample: mask zeros, compute the median of non-zero CPMs, set gene = 1 if CPM > median else 0. Score =  $\Sigma$  (binary  $\times$  elastic-net coef)
  - intercept. Intercept 103.54631743289005 for *C. elegans* (hours); 59.626 for human fibroblasts. 576 predictor genes. [AgingClock biological\\_age\\_prediction.py](#).

- **CellBiAge** (mouse brain single cell, Yu et al., *Cell Reports* 42:113500, Dec 2023). Pipeline: log-normalise → CCA batch integration → scale (mean 0, SD 1 per gene) → **binarise at 0** (scaled value > 0 → 1). Feature space: top 2,000 highly variable genes (HVGs beat highly expressed genes decisively; 1,413 shared between train/test batches in the hypothalamus data). The mean threshold beat BiT age’s median threshold on their test data.

CellBiAge numbers worth carrying into a benchmark suite. Metric is AUPRC (test sets are class-imbalanced), models are elastic-net logistic regression, random forest, gradient boosting, SVM, XGBoost and an MLP, tuned with GridSearchCV / KerasTuner under group CV:

| Setting                       | Before binarisation | After |
|-------------------------------|---------------------|-------|
| Hypothalamus, ElasticNet      | 0.60                | 0.94  |
| Hypothalamus, all five models | ~0.65               | ~0.96 |
| SVZ control, MLP and ELN      | 0.67                | 0.90  |
| Train/test batch swapped      | -                   | 0.94  |
| Top 80 genes only             | -                   | ~0.95 |
| Top 40 genes only             | -                   | 0.90  |

Data: female C57BL/6 hypothalamus snRNA-seq, 40,064 nuclei, 3 vs 24 months, two batches on different chemistry (10x v3 + NovaSeq for train, v2 + HiSeq for test); male SVZ scRNA-seq, 79,123 cells, 15 animals, 6 vs 23 months, sedentary vs 5 weeks voluntary wheel running. Exercise significantly reduced predicted-aged probability in young-animal neuroblasts ( $p = 0.027$ ) and activated NSC/NPCs ( $p = 0.031$ ); no effect in aged animals or in the all-cell pooled analysis. Only 4 genes (*Nes*, *Ly6a*, *Slc5a4b*, *B230312C02Rik*) were shared among the top 200 features across all cell types. CellBiAge slightly outperformed the Buckley 2023 bootstrap clock on the three shared cell types while avoiding the 100-pseudo-cell bootstrap.

#### 4.12 Rank-based transcriptomic clocks

Pasta and REG (2025) fill NaNs with the global median, then rank-normalise **within each sample** with average ranks for ties, then apply a ridge-logistic (Pasta) or ridge (REG) linear layer over 8,113 genes. Pasta’s output transform is  $x \cdot \text{scale} + \text{offset} \cdot \text{scale}$ ; REG’s is  $x + \text{intercept}$ . Rank normalisation makes the model invariant to any monotone per-sample transformation, which is why these clocks work across RNA-seq and microarray without re-normalising. [pyaging\\_models.py:1037–1189](#).

#### 4.13 Spatial single-cell clocks

Sun et al., *Nature* 638:160–171 (2025). Per cell type (18 brain cell types), a LASSO model with 5-fold CV over 20 alphas, fitted on spatially smoothed expression.

**SpatialSmooth.** Build a spatial graph per animal (fixed  $k = 20$  neighbours, binary adjacency  $S$ ), then iterate to convergence:

$$X_{t+1} = (1 - \alpha) X_0 + \alpha S X_t, \quad \alpha = 0.8, \text{ max\_iter} = 30, \text{ tol} = 1e-2$$

This is label propagation applied to expression. Missing genes are imputed with per-gene training means (or SpaGE, which projects from a reference scRNA-seq atlas via shared principal vectors). Age acceleration is defined **within animal and cell type**, as prediction minus the mean prediction for that (animal, cell type) pair, which cancels the animal-level batch and calendar age. The package also computes **cell proximity effects**: for each effector cell type, split target cells into “near” and “far” sets by distance cutoff



(optionally a ring of given width), then t-test the age acceleration difference. Outputs a data frame with `Near Cell`, `AgeAccel Cell`, cutoff multiplier `n`, `t`, `p`, `Aging Effect`, `Near Freq`, `Near Num`.

#### 4.14 Single-cell methylation clocks

- **scAge** (Trapp 2021, Python). Uses the bulk CpG–age linear relationships as priors and computes, for each cell, the age maximising the product of per-CpG likelihoods given the binary methylation call. Handles arbitrary, non-overlapping CpG coverage per cell.
- **scEpiAge** (Bonder 2024, *Nat Commun* 15:7567). Inverts the regression: predicts methylation *from* age, so cells need not share CpGs. Mouse blood 10–101 weeks; DNAm-age distribution is wider than technical noise predicts, and CD4+/CD8+ T cells read younger than B cells. Cell-type effect exceeds age effect, a warning for any single-cell aging score.

#### 4.15 Deep learning

- **AltumAge** (2022) - MLP over 20,318 CpGs shared across 27K/450K/EPIC. Architecture from `pyaging_base_models.py:89`: `BatchNorm(20318) → Linear(20318,32) → SELU → [BatchNorm(32) → Linear(32,32) → SELU] × 4 → BatchNorm(32) → Linear(32,1)`. `BatchNorm` eps 0.001, momentum 0.99 (Keras defaults, preserved on port). Preprocessing is a **robust scale**:  $(x - \text{median}) / \text{std}$  with training-set statistics.
- **CpGPT** (Camillo 2024), transformer pretrained on >1,500 datasets, >100,000 samples. Encodes sequence context, genomic position and epigenetic state per CpG; imputes and reconstructs genome-wide profiles from partial input; transfers across platforms, tissues, mammalian species and single-cell data. Fine-tuned heads give SOTA mortality and chronological-age prediction. Code: [github.com/lucascamillomd/CpGPT](https://github.com/lucascamillomd/CpGPT).
- **GraphAge** (PNAS Nexus 2025) - GNN over a CpG regulatory graph; interprets which CpG interactions matter and how their importance shifts with age.
- **DeepStrataAge** (npj Aging 2026) - DNN on 12,234 CpGs, MAE 1.89 y across >20,000 blood methylomes, with SHAP attributions revealing sex-specific “epigenetic waves”.
- **EpInflammAge** - epigenetic + inflammatory markers, MAE ~7 y,  $r = 0.85$  in healthy controls.
- **XAI-AGE** - pathway-structured network giving pathway-level activity as an interpretable intermediate.
- **PointNet** - used by `scImmuAging` alongside LASSO and random forest for PBMC cell-type clocks.

#### 4.16 SystemsAge

Nature Aging, September 2025 (Higgins-Chen & Levine). One blood methylome → 11 system-specific ages plus a composite. Forward pass (`pyaging_models.py:2617`):

1. Centre 125,175 CpGs and rotate to DNAm PCs.
2. Project PCs onto a `system_vector` to obtain system components.
3. Per system, a linear module over its component indices → 11 system scores.
4. Separately, a PC → predicted-age model, with a quadratic polynomial correction  $a + b \cdot \hat{y} + c \cdot \hat{y}^2$ , then divide by 12 (months → years).
5. Concatenate the 11 system scores with predicted age; for the composite, run a second PCA-based model over that 12-vector.
6. Affine calibration per output:  $((\text{raw} - c_0)/c_1) \times c_3 + c_2$ , then /12.

Systems: Blood, Brain, Inflammation, Heart, Hormone, Immune, Kidney, Liver, Metabolic, Lung, MusculoSkeletal (indices 0–10 in the `pyaging` classes), with index –1 for the composite. Benchmarking by Sehgal et al. found SystemsAge held the highest technical reliability across array positions and DNA extraction protocols among 18 clocks.

#### 4.17 Ensembling and stacking

- **EnsembleAgeStatic / EnsembleAgeStaticTop** (mouse, 2025), elastic net over 288 / 431 CpGs designed to be responsive to lifespan-modulating interventions.
- **EnsembleAgeHumanMouse** - 100 CpGs on the mammalian array, cross-species relative age.
- **StackAge** (2026), ensemble over UK Biobank plasma proteomics + metabolomics,  $n = 30,376$ ; improves risk prediction for 12 chronic diseases over single-omic clocks.
- **DNAmFitAge** - a Kleméra–Doubal composite over four DNAm-predicted fitness measures. It fits sex-specific models. The four inputs are standardised with published slopes and intercepts ([pyaging\\_models.py:1875](#)):

| Input   | Female (offset, slope) | Male (offset, slope)    |
|---------|------------------------|-------------------------|
| VO2max  | 46.825091, -0.13620215 | 49.836389, -0.141862925 |
| Grip    | 39.857718, -0.22074456 | 57.514016, -0.253179827 |
| Gait    | 2.508547, -0.01245682  | 2.349080, -0.009380061  |
| GrimAge | 7.978487, 0.80928530   | 9.549733, 0.835120557   |

Each is transformed  $(\text{value} - \text{offset})/\text{slope}$ , then a sex-specific KDM layer produces the age.

#### 4.18 tAge, relative transcriptomic clocks

Tyshkovskiy et al., *Nature* 654:173–188 (2026), from >11,000 transcriptomes across >25 tissues in mouse, rat, macaque and human. 60 distributed models:  $\{\text{EN, BR}\} \times \{\text{Chronological, Mortality, Normalized age}\} \times \{\text{Mouse, Multispecies, Rodents}\} \times \{\text{Multi-Tissue, Brain, Kidney, Liver, Skeletal Muscle}\} \times \{\text{Scaled, YuGene}\}$ .

Preprocessing chain ([tAge preprocessing.R](#)):

1. `filter_genes` - keep genes with count  $\geq 10$  in  $\geq 20\%$  of samples.
2. `map_genes` - to mouse Entrez space via ortholog tables.
3. `RLE_normalization` - `sweep(counts, 2, libsize × calcNormFactors(RLE), "/") × 1e7`.
4. `log_transform` - `log10(x + 1)`.
5. `scale_eset` - **column-wise** z-score, i.e. per sample across genes.
6. `YuGene` - per sample: shift to non-negative, sort descending, cumulative proportion, carry forward for ties, `value = 1 - cumprop`. A non-parametric rank/uniformisation.
7. `.align_to_gene_list` - reorder to the reference gene list, pad missing genes with NA (not zero) so the model's own `SimpleImputer` fills them with training medians.
8. `control_subtraction` - subtract the per-gene median of a reference group, or of all samples when no reference is given. **All distributed clocks are relative (`_diff`) models; centring is never skipped.**

Output units, from the clock registry ([tAge clocks\\_metadata.csv](#)):

| Outcome        | Units                            | Rescaled by max lifespan?    |
|----------------|----------------------------------|------------------------------|
| Chronological  | years (human) / months (rodents) | Yes - × species max lifespan |
| Mortality      | $\log_{10}(\text{hazard ratio})$ | No                           |
| Normalized age | fraction of max lifespan         | No                           |

Species maximum-lifespan factors: human 122.5, mouse 48, rat 50.4, monkey 39 ([tage\\_predict.py](#)). Mortality clocks must never be multiplied by these, a mistake the registry flag `lifespan_scaled` exists to prevent.



The paper’s biology: genes for senescence, inflammation and apoptosis go up with age across species and cell types; wound healing, differentiation and ECM synthesis go down. CDKN1A and LGALS3 are universal markers whose protein levels also track mortality and multimorbidity in UK Biobank. Modules (inflammation, energy production, ECM organisation) get their own clocks so an intervention’s primary axis of action can be read off.

#### 4.19 SCALE - literature-informed single-cell aging score

Mao et al. A weighted, expression-detection-frequency-scaled z-score over a curated aging gene set (`4_SCALE_score.R`):

```
weights = c(rep(1,50), rep(-1,50)) *
            rowSums(counts[aging_genes,] > 0) / ncol(counts) # detection frequency
z        = t(scale(t(logdata[aging_genes,])))                # gene-wise z across cells
score    = t(z) %*% weights
```

The 50 up / 50 down genes per tissue come from glmnet selection over a union of curated aging resources: GenAge, CSGene, AgeFactDB, Aging Atlas, GO:aging, GO:CellAge, plus manual curation. Applied to Tabula Muris Senis across 20+ tissues. Validated against single-cell entropy and somatic mutation accumulation, and against transcriptomic and methylation clocks in muscle stem cells, Alzheimer’s, and heterochronic parabiosis. Correlation with age is assessed four ways: Pearson, distance correlation (`energy::dcor`), MIC (`minerva`), and nonlinear correlation (`nlcor`), a good template for reporting monotone-but-nonlinear age relationships.

#### 4.20 Transform reference table

Every clock’s output transform, extracted from `pyaging_postprocessing.py` and `pyaging_models.py`:

| Name                                   | Formula                                                                               | Used by                                                                                                      |
|----------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| identity                               | $y$                                                                                   | most elastic-net clocks                                                                                      |
| <code>anti_log_linear</code> (Horvath) | $y < 0 : 21 \cdot e^y - 1; y \geq 0 : 21 \cdot y + 20$                                | Horvath2013, SkinAndBlood, PedBE, Han, PCHorvath2013, PCSkinAndBlood, IntrinsicClock, CorticalClock, Pipek×3 |
| Horvath with adult_age 48 → years      | same with 49/48, then /12                                                             | Wu (children)                                                                                                |
| <code>anti_logp2</code>                | $e^y - 2$                                                                             | Mammalian1                                                                                                   |
| <code>anti_log</code>                  | $e^y$                                                                                 | MammalianLifespan                                                                                            |
| <code>anti_log_log</code>              | $e^{(-e^{-y})}$                                                                       | Mammalian2 relative-age step                                                                                 |
| <code>mortality_to_phenoage</code>     | see §5.1                                                                              | PhenoAge                                                                                                     |
| <code>cox_to_years</code>              | $((y - \text{cox\_mean}) / \text{cox\_std}) \cdot \text{age\_std} + \text{age\_mean}$ | GrimAge, GrimAge2, CpGPTGrimAge3, CpGPTPCGrimAge3                                                            |
| <code>petkovichblood</code>            | $\text{age} = ((y + 1.712) / 0.1666)^{(1/0.4185)} / 30.5$                             | Petkovich (mouse, months)                                                                                    |
| <code>stubbsmultitissue</code>         | $\text{age} = (\exp(0.1207y^2 + 1.2424y + 2.5440) - 3) \times 7/30.5$                 | Stubbs (mouse, months)                                                                                       |
| <code>days_to_months</code>            | $y / 30.5$                                                                            | Meer                                                                                                         |
| <code>days_to_weeks</code>             | $y / 7$                                                                               | Bohlin, EPICGA                                                                                               |

| Name            | Formula                 | Used by                                                                                                                                       |
|-----------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| sigmoid         | $1/(1+e^{(-y)})$        | McCartney BMI/education/cholesterol/body-fat, MammalianFemale, CVD-Westerman, ADBahadoSingh                                                   |
| one_minus       | $1 - y$                 | HypoClock                                                                                                                                     |
| add_constant    | $y + c$                 | REG, Lin (+12.2169841), PhenoAgeV1 in ComputAge (+60.664), YingCausAge (+86.80816381), YingDamAge (+543.4315887), YingAdaptAge (-511.9742762) |
| scale_and_shift | $y \cdot s + o \cdot s$ | Pasta, PastaMouse                                                                                                                             |

Note the intercept convention differs between packages: ComputAge/biolearn carry the intercept in the `transform` lambda, pyaging bakes it into the `nn.Linear` bias. A merged tool must pick one and normalise on import.

## 5. Exact published formulas and constants

### 5.1 PhenoAge (Levine 2018), clinical-biomarker version

Two published parameterisations circulate. They are not the same and give different answers.

**Variant A - SI-ish units (used by BioAge `orig=TRUE`, BioAge `phenoage_calc.R`):**

```
xb = -19.90667
  - 0.03359355 · albumin      (g/L)
  + 0.009506491 · creatinine (μmol/L)
  + 0.1953192   · glucose     (mmol/L)
  + 0.09536762  · ln(CRP)     (CRP in mg/dL)
  - 0.01199984  · lymphocyte% (%)
  + 0.02676401  · MCV         (fL)
  + 0.3306156   · RDW         (%)
  + 0.001868778 · ALP         (U/L)
  + 0.05542406  · WBC         (103/μL)
  + 0.08035356  · age         (years)

m      = 1 - exp( (-1.51714 · exp(xb)) / 0.0076926904 )
PhenoAge = 141.50225 + ln( -0.0055305 · ln(1 - m) ) / 0.090165
```

**Variant B - conventional US units (used by `mcp-phenoage-clock`, `mcp-phenoage-clock phenoage.ts`):**

```
xb = -19.907
  - 0.0336 · albumin      (g/dL)
  + 0.0095 · creatinine   (mg/dL)
  + 0.1953 · glucose      (mg/dL)
  + 0.0954 · ln(CRP)      (CRP in mg/L)
  - 0.0120 · lymphocyte%
  + 0.0268 · MCV
```

```

+ 0.3306 · RDW
+ 0.00188 · ALP
+ 0.0554 · WBC
+ 0.0804 · age

γ = 0.0076927
m = 1 - exp( -exp(xb) · (exp(120γ) - 1) / γ )
PhenoAge = 141.50225 + ln( -0.00553 · ln(1 - m) ) / 0.090165

```

Both reduce to the same Gompertz inversion. The `0.090165` constant is the one BioAge corrected in its April 2026 commit (“Update PhenoAge constant to 0.090165 and recompute data”), earlier code in the wild uses `0.09165`, which shifts results by several years. Guard `m` away from 0 and 1 (mcp clamps to `[1e-6, 1 - 1e-6]`) or `ln(1 - m)` blows up.

**DNAm PhenoAge** is the methylation surrogate of this quantity: 513 CpGs, elastic net trained to predict the clinical PhenoAge. `pyaging` exposes both, `phenoage` (clinical, blood chemistry) and `dnamphenoage` (methylation).

## 5.2 pyaging generalised PhenoAge postprocess

The torch implementation with  $\lambda = 0.0192$  (`pyaging_models.py:1456`):

```

mortality = 1 - exp( -exp(x) · (exp(120λ) - 1) / λ )
age       = 141.50225 + ln( -0.00553 · ln(1 - mortality) ) / 0.090165

```

## 5.3 BiT age (*C. elegans*) second correction

The published biological ages are rescaled to account for truncation of the lifespan distribution. With median lifespan 372 h (15.5 d) and SD 192/3 h (8/3 d) (`AgingClock biological_age_correction.py`):

```

corrected(y) = y - erfinv( 0.5 - ( ( erf(((372 - y)/sd)/√2)/2 ) + 0.5 ) / 2 ) · √2 · 2 · sd

```

Lifespan fold-change with error propagation uses `uncertainties.ufloat` with a 5% lifespan-assay bias and 0.5 d age-reporting bias; biological age is rescaled as `chronological_age × 15.5 / reported_median_lifespan`.

## 5.4 Zhang 2019 predictors

514 elastic-net probes or 319,607 BLUP probes, trained on 13,566 samples. Per-sample z-scoring of probes is mandatory (§4.1). Missing values are filled with the per-probe mean across individuals; probes that are NA across all individuals are dropped (`DNAm-based-age-predictor pred.R`). BLUP degrades more gracefully with missing probes than EN. RMSE reported as 2.04 y, the most accurate chronological clock, and the one whose acceleration stopped predicting mortality.

# 6. Preprocessing, imputation, missing features

## 6.1 The canonical input contract

Rows = samples, columns = features. This is the `pyaging` convention and it should be the new tool’s too. Feature names must be the exact identifiers the clock coefficients use: `cg00000029` for human methylation, Ensembl or Entrez for expression, UniProt/Olink assay names for proteins.

pyaging wraps this in AnnData: `adata.X` is the feature matrix, `adata.obs` holds sample meta-data and later the predictions, `adata.uns` holds per-clock metadata and missingness diagnostics, `adata.obsm["X_<clock>"]` is the transient per-clock feature slice ([pyaging\\_pred.py](#)).

## 6.2 EPIC v2 probe aggregation

EPIC v2 column names look like `cg00000029_TC21`. Split on `_`, group by CpG prefix, and average duplicated probes before anything else ([pyaging\\_epicv2\\_probe\\_aggregation](#)). Skipping this means every EPIC v2 clock silently sees zero overlapping features.

## 6.3 Two distinct imputation stages

**Stage 1 - dataset-level, before the clock.** `df_to_adata(imputer_strategy=...)` with `mean`, `median`, `constant` (0), or `knn` (default). KNN is expensive on wide matrices and is the first thing to move to GPU.

**Stage 2 - clock-level, for features the dataset does not contain at all.** pyaging preallocates the clock's feature matrix, fills present features by index lookup, and fills absent features with the model's `reference_values` (typically the training-set mean or a gold-standard median) or with zero if no reference vector was shipped ([pyaging\\_check\\_features\\_in\\_adata](#)). It records `adata.uns["<clock>_percent_na"]` and `adata.uns["<clock>_missing_features"]`, and raises if 100% of features are missing.

Filling with zero in  $\beta$  space means “fully unmethylated”, which is biologically wrong for most sites, always ship a reference vector.

**Third stage in tAge:** pad missing genes with NA and let the model's own sklearn `SimpleImputer` insert training medians. This is the cleanest of the three because the imputation statistic travels with the model.

## 6.4 Imputation options used for benchmarking

ComputAge exposes three and treats the choice as part of the model config ([ComputAge\\_benchmarking.py:323](#)):

- `none` - reindex and fill 0.
- `sesame_450k` - fill with the sesame 450K per-probe median.
- `average` - fill with the column mean of the current dataset (0 if the column is empty).

Reporting the imputation method alongside a score is not optional; it changes the answer.

## 6.5 Preprocessing steps that live inside a model

These are per-clock and must run *after* feature selection, not before:

| Step                                          | Clocks                      |
|-----------------------------------------------|-----------------------------|
| robust scale $(x - \text{median})/\text{std}$ | AltumAge, CpGPTGrimAge3     |
| standard scale $(x - \text{mean})/\text{std}$ | CpGPTPCGrimAge3             |
| per-sample z-score across features            | ZhangEN, ZhangBLUP          |
| quantile normalise to gold standard           | DunedinPACE, Stubbs         |
| quantile normalise then scale                 | Stubbs                      |
| TPM by feature length then log1p              | OcampoATAC1/2               |
| binarise at per-row non-zero median           | BiTAge                      |
| median-fill then per-sample average rank      | Pasta, PastaMouse, REG      |
| mean over non-missing (−1 sentinel)           | epiTOC1, HypoClock, epiCMIT |
| 0.95 quantile over non-missing                | stemTOC, stemTOCvitro       |

| Step                  | Clocks                                                     |
|-----------------------|------------------------------------------------------------|
| nan → 0               | epiTOC2, epiTOC3                                           |
| autosomal z-score     | XChrom, YChrom                                             |
| NaN → reference value | ~90 clocks (the <a href="#">LinearReferenceClock</a> base) |

## 7. The clock catalogue

pyaging's registry is the most complete machine-readable inventory in existence: **173 clocks** with curated provenance ([pyaging clock\\_metadata.json](#), audited 2026-07 with an evidence ledger of 6 author-confirmed, 313 code-confirmed, 2,390 paper-confirmed and 59 supplement-confirmed field assignments, 0 unresolved).

### 7.1 Distributions

**Data type** - DNA methylation 158, histone modification 8, transcriptomics 4, chromatin accessibility 2, clinical biomarkers 1.

**Species** - *Homo sapiens* 155, multiple species 9, *Mus musculus* 7, *C. elegans* 1, human+mouse 1.

**Platform** - Illumina 450K 94, EPIC 87, 27K 13, MammalMethylChip40 10, ChIP-seq 8, RNA-seq 4, RRBS 3, expression microarray 3, mammalian methylation array 2, MammalMethylChip320 2, ATAC-seq 2, bisulfite sequencing 2, targeted bisulfite 1, clinical lab assays 1.

**Model type** - elastic net 64, LASSO 20, reference-based constrained deconvolution 18, PCA + elastic net 18, PCA + elastic net + ARD 8, linear regression 5, weighted linear score 3, causality-weighted elastic net 3, logistic regression 2, single-CpG score 2, mean methylation aggregation 2, dynamic methylation transmission 2, elastic net Cox 2, 95th-percentile aggregation 2, PCA 2, and one each of: deep neural network, Cox PH, PCA + Cox, elastic net Cox ensemble, Klemmera–Doubal composite, bootstrap-stabilised elastic net, complement of mean methylation, two-stage elastic net + Cox, feature-selected logistic, mean aggregation, two-stage elastic net, ridge logistic, orthologue-transferred ridge logistic, penalised hazards with Gompertz calibration, weighted methylation aggregation, ridge, elastic net logistic, quadratically calibrated elastic net, screened elastic net, BLUP.

**Unit** - years 83, proportion 20, unitless 17, pg/mL 7, beta value 6, weeks 6, probability 3, months 3, m/s 2, kg 2, biological years per chronological year 2, days 2, cell divisions per stem cell 2, pack-years 2, ratio 2, log odds 2, population doublings 2, and one each of hours, passages, mL/kg/min, kb, relative age, ln(%), ln(mg/L), units/week, bp, log hazard.

**Population** - adults 72, older adults 29, all ages 27, multiple mammalian species 9, human age-unspecified 7, mice 6, newborns 4, human cell cultures 4, pregnancies 4, adult men 3, adult women 2, and one each of *C. elegans*, centenarians, humans and mice, children and adolescents, prenatal and newborn, children.

## 7.2 Selected clocks with their key parameters

The full 173-row table is in the registry; this is the subset a new tool should support first, sorted by likely usage.

| Clock                                                                         | Year | Layer                      | Features                      | Model                                 | Predicts                 | Unit          | Notes                                  |
|-------------------------------------------------------------------------------|------|----------------------------|-------------------------------|---------------------------------------|--------------------------|---------------|----------------------------------------|
| horvath2013                                                                   | 2013 | DNAm multi-tissue          | 353                           | elastic net                           | CA                       | years         | anti_log_linear; the reference clock   |
| hannum                                                                        | 2013 | DNAm blood                 | 71                            | elastic net                           | CA                       | years         |                                        |
| lin                                                                           | 2016 | DNAm blood                 | 99                            | linear                                | CA                       | years         | intercept 12.2169841                   |
| skinandblood                                                                  | 2018 | DNAm skin/blood/fibroblast | 391                           | elastic net                           | CA                       | years         | most accurate first-gen across tissues |
| phenoage                                                                      | 2018 | clinical chem              | 10                            | Gompertz-calibrated penalised hazards | phenotypic age           | years         | see §5.1                               |
| dnamphenoage                                                                  | 2018 | DNAm blood                 | 513                           | elastic net                           | phenotypic age           | years         |                                        |
| zhangen                                                                       | 2019 | DNAm blood/saliva          | 514                           | elastic net                           | CA                       | years         | per-sample z-score                     |
| zhangblup                                                                     | 2019 | DNAm                       | 319,607                       | BLUP                                  | CA                       | years         | RMSE 2.04 y                            |
| grimage                                                                       | 2019 | DNAm blood                 | 1,032                         | 2-stage EN + Cox                      | mortality                | years         | 8 sub-clocks                           |
| grimage2                                                                      | 2022 | DNAm blood                 | 1,032                         | EN Cox                                | mortality                | years         | 10 sub-clocks (+logCRP, +logA1C)       |
| dunedinpoam38                                                                 | 2020 | DNAm blood                 | 46                            | elastic net                           | pace                     | bio-y/chron-y |                                        |
| dunedinpace                                                                   | 2022 | DNAm blood                 | 20,000 (173 scoring)          | elastic net                           | pace                     | bio-y/chron-y | quantile norm required                 |
| altumage                                                                      | 2022 | DNAm multi-tissue          | 20,318                        | DNN                                   | CA                       | years         | 5 hidden layers, SELU                  |
| pchorvath2013 / pchannum / pcphenoage / pcgrimage / pcskinandblood / pcdnamtl | 2022 | DNAm                       | 78,464 (78,466 for PCGrimAge) | PCA + EN                              | various                  | years / bp    | high test-retest ICC                   |
| hrsinchphenoage                                                               | 2022 | DNAm blood                 | 959                           | weighted linear score                 | phenotypic age           | years         |                                        |
| yingcausage / yingdamage / yingadaptage                                       | 2024 | DNAm blood                 | 585 / 1,089 / 999             | causality-weighted EN                 | CA / damage / adaptation | years         | MR-filtered CpGs                       |
| dnamfitage                                                                    | 2023 | DNAm blood                 | 630                           | Klemera–Doubal composite              | fitness bio-age          | years         | sex-specific                           |

| Clock                                          | Year           | Layer                             | Features                | Model                     | Predicts                    | Unit                           | Notes                               |
|------------------------------------------------|----------------|-----------------------------------|-------------------------|---------------------------|-----------------------------|--------------------------------|-------------------------------------|
| dnamtl                                         | 2019           | DNAm blood                        | 140                     | elastic net               | telomere length             | kb                             |                                     |
| stemtoc                                        | 2024           | DNAm multi-tissue                 | 371                     | 0.95-quantile aggregation | mitotic age                 | beta                           |                                     |
| epitoc1 / epitoc2 / epitoc3                    | 2016/2020/2020 | DNAm                              | 385 / 163 / 170         | mean / transmission model | mitotic age                 | beta / divisions per stem cell |                                     |
| systemsage + 11 system variants                | 2025           | DNAm blood                        | 125,175                 | PCA + EN                  | 11 system ages + composite  | years                          |                                     |
| cpgptgrimage3                                  | 2024           | DNAm-derived proxies              | 24                      | Cox                       | mortality/bio-age           | years                          | needs CpGPT + GrimAge2 proxies      |
| cpgptpcgrimage3                                | 2024           | DNAm-derived proxies              | 31                      | PCA + Cox                 | mortality/bio-age           | years                          |                                     |
| intrinclock                                    | 2024           | DNAm multi-tissue                 | 380                     | 2-stage EN                | CA                          | years                          | stable across immune cell types     |
| encen40 / encen100                             | 2023           | DNAm blood/saliva/buccal          | 559 / 198               | elastic net               | CA                          | years                          | centenarian-tuned                   |
| retroelementagev1/v2                           | 2024           | DNAm blood                        | 1,317 / 1,378           | elastic net               | CA                          | years                          | retroelement-focused                |
| senchronoage / sencultureage / senmortalityage | 2026           | DNAm                              | 187 / 142 / 91          | EN / EN-logistic / EN-Cox | CA / senescence / mortality | years / log-odds / log-hazard  | newest human clocks in the registry |
| pasta / reg                                    | 2025           | transcriptomics multi-tissue      | 8,113                   | ridge-logistic / ridge    | transcriptomic age          | years                          | rank-normalised                     |
| bitage                                         | 2021           | transcriptomics <i>C. elegans</i> | 576                     | elastic net               | bio age                     | hours                          | binarised                           |
| ocampoatac1/2                                  | 2023           | ATAC-seq PBMC                     | 228 / 380               | elastic net               | CA                          | years                          |                                     |
| camillo* (8 marks)                             | 2025           | histone ChIP-seq                  | 102–3,739               | PCA + EN + ARD            | CA                          | years                          |                                     |
| mammalian1/2/3 + blood/skin variants           | 2023           | DNAm mammalian array              | 335–2,572               | elastic net               | CA                          | years                          | species lookup required             |
| petkovich / stubbs / meer / thompson           | 2017–18        | DNAm mouse                        | 90 / 17,992 / 435 / 582 | elastic net               | CA                          | months / days                  | RRBS or bisulfite-seq               |
| deconvolute* (6 + 12 cell)                     | 2018 / 2022    | DNAm EPIC                         | 600 / 240               | constrained deconvolution | cell proportions            | proportion                     |                                     |
| xchrom / ychrom                                | 2021           | DNAm                              | 4,047 / 284             | PCA                       | sex chromosome dosage       | unitless                       |                                     |
| mccartney* (10)                                | 2018           | DNAm blood                        | 204–1,109               | LASSO                     | lifestyle/clinical traits   | various                        | smoking, BMI, alcohol, lipids, edu- |

| Clock                                                                                    | Year        | Layer                     | Features                | Model               | Predicts                              | Unit                              | Notes                 |
|------------------------------------------------------------------------------------------|-------------|---------------------------|-------------------------|---------------------|---------------------------------------|-----------------------------------|-----------------------|
|                                                                                          |             |                           |                         |                     |                                       |                                   | cation, WHR, body fat |
| knight / bohlin / mayne / lee×3 / epicga                                                 | 2016–2021   | DNAm cord blood/ placenta | 62–558                  | EN / LASSO          | gestational age                       | weeks                             |                       |
| pedbe / wu                                                                               | 2019–2020   | DNAm buccal/ blood        | 94 / 111                | EN                  | CA in children                        | years                             |                       |
| corticalclock / neusin / gliasin                                                         | 2020 / 2024 | DNAm brain cortex         | 347 / 672 / 220         | EN                  | CA                                    | years                             |                       |
| ctsliver / hep / hepatoxu                                                                | 2017–2024   | DNAm liver / cfDNA        | 90 / 70 / 10            | LASSO / logistic    | CA / HCC                              | years / unitless                  |                       |
| cvdwesterman / adbahados- ingh / prostateca ncerkirby / depressionbarbu / down- syndrome | 2017–2021   | DNAm blood                | 235 / 4 / 3 / 196 / 652 | disease risk models | disease probability or score          | probability / log odds / unitless |                       |
| dnamfili / dnamic / dnamstress                                                           | 2022–2025   | DNAm blood                | 20 / 91 / 211           | LASSO / EN          | frailty / intrinsic capacity / stress | unitless                          |                       |
| replitali / replitali- norm / cellpopage                                                 | 2022–2024   | DNAm cultured cells       | 87 / 218 / 42           | EN                  | replicative history                   | population doublings / passages   |                       |



### 7.3 Clocks outside pyaging worth supporting

- **organAging** - 24 organ panels × {chronological, mortality} × {full Olink 3072, feature-reduced Olink 1536, CSF}. Coefficients ship as CSVs in the repo.
- **tAge** - 60 transcriptomic models (§4.18).
- **SpatialAgingClock** - 18 mouse brain cell types (§4.13).
- **scImmuAging** - 5 human PBMC cell types (CD4T, CD8T, MONO, NK, B), glmnet objects with `lambda.min`, applied to pseudocells.
- **SCALE** - >20 mouse tissues, Tabula Muris Senis.
- **CellBiAge** - mouse brain snRNA/scRNA binary classifiers.
- **scAge** / **scEpiAge** - single-cell methylation.
- **OMICmAge** - 421 epigenetic biomarker proxies (110 protein, 286 metabolite, 25 clinical) distilled into a DNAm-only predictor of EMRAge. Discovery MGB Aging Biobank n = 3,451; validation TruDiagnostic n = 14,213 and Generation Scotland n = 18,672.
- **GOLD BioAge** / **GOLD ProtAge** / **GOLD MetAge** / **Light BioAge** - Gompertz-law clinical, proteomic, metabolomic, and 3-marker variants (Adv Sci 12(32), 2025); benchmarked on NHANES and UK Biobank.
- **StackAge** - UKB proteomics + metabolomics ensemble, n = 30,376.
- **Oh 2023 organ clocks** - SomaScan, ~5,700 people, `organage` Python package.
- **GP-age** - Gaussian process regression on 30 CpGs selected by k-means over 80 age-correlated sites; MedAE 2.1 y, 11,910 blood samples.
- **RNAAgeCalc** - cross-tissue and tissue-specific GTEx transcriptomic ages (R/Bioconductor, Python `racpy`, Shiny).
- **MEAT** - muscle epigenetic age test (Bioconductor).
- **MiAge** - mitotic age; coefficients ship in `ComputAge`.
- **GlycanAge**, **MetaboAge**, **IMM-AGE**, microbiome clocks.

## 8. Output types and units

A tool that returns a bare number is dangerous. Every prediction needs at least six attached fields.

### 8.1 The output record

```
sample_id, clock_name, clock_version, value, unit, scale_type,
chronological_age (if supplied), age_acceleration_absolute, age_acceleration_relative,
percent_features_missing, missing_features, imputation_method,
preprocess_applied, postprocess_applied, species, tissue, platform,
research_only_flag, citation, doi
```

`pyaging` writes value into `adata.obs[clock]`, metadata into `adata.uns[f"{clock}_metadata"]`, missingness into `adata.uns[f"{clock}_percent_na"]` and `adata.uns[f"{clock}_missing_features"]`. That is close to the right shape; add explicit units and the two acceleration flavours.

### 8.2 Scale types and what you may do with them

| Scale type                   | Example clocks                                          | Legal operations                                                           |
|------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------|
| Age in years                 | Horvath, GrimAge, PhenoAge, SystemsAge                  | subtract CA; residualise on CA; average across clocks only after z-scoring |
| Age in months / days / hours | Petkovich, Stubbs (months), Meer (days), BiTAge (hours) | convert before comparing to human clocks                                   |

| Scale type                      | Example clocks                                                        | Legal operations                                                                |
|---------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Gestational age in weeks        | Knight, Bohlin, Mayne, Lee, EPICGA                                    | compare to ultrasound GA, not to adult clocks                                   |
| Rate (bio-y per chron-y)        | DunedinPACE, DunedinPoAm38                                            | centred at 1; do <b>not</b> subtract CA                                         |
| Log hazard                      | SenMortalityAge, organAging mortality, raw Cox scores                 | exponentiate for HR; convert to years only with an explicit Gompertz assumption |
| log10(hazard ratio)             | tAge mortality clocks                                                 | never multiply by species lifespan                                              |
| Probability                     | CVDWesterman, ADBahadoSingh, MammalianFemale                          | ROC/AUC, calibration                                                            |
| Log odds                        | ProstateCancerKirby, SenCultureAge                                    | logistic transform before interpretation                                        |
| Proportion                      | deconvolution clocks                                                  | sums to 1 across the cell-type family                                           |
| Beta value                      | epiTOC1, stemTOC, HypoClock, Garagnani, Bocklandt                     | ordinal within a study; not comparable across platforms                         |
| Cell divisions per stem cell    | epiTOC2, epiTOC3                                                      |                                                                                 |
| Population doublings / passages | RepliTali, CellPopAge                                                 |                                                                                 |
| Physical units                  | DNAmTL (kb), PCDNAmTL (bp), gait (m/s), grip (kg), VO2max (mL/kg/min) | compare to measured values                                                      |
| Concentration                   | GrimAge2 sub-clocks (pg/mL), logCRP (ln mg/L), logA1C (ln %)          | intermediate quantities, not ages                                               |
| Unitless score                  | ZhangMortality, DNAmFILI, DNAmIC, DNAmStress, ReedBMI, DownSyndrome   | rank or z-score within cohort                                                   |
| Relative age                    | EnsembleAgeHumanMouse, tAge normalized age                            | fraction of maximum lifespan                                                    |

### 8.3 Aggregate outputs a tool should offer

- Per-sample × per-clock long table (tidy) and wide matrix.
- Age-acceleration matrix (both AAA and RAA).
- Composite score: z-score each clock within cohort, then average within generation class, and report the classes separately rather than mixing generations.
- Per-system / per-organ profile (SystemsAge, organAging, tAge tissue clocks) with a radar or heatmap layout.
- QC block: missingness per clock, platform detected, imputation used, warnings.

## 9. Downstream statistics on an aging score

### 9.1 Association with chronological age

Pearson  $r$ , Spearman  $\rho$ , MAE, MedAE, RMSE, and the regression slope of BA on CA. The slope is the diagnostic: penalised regression compresses predictions toward the training mean, so slopes  $< 1$  are normal and produce the well-documented systematic underestimation in older subjects. Report the slope, not just  $r$ .

### 9.2 Mortality

Cox proportional hazards with the standardised score plus CA and sex as covariates:

```
coxph(Surv(time, status) ~ scale(BA_advance) + age + gender, data)
```

Report the HR per SD with 95% CI. BioAge standardises the advance within sex so the effect size means “one sex-specific SD of biological age advancement” ([BioAge table\\_surv.R](#)). Observed HRs from BioAge’s NHANES IV tables run 1.21–1.61 per SD across measures and subgroups, with KDM advance, PhenoAge advance and homeostatic dysregulation broadly comparable ([BioAge README](#)).

Additional model families worth supporting: time-dependent ROC/AUC, Harrell’s C-index, competing risks (Fine–Gray), and SuperLearner ensembles with out-of-fold predictions (the TwinGene/UK Biobank pipeline uses 10-fold CV, optionally repeated across seeds).

### 9.3 Morbidity and phenotype

Logistic or Poisson models for prevalent and incident disease; multimorbidity counts; frailty index; intrinsic capacity; grip strength and gait speed; cognitive scores. Interaction tests for BA × (BMI, sex, smoking, education) are standard in the epidemiological literature.

### 9.4 Group comparison and intervention response

The ComputAgeBench formulation, which is the cleanest published statement of the problem ([ComputAge benchmarking.py:339–397](#)):

- **AA2** - is the age acceleration in an aging-accelerating condition (AAC) cohort greater than in healthy controls from the same study? One-sided Welch’s t-test if  $\Delta$  is assumed normal, else one-sided Mann–Whitney U.
- **AA1** - is the age acceleration in an AAC cohort greater than zero, with no control group? One-sided one-sample t-test, else one-sided Wilcoxon signed-rank.
- Benjamini–Hochberg FDR correction is applied **per model across datasets**, then thresholded at 0.05.

For paired intervention designs, use paired tests on within-subject deltas and report both the absolute change and the change relative to the measurement’s own ICC.

### 9.5 Reliability

Intraclass correlation coefficient on technical replicates (same DNA, two arrays) and on biological replicates (same person, short interval). Pool across studies with Fisher-Z. The TransLAGE 2025 result is the one to remember: **technical reproducibility does not predict biological reliability**, and PCGrimAge was the only clock reaching ICC > 0.75 for biological reliability. A tool should surface each clock’s published ICC next to its prediction.

## 10. Plots

### 10.1 The standard set

1. **BA vs CA scatter, faceted by clock, coloured by sex**, with a fitted line, the identity line, and Pearson r annotated in the corner. This is `plot_ba` ([BioAge plot\\_ba.R](#)), `ggplot2`, `geom_point(shape = 1)`, `geom_smooth(method = lm)`, `facet_wrap(~ method, scales = "free")`, colours #7294D4 (men) and #E6A0C4 (women).
2. **Pairs / correlation matrix of age accelerations**. Scatter plots below the diagonal, Pearson r above the diagonal colour-mapped, clock labels on the diagonal. `plot_baa` ([BioAge plot\\_baa.R](#)) implements this in base graphics with a custom `layout()` matrix so the diagonal cells can hold text.

3. **Hierarchical clustering + Spearman heatmap of clocks.** pyaging Figure 1a - 38 methylation clocks over ~13,000 samples; telomere predictors cluster together, chronological predictors separate from mortality predictors.
4. **UMAP of samples in clock space.** pyaging Figures 1b–e; reveals “ageotypes” where AltumAge (chronological) and GrimAge2 (mortality) disagree.
5. **Reprogramming / intervention timecourse.** pyaging Figure 1f, GSE54848, 39 clocks over OSKM days; most rejuvenate between days 10 and 20, telomere predictors go *up*, ENCCen100 is flat. Any tool should make “run N clocks across a timepoint variable and plot trajectories” a one-liner.
6. **Benchmark grids.** `ComputAgeBench` `plot_class_bench` - a boolean matrix of clock × (dataset:condition:test) with per-condition-class column groups and a total column; `plot_medae` (MedAE bar chart, upper bound 18 y); `plot_bias` (median error, x-limits ±20 y). Saved as PDF at 180 dpi ([ComputAge benchmarking.py:451](#)).
7. **Forest plot of hazard ratios** per clock, per outcome, per cohort.
8. **Density / violin of age acceleration by group** (disease vs control, treated vs untreated).
9. **Organ or system radar/heatmap.** `organAging` uses heatmaps of organ ages against traits, medications, foods, professions, diseases; `SystemsAge` gives an 11-spoke profile.
10. **Expression-density QC plots at each preprocessing stage.** `tAge`’s `plot_eset_density` is called after raw, RLE, log, scale, `YuGene` and centring - a cheap way to catch a broken normalisation.
11. **Volcano / GSEA plots** when linking clock features to biology.
12. **Spatial maps** of predicted age or age acceleration over tissue coordinates, plus proximity effect matrices (effector cell type × target cell type, coloured by t-statistic).
13. **Calibration plot** for probability-output clocks; **precision–recall curve** for classifiers on imbalanced test sets (`CellBiAge`’s choice of AUPRC over AUROC).
14. **Missingness bar chart** per clock, percent of features absent, drawn before the results so the user sees which clocks are untrustworthy for their platform.

## 10.2 Plotting conventions worth adopting

- Always draw the identity line on BA-vs-CA plots.
- Never plot rate clocks (`DunedinPACE`) on the same axis as age clocks.
- Facet by clock generation, not alphabetically.
- Annotate every panel with n, r and the missingness fraction.
- Colour by group, shape by platform; keep the palette stable across the whole report.

## 10.3 What FALCONAge draws, and against which of the above

Twenty-five figures, identical in Python and R because both read their colours, titles, subtitles and one-line descriptions from the same `colorscheme.yaml`. Every function returns its data frame as well as its figure, which is how two rendering engines draw the same numbers.

| Figure                             | Answers                                                       | From §10.1 |
|------------------------------------|---------------------------------------------------------------|------------|
| <code>ba_vs_ca</code>              | does this clock track age here, and with what bias            | 1          |
| <code>bland_altman</code>          | is the error age-dependent (regression dilution made visible) | -          |
| <code>calibration</code>           | is the predicted age calibrated across the age range          | 13         |
| <code>acceleration_by_group</code> | do cases accelerate relative to controls                      | 8          |

| Figure                                                               | Answers                                                             | From §10.1 |
|----------------------------------------------------------------------|---------------------------------------------------------------------|------------|
| <code>acceleration_density</code>                                    | the same, as distributions rather than summaries                    | 8          |
| <code>acceleration_heatmap</code>                                    | clock × study, the whole pooled picture in one panel                | -          |
| <code>forest</code>                                                  | effect size with an interval, per clock                             | 7          |
| <code>clock_corr</code>                                              | which clocks agree, clustered                                       | 3          |
| <code>clock_chord</code>                                             | how much of that agreement is shared CpGs rather than shared signal | -          |
| <code>clock_radar</code>                                             | one profile per group across clocks, on a common z scale            | 9          |
| <code>clock_pca, clock_embedding</code>                              | sample structure in clock space; ageotypes                          | 4          |
| <code>clock_scatter_matrix</code>                                    | pairwise acceleration, the <code>plot_baa</code> layout             | 2          |
| <code>coverage_bar, missingness</code>                               | which clocks are scoring imputed values, read before the scores     | 14         |
| <code>beta_density</code>                                            | is the beta distribution bimodal, or did normalisation break        | 10         |
| <code>sex_check</code>                                               | does the X/Y dosage match the recorded sex                          | -          |
| <code>platform_comparison, study_comparison</code>                   | is the batch bigger than the biology                                | -          |
| <code>benchmark_bars, benchmark_error_bias, benchmark_heatmap</code> | AA1/AA2, the MedAE–MedE trade-off, clock × dataset                  | 6          |
| <code>timecourse, trajectory</code>                                  | clocks across a timepoint variable                                  | 5          |
| <code>clock_atlas</code>                                             | every clock, every pooled study, one figure                         | -          |

`clock_chord` is the one with no precedent in the list above and it earns its place: the correlation heatmap says two clocks agree, and the chord says how much of that agreement was built in by construction. A pair with a thick chord and a high correlation has told you much less than a pair with a high correlation and no chord at all.

Not implemented, and honestly out of scope for v1.0: volcano and GSEA plots (11) belong to feature-level biology rather than to scoring, and spatial maps (12) wait on the spatial modality.

Two rules the code enforces rather than recommends. A figure whose data is empty, flat, or all zeros raises `NothingToPlot` and `save_all` records the reason, because an axis-free rectangle in a report reads as a measurement of nothing rather than as an absence of measurement. And the scale-type gate applies to figures too: a `pace_ratio` clock never reaches an acceleration panel.

## 11. Benchmarking and validation

### 11.1 ComputAgeBench, the reference methodology

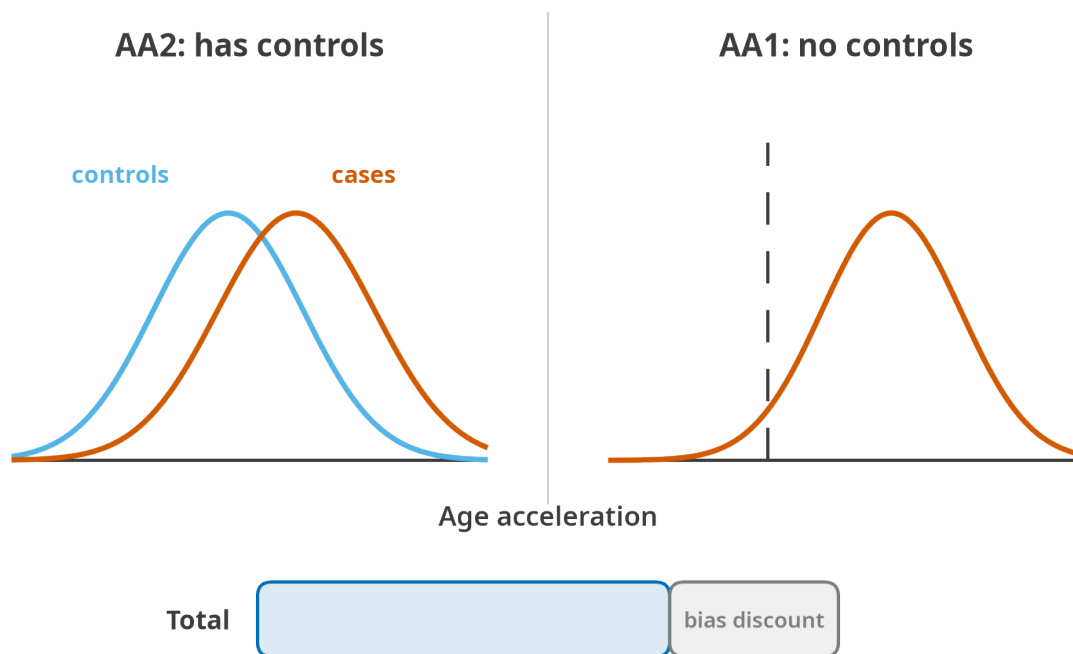


Figure 10: The two tests, and why the one without controls has its credit discounted

AA2 compares a condition group against its own healthy controls from the same study, so a clock that over-predicts everybody gains nothing from it. AA1 has no controls to compare against and asks only whether the condition group’s acceleration sits above zero, which a uniformly over-predicting clock passes everywhere. The total discounts the AA1 credit by the clock’s own median signed error on healthy controls, measured on the same data, which is the correction that separates a clock that detects disease from a clock with a broken intercept.

`computage` (Kriukov, Efimov, Kuzmina, Khrameeva, Dylov; bioRxiv 2024.06.06.597715; KDD ‘25) [36].

**Premise.** MAE against chronological age is the wrong metric because a perfect chronological oracle is useless. Instead, a good clock must show higher age acceleration in people with *pre-defined aging-accelerating conditions* (AACs) than in healthy controls.

**Data.** Benchmark split: **10,404 samples, 900,449 CpGs, 65 studies**. Training split: **7,419 samples, 907,766 CpGs, 46 studies** (healthy people only). Hosted on Hugging Face at [computage/computage\\_bench](https://huggingface.co/computage/computage_bench) as per-study parquet plus a single `computage_bench_meta.tsv`.

**Condition classes** ([ComputAge data\\_utils.py](#)):

| Class | Conditions                              |
|-------|-----------------------------------------|
| HC    | healthy control                         |
| CVD   | AS (atherosclerosis), IHD, CVA (stroke) |

| Class | Conditions                                                               |
|-------|--------------------------------------------------------------------------|
| ISD   | IBD, HIV, HIV_TB                                                         |
| MBD   | XOB (extreme obesity), T1D, T2D                                          |
| MSD   | OP (osteoporosis), RA                                                    |
| NDD   | MS, PD, AD, CJD                                                          |
| RSD   | TB, COPD                                                                 |
| PGS   | CGL (congenital generalised lipodystrophy), WS (Werner), HGPS (progeria) |

The full study table with GSE accessions, PMIDs, first authors, tissues, sample counts and HC availability is [ComputAge BenchBSB\\_table.csv](#) - 66 rows. Highlights: GSE56046 (n = 1,202, atherosclerosis), GSE42861 (n = 689, RA), GSE117859 (n = 608, HIV), GSE111629 (n = 572, PD), GSE59685 (n = 531, AD), GSE131752 (Werner, n = 48), GSE182991 (HGPS, n = 27), GSE214297 (CGL, n = 16).

**Filters.** Tissues restricted to Blood, Saliva, Buccal (“BSB”). Age limited to [18, 90], except progeria (PGS) which is exempt so that young progeria patients are retained.

### Scoring.

```

AA2_score = number of AA2 datasets where FDR-adjusted p < 0.05
AA1_score = number of AA1 datasets where FDR-adjusted p < 0.05
MedAE     = median |prediction - age| over all healthy-control samples
MedE      = median (prediction - age) over healthy controls           # the bias
total     = AA2_score + AA1_score × (1 - max(0, MedE) / MedAE)

```

The bias penalty is the clever part: a clock that simply over-predicts everyone’s age will pass AA1 trivially, so AA1 credit is discounted by how much of the clock’s error is systematic positive bias ([ComputAge benchmarking.py:427](#)). GrimAge2 leads AA1 but carries a large positive HC bias, which the formula penalises.

**Running it.** `run_benchmark(models_config, experiment_prefix, output_folder)` where `models_config` has an `in_library` dict (clock name → params, notably `imputation`) and a `new_models` dict (name → path to a pickled sklearn estimator with `feature_names_in_`). Outputs: AA2 and AA1 p-value CSVs, FDR-corrected boolean CSVs, MedAE CSV, bias CSV, four PDFs, and a pickled benchmark object.

## 11.2 Biolearn and the Biomarkers of Aging Consortium

Open-source Python library providing a unified interface over aging biomarkers plus a data library that loads GEO, NHANES and Framingham. Reference implementations for 39 biomarkers evaluated over >20,000 individuals; the framework paper reports integration of 17 additional GEO datasets covering 12,463 samples across tissues and age ranges. Version 0.7.0 released January 2025. The consortium runs a Synapse challenge series (chronological age, mortality, multimorbidity) with >\$200k in prizes. ComputAge’s `model_definitions` are explicitly ported from biolearn ([ComputAge definitions.py](#)).

## 11.3 TransLAGE - reliability validation

bioRxiv 2025.10.16.682960. Computes ICCs from technical replicates and from repeated biological measures, pools them with Fisher-Z, and compares. Headline findings: technical reproducibility does not predict biological reliability; PCGrimAge is the only clock with biological ICC > 0.75. Sehgal et al.’s



companion benchmark of 18 clocks found SystemsAge most technically reliable across array positions and DNA extraction protocols.

### 11.4 Intervention responsiveness

From the 2026 Frontiers in Genetics review of interventions that lower next-generation clocks: GrimAge responded in 38.1% of interventions, DunedinPACE 36%, DNAm PhenoAge 33.3%. Rapamycin and senolytics showed no detectable effect. Specific effects: omega-3 alone  $-0.0022$  units/year on DunedinPACE and  $-0.64$  years on GrimAge2; omega-3 + vitamin D + exercise  $-0.32$  years on PC PhenoAge additively. CALERIE (12% calorie restriction, 2 years) slowed DunedinPACE by 2–3%, the first RCT to move a methylation clock. Ordering: DunedinPACE responds fastest, GrimAge needs longer or larger interventions, first-generation Horvath barely moves.

### 11.5 What a benchmarking module in the new tool should do

1. Chronological accuracy on healthy controls: MedAE, MAE, RMSE,  $r$ ,  $\rho$ , slope, bias (MedE).
2. AA1 and AA2 with configurable distributional assumption and FDR correction.
3. Mortality: HR per SD, C-index, time-dependent AUC.
4. Reliability: ICC from user-supplied replicate structure, with Fisher-Z pooling.
5. Responsiveness: paired-design effect sizes with the measurement ICC reported alongside.
6. Cross-clock agreement: Spearman matrix, hierarchical clustering, UMAP.
7. A composite leaderboard using the ComputAgeBench total-score formula, with the components shown, never just the total.

## 12. Existing software landscape

### 12.1 Python

| Package                    | Scope                                                | GPU                                           | Notes                                                                                                                       |
|----------------------------|------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>pyaging</b>             | 173 clocks, 5 data types, 5 species                  | Yes. PyTorch, CUDA auto-detect, optional CuPy | The most complete. Weights on Hugging Face <a href="#">lucascamillomd/pyaging-data</a> , one .pt per clock. AnnData-native. |
| <b>biolearn</b>            | 39 biomarkers + data library (GEO/NHANES/Framingham) | No                                            | The consortium's reference implementation; strongest on data harmonisation.                                                 |
| <b>computage</b>           | Benchmarking + de novo clock training (KDM, PLS)     | No                                            | The AA1/AA2 methodology and the 10,404-sample benchmark set.                                                                |
| <b>organage</b>            | Oh 2023 organ proteomic clocks (SomaScan)            | No                                            |                                                                                                                             |
| <b>spatial-aging-clock</b> | 18 mouse brain cell-type spatial clocks              | No                                            | SquidPy-based.                                                                                                              |
| <b>scAge</b>               | Single-cell methylation age                          | No                                            |                                                                                                                             |
| <b>CpGPT</b>               | Foundation model, imputation + fine-tuned heads      | Yes                                           |                                                                                                                             |
| <b>racpy</b>               | RNAAgeCalc port                                      | No                                            |                                                                                                                             |



## 12.2 R

| Package                              | Scope                                                                                                                                                                   | Notes                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>methylock</b> (Bioconductor)      | Chronological clocks, gestational clocks, DNAmTL, PhenoAge                                                                                                              | Limited scope but well-maintained.                                                  |
| <b>methyICIPHER</b> (HigginsChenLab) | PC clocks, SystemsAge, CausalAge, DunedinPACE and many more                                                                                                             | The broadest R clock library.                                                       |
| <b>dnaMethyAge</b>                   | Common clocks with a tidy interface                                                                                                                                     |                                                                                     |
| <b>BioAge</b> (dayoonkwon)           | KDM, PhenoAge, homeostatic dysregulation on NHANES III/IV                                                                                                               | The reference clinical-chemistry implementation, with plotting and survival tables. |
| <b>MEAT</b> (Bioconductor)           | Muscle epigenetic age test                                                                                                                                              |                                                                                     |
| <b>**MammalMethylClock</b>           | Mammalian clock construction and application; SeSaMe normalisation, filtering, annotation, age/sex/species predictors; 11K+ samples, 59 tissues, 185 species, 19 orders |                                                                                     |
| <b>CorticalClock</b>                 | Brain cortex DNAm age                                                                                                                                                   |                                                                                     |
| <b>EpiMitClocks</b>                  | epiTOC1/2/3, epiCMIT, stemTOC, HypoClock                                                                                                                                | The authoritative mitotic clock source.                                             |
| <b>tAge</b> (Gladyshev-Lab)          | 60 transcriptomic clocks; calls Python through reticulate                                                                                                               | MGB Open Access License, non-commercial.                                            |
| <b>scImmuAging</b>                   | 5 PBMC cell-type clocks                                                                                                                                                 |                                                                                     |
| <b>RNAAgeCalc</b> (Bioconductor)     | GTEx transcriptomic age                                                                                                                                                 |                                                                                     |

## 12.3 Web tools and databases

- **ClockBase** (clockbase.org) - 11 clock models reprocessed over >300,000 GEO samples, with an R package and interactive analysis.
- **TACO** (app.gladyshevlab.org/TACO). Transcriptomic Age Calculator Online; the reference application whose behaviour tAge mirrors.
- **DNA Methylation Age Calculator** (dnamage.clockfoundation.org). Horvath, SkinBlood, GrimAge.
- **Open Genes** - 2,402 age-related genes from >1,700 publications, 6 study types, 12 criteria, API + download.
- **Aging Atlas**, **AGEMAP**, **Digital Ageing Atlas**, **Human Ageing Genomic Resources** (GenAge, AnAge, LongevityMap, DrugAge, GenDR), **JenAge**.
- **aging.ai** - blood-biomarker DNN age.
- **Longevity World Cup** - open-source PhenoAge calculator and leaderboard.

## 12.4 The gap this project fills

No package today does all of: (a) methylation *and* transcriptomics *and* proteomics *and* clinical chemistry in one API; (b) Python *and* R with identical numerics; (c) GPU acceleration beyond pyaging's methylation focus; (d) built-in benchmarking against ComputAgeBench-style AAC data; (e) reliability-aware reporting. pyaging is closest on (a) and (c); computage owns (d); methyICIPHER and BioAge own the R side.

### 13. Prior implementations

Sixteen public codebases were read against the papers while this package was written. They are listed because a citation to a paper says what a method *is*, and a citation to an implementation says what its authors actually *did*, and where the two disagree, which they do in eleven documented cases: the disagreement is itself the finding.

None of it is vendored. Every clock architecture in FALCONAge is written from its published description. What these settled was the question a paper cannot answer on its own, when the text is ambiguous, which reading did the authors implement, together with the coefficient sets whose provenance could be traced back to a primary source.

| Project                                    | Origin                | Language     | What it settles                                                                                                                                                                                                                                      |
|--------------------------------------------|-----------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">pyaging</a>                    | lucascamillomd        | Python       | The broadest existing clock registry (173 entries with curated metadata and an evidence ledger), the model base classes, the preprocess and postprocess transform set, the AnnData contract, weight distribution through Hugging Face, GPU dispatch. |
| <a href="#">ComputAge</a>                  | ComputationalAgingLab | Python       | AA1/AA2 benchmarking, the 65-study AAC dataset table, sesame imputation vector, <a href="#">LinearMethylationModel</a> / <a href="#">GrimAgeModel</a> , de novo KDM and PLS estimators, biolearn-derived model definitions.                          |
| <a href="#">computational_aging_course</a> | ComputationalAgingLab | Jupyter Book | Teaching-grade chapters on aging biology (12 hallmarks), aging clocks, methylomics, survival analysis, complex systems and resilience.                                                                                                               |
| <a href="#">BioAge</a>                     | dayoonkwon            | R            | KDM, PhenoAge (original and modified), homeostatic dysregulation, NHANES III/IV data, plotting and survival/health/SES tables. Its correction to the PhenoAge 0.090165 constant is the reading FALCONAge follows.                                    |
| <a href="#">tAge</a>                       | Gladyshev-Lab         | R + Python   | The 2026 Nature transcriptomic clocks, full preprocessing chain, YuGene, ortholog mapping, pseudobulk, outlier removal, relative/differential clock semantics, unit discipline for mortality vs chronological.                                       |

| Project                                           | Origin                    | Language   | What it settles                                                                                                                                                                                                        |
|---------------------------------------------------|---------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organAging                                        | ludgergoeminne            | R + Python | Organ-specific plasma proteomic clocks (Cell Metabolism 2025), GTEx 4× fold-change organ definitions, Cox elastic net and Weibull AFT training code, log-hazard → years Gompertz conversion, ~50 plotting scripts.     |
| SpatialAgingClock                                 | sunericd                  | Python     | Spatial smoothing, cell-type-specific LASSO clocks, within-animal age acceleration, cell proximity effect analysis (Nature 2025).                                                                                      |
| scImmuAging                                       | CiiM-Bioinformatics-group | R          | PBMC cell-type clocks from 1,081 donors aged 18–97 (Nat Aging 2025); pseudocell construction; LASSO/RF/PointNet training scripts.                                                                                      |
| SCALE                                             | ChengLiLab                | R          | Literature-informed aging gene sets, detection-frequency-weighted z-score, single-cell entropy, four correlation measures (Pearson/dcor/MIC/nlcor).                                                                    |
| AgingClock                                        | Meyer-DH                  | Python     | BiT age binarisation, the truncated-Gaussian second correction, lifespan fold-change with error propagation, v2 coefficients.                                                                                          |
| DNAm-based-age-predictor                          | qzhang314                 | R          | Zhang 2019 EN (514 probes) and BLUP (319,607 probes) coefficients and the exact per-sample standardisation.                                                                                                            |
| mcp-phenoage-clock                                | 199-mcp                   | TypeScript | A clean, validated PhenoAge implementation with reference ranges, input bounds, mortality clamping, and interpretation strings. Good template for user-facing validation.                                              |
| phenotypic-age-calc                               | KyteProject               | Python     | Minimal PhenoAge with explicit unit conversions.                                                                                                                                                                       |
| Biological-Aging-and-PRS-for-Mortality-Prediction | shayanmostafaei           | R          | Multi-cohort mortality analysis design: TwinGene n = 9,617 (16.7 y median follow-up) discovery, UK Biobank n = 179,504 (11.83 y) validation; PhenoAge, KDM, HD, frailty index, telomere length, mvAge PRS; Cox + ROC + |

| Project                                 | Origin     | Language        | What it settles                                                                                                                            |
|-----------------------------------------|------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |            |                 | SuperLearner + subgroup + interaction analyses.                                                                                            |
| <a href="#">Aging_clock</a>             | mdozmorov  | Markdown + data | Curated literature index with clock CpG data (Levine 2018, Petkovich 2017, Wang 2017, Yang 2016 epiTOC) and Horvath's original R tutorial. |
| <a href="#">Ageing-analysis-using-R</a> | SWAROOP006 | R               | Telomere repeat (TTAGGG and variants) detection across 28 genomes with seqinr. Peripheral to clock work.                                   |

## 14. GPU computing

### 14.1 What actually benefits

Profile before optimising. The expensive operations, in order:

1. **KNN imputation** on wide matrices ( $n_{\text{samples}} \times 800\text{K}$  probes). Dominant cost for a single dataset. `cuML.KNNImputer` or a chunked CuPy distance computation.
2. **PC-clock rotation** - a dense  $78,464 \times 1,933$  matmul per clock,  $125,175 \times k$  for SystemsAge. Trivially GPU-bound and highly parallel across samples.
3. **Quantile normalisation** for DunedinPACE - a sort per sample over 20,000 probes. `torch.sort` over a batch is one kernel.
4. **Rank normalisation** for Pasta/REG - average ranks over 8,113 genes per sample. pyaging's current implementation is a Python `while` loop per row and is the slowest path in the package; vectorise with `torch.argsort` twice plus a scatter-mean for ties.
5. **Deconvolution simplex projection** - already vectorised in pyaging; keep that pattern.
6. **Running many clocks over many samples** - the embarrassingly parallel case, and the one pyaging's paper benchmarks.
7. **Spatial smoothing propagation** - sparse matmul iterated to convergence; cuSPARSE.
8. **Elastic net training** (for users fitting their own clocks), `cuML.ElasticNet`, `cuML.LogisticRegression`, or a torch implementation with proximal gradient.

### 14.2 Published speedups, and what we measured instead

pyaging vs biolearn on CPU (Bioinformatics 2024, Figure 1g–h):

| Samples | pyaging (s) | biolearn CPU (s) | Ratio |
|---------|-------------|------------------|-------|
| 1,024   | 0.233       | 0.386            | 1.7×  |
| 32,768  | 0.642       | 76.608           | ~120× |
| 65,536  | runs on GPU | out of memory    | -     |

That comparison is GPU-pyaging against CPU-biolearn, so it measures two packages as much as two devices. Measuring one package on both devices gives the opposite answer for the clocks that ship here, eight tier A linear clocks over 2,340 features, on an RTX 4060, best of three:

| Samples | CPU float64    | CUDA float64 | CUDA float32 |
|---------|----------------|--------------|--------------|
| 128     | <b>0.009 s</b> | 0.011 s      | 0.013 s      |
| 1,024   | <b>0.032 s</b> | 0.053 s      | 0.051 s      |
| 4,096   | <b>0.143 s</b> | 0.307 s      | 0.190 s      |
| 16,384  | <b>0.506 s</b> | 2.328 s      | 1.726 s      |

The gap widens with size, which is the reverse of the usual shape, and profiling says why: at 4,096 samples the aligned matrix takes 134 ms to build and the dot products take 5 ms on the CPU against 10 ms on the GPU, most of which is transfer rather than multiply. A dot product over a few thousand features is too small a matrix multiplication to be worth a device, and the list in §14.1 is a list of things that are *not* what a first-generation linear clock does.

So the expected pattern holds only where the list in §14.1 does: the PC rotations (item 2, thirty times the feature count), neural architectures, and the single-cell paths where rapids-singlecell reports ~15× over

the best CPU path and ScaleSC 20–100× over Scanpy. For the 2,340-feature case, `device="auto"` resolving to CUDA would make the common case several times slower on every machine that has a card.

Two other findings worth recording. **CPU and GPU do not agree bit for bit** - up to  $1.3e-13$  years, 14 ulps, because numpy's BLAS and cuBLAS sum in different orders and floating-point addition is not associative. And **single precision costs 7e-5 years** on the same device: four orders of magnitude more than the device difference, still negligible against effects measured in single-digit years, and the right way round.

### 14.3 Implementation pattern from pyaging

- Detect once: `torch.device("cuda" if torch.cuda.is_available() else "cpu")`.
- Optional CuPy for the AnnData matrix itself: `adata.X = cp.array(adata.X) if CUPY_AVAILABLE else np.asfortranarray(adata.X)`.
- Load every model in `float64` and `.eval()`; run under `torch.inference_mode()`.
- Batch with `anndata.experimental.pytorch.AnnLoader`, default `batch_size = 1024`.
- After each clock: `del adata.obsm[f"X_{clock}"]`, `gc.collect()`, `torch.cuda.empty_cache()`.
- Preallocate the per-clock feature matrix with CuPy when available, NumPy Fortran-order otherwise.

Caveats to fix in the new tool: `float64` doubles memory and halves throughput on consumer GPUs with poor FP64 rates, offer `float32` with a documented numerical tolerance, and keep FP64 for the PC clocks where the rotation is ill-conditioned. Several pyaging preprocess functions loop over rows in Python (`stemT0C`, `epiT0C1`, `HypoClock`, `BiTAge`, `Pasta`, `DunedinPACE`, `Stubbs`); all are vectorisable.

### 14.4 R side

R has no comparable GPU story for this workload. The pragmatic answer, which tAge already uses, is **reticulate**: the R package is the interface and the numerics happen in Python. Alternatives if a pure-R path is needed: `gpuR/torch` for R for matmuls, or shell out to a small Python worker over Arrow IPC. Whatever the mechanism, the R and Python front ends must call the same weights and the same transform code so results are bit-comparable.

## 15. Design blueprint

The requirement was one tool, Python and R, multiomic input, optional GPU, and correct, meaning it reproduces published clock values on published test vectors.

This section was the sketch that turned that into a design. It is now documented properly elsewhere, and the sketch has been cut rather than kept in parallel: it used the abandoned working name `agingscore` and an API that no longer exists, so anyone who copied from it got nothing that runs.

**Architecture** is the design record for the built system, and it is checked against the running package rather than remembered. Where to look:

| What the blueprint specified                                                                           | Where it is documented now                                                                        |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Layered architecture, module boundaries                                                                | <a href="#">Architecture §1.4</a> , the file tree                                                 |
| Data model ( <code>X</code> , <code>obs</code> , <code>var</code> , <code>uns</code> , multi-modality) | <a href="#">Architecture §2</a> , the shipped <code>FalconData</code> , AnnData-backed            |
| Clock registry schema                                                                                  | <a href="#">Architecture §3</a> , the schema as validated, field by field                         |
| API surface, Python and R                                                                              | <a href="#">Getting started</a> , and the <a href="#">Python</a> and <a href="#">R</a> references |
| Correctness strategy                                                                                   | §15.1 below, which is the part that did not reduce to an implementation detail                    |

| What the blueprint specified | Where it is documented now                                                            |
|------------------------------|---------------------------------------------------------------------------------------|
| Packaging, distribution, CI  | <a href="#">Architecture §9, §12</a>                                                  |
| Build order and milestones   | <a href="#">Architecture §13</a> , with the exit criterion each milestone was held to |

Two decisions from the blueprint are worth restating because they are about the field rather than about this package. Weights belong outside the git repository, in [safetensors](#) rather than pickles, `torch.load(weights_only=False)` executes arbitrary code while unpickling, and it is what several packages still accept. And a [license](#) field has to gate use and print at score time: tAge models are MGB Open Access (non-commercial), organAging is Academic Non-Commercial, and a tool that redistributes weights must surface that rather than bury it.

### 15.1 Correctness strategy

This is the part that decides whether a clock implementation is trusted, and it is the reason the next subsection exists at all.

1. **Gold-standard vectors.** For every clock, an input matrix and the expected output from the original implementation, asserted to 1e-6 in FP64.
2. **Cross-implementation agreement.** Where two packages implement the same clock, compare, and record disagreements in the registry rather than picking a side silently. §15.2 is that record.
3. **Registry validation** against a controlled vocabulary for tissue, platform, predicts, training target, unit, data type, species, model type and population. With 161 entries there is no other way to keep them coherent.
4. **Property tests.** Proportions sum to 1; probabilities in [0,1]; monotone response to a synthetic age gradient; permuting feature order must not change results; adding a duplicated sample must not change either sample's score.
5. **Numerical dtype tests.** FP32-versus-FP64 divergence held under a documented per-clock tolerance. The PC clocks are the ones to watch.

### 15.2 Where the paper and the coefficients disagree

Eleven catalogued clocks are scored from coefficient sets that do not match what their paper describes. Each carries a [known\\_discrepancies](#) note in the registry and raises it as a warning at score time:

| Clock                                                                                         | The disagreement                                                                                                          |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <a href="#">bocklandt</a>                                                                     | Paper describes 2 CpGs; the circulating coefficient set carries 1                                                         |
| <a href="#">bohlin</a>                                                                        | Paper reports 96 CpGs; the circulating set carries 251                                                                    |
| <a href="#">cvdwesterman</a>                                                                  | Paper reports 1,305 CpGs; the circulating set carries 235                                                                 |
| <a href="#">zhangmortality</a>                                                                | Paper defines an integer count of markers past quartile thresholds; the implementations compute a continuous weighted sum |
| <a href="#">yingcausage</a> , <a href="#">yingdamage</a> , <a href="#">yingadaptage</a>       | Paper reports 586 / 1,090 / 1,000 CpGs; the sets carry 585 / 1,089 / 999, the same off-by-one in all three                |
| <a href="#">senchronage</a> , <a href="#">sencultureage</a> , <a href="#">senmortalityage</a> | The upstream README's feature counts disagree with the coefficient objects it ships                                       |
| <a href="#">phenoage</a>                                                                      | Two constants circulate in the divisor of the Gompertz inversion, 0.090165 and 0.09165                                    |

The list is generated from the registry, not maintained by hand: `reg.get(id).known_discrepancies` returns the full text, and `res.manifest["warnings"]` carries whichever of them fired on your run. The Ying clocks additionally carry a measured cohort-offset note, which is why they sit on `age_years_relative` rather than `age_years`, §19.3.1.

## 16. Failure modes

**Units.** PhenoAge in SI vs conventional units gives different answers with the same biomarker draw. Albumin g/L vs g/dL is a 10× error; creatinine μmol/L vs mg/dL is ~88×; CRP mg/L vs mg/dL is 10× inside a log. Force explicit units at the API boundary and refuse to guess.

**The 0.090165 constant.** Some published code uses 0.09165. BioAge fixed this in April 2026. Check any inherited implementation.

**EPIC v2 suffixes.** Aggregate before scoring or every clock sees zero features.

**Dropping DunedinPACE background probes.** The 19,827 non-scoring probes define the quantile mapping. Subsetting to the 173 scoring CpGs produces a plausible-looking wrong number.

**Zero-filling missing methylation features.**  $\beta = 0$  means fully unmethylated. Use reference values. pyaging warns; make it an error above a configurable missingness threshold.

**Mixing generations in a composite.** Averaging Horvath (chronological) with GrimAge (mortality) produces a number with no interpretation. Composite only within generation, after z-scoring.

**Subtracting CA from a rate clock.** DunedinPACE is not an age.

**Multiplying a mortality clock by species maximum lifespan.** tAge guards this with a registry flag precisely because it is an easy mistake.

**Per-sample vs per-feature scaling.** Zhang z-scores across probes *within* a sample; AltumAge scales each feature by training median/std *across* samples; tAge's `scale_eset` is column-wise, i.e. per sample. Getting the axis wrong is silent.

**Regression dilution.** All penalised clocks compress toward the training mean, so BA is biased low in the old and high in the young, and the naive `BA - CA` correlates with CA. Report the RAA residual too.

**Cell composition confounding.** Second-generation and single-tissue clocks partly read cell proportion. Run deconvolution alongside and either adjust or report both.

**Cell type dominates age in single cells.** scEpiAge found cell-type effect > age effect. Always model within cell type.

**Batch and array position.** Chip position and DNA extraction protocol move clock values by years. PC clocks and SystemsAge are the mitigations; randomising sample layout is the prevention.

**Technical ICC is not biological ICC.** A clock can be perfectly reproducible on a re-run of the same DNA and still be unstable across two blood draws a week apart.

**Age-range extrapolation.** Models trained on adults 20–70 do not extrapolate; gestational and paediatric clocks are separate for a reason. Enforce the registry `population` field.

**`torch.load(weights_only=False)`.** Arbitrary code execution from a downloaded file. Move to safetensors.



**Sex encoding.** GrimAge takes `Female` as 1/0; ComputAge maps unknown sex to 0.5, which is a silent half-male half-female fudge. Make unknown sex an explicit user decision.

**Intercept placement.** Some packages put the clock intercept in the transform lambda, others in the linear layer bias. Normalise at import or double-count.

**FP32 on PC clocks.** The 78,464-dimensional centring and rotation is ill-conditioned; validate before defaulting to FP32.

**Sample size for HD.** The Mahalanobis covariance needs more reference samples than biomarkers, well-conditioned. BioAge warns on single-row inputs; a real tool should check the condition number.

## 17. The frontier, and what FALCONAge does not yet compute

Reviewed to August 2026. This section exists because a tool that silently omits half a field reads as though the field ends where it does. Each item states the science, the blocker, and whether the blocker is technical or legal.

### 17.1 Array normalisation, and what it corrects

FALCONAge takes a beta matrix. It does not produce one from raw intensities, and the chain it skips exists to correct specific instrument artefacts rather than to tidy data.

An Infinium array carries two probe chemistries. **Type I** uses two beads per CpG read in one colour channel, which leaves the other channel at that address carrying no designed signal – the **out-of-band** readings. **Type II** uses one bead, methylated in green and unmethylated in red, and covers roughly 84% of probes from 450K onward. The two designs have different beta distributions, so a raw array is bimodal for reasons that are chemistry rather than biology.

| Step               | What it removes                                | How                                                                                                                                                                                             |
|--------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>noob</b>        | background fluorescence, cross-channel bleed   | normal-exponential deconvolution using the out-of-band Type I readings as the background distribution – the background estimate comes free from probes never meant to fluoresce in that channel |
| <b>dye bias</b>    | red/green intensity difference                 | quantile-matching the channels on Type I probes measured in both                                                                                                                                |
| <b>pOOBAH</b>      | measurements indistinguishable from background | detection p-value from the empirical out-of-band distribution; failures are masked to NA, not deleted                                                                                           |
| <b>BMIQ</b>        | the Type I/Type II design difference           | three-state beta mixture fitted to Type II, Type I quantile-mapped onto it within each state                                                                                                    |
| <b>probe masks</b> | probes untrustworthy anywhere                  | cross-reactive probes, probes overlapping common SNPs at the interrogated base, sex chromosomes                                                                                                 |

The order is load-bearing – noob, dye bias, pOOBAH, then BMIQ – and it matters because **clock coefficients were fitted on the output of a particular chain**. Applying Horvath’s weights to raw betas is not a smaller version of the right answer; it is the right weights on the wrong scale.

Until this ships, normalise with `sesame` or `minfi` and bring the beta matrix here.

## 17.2 Cross-platform harmonisation

Probe content changed across 27K, 450K, EPIC v1 and EPIC v2. EPIC v2 both dropped probes earlier clocks depend on and renamed others with replicate suffixes. FALCONAge handles the suffixes and *measures* the damage from the drops, by count and by coefficient mass. It cannot yet repair them.

`mLiftOver` is the published answer, and it is four things rather than an ID lookup: probe-ID conversion, replicate management, optional imputation with a confidence filter, and – the part that cannot be reconstructed – a mapping table carrying empirical evidence of whether each probe’s cross-platform mapping is reliable. Those tables are measured on paired samples and distributed through Bioconductor’s ExperimentHub, so this is blocked on data rather than effort.

Two mitigations are complementary: principal-component clocks absorb probe loss because a missing probe perturbs a component instead of deleting a term, which is why EPIC v2 disrupts the first-generation clocks and barely moves the PC ones.

## 17.3 Proteomic organ clocks

If a protein is enriched at least fourfold in one organ relative to all others, its plasma level proxies that organ’s state – so one blood draw supports an age estimate per organ. Ten organ clocks plus an organismal one were trained in UK Biobank ( $n = 43,616$ ) and validated in China ( $n = 3,977$ ) and the USA ( $n = 800$ ) at cross-cohort  $r = 0.98$  and  $0.93$ , with brain age most strongly linked to mortality.

Two properties matter for anyone implementing this. **Olink NPX is log2 and relative already**, so re-logging it is an error and comparing across panel versions without bridging is another. And these clocks are **z-scored against a training cohort, not against your own samples** – using your cohort’s mean silently recentres everyone to zero acceleration, the same class of mistake as filling an absent probe with zero. About half the ageing predictors in Olink-based clocks are absent from SomaScan-based ones, so the platforms are not interchangeable.

## 17.4 Transcriptomic clocks

Six clocks from two normalisations (YuGene-diff, Scaling-diff) and three targets (chronological age, normalised age, mortality), trained as multi-species multi-tissue elastic nets over more than 11,000 transcriptomes.

The chain is: filter low-count genes, map to one-to-one orthologues only, RLE-normalise, log10, per-sample z-score or YuGene, then **median-centre across all samples within each dataset**.

That last step is why these are not simply another modality. A transcriptomic clock is **undefined for a single sample in isolation** – the centring is a property of the cohort. An implementation has to refuse a one-sample run rather than warn about it.

Accuracy runs 5-7 years MAE against methylation’s 3-5. The honest reading is that they answer a different question, not that they are worse.

## 17.5 Methylation foundation models

**CpGPT** (pretrained on 2,000+ datasets, 150,000+ samples) and **MethylGPT** (226,555 profiles from 5,281 datasets) are transformers over CpG sites taking sequence, position and epigenetic context, so they can reason about a site they have not seen on an array.

The interesting capability is not another age predictor. It is **zero-shot imputation and cross-array conversion**, which attacks EPIC v2 probe loss, platform harmonisation and single-cell sparsity at once.

CpGPT placed second for age and first for methylation-based mortality in the Biomarkers of Aging Challenge, and its attention weights give per-sample interpretability – which sites drove *this* prediction.

This would also be the first model here where a GPU is the right default. Everything currently shipped is a dot product over a few thousand features, and **GPU support** measures CUDA as *slower* for exactly that reason; a transformer inverts the arithmetic.

## 17.6 Single-cell

Single-cell methylation is binary and sparse – a site is covered or not, methylated or not – so a linear clock cannot be applied directly. **scAge** answers with a per-cell maximum likelihood over the covered subset, closer in spirit to conditioning on coverage than to taking a dot product. On the transcriptomic side **scAgeClock** reaches about 2 years MAE for the best cell types using gated multi-head attention, and single-cell profiles are pooled to pseudobulk before a bulk transcriptomic clock is applied.

## 17.7 Reliability, and why it governs everything above

Technical and biological reliability come apart. Nearly every clock is technically excellent – the same sample assayed twice agrees closely – while biological ICCs, the same person sampled again days later, are substantially lower, with no clear relationship between the two. **GrimAge2 and DunedinPACE are among the most biologically fragile**, and they are the two most often used to claim an intervention worked.

The registry carries both figures where a source exists and **None** where none does, because an unpublished ICC is not a good ICC. The Biomarkers of Aging Consortium’s own stated bottleneck is not the models but the absence of standard procedures for collecting specimens and applying biomarkers.

## 18. Datasets, registries, reference resources

**Benchmark and training data** - [computage/computage\\_bench](#) on Hugging Face - 10,404 benchmark samples / 65 studies and 7,419 training samples / 46 studies, parquet + TSV metadata. - [lucascamillomd/pyaging-data](#) on Hugging Face, clock weights, reference vectors, Ensembl gene annotation, example datasets per modality. - GEO series listed in [ComputAge BenchBSB\\_table.csv](#). - GSE54848 - OSKM reprogramming timecourse used in the pyaging figure. - NHANES III and NHANES IV - shipped as **.rda** in BioAge with a codebook. - UK Biobank, Framingham Heart Study, Generation Scotland, MESA, Health and Retirement Study, Dunedin Study, Tabula Muris Senis, GTEx.

**Gene and biomarker resources** - Open Genes (2,402 genes), GenAge, AnAge (the mammalian clock lifespan lookup), CellAge, CSGene, AgeFactDB, Aging Atlas, GO aging terms, Digital Ageing Atlas, AGEMAP.

**Reference matrices** - [sesame\\_450k\\_median.csv](#) - NA-fill medians. - [450K\\_reinius\\_12\\_reference.csv](#), [EPIC\\_salas\\_18\\_reference.csv](#) - deconvolution signatures. - [biolearn\\_averages\\_450k.csv](#) - alternative fill vector. - [DunedinPACE\\_Gold\\_Means.csv](#) - the 20,000-probe gold standard.

**Model registries** - [pyaging\\_clock\\_metadata.json](#) (173 clocks) + [evidence\\_ledger.jsonl](#) (894 KB of per-field provenance) + [controlled\\_vocabulary.json](#) + [validate\\_metadata.py](#). - [tAge\\_clocks\\_metadata.csv](#) (60 models) + Zenodo record 18763485. - [ComputAge\\_model\\_definitions](#) + [library.yaml](#) (GEO dataset parsers).

## 19. Uncertainty, specimen and platform

Added in this release. Sections 1–19 are about computing the number correctly. This one is about the fact that a correctly computed number is still not interpretable on its own, which is what the field’s own literature says and what no implementation, including this one until now, did anything about.

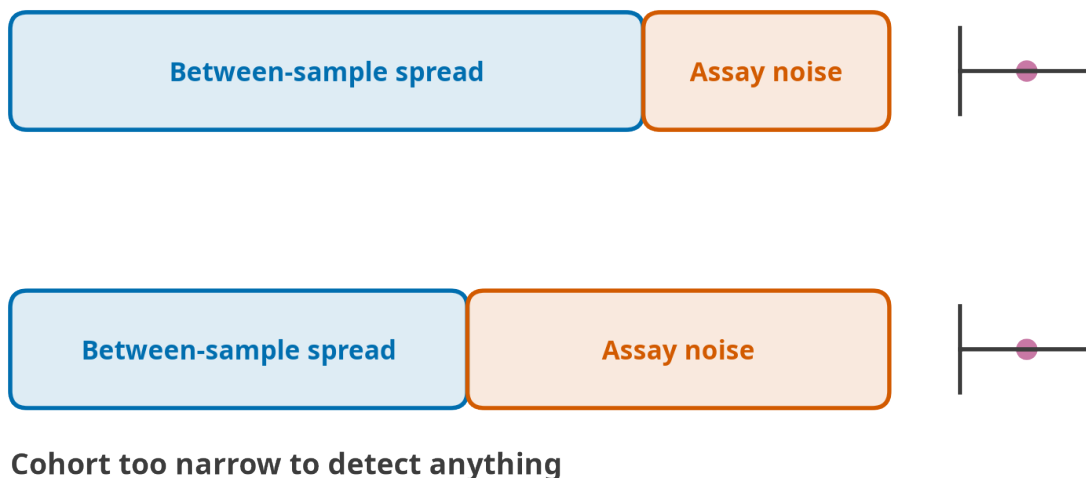


Figure 11: The same assay noise against a wide cohort and a narrow one

Reliability is not a property of a clock alone. The intraclass correlation is  $1 - SE^2 / SD^2$ , so the same technical error divided by a narrower between-sample spread gives a worse coefficient without anything about the assay having changed. A clock with an excellent published ICC, measured on a cohort spanning sixty years of age, can be unable to resolve anything in a trial whose participants are all within a decade of each other. That is the calculation to run before recruiting, not after.

### 20.1 The point estimate problem

Split one DNA sample, run both halves on the same array, and six prominent clocks disagree with themselves by **up to nine years** (Nat Aging 2022). Only **18 % of CpGs on the 450K array reach ICC  $\geq 0.5$**  in adult whole blood (Sugden 2020): for most probes, the technical variance exceeds the biological variance. A reader handed “DNAMAge = 54.2” cannot tell whether a 1.5-year change between two draws is biology or the assay.

Every package in the field emits that bare number. FALCONAge emits an interval, and the arithmetic behind it is short enough to check by hand.

### 20.2 Propagating measurement error

A linear clock is  $score = f(b + w^T\beta)$ . Give each probe a measurement variance  $\sigma^2_j$  and the score’s variance follows directly:

$$\text{Var} = f'(\text{raw})^2 \sum_j w_j^2 \sigma_j^2$$

$$\sigma_j^2 = s_j^2 (1 - \text{ICC}_j)$$

with  $s_j^2$  the spread of probe  $j$  across the cohort in front of you and  $\text{ICC}_j$  its published test-retest reliability. One matrix–vector product per clock.

Three parts of that are load-bearing:

**The ICC table is measured, not assumed.** `registry/data/probe_icc.csv.gz` carries 283,860 probes, derived from 69 blood samples each assayed twice on EPIC v1 and preprocessed with SeSAMe 2, using the two-way random-effects, absolute-agreement, single-rating model, ICC(2,1) (Epigenetics 2024, CC BY 4.0). Its citation, licence and SHA-256 go into the run manifest, because an interval whose provenance is not recorded is worse than no interval.

**An imputed feature contributes its full variance.** Not the reduced  $s^2(1-\text{ICC})$ . An imputed probe is the cohort mean or a published reference; it carries no information about the sample in front of you. Treating it as well-measured would make *worse data produce a narrower interval*, which is the wrong direction and the part of the calculation most likely to look like a bug.

**The output transform is differentiated, not linearised once.** Horvath's `anti_log_linear` has slope  $21 \cdot e^x$  below zero and 21 above it, so a child's uncertainty and an adult's scale differently from the same raw error. Every op in the catalogue has an analytic derivative; the four that destroy the information (`binarize`, `rank_normalize`, `quantile_normalize`, `simplex_projection`) have none, and a chain containing one **refuses** rather than skipping the step and returning a confidently small interval.

**The sum assumes probe errors are independent, and they are not.** Writing  $\sum_j w_j^2 \sigma_j^2$  drops the cross term  $2 \sum_{j < k} w_j w_k \text{Cov}(\varepsilon_j, \varepsilon_k)$ . Chip position, plate and scanner drift move many probes together, and those covariances are positive far more often than negative, so **the SE reported here is a lower bound**, and not necessarily a tight one, since a clock with hundreds of positively correlated terms can have a true SE several times this.

The honest alternative does not exist: a  $319,607 \times 319,607$  error covariance would need replicate designs nobody publishes. What exists is a cross-check. `source="clock"` derives the same quantity from a published clock-level ICC measured on real technical replicates, which *does* contain the correlated part; when the two disagree substantially, the clock-level figure is the one to believe and the gap is the correlated component the probe path cannot see. `icc_from_replicates()` closes it properly on your own duplicates. This is stated rather than buried because an interval presented without its assumption is exactly the false precision the section exists to argue against.

On the corpus, Horvath 2013 comes out at  $\pm 1.58$  years of technical error with an implied cohort ICC of 0.98; DunedinPoAm38 is the least repeatable at 0.72; the 319,607-probe BLUP clock reaches 1.00, because a rotation spread over the whole array averages the noise away. That last result is the same mechanism that makes PC clocks robust to probe loss, arrived at from the other direction. Read all three as lower bounds, for the reason just given.

### 20.3 Prediction error is a different, larger question

`technical_se()` asks what a repeat of the same DNA would do. `conformal_interval()` asks how far the number is from the truth.

Split conformal: take healthy blood samples with known ages, compute the absolute residuals, and the  $[(n+1)(1-\alpha)]/n$  quantile is a half-width that covers the truth at the stated rate on any exchangeable sample. No distributional assumption is made, and the guarantee is finite-sample. Conformalised quantile regression (Genet Epidemiol 2025) is the better version, widening the interval where the training data were sparse; it needs fitted quantile models, so split conformal ships and the calibration table reports the width per age band so a reader can see where one width is lying.

The limitation is not small and is stated in the output: every row carries `exchangeable = False`. The calibration cohort is adult, blood, and overwhelmingly of European ancestry. On a paediatric or non-European cohort the guarantee does not transfer, and no widening factor would make it.

### 20.3.1 What the calibration found, and what it turned into

The first calibration put Ying’s DamAge and AdaptAge 37–150 years from chronological age. The obvious reading (“these are not age predictors”) turned out to be wrong, and the measurement that settled it is worth reproducing because the distinction it draws is a general one.

Regress predicted on chronological across the corpus’s healthy cohorts:

| clock               | r     | slope        | offset swing between datasets |
|---------------------|-------|--------------|-------------------------------|
| horvath2013         | 0.905 | 0.982        | 15 y                          |
| yingcausage         | 0.960 | 0.954        | 15 y                          |
| <b>yingdamage</b>   | 0.909 | <b>0.967</b> | <b>162 y</b>                  |
| <b>yingadaptage</b> | 0.745 | 1.170        | <b>162 y</b>                  |

DamAge’s slope is 0.967, better than DNAmPhenoAge’s 1.199. It tracks age. What it does not have is a **fixed origin**: its offset against chronological age is –84 years in one healthy cohort and +78 in another.

The mechanism is visible in one number. DamAge and AdaptAge move as near mirror images,  $r = -0.975$  across datasets, and the swing in their *sum* is only 33 years against 162 for each alone. Their published intercepts are +543.43 and –511.97; a dataset-level shift in overall methylation pushes one up and the other down by nearly the same amount. One shared offset, hugely amplified, not two broken clocks.

So neither obvious label fits. `age_years` would license `predicted – chronological`, which measures the cohort. `relative_score` would forbid the residual and the group difference, the operations the paper itself reports, and would contradict a unit of years and a training target of chronological age, both of which are accurate.

They are `age_years_relative`: years, slope near one, no fixed zero. LEGAL\_OPS admits `residual`, `difference`, `mean` and `correlate`, and refuses `acceleration`. The vocabulary had listed `acceleration` and `residual` separately since v1.0 and nothing had ever read the difference; this is the first clock that needs it.

### 20.4 Specimen type is not metadata

Same 91 people, three specimens (bioRxiv 2025.09.16.673560, CATSLife1, mean age 30.9):

| Comparison           | Mean difference in clock age | Spearman between the two |
|----------------------|------------------------------|--------------------------|
| saliva vs buffy coat | 3.83 – 16.46 years           | 0.45 – 0.69              |
| saliva vs PBMC       | in the same direction        | 0.25 – 0.55              |
| buffy coat vs PBMC   | –0.06 – 0.39 years           | 0.66 – 0.87              |

Read the two columns together, because separately each is reassuring. Saliva and buffy coat clock ages correlate at up to 0.69, a correlation most people would accept, and are still a decade apart. **Correlation is not agreement**, and the check that gets run is the correlation. The blood pair is the control: same study, same pipeline, difference under half a year, which is what makes the saliva number a specimen effect rather than noise.

Which clock matters as much as which specimen. On those samples DunedinPACE differed by  $-0.007$  years ( $p = 0.486$ ) where Hannum differed by 12.5, so “this clock transfers” is a per-clock fact, not a per-tissue one, and FALCONAge therefore holds the policy per clock. Across unrelated tissues the published figure is 20–30 years (PMC12714307).

FALCONAge reads `obs["tissue"]` against each clock’s declared training tissue and takes one of three positions:

#### allow

the genuinely pan-tissue clocks. Nothing to say.

#### warn

the default. Says what the clock was fitted on, what it was given, and the measured discordance where one is published.

#### refuse

twelve clocks whose training tissue has no counterpart elsewhere: three placenta clocks, cord blood and neonatal blood spots, buccal, brain cortex. A placenta clock on blood is a category error, not an offset.

Cell-free DNA is refused by **every** clock regardless of policy. It is not a tissue a clock generalises to; it is a fragment population shed from many of them, and array clocks applied to it directly perform poorly ([bioRxiv 2025.11.27.690895](#)).

And when `obs` has no `tissue` column at all, the run says so once, because silence was the previous behaviour, and silence is the bug.

#### **i** This check found a real error in our own test suite

At first the integration tests scored the three Lee **placenta** clocks on GSE66459, which is umbilical **cord blood**, and asserted only that the answers looked like gestational weeks. They did, because gestational age is gestational age whatever tissue you read it from. That is exactly the failure the check exists to catch, and it caught it here first.

## 20.5 Probe loss, priced

`probe_loss()` has always said how much of a model is absent: count and coefficient mass. It could not say what that costs. Now it can, because the cost was measured: every full-450K matrix in the corpus, masked down to each platform’s probe set and rescored, 109 samples across five datasets.

| clock           | platform | probes kept | median shift   |
|-----------------|----------|-------------|----------------|
| hrsinchphenoage | EPICv2   | 874 / 959   | <b>+16.7 y</b> |
| yingcausage     | EPICv2   | 492 / 585   | +13.9 y        |
| lin             | EPICv2   | 96 / 99     | −9.4 y         |



| clock       | platform | probes kept       | median shift   |
|-------------|----------|-------------------|----------------|
| hannum      | EPICv2   | 64 / 71           | -7.7 y         |
| horvath2013 | EPICv2   | 339 / 353         | -2.6 y         |
| zhangblup   | EPICv2   | 286,843 / 319,607 | <b>-0.23 y</b> |
| yingdamage  | EPICv1   | 1089 / 1089       | <b>0.00 y</b>  |
| vidalbralo  | either   | 8 / 8             | <b>0.00 y</b>  |

Three things to read out of that table. `lin` loses three probes of ninety-nine and moves nine years, the coefficient-mass argument in its most extreme form. `zhangblup` loses ten percent of a 319,607-probe model and moves a quarter of a year, which is the graceful-degradation property published for PC clocks, reproduced here in a different architecture. And the two rows at exactly zero are the internal check: a clock that loses nothing must shift by nothing, and does.

**What the table does not cover, so it is not read as covering it.** Thirty rows: fifteen clocks  $\times$  two targets, EPIC v1 and EPIC v2. 450K is the *reference* the shift is measured from, so it cannot appear, its shift is zero by construction, not by measurement. 27K is absent for a different reason: masking a 450K matrix to 27,578 probes leaves most of these clocks below the coverage floor, so what would be measured is the shift of a clock that would have been refused anyway. And the reference cohort is 109 blood samples of adults; a shift measured there is a reasonable prior for another blood cohort and not a correction factor for a different tissue.

The shift is **reported and never applied**. An automatic offset would be a second number nobody can trace, which is the failure the coefficient digests exist to prevent.

## 20.6 A reported score must not move when the next plate arrives

ComBat estimates its location and scale from every sample at once. Add a batch, re-run, and every previously corrected value changes, measured directly on epigenetic ages as a mean shift of 0.077 to 0.39 years with a **maximum of 2.20 years** ([iComBat](#), [PMC12495439](#)).

A 2.2-year retrospective change in someone’s reported biological age, caused by a stranger’s sample being run three months later, is not a rounding difference. The individual-level critique of these measures lists “longitudinal comparability without re-running all samples” as a prerequisite for any clinical use, and this is precisely it.

`fit_batch_reference()` freezes the per-probe grand mean, covariate effects, pooled variance **and the empirical-Bayes hyperparameters** on a reference cohort. `apply_batch_reference()` standardises each later batch against those frozen values and estimates only that batch’s own additive and multiplicative effects. A batch’s corrected values then depend on that batch and the reference alone. There is no path through the arithmetic by which a new plate could move an old one, and the test asserts bit equality rather than closeness.

Freezing the hyperparameters is the part that is easy to get wrong. Leave them free and adding a batch re-estimates the prior, which moves every other batch’s shrunk estimate, and the whole exercise is ComBat with extra steps.

Two designs are refused: a batch with fewer than eight samples, and one where the variable of interest is nested inside batch. The second matters because correcting a confounded design removes the effect along with the artefact and returns a clean-looking null, the one failure mode of batch correction that looks like success.



## 20.7 Before the first array, and after the last

**power()** answers “how many samples to see this effect on this clock”, and needs no data file. It refuses to default the standard deviation, because  $n$  scales with its square and a guessed SD is a guessed answer reported to three significant figures. Where a reliability figure is available it splits the answer: at ICC 0.9, a tenth of the sample size exists only to average out the instrument. That is the arithmetic behind the published finding that the original clocks need 3–16 replicates per condition where their PC versions need 1–2.

**consensus()** implements the decision rule from *When to Trust Epigenetic Clocks* [77]. Re-analysing six intervention datasets, the authors found that in **five of six** exactly one clock reached significance, a first-generation clock every time, and **four of those five lost it** under multiple-testing correction. In no case did the PC version of the same clock corroborate. Their conclusion, in their words: *a single significant clock after an intervention is likely a false positive*. The same group later widened this to 16 clocks and 94 further biomarkers across the 51 longitudinal studies of TranslAGE [78], which is where the clock grouping the FALCONAge report uses comes from.

**That PC clause is the sharpest test in the paper, and by default FALCONAge cannot run it.** Every principal-component clock in the registry is catalogued as untraced or licensed, so their weights are not ours to distribute, and a corroboration check with nothing to check against would be inert. **consensus()** therefore names each clock’s counterpart in **high\_reliability\_partner**, reports agreement in **partner\_corroborates**, and where the counterpart was not scored it says so in the verdict string rather than dropping the clause. Registering the weights with **fa.registry.register\_local\_weights()** turns the check on.

So **consensus()** tests every scored clock, corrects across the whole set actually run, and returns a verdict rather than a table to pick from: **supported** needs significance across at least two generations including an outcome-trained clock and survival of Bonferroni; one lit clock is **unsupported** whatever its p-value. The verdict string always carries the counts it was computed from, a verdict without its arithmetic is an oracle, and that is the opposite of the posture of this package.

## 20.8 Evidence, attached to the score

**result.evidence()** returns what a value on this clock has been *shown* to predict, with effect sizes and DOIs. Seeded from studies that tested many clocks on one cohort under one protocol, so the rows are comparable to each other: GrimAge v2 at hazard ratio 1.54 per SD for all-cause mortality and 1.86 for cirrhosis, DunedinPACE at 1.44 for diabetes, and the observation that first-generation clocks account for about 5 % of significant disease associations across 174 outcomes in 18,859 people (**Nat Commun 2025**).

A row without a DOI is rejected at load time. An uncheckable effect size looks identical to a checkable one in a table, and is worth less than nothing.

The most useful row is not about any clock. Across 39 biomarkers in more than 20,000 people, a clock’s chronological-age accuracy and its mortality prediction are **uncorrelated:  $R = 0.12$ ,  $P = 0.67$**  (**Nat Aging 2025**). That is the empirical justification for **scale\_type** and **LEGAL\_OPS**, and it is the reason “which clock is most accurate” is the wrong question.

## 20.9 What was deliberately not built

Recorded so it does not appear by accident.

- **No silent correction.** Batch reference, platform offset and interval are all reported. A score adjusted by an untraceable factor destroys the provenance discipline that is the reason to use this package.

- **No percentile against an unnamed population.** The same person reads as accelerated or decelerated depending on the comparator; a percentile without a cited reference cohort means different things to different readers.
- **No diagnostic thresholds.** None exist in the literature. Nothing here is coloured red above a cut-off.
- **No LLM dependency.** A scoring library whose output depends on a model that changes weekly is not reproducible.

## 20. From raw arrays to betas

Also new here, and the other half of it. Section 20 is about what a score is worth; this is about being able to produce one from what a laboratory actually has.

At first FALCONAge read beta matrices and told you to normalise elsewhere. That was accurate and it was the single largest reason somebody with their own IDATs could not use the package.

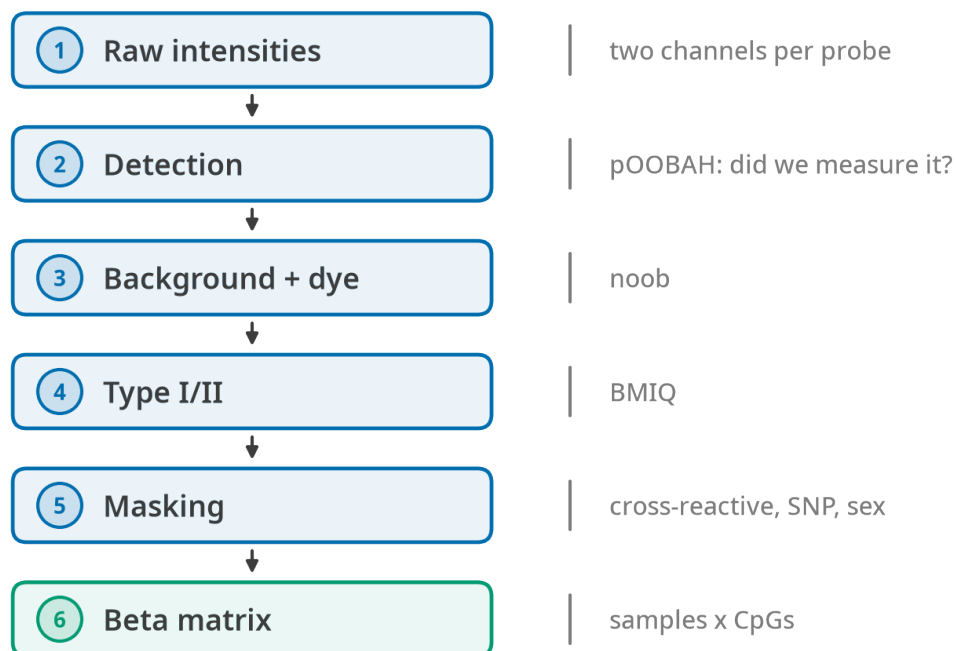


Figure 12: The chain from two channels of raw intensity to a beta matrix

Each step answers a question the next one assumes. Detection asks whether a probe was measured at all, background and dye correction asks what the intensity would have been without the array's own offset, and BMIQ asks how to put two probe chemistries on one scale. Masking removes probes that are known to be unreliable rather than merely absent. Skipping a step does not fail; it produces a beta matrix that is quietly on a different scale from the one the clock's coefficients were fitted on.

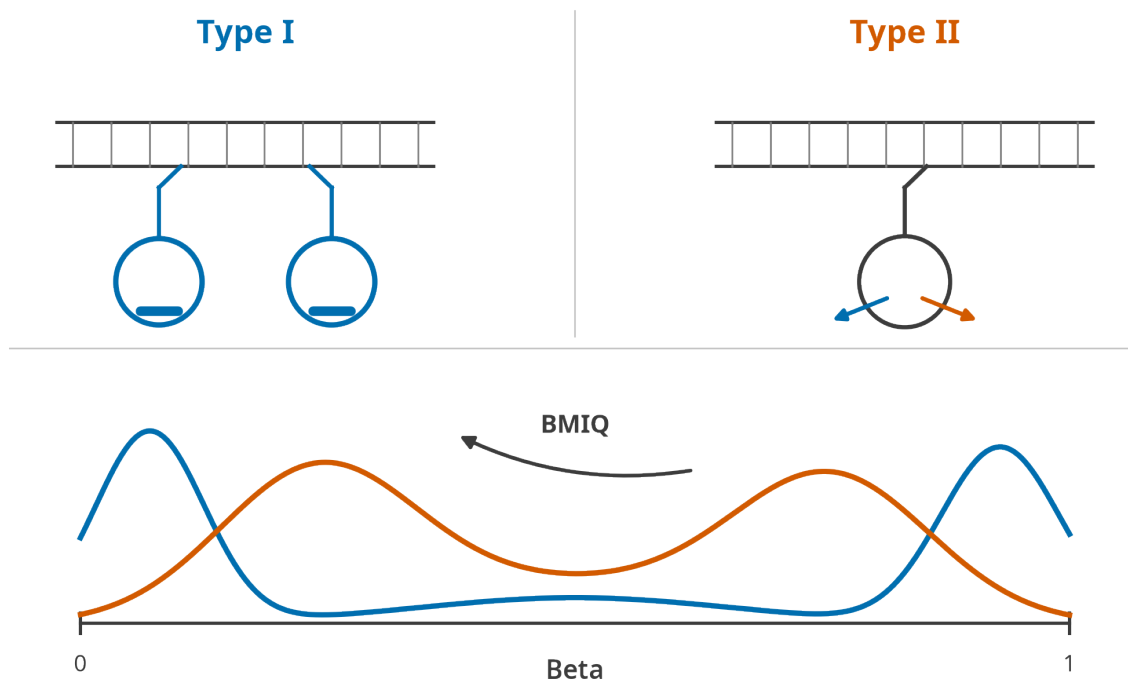


Figure 13: Why the two probe designs need reconciling before a coefficient means anything

A type I probe uses two bead addresses and one colour channel; a type II probe uses one address and both channels. The two designs give measurably different beta distributions, with type II compressed toward the middle of the range, which is what BMIQ corrects by mapping type II onto the type I reference. On EPIC arrays type II probes are the large majority, so leaving this uncorrected shifts most of the matrix a clock will read.

### 21.1 What an IDAT does not contain

An IDAT is keyed by **bead address**, not by probe. A type I probe has two addresses and reads both in one colour; a type II probe has one address and reads methylated in green, unmethylated in red. Which colour a type I probe uses is a property of its chemistry and is written down nowhere in the file.

So the first thing the chain needs is the array manifest, and Illumina's licence does not permit redistributing it. `methyIprep` (MIT) solved this by downloading from a public S3 bucket and caching; FALCONAge uses the same bucket, the same filenames, and its own `download.fetch`. One network round trip per machine, ever. Every other path in the package works offline and this one says so.

The manifest's SHA-256 goes into the run manifest. A beta matrix is a function of which manifest resolved its addresses, in exactly the way a score is a function of which coefficient file produced it, and recording one and not the other would be an odd place to stop.

### 21.2 Out-of-band signal, which everything downstream rests on

A type I probe's *other* colour, at its own two addresses, measures nothing real. It is the array's background, sampled on this chip, in this run, at tens of thousands of addresses. That is what makes it a usable null, and it is why type II probes, which have no out-of-band signal, borrow the type I null.

Two steps use it and they must run in this order:

**pOOBAH** (Zhou et al. 2018) asks, per probe, the probability that a signal this bright could come from background alone; read straight off the empirical out-of-band distribution in the relevant channel. A probe above 0.05 becomes **NaN**, because a beta of 0.43 computed from two noise readings is indistinguishable from a real 0.43 once it is in the matrix.

**noob** (Triche et al. 2013) then deconvolves the observed signal into a normal background plus an exponential true signal, per channel, using the same out-of-band values.

Detection **before** correction, and **noob()** raises if asked to run first. The reason is not stylistic: pOOBAH's null is the *uncorrected* out-of-band distribution, and correcting first subtracts the background from the very numbers that define what background looks like. Every probe then clears a bar that has been lowered underneath it.

Two numerical points worth recording because both were wrong in a first draft:

- The background location and scale use **Huber's M-estimator**, as minfi does. The upper tail of the out-of-band signal is cross-hybridisation, not background. On the corpus's EPIC v1 arrays a plain mean of the contaminated green channel reads about three times the robust estimate.
- The normal-exponential posterior is evaluated through **log\_ndtr**. Written directly as a density over a tail probability it is **0/0** for any probe well below background, which is *every undetected probe on the array*, not an edge case.

### 21.3 The validation, and why it is a real one

The corpus holds raw IDATs for GSM5548192 and GSM5548193, and the betas their authors published for those same two physical samples. The two paths share nothing.

| correlation                | r = 0.99928 |
|----------------------------|-------------|
| median absolute difference | 0.0113      |
| probes within 0.05         | 99.7%       |
| probes compared            | 809,400     |

That is a check on the address decoding, the type I/II split and the channel assignment, where a raw-array reader actually goes wrong, and where an error would be a whole-matrix error rather than a subtle one.

It is asserted on the **uncorrected** betas, deliberately. GSE182991's published matrix agrees with our raw decode at  $r = 0.999$ , which says its own processing was minimal; comparing our background-corrected output against it would be asserting that a correction does nothing.

### 21.4 Two steps that ship switched off, and why

**Dye bias.** Green and red do not fluoresce equally, and a type II probe reads methylated in one and unmethylated in the other, so it carries a tilt that is not methylation. The standard fix matches the two channels using type I probes.

Measured on real IDATs, our implementation of it moves the median beta by **+0.10 to +0.12**: far more than a dye correction should. The cause is visible in the data: on that chip red runs about twice as hot as green (median out-of-band 1,171 against 415), so mapping red onto green halves every type II unmethylated signal.

That is a limitation of the *assumption*, not an arithmetic slip. Matching the two channels' distributions presumes type I green probes and type I red probes measure the same underlying quantity; on this array their median betas are 0.05 and 0.14. A correct correction needs the normalisation control probes, whose

addresses are not in the core-columns manifest we can fetch. So it is left in, opt-in, and documented. Turned on by default it would silently make results worse.

**BMIQ.** Implemented in full, a three-state beta mixture per probe type, fitted by EM, then a quantile map within each state (Teschendorff et al. 2013). It has to be asked for because clocks differ in how much they depend on it, and type I probes come out untouched, so BMIQ never moves a clock built purely on them.

### 21.5 Probe masks: standard for an EWAS, a decision for a clock

Zhou, Laird and Shen catalogued the probes that do not measure what their name says, multi-mapping, common SNP at the interrogated base, repeats, polymorphic extension base, and publish a recommended general mask. On 450K it flags 29,504 probes, six percent of the array.

Masking them is correct for an EWAS. **Scoring a pre-fitted clock is a different situation:** every clock in this registry was trained on unmasked data, before most of these masks existed, and its coefficients were fitted with those probes in, whatever the probes were measuring. Removing them at score time does not recover a truer number: it deletes an input the model expects and hands the gap to the imputer.

So nothing is masked automatically, and `mask_report()` prices the decision:

| clock         | probes masked  | share of model | share of  coefficient |
|---------------|----------------|----------------|-----------------------|
| dunedinpoam38 | 5 / 46         | 10.9%          | <b>6.1%</b>           |
| dnamt1        | 15 / 140       | 10.7%          | 5.4%                  |
| hannum        | 1 / 71         | 1.4%           | 1.6%                  |
| horvath2013   | <b>1 / 353</b> | 0.3%           | <b>0.2%</b>           |

Two clocks, the same published mask, and opposite answers. For Horvath it is free; for DunedinPoAm38 it removes a ninth of the model. The coefficient-mass column is the one to read, and it is why this is a report rather than a step.

### 21.6 What the chain still does not do

- **No control-probe normalisation.** The core-columns manifest carries no control addresses, which is why dye bias is unvalidated (§20.4).
- **No cross-platform liftover.** Unchanged from v1.0, and blocked for the same reason: mLiftOver's tables encode empirical paired-sample concordance.
- **EPIC v1 and v2 cannot be told apart by address count.** They are five percent apart, so detection refuses as ambiguous rather than picking the closer one. Pass `platform=` for a chip near the boundary.

## 21. Genetic variation under the probe

Every section above treats a beta value as a measurement of methylation. It is also, in part, a measurement of genotype, and the two are not separable by looking at the number.

### 21.1 A probe sits at a genomic coordinate

Each probe has a chromosome and a base position, so the probe set can be intersected with any variant catalogue: ClinVar for clinically annotated variants, dbSNP for the general catalogue. The intersection is an ordinary range join, which `GenomicRanges` or `bedtools` will do, and it answers a question that matters before scoring: does this clock depend on probes whose signal a common variant can move.

## 21.2 Three ways a variant reaches the beta value

**A SNP at the interrogated base.** Single-base extension reads the variant rather than the methylation state, so the beta becomes a genotype readout. This is the population of probes the published masks remove, and it is why FALCONAge applies the Zhou masks rather than treating every probe as equally informative.

**A SNP under the probe body.** Hybridisation is weakened, intensity falls, and the beta drifts toward the middle of the range without any obvious signature of failure.

**Cross-reactivity.** A 50-mer against a three-base genome maps to more than one locus more often than the design intends. Chen and colleagues catalogued these for the 450K, and the affected probes carry a mixture of signals from several places in the genome.

## 21.3 meQTL: the variant is somewhere else entirely

A methylation quantitative trait locus is a variant associated with methylation at a CpG, usually nearby but sometimes far away. GoDMC and the EPiGEN meQTL resource map these genome-wide. The consequence for a clock is direct: if a clock CpG is under strong meQTL control, part of the between-person variance in that clock is genetic and fixed at conception rather than acquired, and no intervention will move it.

### i What this changes about a score

A clock's between-person variance splits into an acquired component, which is what the clock is meant to measure, and a genetic component, which is not. The split is not visible in the score. Two people with the same clock age can differ because one has lived differently and the other inherited a different genotype at a handful of high-weight probes.

## 21.4 The arrays carry genotype probes on purpose

450K and EPIC include rsID-labelled SNP probes, 65 on the 450K, for sample identity. They confirm that two arrays came from the same person, which is the check that catches a swapped sample. [MethylToSNP](#) goes further and infers genotypes from the methylation pattern itself, which is useful and is also a reminder that array data is identifiable.

# 22. Structure above the single probe

## 22.1 A clock is a combination score

A clock sums hundreds of weighted features, which makes it the same kind of object as a polygenic risk score, and methylation risk scores for individual traits are built the same way. The TransLAGE panel is largely made of them: 1,694 of its 1,803 biomarkers are proxies for proteins, metabolites and clinical values rather than clocks.

The shared property matters more than the shared arithmetic. A combination score is robust where a single probe is noisy, and it is opaque in the same measure: a change in the total does not say which component moved. That is the argument for the subscore clocks in section 5.

## 22.2 Co-methylation modules

CpGs covary. Weighted correlation network analysis groups them into modules and summarises each by its first principal component, and the module can then be tested as one unit rather than as several thousand

correlated tests. This is where the multiple-testing burden of an epigenome-wide scan becomes tractable, and it recovers structure that a per-probe analysis cannot see.

### 22.3 Allele-specific methylation

The two alleles at a locus can carry different methylation, and the pattern can follow a haplotype block. Array data cannot resolve this because a beta value averages both alleles into one number. Sequencing can, and tools such as MethHaplo combine allele-specific methylation with genotype to identify the regions involved. Imprinted loci are the familiar case and not the only one.

### 22.4 Interactions between variants are real and small

Genome-wide tests for SNP by SNP effects on methylation find them, and find that they explain little variance. The practical reading is that additive models are not leaving much on the table at this level, which is consistent with the observation in section 4 that elastic net remains competitive with deep networks on this problem.

## 23. Building a clock of your own

The catalogue is other people's clocks. This is the arithmetic of fitting one, with the numbers the published clocks actually used.

### 23.1 The objective

With  $n$  samples and  $p$  CpGs,  $p \gg n$  always. Horvath had 8,000 samples against 21,369 candidate probes; a modern EPIC study has 850,000 columns and a few hundred rows. An unpenalised least-squares fit is not identifiable, because  $X^\top X$  is singular whenever  $p > n$ . Penalisation is what makes the problem have an answer, not a refinement on one.

$$\hat{\beta} = \arg \min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - \beta_0 - x_i^\top \beta)^2 + \lambda \left[ \frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right] \right\}$$

The two penalties do different work. The  $\ell_1$  term has a corner at zero, so its subgradient can hold a coefficient exactly at zero, which is what performs selection. The  $\ell_2$  term is smooth and shrinks correlated predictors together rather than choosing between them, which matters because neighbouring CpGs are strongly correlated and pure LASSO picks among them close to arbitrarily.

| $\alpha$ | Name        | Behaviour on correlated CpGs                                           | Used by                       |
|----------|-------------|------------------------------------------------------------------------|-------------------------------|
| 1        | LASSO       | Selects one per correlated group, close to arbitrarily                 | eFRS, the McCartney EpiScores |
| 0.5      | Elastic net | Keeps correlated groups together while still zeroing most coefficients | Horvath 2013                  |
| 0        | Ridge       | Shrinks everything, selects nothing                                    | Rarely used alone             |

Horvath fitted  $\alpha = 0.5$  with  $\lambda = 0.0226$  chosen by cross-validation, over the 21,369 probes shared by the 27K and 450K platforms that had fewer than ten missing values. That produced the 353 sites.



! The selected set is not the biology

Rerun the same fit with a slightly different  $\lambda$ , a different fold split, or a different subset of samples, and a different CpG set comes out with almost identical accuracy. The correlated groups are interchangeable to the optimiser. This is the mechanism behind the eleven discrepancies in section 15, and it is why two faithful implementations of the same clock can disagree on membership.

### 23.2 Choosing the penalty honestly

$\lambda$  comes from  $k$ -fold cross-validation: fit on  $k - 1$  folds, measure on the held-out one, take the  $\lambda$  minimising mean cross-validated error, or the largest  $\lambda$  within one standard error of it for a sparser model.

Any step that looked at the outcome must sit inside the loop. Selecting CpGs by their correlation with age on the full dataset and then cross-validating the fit on those CpGs leaks the held-out fold into the selection, and the reported error is optimistic by an amount that cannot be recovered afterwards.

### 23.3 The age transform

Methylation changes quickly in childhood and slowly afterwards, so a model linear in age fits adults and misses the first two decades. Horvath regresses a transformed age and inverts afterwards, with  $A = 20$ :

$$F(\text{age}) = \begin{cases} \log(\text{age} + 1) - \log(A + 1) & \text{if } \text{age} \leq A \\ \frac{\text{age} - A}{A + 1} & \text{if } \text{age} > A \end{cases}$$

Both branches give 0 at  $\text{age} = A$  and both have derivative  $1/(A + 1)$  there, so  $F$  is continuous and differentiable at the join. The model predicts  $F$ ; the reported age is  $F^{-1}$  of the linear predictor. Applying the inverse twice, or omitting it, is invisible in an adult cohort because the branches nearly agree there, and badly wrong in a paediatric one.

### 23.4 Feature selection is not settled

Elastic net is the default rather than the optimum. Benchmarked on the Hannum blood data (GSE40279, 656 samples, 473,035 CpGs, 10-fold cross-validation, 80/20 split):

| Method                                            | CpGs kept | R <sup>2</sup>          | Median error                         |
|---------------------------------------------------|-----------|-------------------------|--------------------------------------|
| SelectKBest (2,000)<br>then Boruta                | 35        | 0.873                   | 3.08 years                           |
| Boruta alone, or KBest<br>alone                   | varies    | 0.861 to 0.862          |                                      |
| Recursive elimination,<br>1% per iteration        | varies    | ≥ 0.81                  |                                      |
| Genetic algorithm,<br>most threshold meth-<br>ods | varies    | 0.77 to 0.81            |                                      |
| Neural network on 65<br>features                  | 65        | 0.76                    | found 61 CpGs no<br>other method did |
| Elastic net, no prior<br>selection                | 276       | below five of the above |                                      |
| The five most-selected<br>CpGs overall            | 5         | 0.83                    | 3.79 years                           |
| cg16867657 (ELOVL2)<br>alone                      | 1         | 0.73                    |                                      |



Two rows there are worth sitting with. A single CpG in ELOVL2 reaches  $R^2 = 0.73$ , so most of what a 353-CpG clock knows about chronological age is available from a handful of sites. And an unfiltered elastic net over all 450,000 probes was beaten by five selection strategies, which is the opposite of the usual intuition: past a point the extra probes contribute noise rather than signal.

### 23.5 Two metrics that disagree, and both are needed

$$\text{MAE} = \text{median}_i |\hat{y}_i - y_i|, \quad r = \text{corr}(\hat{y}, y)$$

Correlation is inflated by a wide age range and says nothing about bias. Median absolute error is in years and says nothing about ordering. A clock at  $r = 0.95$  on a cohort spanning 20 to 90 may have an MAE of six years; the same clock on a cohort spanning 55 to 65 may have  $r$  near zero and the same MAE. Neither number is wrong and neither alone is informative.

Expect the fitted slope of  $\hat{y}$  on  $y$  to fall below one. Penalisation shrinks coefficients toward zero, which compresses predictions toward the training mean, so the old read young and the young read old. That is regression to the mean, a property of the estimator rather than a defect of the cohort, and it leaves the residual correlated with age unless the slope is corrected. It is why section 19 residualises before comparing anything.

### 23.6 The step that gets skipped

A model fitted to chronological age will correlate with chronological age. Reporting that as validation is circular. The claim worth making is about the residual: whether age acceleration associates with disease, mortality or an intervention, on a cohort that was not used to fit the model. Everything in section 11 applies to a clock you fitted exactly as it applies to a published one.

### 23.7 What to publish

Feature list, weights, intercept, the preprocessing chain and the postprocess, together and in the order they apply. A clock published without its postprocess is the commonest source of the disagreements in section 15, because the next implementer has to guess whether an anti-log was already applied, and a plausible wrong guess produces plausible wrong ages.

## References

The bibliography moved to [its own page](#), where it is numbered once and cited by number from every page of this site. It reads the same in the PDF, where it is bound as the last chapter.

Citations in this document are the bracketed numbers, for example [5] for Horvath 2013. The numbering is global, so a source keeps its number wherever it appears.

## Appendix A - Full pyaging clock registry (173 clocks)

Generated from `pyaging clock_metadata.json`. Columns: clock id, year, data type, species, tissue, predicts, training target, unit, model type, platform, n features, population, in-model preprocess, in-model postprocess, ships reference values.

| clock           | yr   | data                 | species                | tissue         | predicts                     | trained_on                   | unit        | model                              | platform                     | n_feat | pop                    | pre      | post          | ref |
|-----------------|------|----------------------|------------------------|----------------|------------------------------|------------------------------|-------------|------------------------------------|------------------------------|--------|------------------------|----------|---------------|-----|
| abec            | 2020 | DNA methylation      | Homo sapiens           | whole blood    | chronological age            | chronological age            | years       | elastic net regression             | Illumina EPIC                | 1695   | adults                 |          |               |     |
| adbahadosingh   | 2021 | DNA methylation      | Homo sapiens           | whole blood    | late-onset Alzheimer disease | late-onset Alzheimer disease | probability | logistic regression                | Illumina EPIC                | 4      | older adults           |          | sigmoid       |     |
| altumage        | 2022 | DNA methylation      | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | deep neural network                | Illumina 27K; Illumina 450K  | 20318  | all ages               | scale    |               | ref |
| bitage          | 2021 | transcriptomics      | Caenorhabditis elegans | whole organism | biological age               | biological age               | hours       | elastic net regression             | RNA-seq                      | 576    | Caenorhabditis elegans | binarize |               |     |
| bocklandt       | 2011 | DNA methylation      | Homo sapiens           | saliva         | EDARADD methylation          | chronological age            | beta value  | single-CpG score                   | Illumina 27K                 | 1      | adults                 |          |               |     |
| bohlin          | 2016 | DNA methylation      | Homo sapiens           | cord blood     | gestational age              | gestational age              | weeks       | LASSO regression                   | Illumina 450K                | 251    | newborns               |          | days_to_weeks |     |
| cabec           | 2020 | DNA methylation      | Homo sapiens           | whole blood    | chronological age            | chronological age            | years       | elastic net regression             | Illumina 450K; Illumina EPIC | 1892   | adults                 |          |               |     |
| camilloh3k27ac  | 2025 | histone modification | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | PCA + elastic net + ARD regression | ChIP-seq                     | 1275   | all ages               |          |               |     |
| camilloh3k27me3 | 2025 | histone modification | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | PCA + elastic net + ARD regression | ChIP-seq                     | 922    | all ages               |          |               |     |
| camilloh3k36me3 | 2025 | histone modification | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | PCA + elastic net + ARD regression | ChIP-seq                     | 870    | all ages               |          |               |     |
| camilloh3k4me1  | 2025 | histone modification | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | PCA + elastic net + ARD regression | ChIP-seq                     | 892    | all ages               |          |               |     |
| camilloh3k4me3  | 2025 | histone modification | Homo sapiens           | multi-tissue   | chronological age            | chronological age            | years       | PCA + elastic net                  | ChIP-seq                     | 1240   | all ages               |          |               |     |

| clock                          | yr   | data                 | species      | tissue                    | predicts                       | trained_on                              | unit        | model                                     | platform      | n_feat | pop                    | pre                       | post            | ref |
|--------------------------------|------|----------------------|--------------|---------------------------|--------------------------------|-----------------------------------------|-------------|-------------------------------------------|---------------|--------|------------------------|---------------------------|-----------------|-----|
|                                |      |                      |              |                           |                                |                                         |             | + ARD regression                          |               |        |                        |                           |                 |     |
| camilloh3k9ac                  | 2025 | histone modification | Homo sapiens | multi-tissue              | chronological age              | chronological age                       | years       | PCA + elastic net + ARD regression        | ChIP-seq      | 102    | all ages               |                           |                 |     |
| camilloh3k9me3                 | 2025 | histone modification | Homo sapiens | multi-tissue              | chronological age              | chronological age                       | years       | PCA + elastic net + ARD regression        | ChIP-seq      | 341    | all ages               |                           |                 |     |
| camil-lpanhistone              | 2025 | histone modification | Homo sapiens | multi-tissue              | chronological age              | chronological age                       | years       | PCA + elastic net + ARD regression        | ChIP-seq      | 3739   | all ages               |                           |                 |     |
| cellpopage                     | 2024 | DNA methylation      | Homo sapiens | cultured fibroblasts      | cell-population passage age    | cell passage number                     | passages    | elastic net regression                    | Illumina EPIC | 42     | human cell cultures    |                           |                 |     |
| compil6                        | 2021 | DNA methylation      | Homo sapiens | whole blood               | interleukin-6                  | interleukin-6                           | unitless    | elastic net regression                    | Illumina 450K | 35     | older adults           |                           |                 |     |
| cortical-clock                 | 2020 | DNA methylation      | Homo sapiens | brain cortex              | chronological age              | chronological age                       | years       | elastic net regression                    | Illumina 450K | 347    | human, age unspecified |                           | anti_log_linear | ref |
| cpgpt-grimage3                 | 2024 | DNA methylation      | Homo sapiens | whole blood               | biological age; mortality risk | mortality                               | years       | Cox proportional hazards regression       | Illumina 450K | 24     | adults                 | scale                     | cox_to_years    |     |
| cpgptpc-grimage3               | 2024 | DNA methylation      | Homo sapiens | whole blood               | biological age; mortality risk | mortality                               | years       | PCA + Cox regression                      | Illumina 450K | 31     | adults                 | scale                     | cox_to_years    |     |
| ctsliver                       | 2024 | DNA methylation      | Homo sapiens | liver                     | chronological age              | chronological age                       | years       | LASSO regression                          | Illumina EPIC | 90     | adults                 |                           |                 |     |
| cvdwesterman                   | 2020 | DNA methylation      | Homo sapiens | whole blood               | cardiovascular disease risk    | cardiovascular disease                  | probability | elastic net Cox ensemble                  | Illumina 450K | 235    | adults                 |                           | sigmoid         |     |
| deconvoluteblood-epicbcell     | 2018 | DNA methylation      | Homo sapiens | purified blood leukocytes | B cell proportion              | cell-type-specific methylation contrast | proportion  | reference-based constrained deconvolution | Illumina EPIC | 600    | adults                 | fill_with_reference_means |                 | ref |
| deconvoluteblood-epic-cd4tcell | 2018 | DNA methylation      | Homo sapiens | purified blood leukocytes | CD4+ T cell proportion         | cell-type-specific methylation          | proportion  | reference-based constrained               | Illumina EPIC | 600    | adults                 | fill_with_reference_means |                 | ref |

| clock                           | yr   | data            | species      | tissue                    | predicts                        | trained_on                              | unit              | model                                     | platform                     | n_feat | pop          | pre                       | post | ref |
|---------------------------------|------|-----------------|--------------|---------------------------|---------------------------------|-----------------------------------------|-------------------|-------------------------------------------|------------------------------|--------|--------------|---------------------------|------|-----|
| deconvoluteblood-epic-cd8tcell  | 2018 | DNA methylation | Homo sapiens | purified blood leukocytes | CD8+ T cell proportion          | cell-type-specific methylation contrast | proportion        | reference-based constrained deconvolution | Illumina EPIC                | 600    | adults       | fill_with_reference_means |      | ref |
| deconvoluteblood-epicmonocyte   | 2018 | DNA methylation | Homo sapiens | purified blood leukocytes | monocyte proportion             | cell-type-specific methylation contrast | proportion        | reference-based constrained deconvolution | Illumina EPIC                | 600    | adults       | fill_with_reference_means |      | ref |
| deconvoluteblood-epicneutrophil | 2018 | DNA methylation | Homo sapiens | purified blood leukocytes | neutrophil proportion           | cell-type-specific methylation contrast | proportion        | reference-based constrained deconvolution | Illumina EPIC                | 600    | adults       | fill_with_reference_means |      | ref |
| deconvoluteblood-epicnkcell     | 2018 | DNA methylation | Homo sapiens | purified blood leukocytes | natural killer cell proportion  | cell-type-specific methylation contrast | proportion        | reference-based constrained deconvolution | Illumina EPIC                | 600    | adults       | fill_with_reference_means |      | ref |
| depressionbarbu                 | 2021 | DNA methylation | Homo sapiens | whole blood               | major depressive disorder       | major depressive disorder               | unitless          | elastic net regression                    | Illumina EPIC                | 196    | adults       |                           |      |     |
| dnamfili                        | 2022 | DNA methylation | Homo sapiens | whole blood               | frailty risk                    | frailty                                 | unitless          | LASSO regression                          | Illumina EPIC; Illumina 450K | 20     | older adults |                           |      |     |
| dnamfitage                      | 2023 | DNA methylation | Homo sapiens | whole blood               | physical-fitness biological age | biological age                          | years             | Klemera–Doubal composite                  | Illumina 450K                | 630    | adults       |                           |      | ref |
| dnamfitagegaitf                 | 2023 | DNA methylation | Homo sapiens | whole blood               | gait speed                      | gait speed                              | meters per second | LASSO regression                          | Illumina 450K                | 53     | adult women  |                           |      | ref |
| dnamfitagegaitm                 | 2023 | DNA methylation | Homo sapiens | whole blood               | gait speed                      | gait speed                              | meters per second | LASSO regression                          | Illumina 450K                | 59     | adult men    |                           |      | ref |
| dnamfitagegripf                 | 2023 | DNA methylation | Homo sapiens | whole blood               | grip strength                   | grip strength                           | kilograms         | LASSO regression                          | Illumina 450K                | 91     | adult women  |                           |      | ref |
| dnamfitagegripm                 | 2023 | DNA methylation | Homo sapiens | whole blood               | grip strength                   | grip strength                           | kilograms         | LASSO regression                          | Illumina 450K                | 93     | adult men    |                           |      | ref |

| clock                     | yr   | data                    | species         | tissue                                               | predicts                                   | trained_on                      | unit                                                  | model                                                            | platform                                                      | n_feat | pop               | pre                                                            | post | ref |
|---------------------------|------|-------------------------|-----------------|------------------------------------------------------|--------------------------------------------|---------------------------------|-------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|--------|-------------------|----------------------------------------------------------------|------|-----|
| dnam-<br>fitage<br>vo2max | 2023 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | VO2max                                     | VO2max                          | milliliters<br>per kilo-<br>gram per<br>minute        | LASSO<br>regres-<br>sion                                         | Illumina<br>450K                                              | 41     | adults            |                                                                |      | ref |
| dnamic                    | 2025 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | intrinsic<br>capacity                      | intrinsic<br>capacity           | unitless                                              | elastic net<br>regres-<br>sion                                   | Illumina<br>EPIC                                              | 91     | older<br>adults   |                                                                |      |     |
| dnamphe-<br>noage         | 2018 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | pheno-<br>typic age                        | pheno-<br>typic age             | years                                                 | elastic net<br>regres-<br>sion                                   | Illumina<br>27K; Il-<br>lumina<br>450K; Il-<br>lumina<br>EPIC | 513    | adults            |                                                                |      |     |
| dnam-<br>stress           | 2023 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | stress ex-<br>posure                       | stress ex-<br>posure            | unitless                                              | boot-<br>strap-sta-<br>bilized<br>elastic net<br>regres-<br>sion | Illumina<br>EPIC                                              | 211    | adults            |                                                                |      |     |
| dnamtl                    | 2019 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | leukocyte<br>telomere<br>length            | leukocyte<br>telomere<br>length | kilobases                                             | elastic net<br>regres-<br>sion                                   | Illumina<br>450K; Il-<br>lumina<br>EPIC                       | 140    | adults            |                                                                |      |     |
| downsyn-<br>drome         | 2021 | DNA<br>methyla-<br>tion | Homo<br>sapiens | neona-<br>tal blood<br>spots                         | Down<br>syndrome<br>methyla-<br>tion score | not ap-<br>plicable             | unitless                                              | weighted<br>linear<br>score                                      | Illumina<br>EPIC                                              | 652    | newborns          |                                                                |      |     |
| dunedin-<br>pace          | 2022 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | pace of<br>aging                           | pace of<br>aging                | biological<br>years per<br>chronolo-<br>gical<br>year | elastic net<br>regres-<br>sion                                   | Illumina<br>EPIC                                              | 20000  | adults            | quantile_<br>normal-<br>ization_<br>with_<br>gold_<br>standard |      | ref |
| dunedin-<br>poam38        | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | pace of<br>aging                           | pace of<br>aging                | biological<br>years per<br>chronolo-<br>gical<br>year | elastic net<br>regres-<br>sion                                   | Illumina<br>450K                                              | 46     | adults            |                                                                |      |     |
| eabec                     | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood                                       | chronolo-<br>gical<br>age                  | chronolo-<br>gical<br>age       | years                                                 | elastic net<br>regres-<br>sion                                   | Illumina<br>EPIC                                              | 1791   | adults            |                                                                |      |     |
| encen100                  | 2023 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood;<br>saliva;<br>buccal ep-<br>ithelium | chronolo-<br>gical<br>age                  | chronolo-<br>gical<br>age       | years                                                 | elastic net<br>regres-<br>sion                                   | Illumina<br>450K; Il-<br>lumina<br>EPIC                       | 198    | centenari-<br>ans |                                                                |      |     |
| encen40                   | 2023 | DNA<br>methyla-<br>tion | Homo<br>sapiens | whole<br>blood;<br>saliva;                           | chronolo-<br>gical<br>age                  | chronolo-<br>gical<br>age       | years                                                 | elastic net<br>regres-<br>sion                                   | Illumina<br>450K; Il-<br>lumina<br>EPIC                       | 559    | older<br>adults   |                                                                |      |     |

| clock                               | yr   | data                    | species                                | tissue                                                                  | predicts                                              | trained_on                                            | unit                                 | model                                                     | platform                                                                           | n_feat | pop                            | pre         | post              | ref |
|-------------------------------------|------|-------------------------|----------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|--------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------------------------------|-------------|-------------------|-----|
|                                     |      |                         |                                        | buccal ep-<br>ithelium                                                  |                                                       |                                                       |                                      |                                                           |                                                                                    |        |                                |             |                   |     |
| ensem-<br>bleagehu-<br>manmous<br>e | 2025 | DNA<br>methyla-<br>tion | Homo<br>sapiens<br>and Mus<br>musculus | multi-<br>tissue;<br>whole<br>blood                                     | relative<br>age                                       | relative<br>age                                       | relative<br>age                      | elastic net<br>regres-<br>sion                            | mam-<br>malian<br>methyla-<br>tion array                                           | 100    | humans<br>and mice             |             |                   |     |
| ensem-<br>bleagesta-<br>tic         | 2025 | DNA<br>methyla-<br>tion | Mus mus-<br>culus                      | multi-tis-<br>sue                                                       | interventi<br>on-re-<br>sponsive<br>epigenetic<br>age | interventi<br>on-re-<br>sponsive<br>epigenetic<br>age | years                                | elastic net<br>regres-<br>sion                            | Horvath<br>Mam-<br>malMeth<br>ylChip40;<br>Horvath<br>Mam-<br>malMeth<br>ylChip320 | 288    | mice                           |             |                   |     |
| ensem-<br>bleagesta-<br>tictop      | 2025 | DNA<br>methyla-<br>tion | Mus mus-<br>culus                      | multi-tis-<br>sue                                                       | interventi<br>on-re-<br>sponsive<br>epigenetic<br>age | interventi<br>on-re-<br>sponsive<br>epigenetic<br>age | years                                | elastic net<br>regres-<br>sion                            | Horvath<br>Mam-<br>malMeth<br>ylChip40;<br>Horvath<br>Mam-<br>malMeth<br>ylChip320 | 431    | mice                           |             |                   |     |
| epicga                              | 2021 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | cord<br>blood                                                           | gesta-<br>tional age                                  | gesta-<br>tional age                                  | days                                 | LASSO<br>regres-<br>sion                                  | Illumina<br>EPIC                                                                   | 176    | newborns                       |             | days_to_<br>weeks |     |
| epicmithy-<br>per                   | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | B cells                                                                 | mitotic<br>age                                        | replica-<br>tive his-<br>tory                         | propor-<br>tion                      | mean<br>methyla-<br>tion ag-<br>gregation                 | Illumina<br>450K; Il-<br>lumina<br>EPIC                                            | 184    | human,<br>age un-<br>specified | mean        |                   | ref |
| epicmithypo                         | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | B cells                                                                 | mitotic<br>age                                        | replica-<br>tive his-<br>tory                         | propor-<br>tion                      | comple-<br>ment of<br>mean<br>methyla-<br>tion            | Illumina<br>450K; Il-<br>lumina<br>EPIC                                            | 1164   | human,<br>age un-<br>specified | mean        |                   | ref |
| epitoc1                             | 2016 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | multi-<br>tissue;<br>whole<br>blood                                     | mitotic<br>age                                        | chrono-<br>logical<br>age                             | beta value                           | mean<br>methyla-<br>tion ag-<br>gregation                 | Illumina<br>450K                                                                   | 385    | all ages                       | mean        |                   | ref |
| epitoc2                             | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | whole<br>blood                                                          | mitotic<br>age                                        | chrono-<br>logical<br>age                             | cell divi-<br>sions per<br>stem cell | dynamic<br>methyla-<br>tion<br>trans-<br>mission<br>model | Illumina<br>450K                                                                   | 163    | adults                         | nan_to_zero |                   | ref |
| epitoc3                             | 2020 | DNA<br>methyla-<br>tion | Homo<br>sapiens                        | cultured<br>primary<br>human<br>cells;<br>whole<br>blood;<br>multi-tis- | mitotic<br>age                                        | popula-<br>tion dou-<br>blings                        | cell divi-<br>sions per<br>stem cell | dynamic<br>methyla-<br>tion<br>trans-<br>mission<br>model | Illumina<br>450K; Il-<br>lumina<br>EPIC                                            | 170    | all ages                       | nan_to_zero |                   | ref |

| clock            | yr   | data            | species      | tissue             | predicts             | trained_on           | unit                             | model                                  | platform      | n_feat | pop          | pre | post         | ref |
|------------------|------|-----------------|--------------|--------------------|----------------------|----------------------|----------------------------------|----------------------------------------|---------------|--------|--------------|-----|--------------|-----|
|                  |      |                 |              | sue; cord<br>blood |                      |                      |                                  |                                        |               |        |              |     |              |     |
| garagnani        | 2012 | DNA methylation | Homo sapiens | whole blood        | ELOVL2 methylation   | not applicable       | beta value                       | single-CpG score                       | Illumina 450K | 1      | all ages     |     |              |     |
| gliasin          | 2024 | DNA methylation | Homo sapiens | brain cortex       | chronological age    | chronological age    | years                            | elastic net regression                 | Illumina 450K | 220    | adults       |     |              |     |
| grimage          | 2019 | DNA methylation | Homo sapiens | whole blood        | mortality risk       | mortality            | years                            | two-stage elastic net + Cox regression | Illumina 450K | 1032   | adults       |     | cox_to_years | ref |
| grimage2         | 2022 | DNA methylation | Homo sapiens | whole blood        | mortality risk       | mortality            | years                            | elastic net Cox regression             | Illumina 450K | 1032   | older adults |     | cox_to_years | ref |
| grimage2adm      | 2022 | DNA methylation | Homo sapiens | whole blood        | adrenomedullin       | adrenomedullin       | picograms per milliliter         | elastic net regression                 | Illumina 450K | 187    | older adults |     | cox_to_years | ref |
| grimage2b2m      | 2022 | DNA methylation | Homo sapiens | whole blood        | beta-2-microglobulin | beta-2-microglobulin | picograms per milliliter         | elastic net regression                 | Illumina 450K | 92     | older adults |     | cox_to_years | ref |
| grimage2cystatin | 2022 | DNA methylation | Homo sapiens | whole blood        | cystatin C           | cystatin C           | picograms per milliliter         | elastic net regression                 | Illumina 450K | 88     | older adults |     | cox_to_years | ref |
| grimage2gdf15    | 2022 | DNA methylation | Homo sapiens | whole blood        | GDF-15               | GDF-15               | picograms per milliliter         | elastic net regression                 | Illumina 450K | 138    | older adults |     | cox_to_years | ref |
| grimage2leptin   | 2022 | DNA methylation | Homo sapiens | whole blood        | leptin               | leptin               | picograms per milliliter         | elastic net regression                 | Illumina 450K | 187    | older adults |     | cox_to_years | ref |
| grimage2loga1c   | 2022 | DNA methylation | Homo sapiens | whole blood        | hemoglobin A1c       | hemoglobin A1c       | natural-log percent              | elastic net regression                 | Illumina 450K | 87     | older adults |     | cox_to_years | ref |
| grimage2logcrp   | 2022 | DNA methylation | Homo sapiens | whole blood        | C-reactive protein   | C-reactive protein   | natural-log milligrams per liter | elastic net regression                 | Illumina 450K | 132    | older adults |     | cox_to_years | ref |
| grimage2packyrs  | 2022 | DNA methylation | Homo sapiens | whole blood        | smoking exposure     | smoking exposure     | pack-years                       | elastic net regression                 | Illumina 450K | 173    | older adults |     | cox_to_years | ref |
| grimage2pai1     | 2022 | DNA methylation | Homo sapiens | whole blood        | PAI-1                | PAI-1                | picograms per milliliter         | elastic net regression                 | Illumina 450K | 211    | older adults |     |              | ref |
| grimage2timp1    | 2022 | DNA methylation | Homo sapiens | whole blood        | TIMP-1               | TIMP-1               | picograms per milliliter         | elastic net regression                 | Illumina 450K | 43     | older adults |     | cox_to_years | ref |

| clock             | yr   | data            | species      | tissue                           | predicts                 | trained_on               | unit       | model                                | platform                      | n_feat | pop                    | pre  | post            | ref |
|-------------------|------|-----------------|--------------|----------------------------------|--------------------------|--------------------------|------------|--------------------------------------|-------------------------------|--------|------------------------|------|-----------------|-----|
| han               | 2020 | DNA methylation | Homo sapiens | whole blood                      | chronological age        | chronological age        | years      | linear regression                    | Illumina 450K                 | 65     | all ages               |      | anti_log_linear |     |
| hannum            | 2013 | DNA methylation | Homo sapiens | whole blood                      | chronological age        | chronological age        | years      | elastic net regression               | Illumina 450K                 | 71     | adults                 |      |                 |     |
| hep               | 2024 | DNA methylation | Homo sapiens | liver                            | chronological age        | chronological age        | years      | LASSO regression                     | Illumina EPIC                 | 70     | adults                 |      |                 |     |
| hepatoxu          | 2017 | DNA methylation | Homo sapiens | plasma cell-free DNA             | hepatocellular carcinoma | hepatocellular carcinoma | unitless   | feature-selected logistic regression | targeted bisulfite sequencing | 10     | adults                 |      |                 |     |
| horvath2013       | 2013 | DNA methylation | Homo sapiens | multi-tissue                     | chronological age        | chronological age        | years      | elastic net regression               | Illumina 27K; Illumina 450K   | 353    | all ages               |      | anti_log_linear | ref |
| hrsinch-phenoage  | 2022 | DNA methylation | Homo sapiens | whole blood                      | phenotypic age           | phenotypic age           | years      | weighted linear score                | Illumina 450K; Illumina EPIC  | 959    | adults                 |      |                 |     |
| hypoclock         | 2020 | DNA methylation | Homo sapiens | multi-tissue                     | mitotic age              | not applicable           | beta value | mean aggregation                     | Illumina 450K                 | 678    | human, age unspecified | mean | one_minus       | ref |
| intrinsic-clock   | 2024 | DNA methylation | Homo sapiens | multi-tissue                     | chronological age        | chronological age        | years      | two-stage elastic net regression     | Illumina 450K; Illumina EPIC  | 380    | all ages               |      | anti_log_linear |     |
| knight            | 2016 | DNA methylation | Homo sapiens | cord blood; neonatal blood spots | gestational age          | gestational age          | weeks      | elastic net regression               | Illumina 27K; Illumina 450K   | 148    | newborns               |      |                 | ref |
| leecontrol        | 2019 | DNA methylation | Homo sapiens | placenta                         | gestational age          | gestational age          | weeks      | elastic net regression               | Illumina 450K; Illumina EPIC  | 546    | pregnancies            |      |                 |     |
| leerefine-drobust | 2019 | DNA methylation | Homo sapiens | placenta                         | gestational age          | gestational age          | weeks      | elastic net regression               | Illumina 450K; Illumina EPIC  | 395    | pregnancies            |      |                 |     |
| leerobust         | 2019 | DNA methylation | Homo sapiens | placenta                         | gestational age          | gestational age          | weeks      | elastic net regression               | Illumina 450K; Illumina EPIC  | 558    | pregnancies            |      |                 |     |



| clock             | yr   | data            | species          | tissue       | predicts                 | trained_on               | unit           | model                  | platform                    | n_feat | pop                        | pre | post       | ref |
|-------------------|------|-----------------|------------------|--------------|--------------------------|--------------------------|----------------|------------------------|-----------------------------|--------|----------------------------|-----|------------|-----|
| lin               | 2016 | DNA methylation | Homo sapiens     | whole blood  | chronological age        | chronological age        | years          | linear regression      | Illumina 27K; Illumina 450K | 99     | adults                     |     |            |     |
| mammalian1        | 2023 | DNA methylation | multiple species | multi-tissue | chronological age        | chronological age        | years          | elastic net regression | Horvath MammalMeth ylChip40 | 335    | multiple mammalian species |     | anti_logp2 |     |
| mammalian2        | 2023 | DNA methylation | multiple species | multi-tissue | chronological age        | relative age             | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2572   | multiple mammalian species |     | mammalian2 | ref |
| mammalian3        | 2023 | DNA methylation | multiple species | multi-tissue | chronological age        | chronological age        | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2467   | multiple mammalian species |     | mammalian3 | ref |
| mammalianblood2   | 2023 | DNA methylation | multiple species | blood        | chronological age        | relative age             | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2257   | multiple mammalian species |     | mammalian2 | ref |
| mammalianblood3   | 2023 | DNA methylation | multiple species | blood        | chronological age        | chronological age        | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2097   | multiple mammalian species |     | mammalian3 | ref |
| mammalianfemale   | 2023 | DNA methylation | multiple species | multi-tissue | sex                      | sex                      | probability    | elastic net regression | mammalian methylation array | 101    | multiple mammalian species |     | sigmoid    |     |
| mammalianlifespan | 2024 | DNA methylation | multiple species | multi-tissue | species maximum lifespan | species maximum lifespan | years          | elastic net regression | Horvath MammalMeth ylChip40 | 152    | multiple mammalian species |     | anti_log   | ref |
| mammaliankin2     | 2023 | DNA methylation | multiple species | skin         | chronological age        | relative age             | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2240   | multiple mammalian species |     | mammalian2 | ref |
| mammaliankin3     | 2023 | DNA methylation | multiple species | skin         | chronological age        | chronological age        | years          | elastic net regression | Horvath MammalMeth ylChip40 | 2055   | multiple mammalian species |     | mammalian3 | ref |
| mayne             | 2017 | DNA methylation | Homo sapiens     | placenta     | gestational age          | gestational age          | weeks          | elastic net regression | Illumina 27K; Illumina 450K | 62     | pregnancies                |     |            |     |
| mccartneyalcohol  | 2018 | DNA methylation | Homo sapiens     | whole blood  | alcohol consumption      | alcohol consumption      | units per week | LASSO regression       | Illumina EPIC               | 450    | adults                     |     |            |     |

| clock                      | yr   | data                    | species      | tissue                             | predicts                       | trained_on                     | unit       | model                     | platform                            | n_feat | pop                    | pre                                | post            | ref |
|----------------------------|------|-------------------------|--------------|------------------------------------|--------------------------------|--------------------------------|------------|---------------------------|-------------------------------------|--------|------------------------|------------------------------------|-----------------|-----|
| mccart-neybmi              | 2018 | DNA methylation         | Homo sapiens | whole blood                        | body mass index                | body mass index                | unitless   | LASSO regression          | Illumina EPIC                       | 1109   | adults                 |                                    | sigmoid         |     |
| mccart-neybodyfat          | 2018 | DNA methylation         | Homo sapiens | whole blood                        | body fat percentage            | body fat percentage            | unitless   | LASSO regression          | Illumina EPIC                       | 968    | adults                 |                                    | sigmoid         |     |
| mccartn-eyeducation        | 2018 | DNA methylation         | Homo sapiens | whole blood                        | educational attainment         | educational attainment         | unitless   | LASSO regression          | Illumina EPIC                       | 373    | adults                 |                                    | sigmoid         |     |
| mccart-neyhdlcholesterol   | 2018 | DNA methylation         | Homo sapiens | whole blood                        | HDL cholesterol                | HDL cholesterol                | unitless   | LASSO regression          | Illumina EPIC                       | 737    | adults                 |                                    | sigmoid         |     |
| mccart-neyldlcholesterol   | 2018 | DNA methylation         | Homo sapiens | whole blood                        | LDL cholesterol                | LDL cholesterol                | unitless   | LASSO regression          | Illumina EPIC                       | 233    | adults                 |                                    | sigmoid         |     |
| mccart-neysmoking          | 2018 | DNA methylation         | Homo sapiens | whole blood                        | smoking exposure               | smoking exposure               | pack-years | LASSO regression          | Illumina EPIC                       | 233    | adults                 |                                    |                 |     |
| mccart-neytotalcholesterol | 2018 | DNA methylation         | Homo sapiens | whole blood                        | total cholesterol              | total cholesterol              | unitless   | LASSO regression          | Illumina EPIC                       | 204    | adults                 |                                    | sigmoid         |     |
| mccart-neytotalhdlratio    | 2018 | DNA methylation         | Homo sapiens | whole blood                        | total-to-HDL cholesterol ratio | total-to-HDL cholesterol ratio | ratio      | LASSO regression          | Illumina EPIC                       | 412    | adults                 |                                    |                 |     |
| mccart-neywhr              | 2018 | DNA methylation         | Homo sapiens | whole blood                        | waist-to-hip ratio             | waist-to-hip ratio             | ratio      | LASSO regression          | Illumina EPIC                       | 226    | adults                 |                                    |                 |     |
| meer                       | 2018 | DNA methylation         | Mus musculus | multi-tissue                       | chronological age              | chronological age              | days       | elastic net regression    | RRBS                                | 435    | mice                   |                                    |                 |     |
| neusin                     | 2024 | DNA methylation         | Homo sapiens | brain cortex                       | chronological age              | chronological age              | years      | elastic net regression    | Illumina 450K                       | 672    | adults                 |                                    |                 |     |
| ocampoatac1                | 2023 | chromatin accessibility | Homo sapiens | peripheral blood mononuclear cells | chronological age              | chronological age              | years      | elastic net regression    | ATAC-seq                            | 228    | adults                 | tpm_norm_log1p                     |                 |     |
| ocampoatac2                | 2023 | chromatin accessibility | Homo sapiens | peripheral blood mononuclear cells | chronological age              | chronological age              | years      | elastic net regression    | ATAC-seq                            | 380    | adults                 | tpm_norm_log1p                     |                 |     |
| pasta                      | 2025 | transcriptomics         | Homo sapiens | multi-tissue                       | transcriptomic age             | age ordering                   | years      | ridge logistic regression | RNA-seq; gene expression microarray | 8113   | human, age unspecified | median_fill_and_rank_normalization | scale_and_shift | ref |

| clock           | yr   | data                | species      | tissue                                  | predicts                  | trained_on                  | unit       | model                                                  | platform                            | n_feat | pop                      | pre                                | post                  | ref |
|-----------------|------|---------------------|--------------|-----------------------------------------|---------------------------|-----------------------------|------------|--------------------------------------------------------|-------------------------------------|--------|--------------------------|------------------------------------|-----------------------|-----|
| pasta-mouse     | 2025 | transcriptomics     | Mus musculus | multi-tissue                            | transcriptomic age        | age ordering                | years      | orthologue-transferred ridge logistic regression       | RNA-seq; gene expression microarray | 1600   | human, age unspecified   | median_fill_and_rank_normalization | scale_and_shift       | ref |
| pcdnmtl         | 2022 | DNA methylation     | Homo sapiens | whole blood                             | leukocyte telomere length | DNAmTL output               | base pairs | PCA + elastic net regression                           | Illumina 450K                       | 78464  | adults                   |                                    |                       | ref |
| pcgrim-age      | 2022 | DNA methylation     | Homo sapiens | whole blood                             | mortality risk            | DNAm GrimAge output         | years      | PCA + elastic net regression                           | Illumina 450K                       | 78466  | adults                   |                                    |                       | ref |
| pchan-num       | 2022 | DNA methylation     | Homo sapiens | whole blood                             | chronological age         | Hannum clock output         | years      | PCA + elastic net regression                           | Illumina 450K                       | 78464  | adults                   |                                    |                       | ref |
| pchorvath2013   | 2022 | DNA methylation     | Homo sapiens | multi-tissue                            | chronological age         | Horvath clock output        | years      | PCA + elastic net regression                           | Illumina 450K; Illumina EPIC        | 78464  | all ages                 |                                    | anti_log_linear       | ref |
| pcphe-noage     | 2022 | DNA methylation     | Homo sapiens | whole blood                             | phenotypic age            | phenotypic age              | years      | PCA + elastic net regression                           | Illumina 450K; Illumina EPIC        | 78464  | adults                   |                                    |                       | ref |
| pskin-and-blood | 2022 | DNA methylation     | Homo sapiens | skin; whole blood; cultured fibroblasts | chronological age         | skin-and-blood clock output | years      | PCA + elastic net regression                           | Illumina 450K; Illumina EPIC        | 78464  | all ages                 |                                    | anti_log_linear       | ref |
| pedbe           | 2020 | DNA methylation     | Homo sapiens | buccal epithelium                       | chronological age         | chronological age           | years      | elastic net regression                                 | Illumina 450K; Illumina EPIC        | 94     | children and adolescents |                                    | anti_log_linear       |     |
| petkovich       | 2017 | DNA methylation     | Mus musculus | whole blood                             | chronological age         | chronological age           | months     | elastic net regression                                 | bisulfite sequencing                | 90     | mice                     |                                    | petkovich             |     |
| phenoage        | 2018 | clinical biomarkers | Homo sapiens | blood                                   | phenotypic age            | mortality                   | years      | penalized hazards regression with Gompertz calibration | clinical laboratory assays          | 10     | adults                   |                                    | mortality_to_phenoage |     |
| pipekelasticnet | 2022 | DNA methylation     | Homo sapiens | multi-tissue                            | chronological age         | chronological age           | years      | elastic net regression                                 | Illumina 27K; Illumina 450K; Il-    | 239    | all ages                 |                                    | anti_log_linear       |     |

| clock              | yr   | data            | species      | tissue                                     | predicts              | trained_on           | unit                 | model                            | platform                                   | n_feat | pop                    | pre                                | post            | ref |
|--------------------|------|-----------------|--------------|--------------------------------------------|-----------------------|----------------------|----------------------|----------------------------------|--------------------------------------------|--------|------------------------|------------------------------------|-----------------|-----|
|                    |      |                 |              |                                            |                       |                      |                      |                                  | lumina<br>EPIC                             |        |                        |                                    |                 |     |
| pipekfilteredh     | 2022 | DNA methylation | Homo sapiens | multi-tissue                               | chronological age     | chronological age    | years                | elastic net regression           | Illumina 27K; Illumina 450K; Illumina EPIC | 272    | all ages               |                                    | anti_log_linear |     |
| pipekretrainedh    | 2022 | DNA methylation | Homo sapiens | multi-tissue                               | chronological age     | chronological age    | years                | linear regression                | Illumina 27K; Illumina 450K; Illumina EPIC | 308    | all ages               |                                    | anti_log_linear |     |
| prostata ncerkirby | 2017 | DNA methylation | Homo sapiens | prostate                                   | prostate cancer       | prostate cancer      | log odds             | logistic regression              | Illumina 450K                              | 3      | adult men              |                                    |                 |     |
| reedbmi            | 2020 | DNA methylation | Homo sapiens | whole blood; cord blood                    | BMI methylation score | body mass index      | unitless             | weighted methylation aggregation | Illumina 450K                              | 135    | all ages               |                                    |                 |     |
| reg                | 2025 | transcriptomics | Homo sapiens | multi-tissue                               | chronological age     | chronological age    | years                | ridge regression                 | RNA-seq; gene expression microarray        | 8113   | human, age unspecified | median_fill_and_rank_normalization | add_constant    | ref |
| replitali          | 2022 | DNA methylation | Homo sapiens | cultured primary human cells               | replicative history   | population doublings | population doublings | elastic net regression           | Illumina EPIC                              | 87     | human cell cultures    |                                    |                 |     |
| replitali-norm     | 2022 | DNA methylation | Homo sapiens | cultured fibroblasts                       | replicative history   | population doublings | population doublings | elastic net regression           | Illumina EPIC                              | 218    | human cell cultures    |                                    |                 |     |
| retroelementagev1  | 2024 | DNA methylation | Homo sapiens | whole blood                                | chronological age     | chronological age    | years                | elastic net regression           | Illumina EPIC                              | 1317   | all ages               |                                    |                 |     |
| retroelementagev2  | 2024 | DNA methylation | Homo sapiens | whole blood                                | chronological age     | chronological age    | years                | elastic net regression           | Illumina EPIC                              | 1378   | all ages               |                                    |                 |     |
| senchronorage      | 2026 | DNA methylation | Homo sapiens | whole blood                                | chronological age     | chronological age    | years                | elastic net regression           | Illumina 450K; Illumina EPIC               | 187    | adults                 |                                    |                 |     |
| sencultureage      | 2026 | DNA methylation | Homo sapiens | cultured fibroblasts; cultured mesenchymal | cellular senescence   | cellular senescence  | log odds             | elastic net logistic regression  | Illumina 450K; Illumina EPIC               | 142    | human cell cultures    |                                    |                 |     |

| clock           | yr   | data            | species      | tissue                                                                              | predicts                   | trained_on                              | unit       | model                                           | platform                     | n_feat | pop                  | pre                                                 | post            | ref |
|-----------------|------|-----------------|--------------|-------------------------------------------------------------------------------------|----------------------------|-----------------------------------------|------------|-------------------------------------------------|------------------------------|--------|----------------------|-----------------------------------------------------|-----------------|-----|
|                 |      |                 |              | stromal cells                                                                       |                            |                                         |            |                                                 |                              |        |                      |                                                     |                 |     |
| senmortalityage | 2026 | DNA methylation | Homo sapiens | whole blood                                                                         | mortality risk             | mortality                               | log hazard | elastic net Cox regression                      | Illumina 450K; Illumina EPIC | 91     | adults               |                                                     |                 |     |
| skinand-blood   | 2018 | DNA methylation | Homo sapiens | buccal epithelium; whole blood; epithelium; cultured fibroblasts; skin; cord blood  | chronological age          | chronological age                       | years      | elastic net regression                          | Illumina 450K; Illumina EPIC | 391    | all ages             |                                                     | anti_log_linear |     |
| stemtoc         | 2024 | DNA methylation | Homo sapiens | multi-tissue; cultured human cells; whole blood                                     | mitotic age                | population doublings; chronological age | beta value | 95th-percentile methylation aggregation         | Illumina 450K; Illumina EPIC | 371    | all ages             | 0.95 quantile                                       |                 | ref |
| stem-tocvitro   | 2024 | DNA methylation | Homo sapiens | multi-tissue; cultured human cells                                                  | mitotic age                | population doublings                    | beta value | 95th-percentile methylation aggregation         | Illumina 450K; Illumina EPIC | 629    | prenatal and newborn | 0.95 quantile                                       |                 | ref |
| stoch           | 2024 | DNA methylation | Homo sapiens | sorted monocytes                                                                    | chronological age          | chronological age                       | years      | elastic net regression                          | Illumina 450K                | 353    | adults               |                                                     |                 |     |
| stocp           | 2024 | DNA methylation | Homo sapiens | sorted monocytes                                                                    | chronological age          | chronological age                       | years      | elastic net regression                          | Illumina 450K                | 513    | adults               |                                                     |                 |     |
| stocz           | 2024 | DNA methylation | Homo sapiens | sorted monocytes                                                                    | chronological age          | chronological age                       | years      | elastic net regression                          | Illumina 450K                | 514    | adults               |                                                     |                 |     |
| stubbs          | 2017 | DNA methylation | Mus musculus | multi-tissue; liver; lung; heart; brain cortex; skeletal muscle; cerebellum; spleen | chronological age          | chronological age                       | months     | quadratically calibrated elastic net regression | RRBS                         | 17992  | mice                 | quantile_normalization_and_scale_with_gold_standard | stubbs          | ref |
| system-sage     | 2025 | DNA methylation | Homo sapiens | whole blood                                                                         | multisystem biological age | mortality                               | years      | PCA + elastic net regression                    | Illumina 450K; Illumina EPIC | 125175 | older adults         |                                                     |                 | ref |

| clock                      | yr   | data            | species      | tissue      | predicts                              | trained_on | unit  | model                        | platform                     | n_feat | pop          | pre | post | ref |
|----------------------------|------|-----------------|--------------|-------------|---------------------------------------|------------|-------|------------------------------|------------------------------|--------|--------------|-----|------|-----|
| system-sageblood           | 2025 | DNA methylation | Homo sapiens | whole blood | blood-system biological age           | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagebrain           | 2025 | DNA methylation | Homo sapiens | whole blood | brain-system biological age           | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sageheart           | 2025 | DNA methylation | Homo sapiens | whole blood | cardio-vascular-system biological age | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagehormone         | 2025 | DNA methylation | Homo sapiens | whole blood | endocrine-system biological age       | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sageimmune          | 2025 | DNA methylation | Homo sapiens | whole blood | immune-system biological age          | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sageinflammation    | 2025 | DNA methylation | Homo sapiens | whole blood | inflammatory-system biological age    | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagekidney          | 2025 | DNA methylation | Homo sapiens | whole blood | renal-system biological age           | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sageliver           | 2025 | DNA methylation | Homo sapiens | whole blood | hepatic-system biological age         | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagelung            | 2025 | DNA methylation | Homo sapiens | whole blood | pulmonary-system biological age       | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagemetabolic       | 2025 | DNA methylation | Homo sapiens | whole blood | metabolic-system biological age       | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |
| system-sagemusculoskeletal | 2025 | DNA methylation | Homo sapiens | whole blood | musculoskeletal-system biological age | mortality  | years | PCA + elastic net regression | Illumina 450K; Illumina EPIC | 125175 | older adults |     |      | ref |

| clock                                   | yr   | data            | species      | tissue                                                                                               | predicts                      | trained_on            | unit       | model                                     | platform      | n_feat | pop    | pre                       | post | ref |
|-----------------------------------------|------|-----------------|--------------|------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------|------------|-------------------------------------------|---------------|--------|--------|---------------------------|------|-----|
| thompson                                | 2018 | DNA methylation | Mus musculus | adipose tissue; blood; cerebellum; brain cortex; heart; kidney; liver; lung; skeletal muscle; spleen | chronological age             | chronological age     | months     | elastic net regression                    | RRBS          | 582    | mice   |                           |      |     |
| twelvecell deconvoluteblood-epicbas     | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | basophil proportion           | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epicbmem    | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | memory B cell proportion      | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epicbnv     | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | naive B cell proportion       | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epic cd4mem | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | memory CD4+ T cell proportion | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epiccd4nv   | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | naive CD4+ T cell proportion  | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epic cd8mem | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | memory CD8+ T cell proportion | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |
| twelvecell deconvoluteblood-epiccd8nv   | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes                                                                            | naive CD8+ T cell proportion  | cell-type proportions | proportion | reference-based constrained               | Illumina EPIC | 240    | adults | fill_with_reference_means |      | ref |

| clock                                       | yr   | data            | species      | tissue                    | predicts                       | trained_on            | unit       | model                                     | platform                           | n_feat | pop      | pre                                   | post            | ref |
|---------------------------------------------|------|-----------------|--------------|---------------------------|--------------------------------|-----------------------|------------|-------------------------------------------|------------------------------------|--------|----------|---------------------------------------|-----------------|-----|
|                                             |      |                 |              |                           |                                |                       |            | deconvolution                             |                                    |        |          |                                       |                 |     |
| twelvecell<br>deconvoluteblood-<br>epiceos  | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes | eosinophil proportion          | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC                      | 240    | adults   | fill_with_reference_means             |                 | ref |
| twelvecell<br>deconvoluteblood-<br>epicmono | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes | monocyte proportion            | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC                      | 240    | adults   | fill_with_reference_means             |                 | ref |
| twelvecell<br>deconvoluteblood-<br>epicneu  | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes | neutrophil proportion          | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC                      | 240    | adults   | fill_with_reference_means             |                 | ref |
| twelvecell<br>deconvoluteblood-<br>epicnk   | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes | natural killer cell proportion | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC                      | 240    | adults   | fill_with_reference_means             |                 | ref |
| twelvecell<br>deconvoluteblood-<br>epictreg | 2022 | DNA methylation | Homo sapiens | purified blood leukocytes | regulatory T-cell proportion   | cell-type proportions | proportion | reference-based constrained deconvolution | Illumina EPIC                      | 240    | adults   | fill_with_reference_means             |                 | ref |
| vidalbralo                                  | 2016 | DNA methylation | Homo sapiens | whole blood               | chronological age              | chronological age     | years      | linear regression                         | Illumina 27K                       | 8      | adults   |                                       |                 |     |
| weidner                                     | 2014 | DNA methylation | Homo sapiens | whole blood               | biological age                 | chronological age     | years      | linear regression                         | Illumina 27K; bisulfite sequencing | 3      | adults   |                                       |                 |     |
| wu                                          | 2019 | DNA methylation | Homo sapiens | whole blood               | chronological age              | chronological age     | years      | screened elastic net regression           | Illumina 27K; Illumina 450K        | 111    | children |                                       | anti_log_linear |     |
| xchrom                                      | 2021 | DNA methylation | Homo sapiens | whole blood               | X-chromosome dosage            | X-chromosome dosage   | unitless   | principal component analysis              | Illumina 450K; Illumina EPIC       | 4047   | adults   | sex_estimation<br>autosomal<br>zscore |                 | ref |



| clock           | yr   | data            | species      | tissue              | predicts                | trained_on            | unit     | model                                     | platform                     | n_feat | pop          | pre                                          | post | ref |
|-----------------|------|-----------------|--------------|---------------------|-------------------------|-----------------------|----------|-------------------------------------------|------------------------------|--------|--------------|----------------------------------------------|------|-----|
| ychrom          | 2021 | DNA methylation | Homo sapiens | whole blood         | Y-chromosome presence   | Y-chromosome presence | unitless | principal component analysis              | Illumina 450K; Illumina EPIC | 284    | adults       | sex_estimation<br><i>autosomal</i><br>zscore |      | ref |
| yingadap-tage   | 2024 | DNA methylation | Homo sapiens | whole blood         | adaptive epigenetic age | chronological age     | years    | causality-weighted elastic net regression | Illumina 450K                | 999    | adults       |                                              |      |     |
| ying-causage    | 2024 | DNA methylation | Homo sapiens | whole blood         | chronological age       | chronological age     | years    | causality-weighted elastic net regression | Illumina 450K                | 585    | adults       |                                              |      |     |
| yingdam-age     | 2024 | DNA methylation | Homo sapiens | whole blood         | damaging epigenetic age | chronological age     | years    | causality-weighted elastic net regression | Illumina 450K                | 1089   | adults       |                                              |      |     |
| zhang-blup      | 2019 | DNA methylation | Homo sapiens | whole blood; saliva | chronological age       | chronological age     | years    | best linear unbiased prediction           | Illumina 450K; Illumina EPIC | 319607 | all ages     | scale_row                                    |      | ref |
| zhangen         | 2019 | DNA methylation | Homo sapiens | whole blood; saliva | chronological age       | chronological age     | years    | elastic net regression                    | Illumina 450K; Illumina EPIC | 514    | all ages     | scale_row                                    |      | ref |
| zhang-mortality | 2017 | DNA methylation | Homo sapiens | whole blood         | mortality risk          | mortality             | unitless | weighted linear score                     | Illumina 450K                | 10     | older adults |                                              |      |     |

## Appendix B - tAge transcriptomic clock registry (60 models)

From [tAge clocks\\_metadata.csv](#). Weights on Zenodo record 18763485. `lifespan_scaled = TRUE` means the raw prediction is multiplied by the species maximum lifespan (human 122.5, mouse 48, rat 50.4, monkey 39) to reach age units; `FALSE` means the output is already on its native scale,  $\log_{10}(\text{hazard ratio})$  for mortality clocks, or a fraction of maximum lifespan for normalized-age clocks. **BR** is Bayesian Ridge, **EN** is Elastic Net. **Scaled** clocks take per-sample z-scored  $\log_{10}$  RLE expression; **YuGene** clocks take the YuGene uniformised version. All are relative models and require reference centring.

| filename                                             | type | outcome       | species      | tissue       | scaling | lifespan_scaled |
|------------------------------------------------------|------|---------------|--------------|--------------|---------|-----------------|
| BR_Chronoage_Mouse_Multitissue_scaleddiff.pkl        | BR   | Chronological | Mouse        | Multi-Tissue | Scaled  | TRUE            |
| BR_Chronoage_Mouse_Multitissue_yugenediff.pkl        | BR   | Chronological | Mouse        | Multi-Tissue | YuGene  | TRUE            |
| BR_Chronoage_Multispecies_Multitissue_scaleddiff.pkl | BR   | Chronological | Multispecies | Multi-Tissue | Scaled  | TRUE            |
| BR_Chronoage_Multispecies_Multitissue_yugenediff.pkl | BR   | Chronological | Multispecies | Multi-Tissue | YuGene  | TRUE            |
| BR_Chronoage_Rodents_Brain_WT_scaleddiff.pkl         | BR   | Chronological | Rodents      | Brain        | Scaled  | TRUE            |
| BR_Chronoage_Rodents_Brain_WT_yugenediff.pkl         | BR   | Chronological | Rodents      | Brain        | YuGene  | TRUE            |
| BR_Chronoage_Rodents_Kidney_WT_scaleddiff.pkl        | BR   | Chronological | Rodents      | Kidney       | Scaled  | TRUE            |
| BR_Chronoage_Rodents_Kidney_WT_yugenediff.pkl        | BR   | Chronological | Rodents      | Kidney       | YuGene  | TRUE            |
| BR_Chronoage_Rodents_Liver_scaleddiff.pkl            | BR   | Chronological | Rodents      | Liver        | Scaled  | TRUE            |
| BR_Chronoage_Rodents_Liver_yugenediff.pkl            | BR   | Chronological | Rodents      | Liver        | YuGene  | TRUE            |

| filename                                               | type | outcome       | species      | tissue          | scaling | lifespan_scaled |
|--------------------------------------------------------|------|---------------|--------------|-----------------|---------|-----------------|
| BR_Chronoage_Rodents_Multitissue_scaleddiff.pkl        | BR   | Chronological | Rodents      | Multi-Tissue    | Scaled  | TRUE            |
| BR_Chronoage_Rodents_Multitissue_yugenediff.pkl        | BR   | Chronological | Rodents      | Multi-Tissue    | YuGene  | TRUE            |
| BR_Chronoage_Rodents_Skeletal_muscle_WT_yugenediff.pkl | BR   | Chronological | Rodents      | Skeletal Muscle | YuGene  | TRUE            |
| BR_Chronoage_Rodents_Skeletal_muscle_WT_scaleddiff.pkl | BR   | Chronological | Rodents      | Skeletal Muscle | Scaled  | TRUE            |
| BR_Mortality_Mouse_Multitissue_scaleddiff.pkl          | BR   | Mortality     | Mouse        | Multi-Tissue    | Scaled  | FALSE           |
| BR_Mortality_Mouse_Multitissue_yugenediff.pkl          | BR   | Mortality     | Mouse        | Multi-Tissue    | YuGene  | FALSE           |
| BR_Mortality_Multi-species_Multitissue_scaleddiff.pkl  | BR   | Mortality     | Multispecies | Multi-Tissue    | Scaled  | FALSE           |
| BR_Mortality_Multi-species_Multitissue_yugenediff.pkl  | BR   | Mortality     | Multispecies | Multi-Tissue    | YuGene  | FALSE           |
| BR_Mortality_Rodents_Liver_scaleddiff.pkl              | BR   | Mortality     | Rodents      | Liver           | Scaled  | FALSE           |
| BR_Mortality_Rodents_Liver_yugenediff.pkl              | BR   | Mortality     | Rodents      | Liver           | YuGene  | FALSE           |
| BR_Mortality_Rodents_Multitissue_scaleddiff.pkl        | BR   | Mortality     | Rodents      | Multi-Tissue    | Scaled  | FALSE           |
| BR_Mortality_Rodents_Multitissue_yugenediff.pkl        | BR   | Mortality     | Rodents      | Multi-Tissue    | YuGene  | FALSE           |

| filename                                                  | type | outcome        | species      | tissue       | scaling | lifespan_scaled |
|-----------------------------------------------------------|------|----------------|--------------|--------------|---------|-----------------|
| BR_NormalizedAge_Mouse_Multitissue_scaleddiff.pkl         | BR   | Normalized age | Mouse        | Multi-Tissue | Scaled  | FALSE           |
| BR_NormalizedAge_Mouse_Multitissue_yugenediff.pkl         | BR   | Normalized age | Mouse        | Multi-Tissue | YuGene  | FALSE           |
| BR_NormalizedAge_Multispecies_Multi-tissue_scaleddiff.pkl | BR   | Normalized age | Multispecies | Multi-Tissue | Scaled  | FALSE           |
| BR_NormalizedAge_Multispecies_Multi-tissue_yugenediff.pkl | BR   | Normalized age | Multispecies | Multi-Tissue | YuGene  | FALSE           |
| BR_NormalizedAge_Rodents_Liver_scaleddiff.pkl             | BR   | Normalized age | Rodents      | Liver        | Scaled  | FALSE           |
| BR_NormalizedAge_Rodents_Liver_yugenediff.pkl             | BR   | Normalized age | Rodents      | Liver        | YuGene  | FALSE           |
| BR_NormalizedAge_Rodents_Multitissue_scaleddiff.pkl       | BR   | Normalized age | Rodents      | Multi-Tissue | Scaled  | FALSE           |
| BR_NormalizedAge_Rodents_Multitissue_yugenediff.pkl       | BR   | Normalized age | Rodents      | Multi-Tissue | YuGene  | FALSE           |
| EN_Chronoage_Mouse_Multitissue_scaleddiff.pkl             | EN   | Chronological  | Mouse        | Multi-Tissue | Scaled  | TRUE            |
| EN_Chronoage_Mouse_Multitissue_yugenediff.pkl             | EN   | Chronological  | Mouse        | Multi-Tissue | YuGene  | TRUE            |
| EN_Chronoage_Multispecies_Multitissue_scaleddiff.pkl      | EN   | Chronological  | Multispecies | Multi-Tissue | Scaled  | TRUE            |
| EN_Chronoage_Multispecies_Multitissue_yugenediff.pkl      | EN   | Chronological  | Multispecies | Multi-Tissue | YuGene  | TRUE            |
| EN_Chronoage_Rodents_Brain_scaleddiff.pkl                 | EN   | Chronological  | Rodents      | Brain        | Scaled  | TRUE            |

| filename                                              | type | outcome       | species      | tissue          | scaling | lifespan_scaled |
|-------------------------------------------------------|------|---------------|--------------|-----------------|---------|-----------------|
| EN_Chronoage_Rodents_Brain_yugenediff.pkl             | EN   | Chronological | Rodents      | Brain           | YuGene  | TRUE            |
| EN_Chronoage_Rodents_Kidney_scaleddiff.pkl            | EN   | Chronological | Rodents      | Kidney          | Scaled  | TRUE            |
| EN_Chronoage_Rodents_Kidney_yugenediff.pkl            | EN   | Chronological | Rodents      | Kidney          | YuGene  | TRUE            |
| EN_Chronoage_Rodents_Liver_scaleddiff.pkl             | EN   | Chronological | Rodents      | Liver           | Scaled  | TRUE            |
| EN_Chronoage_Rodents_Liver_yugenediff.pkl             | EN   | Chronological | Rodents      | Liver           | YuGene  | TRUE            |
| EN_Chronoage_Rodents_Multitissue_scaleddiff.pkl       | EN   | Chronological | Rodents      | Multi-Tissue    | Scaled  | TRUE            |
| EN_Chronoage_Rodents_Multitissue_yugenediff.pkl       | EN   | Chronological | Rodents      | Multi-Tissue    | YuGene  | TRUE            |
| EN_Chronoage_Rodents_Skeletal_muscle_scaleddiff.pkl   | EN   | Chronological | Rodents      | Skeletal Muscle | Scaled  | TRUE            |
| EN_Chronoage_Rodents_Skeletal_muscle_yugenediff.pkl   | EN   | Chronological | Rodents      | Skeletal Muscle | YuGene  | TRUE            |
| EN_Mortality_Mouse_Multitissue_scaleddiff.pkl         | EN   | Mortality     | Mouse        | Multi-Tissue    | Scaled  | FALSE           |
| EN_Mortality_Mouse_Multitissue_yugenediff.pkl         | EN   | Mortality     | Mouse        | Multi-Tissue    | YuGene  | FALSE           |
| EN_Mortality_Multi-species_Multitissue_scaleddiff.pkl | EN   | Mortality     | Multispecies | Multi-Tissue    | Scaled  | FALSE           |
| EN_Mortality_Multi-species_Multitissue_yugenediff.pkl | EN   | Mortality     | Multispecies | Multi-Tissue    | YuGene  | FALSE           |

| filename                                                          | type | outcome        | species      | tissue       | scaling | lifespan_scaled |
|-------------------------------------------------------------------|------|----------------|--------------|--------------|---------|-----------------|
| EN_Mortality_<br>Rodents_Liver_scaled-<br>diff.pkl                | EN   | Mortality      | Rodents      | Liver        | Scaled  | FALSE           |
| EN_Mortality_<br>Rodents_Liver_yu-<br>genediff.pkl                | EN   | Mortality      | Rodents      | Liver        | YuGene  | FALSE           |
| EN_Mortality_<br>Rodents_Multitissue_<br>scaleddiff.pkl           | EN   | Mortality      | Rodents      | Multi-Tissue | Scaled  | FALSE           |
| EN_Mortality_<br>Rodents_Multitissue_<br>yugenediff.pkl           | EN   | Mortality      | Rodents      | Multi-Tissue | YuGene  | FALSE           |
| EN_NormalizedAge_<br>Mouse_Multitissue_<br>scaleddiff.pkl         | EN   | Normalized age | Mouse        | Multi-Tissue | Scaled  | FALSE           |
| EN_NormalizedAge_<br>Mouse_Multitissue_<br>yugenediff.pkl         | EN   | Normalized age | Mouse        | Multi-Tissue | YuGene  | FALSE           |
| EN_NormalizedAge_<br>Multispecies_Multi-<br>tissue_scaleddiff.pkl | EN   | Normalized age | Multispecies | Multi-Tissue | Scaled  | FALSE           |
| EN_NormalizedAge_<br>Multispecies_Multi-<br>tissue_yugenediff.pkl | EN   | Normalized age | Multispecies | Multi-Tissue | YuGene  | FALSE           |
| EN_NormalizedAge_<br>Rodents_Liver_scaled-<br>diff.pkl            | EN   | Normalized age | Rodents      | Liver        | Scaled  | FALSE           |
| EN_NormalizedAge_<br>Rodents_Liver_yu-<br>genediff.pkl            | EN   | Normalized age | Rodents      | Liver        | YuGene  | FALSE           |
| EN_NormalizedAge_<br>Rodents_Multitissue_<br>scaleddiff.pkl       | EN   | Normalized age | Rodents      | Multi-Tissue | Scaled  | FALSE           |
| EN_NormalizedAge_<br>Rodents_Multitissue_<br>yugenediff.pkl       | EN   | Normalized age | Rodents      | Multi-Tissue | YuGene  | FALSE           |

## Appendix C - ComputAgeBench benchmark studies (65 datasets, blood/saliva/buccal)

From `ComputAge BenchBSB_table.csv`. `HC content = HC` means the study contains its own healthy controls and can run the AA2 test; `No HC` means only the AA1 test is possible. Condition abbreviations: AS atherosclerosis, IHD ischaemic heart disease, CVA cerebrovascular accident, IBD inflammatory bowel disease, CD Crohn's, UC ulcerative colitis, HIV, TB tuberculosis, XOB extreme obesity, OBS obesity, T1D/T2D diabetes, ASD, RA rheumatoid arthritis, PA psoriatic arthritis, UA undifferentiated arthritis, OA osteoarthritis, OP osteoporosis, MS multiple sclerosis, AD Alzheimer's, PD Parkinson's, CJD Creutzfeldt–Jakob, COPD, WS Werner, aWS atypical Werner, MDPS mandibular dysplasia, HGPS Hutchinson–Gilford progeria, nCLMNA non-classical laminopathy, CGL congenital generalised lipodystrophy.

| Class | Condition | Conditions   | Dataset ID | PMID        | Cited | First author, year    | Tissue         | N samples    | HC content | Comment |
|-------|-----------|--------------|------------|-------------|-------|-----------------------|----------------|--------------|------------|---------|
| CVD   | AS        | AS           | GSE56046   | 25404168    | y     | Reynolds, 2014        | Blood          | 1202         | HC         |         |
| CVD   | AS        | AS           | GSE56581   | 25404168    | y     | Reynolds, 2014        | Blood          | 214          | HC         |         |
| CVD   | IHD       | IHD          | GSE62867   | 25856389    | y     | Nazarenko, 2014       | Blood, vessels | 24           | No HC      |         |
| CVD   | CVA       | CVA          | GSE69138   | 26643952    | y     | Soriano-Tárraga, 2016 | Blood          | 185 w/ages   | No HC      |         |
| CVD   | AS        | AS           | GSE107143  | 28698603    | y     | Istas, 2017           | Blood          | 16 (8 HC)    | HC         |         |
| CVD   | CVA       | CVA          | GSE203399  | 36180927    | y     | Cullell, 2022         | Blood          | 121          | No HC      |         |
| ISD   | IBD       | IBD          | GSE32148   | 22467598    | y     | Harris, 2012          | Blood          | 48 (20 HC)   | HC         |         |
| ISD   | HIV       | HIV          | GSE53840   | 25969563    | y     | Horvath, 2015a        | Blood          | 120          | No HC      |         |
| ISD   | HIV       | HIV          | GSE53841   | 25969563    | y     | Horvath, 2015a        | Blood          | 24           | No HC      |         |
| ISD   | HIV       | HIV          | GSE67705   | 27105112    | y     | Gross, 2016           | Blood          | 284          | HC         |         |
| ISD   | HIV       | HIV          | GSE67751   | 25969563    | y     | Horvath, 2015a        | Blood          | 92 (69 HC)   | HC         |         |
| ISD   | HIV       | HIV          | GSE77696   | 27672717    | y     | Zhang, 2016           | Blood          | 378 (117 HC) | HC         |         |
| ISD   | IBD       | IBD (CD)     | GSE81961   | 27279921    | y     | Li Yim, 2016          | Blood          | 40 (25 HC)   | HC         |         |
| ISD   | IBD       | IBD (CD, UC) | GSE87640   | 27886173    | y     | Ventham, 2016         | Blood          | 240 (84 HC)  | HC         |         |
| ISD   | IBD       | IBD (CD, UC) | GSE87648   | 27886173    | y     | Ventham, 2016         | Blood          | 382 (190 HC) | HC         |         |
| ISD   | HIV       | HIV          | GSE100264  | 29269866    | v     | Zhang, 2017           | Blood          | 386          | No HC      |         |
| ISD   | HIV       | HIV          | GSE107080  | 29269866    | v     | Zhang, 2017           | Blood          | 405          | No HC      |         |
| ISD   | HIV       | HIV          | GSE117859  | 30545403    | v     | Zhang, 2018           | Blood          | 608          | No HC      |         |
| ISD   | HIV       | HIV          | GSE117860  | 30545403    | v     | Zhang, 2018           | Blood          | 529          | No HC      |         |
| ISD   | HIV       | HIV          | GSE140800  | 32760119    | v     | Oriol-Tordera, 2020   | Blood          | 70           | No HC      |         |
| ISD   | HIV       | HIV          | GSE143942  | Unpublished |       | Unpublished           | Blood          | 85           | HC         |         |
| ISD   | HIV_TB    | HIV, TB      | GSE145714  | 32125282    | v     | DiNardo, 2020         | Blood          | 32           | HC         |         |
| ISD   | HIV       | HIV          | GSE185389  | 35325780    | v     | Oriol-Tordera, 2022   | Blood          | 56           | No HC      |         |
| ISD   | HIV       | HIV          | GSE185390  | 35325780    | v     | Oriol-Tordera, 2022   | Blood          | 30           | No HC      |         |
| ISD   | HIV       | HIV          | GSE217633  | 36640455    | v     | Esteban-Cantos, 2023  | Blood          | 413 (44 HC)  | HC         |         |



| Class | Condition | Conditions     | Dataset ID | PMID        | Cited | First author, year         | Tissue         | N samples    | HC content | Comment                         |
|-------|-----------|----------------|------------|-------------|-------|----------------------------|----------------|--------------|------------|---------------------------------|
| MBD   | XOB       | XOB, OBS, ASD  | GSE49909   | 24034465    | v     | Day, 2013                  | Blood, brain   | 283          | HC         |                                 |
| MBD   | T1D       | T1D            | GSE56606   | 21980303    | v     | Rakyan, 2011               | Blood          | 100 (68 HC)  | HC         | MZ twins                        |
| MBD   | T2D       | T2D            | GSE62003   | 25585531    | v     | Lunnon, 2015               | Blood          | 70           | No HC      |                                 |
| MBD   | XOB       | XOB            | GSE131461  | 31333715    | v     | Ramos-Molina, 2019         | Blood, adipose | 40           | No HC      |                                 |
| MBD   | XOB       | XOB, OBS       | GSE166611  | Unpublished |       | Unpublished                | Blood          | 32 (17 HC)   | HC         |                                 |
| MBD   | XOB       | XOB            | GSE193836  | 35369101    | v     | Noronha, 2022              | Blood          | 21 (11 HC)   | HC         |                                 |
| MSD   | RA        | RA             | GSE42861   | 23334450    | v     | Liu, 2013                  | Blood          | 689          | HC         |                                 |
| MSD   | RA        | RA             | GSE71841   | Unpublished |       | Unpublished                | Blood          | 24 (12 HC)   | HC         | Unpublished                     |
| MSD   | OP        | OP             | GSE99624   | 28926142    | v     | Fernandez-Rebollo, 2018    | Blood          | 48           | HC         |                                 |
| MSD   | RA        | RA             | GSE131989  | 27723282    | v     | Rhead, 2017                | Blood          | 371          | HC         |                                 |
| MSD   | RA        | RA             | GSE134429  | Unpublished |       | Unpublished                | Blood          | 64 (17 HC)   | HC         | Unpublished                     |
| MSD   | RA        | RA, PA, UA, OA | GSE137593  | 31945409    | v     | Clark, 2020                | Blood          | 103          | No HC      | Early arthritis                 |
| MSD   | RA        | RA, PA, UA, OA | GSE137594  | 31945409    | v     | Clark, 2020                | Blood          | 119          | No HC      | Early arthritis                 |
| MSD   | RA        | RA             | GSE138653  | 32909363    | v     | Tao, 2021                  | Blood          | 80           | No HC      | Before medication               |
| MSD   | RA        | RA, UA         | GSE175364  | 34105306    | v     | de la Calle-Fabregat, 2021 | Blood          | 101          | HC         | UA -> RA/OA                     |
| MSD   | RA        | RA             | GSE176168  | 35576644    | v     | Julià, 2022                | Blood          | 113          | No HC      | Longitudinal (0,12 w), anti-TNF |
| MSD   | RA        | RA             | GSE228104  | 37744384    | v     | Chen, 2023                 | Blood          | 40           | No HC      | Leflunomide                     |
| NDD   | MS        | MS             | GSE43976   | 23422812    | v     | Marabita, 2013             | Blood          | 95 (20 HC)   | No HC      |                                 |
| NDD   | AD        | AD             | GSE59685   | 25129077    | v     | Lunnon, 2014               | Brain, blood   | 531          | HC         |                                 |
| NDD   | PD        | PD             | GSE72774   | 26655927    | v     | Horvath, 2015b             | Blood          | 508 (219 HC) | HC         |                                 |
| NDD   | PD        | PD             | GSE72776   | 26655927    | v     | Horvath, 2015b             | Blood          | 84 (38 HC)   | HC         |                                 |
| NDD   | MS        | MS, OBS        | GSE103929  | 30981648    | v     | Castro, 2019               | Blood          | 49           | No HC      | BMI                             |
| NDD   | MS        | MS             | GSE106648  | 29921915    | v     | Kular, 2019                | Blood          | 279 (139 HC) | HC         |                                 |
| NDD   | PD        | PD             | GSE111223  | 28851441    | v     | Chuang, 2017               | Saliva         | 259 (131 HC) | HC         |                                 |

| Class | Condition | Conditions    | Dataset ID | PMID        | Cited | First author, year | Tissue       | N samples    | HC content | Comment                     |
|-------|-----------|---------------|------------|-------------|-------|--------------------|--------------|--------------|------------|-----------------------------|
| NDD   | PD        | PD            | GSE111629  | 30958317    | v     | Chuang, 2017       | Blood        | 572 (237 HC) | HC         |                             |
| NDD   | MS        | MS            | GSE112596  | 30698680    | v     | Ntranos, 2019      | Blood        | 112          | No HC      |                             |
| NDD   | PD        | PD            | GSE122244  | Unpublished |       | Unpublished        | Blood        | 70           | HC         | Non-Neurological controls   |
| NDD   | MS        | MS            | GSE130029  | 31053557    | v     | Ewing, 2019        | Blood        | 31 (11 HC)   | HC         |                             |
| NDD   | MS        | MS            | GSE130030  | 31053557    | v     | Ewing, 2019        | Blood        | 28 (14 HC)   | HC         |                             |
| NDD   | MS        | MS            | GSE130491  | 31300673    | v     | Carlström, 2019    | Blood        | 82 (19x3x2)  | No HC      |                             |
| NDD   | AD        | AD            | GSE144858  | 32745807    | v     | Roubroeks, 2020    | Blood        | 300          | HC         |                             |
| NDD   | PD        | PD            | GSE151355  | 32650713    | v     | Go, 2020           | Blood, brain | 40           | No HC      |                             |
| NDD   | CJD       | CJD           | GSE156994  | 32918118    | v     | Dabin, 2020        | Blood        | 219 (106 HC) | HC         |                             |
| NDD   | MS        | MS            | GSE219293  | 36685876    | v     | Bingen, 2023       | Blood        | 47 (18 HC)   | HC         |                             |
| RSD   | TB        | TB            | GSE72338   | 26374119    | v     | Esterhuyse, 2015   | Blood        | 38           | HC         | HC = LTBI household concats |
| RSD   | COPD      | COPD          | GSE118468  | 33658578    | v     | Chen, 2021         | Blood        | 21 (6 HC)    | HC         |                             |
| RSD   | TB        | TB            | GSE118469  | 32365959    | v     | Chen, 2020         | Blood        | 21 (6 HC)    | HC         |                             |
| RSD   | TB        | HIV, TB       | GSE145714  | 32125282    | v     | DiNardo, 2020      | Blood        | 32           | HC         |                             |
| PGS   | WS        | WS, aWS, MDPS | GSE131752  | 31259468    | v     | Maierhofer, 2019   | Blood        | 48 (24 HC)   | HC         |                             |
| PGS   | HGPS      | HGPS, ncLMNA  | GSE182991  | 35045206    | v     | Bejaoui, 2022      | Blood        | 27 (12 HC)   | HC         |                             |
| PGS   | CGL       | CGL           | GSE214297  | 36715159    | v     | Qannan, 2022       | Blood        | 16 (9 HC)    | HC         |                             |

## Appendix D - organAging organ protein panels

Organ assignment rule: a plasma protein belongs to an organ if its GTEx expression in that tissue is at least 4× its mean expression across all other tissues. Counts from [organAging GTEx\\_4x\\_FC\\_genes.json](#). Three model families ship per organ: full (Olink Explore 3072), feature-reduced (Olink Explore 1536), and CSF. Each family has first-generation (chronological age, with intercept) and mortality (Cox elastic net, relative log hazard, no intercept) variants, each fitted over five folds, use fold 1 unless working in UK Biobank itself.

| Organ panel                             | Proteins                   |
|-----------------------------------------|----------------------------|
| Adipose                                 | 5                          |
| Adrenal                                 | 4                          |
| Artery                                  | 14                         |
| Bladder                                 | 0 (excluded from training) |
| Brain                                   | 118                        |
| Esophagus                               | 5                          |
| Female (reproductive)                   | 6                          |
| Heart                                   | 7                          |
| Immune                                  | 124                        |
| Intestine                               | 31                         |
| Kidney                                  | 9                          |
| Liver                                   | 79                         |
| Lung                                    | 11                         |
| Male (reproductive)                     | 55                         |
| Muscle                                  | 20                         |
| Pancreas                                | 29                         |
| Pituitary                               | 12                         |
| Salivary                                | 10                         |
| Skin                                    | 17                         |
| Stomach                                 | 12                         |
| Thyroid                                 | 2                          |
| Organismal (any organ-enriched protein) | 2,347                      |
| Multi-organ (enriched in >1 organ)      | 569                        |
| Conventional (all measured proteins)    | 2,916                      |

Applying them: multiply Olink NPX values by the coefficients and sum; add the intercept for first-generation models only. For non-Olink platforms, log-transform to symmetry, z-score each protein, then multiply by the published Olink NPX standard deviations (Table S3 of the paper) before scoring. The authors ignored missing values rather than imputing, and warn that performance degrades with the number of missing high-coefficient proteins.

## Appendix E - Preprocess and postprocess operation inventory

Every distinct operation a clock can request, with the clocks that use it and the shape of the computation. This is the complete op set a new engine must implement to cover the pyaging registry.

**Preprocess (runs on the clock's feature slice, after feature selection)**

| Op                                                 | Signature                                                                           | Dependencies stored with model                                 | Clocks                                                  |
|----------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| <code>identity</code>                              | $x \rightarrow x$                                                                   | -                                                              | most                                                    |
| <code>fill_nan_with_reference</code>               | <code>where(isnan(x), ref, x)</code>                                                | <code>reference_values</code> (len = <code>n_features</code> ) | ~90 ( <code>LinearReferenceClock</code> and subclasses) |
| <code>scale</code>                                 | $(x - \text{median}) / \text{std}$                                                  | median vector, std vector                                      | AltumAge, CpGPTGrimAge3                                 |
| <code>scale_mean</code>                            | $(x - \text{mean}) / \text{std}$                                                    | mean vector, std vector                                        | CpGPTPCGrimAge3                                         |
| <code>scale_row</code>                             | $(x - \text{rowmean}) / \text{rowstd}$                                              | -                                                              | ZhangEN, ZhangBLUP                                      |
| <code>quantile_normalize_with_gold_standard</code> | rank map onto sorted gold means, then subset                                        | 20,000-length gold mean vector + scoring index list            | DunedinPACE                                             |
| <code>quantile_normalize_and_scale</code>          | as above, then $(x - \text{gold\_mean}) / \text{gold\_std}$ , then subset           | gold means, gold stds, index list                              | Stubbs                                                  |
| <code>tpm_norm_log1p</code>                        | $\log_{1p}(1e6 \cdot (1000 \cdot x / \text{len}) / \text{rowsum})$ , then subset    | region lengths, index list                                     | OcampoATAC1/2                                           |
| <code>binarize</code>                              | per row, mask zeros, threshold at non-zero median                                   | -                                                              | BiTAge                                                  |
| <code>median_fill_and_rank</code>                  | fill NaN with global median, then per-row average ranks                             | -                                                              | Pasta, PastaMouse, REG                                  |
| <code>mean_over_present</code>                     | mean of entries $\neq -1$ per row                                                   | -                                                              | epiTOC1, HypoClock, epiCMITHyper, epiCMITHypo           |
| <code>quantile95_over_present</code>               | 0.95 quantile of entries $\neq -1$ per row                                          | -                                                              | stemTOC, stemTOCvitro                                   |
| <code>nan_to_zero</code>                           | <code>nan_to_num(x, 0)</code>                                                       | -                                                              | epiTOC2, epiTOC3                                        |
| <code>autosomal_zscore</code>                      | z-score all probes against the autosomal mean/SD of that sample                     | autosome index list                                            | XChrom, YChrom                                          |
| <code>expand_with_reference</code>                 | insert reference values for out-of-species features, then run the shared preprocess | full feature list, full reference vector, species index list   | PastaMouse                                              |

**Postprocess (runs on the scalar model output)**

Listed in the transform table in §4.20. Fifteen distinct forms, of which `identity`, `anti_log_linear` and `cox_to_years` cover the large majority.

**What FALCONAge implements of this**

Seven preprocess ops and nine postprocess transforms, in `models/ops.py`, which is what the 23 scoring clocks reach for. Several are more general than their row above: `scale` takes centre and spread vectors and so covers `scale`, `scale_mean` and (with row statistics) `scale_row`; `quantile_normalize` takes the gold vector and the scoring index, covering the DunedinPACE and Stubbs rows.

| Shipped                                                                                            | Covers                                                                                                      |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>beta_to_m, m_to_beta</code>                                                                  | the $\beta \leftrightarrow M$ logit pair, which no row above names because pyaging folds it into each clock |
| <code>scale</code>                                                                                 | <code>scale, scale_mean, scale_row</code>                                                                   |
| <code>rank_normalize</code>                                                                        | <code>median_fill_and_rank</code>                                                                           |
| <code>quantile_normalize</code>                                                                    | <code>quantile_normalize_with_gold_standard, quantile_normalize_and_scale</code>                            |
| <code>binarize</code>                                                                              | <code>binarize</code>                                                                                       |
| <code>simplex_projection</code>                                                                    | the deconvolution row of the special passes                                                                 |
| <code>add, multiply, divide_by, clip, exp, expit, log_linear, anti_log_linear, cox_to_years</code> | the postprocess set                                                                                         |

Reference filling is not an op here. `fill_nan_with_reference` is the largest row in the table and FALCONAge does it in `align()` instead, before any op runs, because the imputation choice is a property of the *dataset* rather than of the clock and has to be recorded in the manifest either way. Zero is never a fill value: in  $\beta$  space zero means completely unmethylated, which is a real and extreme measurement, and a clock with a large positive coefficient on a probe the array does not carry will read “totally unmethylated” and shift the prediction by whole years.

Unbuilt: `tpm_norm_loglp`, `mean_over_present`, `quantile95_over_present`, `nan_to_zero`, `autosomal_zscore`, `expand_with_reference`, and all ten special forward passes. Every one belongs to a clock family whose coefficients are not redistributable, so building the op first would produce code with no real answer to test against, see the tier table in [Architecture](#).

### Special forward passes (not expressible as preprocess → linear → postprocess)

| Clock family                          | Structure                                                                                                                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GrimAge / GrimAge2                    | 8 or 10 sub-linear-models on disjoint feature index sets → concat with Age and Female in a fixed order → Cox linear layer → <code>cox_to_years</code>                                         |
| PCGrimAge                             | centre + rotate 78,464 → 1,933 PCs → concat Female, Age → 8 PC sub-models → Cox layer                                                                                                         |
| CpGPTPCGrimAge3                       | scale → split age from proxies → rotate 30 → 29 PCs → concat → rescale PCs → linear → <code>cox_to_years</code>                                                                               |
| DNAmFitAge                            | split by sex → 5 sub-models (GaitF/M, GripF/M, VO2Max) → affine standardisation with sex-specific constants → sex-specific KDM layer → recombine                                              |
| Mammalian 2/3 and blood/skin variants | split CpG block from species one-hot (1,756 or 1,707 wide) → linear on CpGs → species-dependent inverse age transform using AnAge gestation / maturity / max lifespan                         |
| SystemsAge                            | centre + rotate 125,175 CpGs → project to system components → 11 system linear modules + a PC → age model with quadratic correction → optional composite PCA layer → affine calibration → /12 |
| Deconvolution                         | pseudo-inverse of the signature matrix → simplex projection → select one cell-type column                                                                                                     |
| epiTOC2 / epiTOC3                     | $2/k \cdot \sum (\beta - \beta_0) / (\delta(1-\beta_0))$ with per-CpG $\delta$ and $\beta_0$ vectors                                                                                          |

| Clock family    | Structure                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------|
| XChrom / YChrom | autosomal z-score $\rightarrow$ select sex-chromosome probes $\rightarrow \sum (z - \text{mean}) \cdot \text{coef}$ |
| PastaMouse      | expand mouse features into the 8,113-gene human space with reference values $\rightarrow$ shared Pasta forward      |

## Appendix F - Reproducing the four reference computations by hand

Small worked checks to use as smoke tests while building.

**PhenoAge, conventional units.** Inputs: age 50, albumin 4.5 g/dL, creatinine 0.9 mg/dL, glucose 90 mg/dL, CRP 1.0 mg/L, lymphocyte 30%, MCV 90 fL, RDW 13%, ALP 70 U/L, WBC  $6.0 \times 10^3/\mu\text{L}$ .

```
xb = -19.907 - 0.0336(4.5) + 0.0095(0.9) + 0.1953(90) + 0.0954·ln(1.0)
      - 0.0120(30) + 0.0268(90) + 0.3306(13) + 0.00188(70) + 0.0554(6.0) + 0.0804(50)
m   = 1 - exp( -exp(xb) · (exp(120·0.0076927) - 1) / 0.0076927 )
PhenoAge = 141.50225 + ln(-0.00553·ln(1 - m)) / 0.090165
```

**Horvath output transform.** A raw linear score of  $-0.5$  becomes  $21 \cdot \exp(-0.5) - 1 = 11.74$  years; a score of  $1.2$  becomes  $21 \cdot 1.2 + 20 = 45.2$  years.

**GrimAge2 Cox  $\rightarrow$  years.** A Cox linear predictor of  $16.0$  becomes  $((16.0 - 15.370829484122)/1.09534876966487) \times 9.05974444998421 + 66.0943807965085 = 71.30$  years.

**tAge chronological, mouse.** A model output of  $0.35$  (fraction of maximum lifespan) becomes  $0.35 \times 48 = 16.8$  months. The same  $0.35$  from a mortality clock stays as  $\log_{10}(\text{HR}) = 0.35$ , i.e.  $\text{HR} \approx 2.24$  - never 16.8 anything.

## Chapter 7. Architecture

---

Framework for Aging CLocks, Omics Normalisation and Age scoring.

A single-stop toolkit for computing aging scores from multiomic data, usable identically from Python and R, on CPU or GPU, with a download module that takes public-database accessions and a preprocessing module that turns raw or public data into a scoreable matrix.

This document is the design record. It says which file computes what, what each function takes and returns, and why each decision went the way it did. The science it implements is in [The science of aging clocks](#); every algorithmic claim here cites a section there rather than restating it.

### What of this exists

**!** Read this before believing any section below

This document was written **before** the code and is deliberately wider than what ships. Every section whose specification outruns the implementation says so at its head. The table here is the reconciliation, and it is checked against the running package rather than remembered, the counts below come from `len(ops.PREPROCESS)`, `registry.load()` and the container's own `Modality` type.

Current as of v1.0.0.

| Area             | Shipped in v1.0.0                                                                                                                                                                     | Specified, not built                                                                            |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Clock catalogue  | 161 entries. <b>35 tier A score offline</b> : 32 ship a coefficient file, 3 (PhenoAge, KDM, homeostatic dysregulation) are formulas with none to ship. 28 tier C are tested scaffolds | The 98 tier B entries carry metadata but no traced coefficients                                 |
| Modalities       | DNA methylation, clinical chemistry, RRBS, <b>nanopore, transcriptomics, proteomics</b>                                                                                               | Single-cell as a <i>modality</i> (scAge is a model, not a container type); spatial              |
| Inputs           | Beta matrix, GEO series matrix, <b>raw IDATs</b> , RRBS, <b>bedMethyl, targeted panels, Olink NPX, So-maScan RFU, bulk counts</b> , clinical tables, ComputAgeBench                   | —                                                                                               |
| Normalisation    | <b>pOOBAH, noob, BMIQ, probe masks</b> (§20 of the science page)                                                                                                                      | Control-probe dye correction; mLiftOver cross-platform harmonisation                            |
| Model classes    | <b>LinearClock, ClinicalClock, ScaffoldClock, PCLinearClock, AggregationClock, NeuralClock, ScAgeReference</b>                                                                        | <b>MultiStageCoxClock, DeconvolutionClock, CompositeClock</b>                                   |
| Preprocess ops   | 7: <b>beta_to_m, m_to_beta, scale, rank_normalize, quantile_normalize, binarize, simplex_projection</b>                                                                               | The other eight in §4.1                                                                         |
| Postprocess ops  | <b>21</b> , plus <b>24 analytic derivatives</b> for uncertainty propagation and 4 ops explicitly marked non-differentiable                                                            | —                                                                                               |
| Uncertainty      | <b>Technical SE, conformal prediction intervals, per-probe ICC table, power, consensus</b>                                                                                            | Conformalised quantile regression (split conformal ships instead)                               |
| Batch            | <b>Frozen-reference ComBat</b> : adding a plate cannot move an earlier one                                                                                                            | —                                                                                               |
| Download         | GEO series and sample, ArrayExpress, SRA/ENA, Zenodo by DOI, Hugging Face, <b>Figshare, PRIDE, MetaboLights, GDC</b>                                                                  | Credentialed archives (Synapse, EGA, dbGaP, UK Biobank): documented, deliberately not automated |
| Registry storage | One <b>clocks.yaml</b> inside the package, loaded into frozen dataclasses                                                                                                             | The per-clock file tree and pydantic schema of §3                                               |
| Devices          | CPU (numpy) and CUDA (torch), fp64 default                                                                                                                                            | MPS is resolved and refused rather than supported                                               |
| R interface      | S3 classes, ggplot2 figures, bit-identical to Python, <b>including the uncertainty and batch verbs</b>                                                                                | The raw-array chain and the model architectures are Python-only                                 |



Two gaps are worth naming rather than leaving in a cell.

**Tier B coefficient provenance** is not an engineering task. It is a per-clock literature hunt, find the paper, find the supplement, check the coefficient count against what circulates, record the discrepancy where they differ, and some of the 110 have no public supplement at all. `primary_source_traced: false` is the honest record of that debt.

**The proteomic and transcriptomic chains ship without registry entries.** Every published model in both families is licence-restricted, so each would enter as a tier C scaffold. Scoring a proteomic matrix today refuses with “no clocks to score”, which is accurate rather than broken.

## Contents

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## 1. Scope, decisions and repository layout

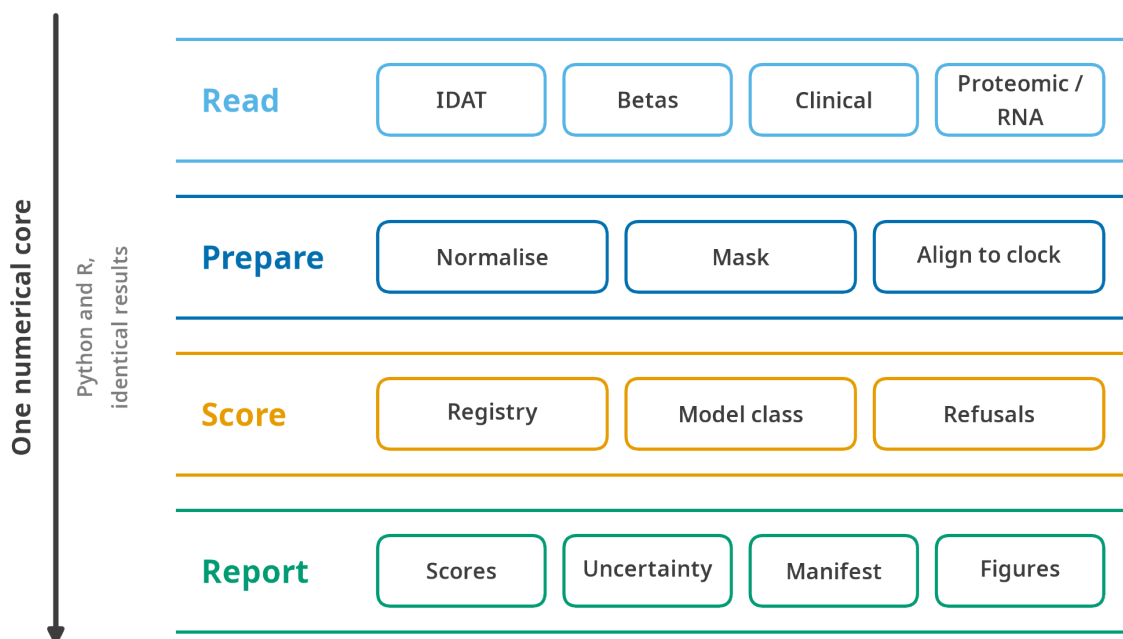


Figure 14: The four stages a dataset passes through, and the one numerical core beneath them

The R package does not reimplement any of this. It calls the same Python functions through reticulate, which is why the two languages return identical numbers rather than numbers that agree to a tolerance. The stages are separable: a beta matrix that is already normalised enters at Prepare, and a clinical table skips the methylation path entirely. What every route shares is the last stage, where a score is never emitted without the manifest recording how it was produced.

### 1.1 What each release delivered

| Capability                   | Ships in 1.0.0                                                                           | Later                                |
|------------------------------|------------------------------------------------------------------------------------------|--------------------------------------|
| DNA methylation clocks       | 158 catalogued: linear, PC, GrimAge family, mitotic, deconvolution, SystemsAge, AltumAge | —                                    |
| Clinical chemistry           | PhenoAge (both unit variants), KDM, homeostatic dysregulation                            | —                                    |
| Clock coefficients           | 32 bundled, 28 scaffold-only, 98 untraced (§3.7, §4.7)                                   | tier B narrows as sources are traced |
| Raw arrays                   | full chain: manifest fetch, pOOBAH, noob, BMIQ, probe masks                              | control-probe dye correction         |
| Uncertainty                  | technical SE, conformal intervals, power, consensus                                      | conformalised quantile regression    |
| Batch                        | frozen-reference ComBat                                                                  | —                                    |
| Specimen and platform checks | species, specimen type, cell-free DNA refusal, platform bias in years                    | —                                    |

| Capability        | Ships in 1.0.0                                                                                              | Later                                              |
|-------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Transcriptomics   | the tAge chain ships; no registry entries                                                                   | tAge’s 60 models, licence permitting               |
| Proteomics        | Olink and SomaScan readers ship; no registry entries                                                        | organAging, Oh 2023 organ clocks                   |
| Single cell       | scAge (per-cell likelihood)                                                                                 | pseudobulk, CellBiAge, SpatialAging-Clock          |
| Foundation models | NeuralClock architecture, safetensors only                                                                  | CpGPT as an <i>imputation</i> backend, not a clock |
| Download          | GEO, ArrayExpress, SRA/ENA, Zenodo, Figshare, Hugging Face, GDC, PRIDE, MetaboLights; ComputAgeBench loader | Metabolomics Workbench                             |
| Preprocess        | IDAT → beta, series matrix, beta CSV/parquet, clinical chemistry, the RNA-seq and Olink/SomaScan chains     | —                                                  |
| Compute           | CPU and CUDA, FP64 default, FP32 opt-in                                                                     | MPS validated, ROCm                                |
| Interfaces        | Python API, R API, CLI, Docker (CPU + CUDA)                                                                 | -                                                  |

The v1.0 line is drawn where it is because methylation plus clinical chemistry covers the overwhelming majority of real use, is fully specified in [the science notes](#) down to the constants, and has published values available to test against.

Every **architecture** is written from scratch, with the equations taken from [the science notes](#), §4 rather than from anybody’s source: 15 preprocess ops, 15 postprocess transforms and 10 special forward passes are specified, of which v1.0 implements the seven, nine and zero that its 23 scoring clocks reach for. No clock implementation is imported from another package.

**Coefficients** are a different matter, because they are fitted data rather than a procedure and cannot be written at all. §3.6 surveys where each set actually comes from and §3.7 sorts them into three tiers by what FALCONAge may distribute. Twenty-eight clocks, the GrimAge and SystemsAge families and DunedinPACE - ship as a complete, tested scaffold with no numbers, because their coefficients are research-use-only or have no traceable public source. §4.7 covers what that means in practice.

## 1.2 The four architectural decisions

**One numerical core, in Python.** The R package is a native-feeling front end that calls the Python core through `reticulate`. This is the design tAge uses (`tAge/R/predict.R`), and the reason is that “identical results in both languages” is either guaranteed by construction or it is a permanent testing liability. Two native implementations means writing 15 preprocess ops, 15 postprocess transforms and 10 special forward passes twice, then proving on every release that they still agree. One core means the R and Python answers are the same bytes because they are the same computation. GPU support in R comes free. The cost is that R users need a Python environment, which `falconage_install()` and the Docker image handle.

**Quarto for both documentation sites.** `quartodoc` renders the Python API, `pkgdown` renders the R API, and one Quarto site wraps both with the same theme and navigation. Tutorials are `.qmd` with Python and R tabs showing the same analysis. The risk is that `quartodoc` is younger than Sphinx and Quarto 2 (a Rust rewrite) lands late 2026, so the Quarto version is pinned in CI and in the Docker image. Docstrings

are numpydoc-format, which means the fallback to Sphinx + `pydata-sphinx-theme` requires no source changes if the toolchain becomes a problem.

### FALCONAge.

| Context            | Name                                                                                        |
|--------------------|---------------------------------------------------------------------------------------------|
| PyPI distribution  | <code>falconage</code>                                                                      |
| Python import      | <code>import falconage as fa</code>                                                         |
| R package          | <code>FALCONAge</code>                                                                      |
| R library call     | <code>library(FALCONAge)</code>                                                             |
| CLI                | <code>falconage &lt;verb&gt;</code>                                                         |
| Docker image       | <code>falconage:VERSION-cpu</code> , <code>falconage:VERSION-cuda</code>                    |
| Docs               | <a href="https://bhagesh-h.github.io/FALCONAge/">https://bhagesh-h.github.io/FALCONAge/</a> |
| Registry namespace | <code>falconage-registry</code> on Hugging Face                                             |

PyPI lowercase follows PEP 8 and PyPI normalisation; R CamelCase follows the `cyRAVEN` convention. `falconage` and `FALCONAge` were both free on PyPI at the time of writing; CRAN availability must still be confirmed with `available::available("FALCONAge")` before the first submission.

**Versioning.** One version number governs `python/pyproject.toml`, `r/DESCRIPTION`, the Docker tag and the git tag; CI refuses a release when they disagree (§12). The clock registry carries its own independent `schema_version` and `registry_version`, because adding a clock should not force a package release and a package release should not silently change which weights a user gets. Every result records both.

### 1.3 Repository layout

```

FALCONAge/
├─ README.md
├─ PUBLISHING.md
├─ CITATION.cff
├─ LICENSE                                # GPL-3
├─ CHANGELOG.md                          # shared; r/NEWS.md is generated from it
├─
├─ python/
│   ├─ pyproject.toml
│   ├─ README.md                        # symlink or short pointer to the root README
│   └─ src/falconage/
│       ├─ __init__.py                  # public API re-exports, __version__
│       └─ _version.py
│       └─
│           └─ core/
│               ├─ device.py            # resolve_device, resolve_dtype, DeviceSpec
│               ├─ container.py         # FalconData
│               ├─ manifest.py          # RunManifest
│               ├─ logging.py           # get_logger, verbosity, progress
│               ├─ config.py            # FalconConfig, from_yaml, from_env
│               ├─ units.py             # UnitRegistry, convert, require_units
│               └─ errors.py            # exception hierarchy

```

```

└─ io/
    │ methylation.py      # read_betas, read_idat_dir, read_series_matrix
    │ clinical.py        # read_clinical
    │ expression.py      # read_counts
    │ proteomics.py      # read_olink, read_somascan (v1.2)
    │ anndata_io.py      # read_h5ad, write_h5ad
    └─ tables.py         # read_sample_table, write_results

└─ download/
    │ base.py            # Downloader ABC, DownloadResult
    │ resolve.py         # accession → Downloader dispatch
    │ cache.py           # content-addressed cache
    │ transfer.py        # resumable HTTP/FTP, checksum, retry
    │ geo.py             # GSE/GSM
    │ arrayexpress.py    # E-MTAB
    │ sra.py             # SRP/SRR/PRJNA
    │ zenodo.py          # DOI
    │ figshare.py
    │ huggingface.py
    │ gdc.py             # TCGA
    │ pride.py           # PXD (v1.2)
    │ metabolights.py   # MTBLS, ST (v1.2)
    │ credentialed.py    # Synapse/EGA/dbGaP/UKB guidance + hooks
    └─ metadata.py       # source metadata → common sample table

└─ preprocess/
    │ methylation/
    │ │ idat.py          # IDAT parse
    │ │ noob.py
    │ │ bmiq.py
    │ │ sesame.py
    │ │ quality.py      # detection p, probe masks
    │ │ platform.py     # detect_platform, harmonise_probe_ids
    │ └─ epicv2.py      # aggregate_replicate_probes
    │ clinical.py       # validate_ranges, convert_units
    │ expression.py
    │ proteomics.py     # v1.2
    │ impute.py         # dataset-level imputation
    └─ qc.py            # QCReport

└─ registry/
    │ schema.py         # pydantic models
    │ registry.py       # ClockRegistry: list/search/get/filter
    │ weights.py        # HF resolution, safetensors load, cache
    │ vocabulary.py     # controlled vocabulary
    └─ data/            # the YAML files (packaged)

```

```

├── models/
│   ├── base.py          # FalconClock ABC
│   ├── ops/
│   │   ├── preprocess.py # the 15 preprocess ops
│   │   ├── postprocess.py # the 15 postprocess transforms
│   │   └── functional.py  # shared kernels (rank, quantile, simplex)
│   ├── linear.py         # LinearClock, PCLinearClock
│   ├── cox.py            # MultiStageCoxClock (GrimAge family)
│   ├── neural.py         # NeuralClock (AltumAge)
│   ├── aggregation.py    # AggregationClock (mitotic clocks)
│   ├── deconvolution.py  # DeconvolutionClock
│   ├── composite.py      # CompositeClock (SystemsAge, DNAmFitAge)
│   ├── clinical.py       # PhenoAge, KDM, HD
│   └── species.py        # mammalian species-lookup transforms
├── score/
│   ├── predict.py        # score()
│   ├── features.py       # feature alignment, clock-level imputation
│   └── result.py         # FalconResult
├── analysis/
│   ├── acceleration.py   # absolute, residual, within-group
│   ├── survival.py       # Cox, C-index, time-dependent AUC
│   ├── association.py     # logistic, linear, interactions
│   ├── reliability.py     # ICC, Fisher-Z pooling
│   ├── benchmark.py       # AA1/AA2, MedAE, bias, total score
│   └── agreement.py       # cross-clock correlation, clustering, UMAP
├── plot/
│   ├── theme.py          # shared palette and rcParams
│   ├── age.py            # ba_vs_ca, acceleration_density
│   ├── agreement.py      # clock_corr, clock_umap
│   ├── qc.py             # missingness, beta_distribution
│   ├── benchmark.py      # class_bench, medae, bias
│   ├── survival.py       # forest
│   ├── systems.py        # organ radar, system heatmap
│   └── trajectory.py     # timecourse
├── report/
│   ├── html.py           # self-contained HTML report
│   └── templates/
├── cli/
│   ├── __main__.py
│   ├── app.py            # typer app
│   └── commands/         # download, preprocess, score, analyse,
│                           # bench, clocks, cache, config, report
└── tests/

```

```

|   |   |— unit/ integration/ property/ conformance/
|   |   |— conftest.py
|
|— r/
|   |— DESCRIPTION  NAMESPACE  NEWS.md  _pkgdown.yml  .Rbuildignore
|   |— R/
|   |   |— FALCONAge-package.R      # @keywords internal "_PACKAGE" + overview
|   |   |— zzz.R                    # .onLoad delayed import, options
|   |   |— install.R                # falconage_install, falconage_config
|   |   |— bridge.R                 # py handle, error translation, marshalling
|   |   |— download.R preprocess.R score.R analysis.R plot.R report.R
|   |   |— classes.R                # falcon_data, falcon_result S3
|   |   |— clocks.R                 # list_clocks, clock_info, cite_clock
|   |   |— cli-options.R            # option parser for inst/scripts
|   |   |— utils.R
|   |— man/                          # roxygen-generated
|   |— vignettes/
|   |   |— FALCONAge.Rmd methylation.Rmd clinical.Rmd
|   |   |— download.Rmd gpu.Rmd benchmarking.Rmd clocks.Rmd
|   |— tests/testthat/
|   |— inst/
|   |   |— CITATION
|   |   |— scripts/falconage.R      # CLI front end
|   |   |— extdata/                 # small example inputs
|
|— registry/                          # source of truth, packaged into both
|   |— schema.yaml
|   |— vocabulary.yaml
|   |— clocks/*.yaml                 # one file per clock
|   |— validate.py
|
|— docs/                              # Quarto site
|   |— _quarto.yml
|   |— index.qmd installation.qmd quickstart.qmd
|   |— guide/*.qmd                  # narrative, Python+R tabs
|   |— reference/                   # quartodoc output (Python)
|   |— r/                           # pkgdown output (R), built in
|   |— _extensions/
|
|— docker/
|   |— Dockerfile.cpu Dockerfile.cuda
|   |— entrypoint.sh install_deps.R
|   |— docker-compose.yml
|
|— tests/vectors/                     # shared gold-standard IO pairs
|   |— manifest.yaml
|   |— <clock>/{input.parquet,expected.parquet}
|

```

```
└─ .github/workflows/
  │ python-test.yaml  r-cmd-check.yaml  docs.yaml
  │ docker.yaml  release-pypi.yaml  release-r.yaml
  └─ registry-validate.yaml
```

### 1.4 What actually got laid down

Flatter than the plan in three places, each for the same reason: a directory with one file in it is a directory you have to open to find out it has one file in it.

```
python/src/falconage/
├─ core/      backend.py config.py container.py errors.py
│             logging.py manifest.py units.py
│             tissue.py preanalytical.py
├─ io/        __init__.py methylation.py          # readers, platform detection
│             bench.py # ComputAgeBench loader
├─ preprocess/ __init__.py methylation.py          # prepare, qc, EPIC v2, units
│             manifest.py idat.py bmiq.py masks.py # raw-array chain
│             batch.py # frozen-reference ComBat
│             proteomic.py transcriptomic.py # modality chains
├─ uncertainty/ __init__.py # SE, conformal, ICC
├─ registry/   registry.py data/clocks.yaml        # all 161 entries, one file
│             data/probe_icc.csv.gz # reliability table
│             data/platform_bias.csv conformal.csv # derived artefacts
│             data/evidence.yaml # curated associations
├─ models/     linear.py clinical.py ops.py pc.py
│             aggregation.py neural.py single_cell.py
├─ score/      __init__.py                        # score(), combine()
├─ analysis/   __init__.py                        # acceleration, cox, icc,
│             # benchmark, power, consensus
├─ plot/       __init__.py                        # public surface + save_all
│             _common.py spec.py colorscheme.yaml # canvas, theme, captions
│             accuracy.py acceleration.py agreement.py
│             qc.py benchmark.py outcomes.py atlas.py
│             uncertainty.py
├─ report/     html.py
├─ cli/        app.py __main__.py                 # + report/power/consensus verbs

python/tools/   build_probe_icc.py build_platform_bias.py # derivations
                build_conformal.py
r/R/           FALCONAge-package.R zzz.R install.R bridge.R
                data.R score.R analysis.R clocks.R download.R
                plot.R atlas.R uncertainty.R
docs/          _quarto.yml reference-groups.yml build_docs.py
                build_downloads.py build_catalogue.py build_gallery.py
                index.qmd clocks.qmd gallery.qmd gpu.md guide/
test/          run_all.py gpu_check.py responsive_check.py
                check_versions.py data/ output/ output_figures/
```



Two collapses and one reversal. `models/ops/{preprocess,postprocess,functional}.py` became one `ops.py`, because the shared kernels the third file was for are four functions. `registry/{schema,weights,vocabulary}.py` became `registry.py` plus the data file: the vocabulary is a frozen set: the schema is the dataclass, and weight resolution is one function because nothing is fetched at run time.

`plot/` went the other way, and the original reasoning was wrong. It was kept as one module plus `atlas.py` on the argument that splitting the figures by subject would make the shared helpers a circular import waiting to happen. It grew to 1,523 lines, an implementation living in a package initialiser, which is the thing an initialiser should not be. The circularity worry was avoidable by naming the problem: the helpers are not shared *between* figure families: they are shared *by* them, so they belong in a `_common.py` that imports from nobody in the package. Each family module imports `_common`, `__init__` imports the families and re-exports, and the dependency graph is a tree. `save_all` stays in `__init__` because orchestrating the families is the one job that legitimately knows about all of them. No public name moved; `fa.plot.ba_vs_ca` is where it was.

There is no top-level `registry/` directory and no `tests/vectors/` tree; the catalogue ships inside the package and the gold-standard vectors live beside the tests that assert on them.

## 2. Core engine

### 2.1 Device and dtype

`falconage/core/device.py`

```
@dataclass(frozen=True)
class DeviceSpec:
    device: torch.device          # cpu | cuda:N | mps
    dtype: torch.dtype            # float64 | float32
    batch_size: int
    backend: Literal["torch", "numpy"]

def resolve_device(
    device: str = "auto",          # "auto" | "cpu" | "cuda" | "cuda:1" | "mps"
    dtype: str = "float64",        # "float64" | "float32"
    batch_size: int | None = None,
) -> DeviceSpec
```

`auto` picks CUDA when `torch.cuda.is_available()`, then MPS, then CPU, and logs the choice, mirroring pyaging's `set_torch_device` (`pyaging/predict/_pred_utils.py:398`). `batch_size` defaults to 1024 on CPU and is raised to fill available VRAM on CUDA, capped so that the largest PC rotation ( $125,175 \times k$  for SystemsAge) fits.

**FALCONAge diverges here in four ways, three of them deliberate.** The module is `core/backend.py`, the function is `resolve()`, and `DeviceSpec` carries plain strings with no `batch_size` field: nothing shipping is large enough to need batching, and a field that is always its default is a field that is wrong the first time it matters.

The substantive difference is `auto`. It resolves to **CPU even when a CUDA device is present**, because on the clocks that ship the card loses: a 2,340-feature dot product is a rounding error against

the pandas alignment that feeds it, and the transfer is pure cost. Measured at  $4.6\times$  slower on an RTX 4060 at 16,384 samples. Picking CUDA because one exists would make the common case several times worse on every machine that has one, silently, so the GPU is opt-in via `device="cuda"` or `FALCONAGE_DEVICE=cuda`. The numbers are in [GPU support](#).

`resolve()` also refuses a device it cannot provide rather than falling back. A run asked for a GPU and quietly given a CPU looks like a very slow success, and the user finds out hours later.

FP64 is the default because that is what pyaging uses and what the gold-standard vectors are generated at. FP32 roughly doubles throughput on consumer GPUs, which have poor FP64 rates, and is opt-in. Clocks whose registry entry sets `requires_fp64: true` are promoted back to FP64 for their own forward pass and a warning is emitted; this covers the PC clocks, whose 78,464- and 125,175-dimensional centring and rotation is ill-conditioned ([the science notes](#), §14.3, §16).

```
def check_dtype_safety(clock: ClockMeta, spec: DeviceSpec) -> DeviceSpec:
    """Promote to FP64 for clocks that need it. Returns a possibly-modified spec."""
```

Memory guard: before a clock runs, estimate `n_samples × n_features × itemsize` plus the rotation matrix, and either lower the batch size or raise `InsufficientMemoryError` with the required figure, rather than letting CUDA OOM mid-run.

## 2.2 The data container

`falconage/core/container.py`

`FalconData` wraps `anndata.AnnData` rather than replacing it, so the on-disk artefact is a plain `.h5ad` that both languages and every scverse tool can read.

```
class FalconData:
    adata: AnnData          # X = primary modality, obs = samples, var = features
    modality: str           # "methylation" | "clinical" | "expression"
    | "proteomics"
    platform: str | None    # "illumina_450k" | "illumina_epic_v2" | ...
    species: str            # "homo_sapiens" | ...
    units: dict[str, str]   # feature → unit, for clinical chemistry
```

Conventions, matching `pyaging`'s `AnnData` slot contract and extending it:

| Slot                                    | Content                                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>X</code>                          | samples × features, the primary modality                                                         |
| <code>obs</code>                        | age, sex, tissue, condition, subject_id, timepoint, time, status, plus predictions after scoring |
| <code>var</code>                        | feature_id, id_type, platform, chromosome, position                                              |
| <code>obsm["X_&lt;clock&gt;"]</code>    | transient per-clock feature slice, deleted after scoring unless <code>keep=True</code>           |
| <code>layers["&lt;modality&gt;"]</code> | a second modality on the same features                                                           |
| <code>uns["falconage"]</code>           | the run manifest                                                                                 |

| Slot                                                                                                               | Content               |
|--------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>uns["&lt;clock&gt;_percent_na"], uns["&lt;clock&gt;_missing_features"], uns["&lt;clock&gt;_metadata"]</code> | per-clock diagnostics |

Multi-modality with different feature spaces (850K methylation probes alongside 3,000 proteins) uses `MuData`, with `FalconData.from_mudata()` / `.to_mudata()`. A single `FalconData` always has one primary feature space.

Constructors:

```
FalconData.from_dataframe(df, modality, platform=None, metadata_cols=(), units=None)
FalconData.from_anndata(adata, modality, ...)
FalconData.read_h5ad(path)
FalconData.write_h5ad(path)
```

`from_dataframe` takes samples in rows and features in columns, splits `metadata_cols` into `obs`, and preserves `df.index` as sample identifiers. This is the same signature as `pyaging's df_to_adata` minus the imputation, which moves to the `preprocess` module where it belongs.

## 2.3 Run manifest

`falconage/core/manifest.py`

Everything needed to reproduce a run, written into `uns["falconage"]` and into a sidecar `run_manifest.json` next to the results.

```
@dataclass
class RunManifest:
    falconage_version: str
    registry_version: str
    registry_schema: str
    python_version: str
    r_version: str | None # set when the call came through reticulate
    platform: str # OS, arch
    device: str
    dtype: str
    torch_version: str
    cuda_version: str | None
    timestamp: str # ISO 8601, UTC
    command: str # the resolved call
    config: dict # fully resolved, defaults expanded
    inputs: list[InputRecord] # path, sha256, n_samples, n_features
    clocks: list[ClockRun] # name, version, weights_sha256, percent_na, warnings
    warnings: list[str]
```

`ClockRun.weights_sha256` is what makes a result auditable: two runs that report the same score for the same clock either used the same weights or the manifest says they did not.

## 2.4 Logging and verbosity

`falconage/core/logging.py`

Three levels, matching cyRAVEN's `cyRAVEN.verbose` option so the two front ends behave the same:

| Level            | Python                                                 | R                                                |
|------------------|--------------------------------------------------------|--------------------------------------------------|
| none             | <code>logging.CRITICAL</code>                          | <code>options(FALCONAge.verbose = "none")</code> |
| inform (default) | <code>logging.INFO</code> , indented progress          | "inform"                                         |
| debug            | <code>logging.DEBUG</code> , per-op timings and shapes | "debug"                                          |

Progress bars via `rich` on a TTY, plain line output otherwise, suppressed entirely under `none`. Every message goes through the logger, so `logging.disable()` in Python and `suppressMessages()` in R both work.

## 2.5 Configuration

`falconage/core/config.py`

Precedence, lowest to highest: packaged defaults → user config file (`~/.config/falconage/config.yaml`) → environment (`FALCONAGE_*`) → explicit function arguments.

```
# ~/.config/falconage/config.yaml
device: auto
dtype: float64
batch_size: 1024
verbose: inform
cache_dir: ~/.cache/falconage
registry_version: "1.1.0"      # pin to make a project reproducible
hf_token: null
max_download_gb: 50
```

## 2.6 Units

`falconage/core/units.py`

The single most common source of wrong answers in this field is a clinical biomarker in the wrong unit ([the science notes](#), §16). The unit registry makes that a hard error.

```
class UnitRegistry:
    def convert(self, value: ArrayLike, marker: str, frm: str, to: str) -> ArrayLike
    def canonical(self, marker: str, variant: str) -> str
    def require(self, markers: Sequence[str], units: dict[str, str] | None) -> dict[str, str]
```

`require` raises `MissingUnitsError` naming every marker whose unit was not declared. There is no default and no inference from magnitude, a creatinine of 80 is plausible as both  $\mu\text{mol/L}$  and an absurd  $\text{mg/dL}$ , and guessing is how the  $88\times$  error happens. Conversions carried:

| Marker     | Units                                          | Factor          |
|------------|------------------------------------------------|-----------------|
| albumin    | $\text{g/L} \leftrightarrow \text{g/dL}$       | $\times 10$     |
| creatinine | $\mu\text{mol/L} \leftrightarrow \text{mg/dL}$ | $\times 88.42$  |
| glucose    | $\text{mmol/L} \leftrightarrow \text{mg/dL}$   | $\times 18.016$ |
| CRP        | $\text{mg/L} \leftrightarrow \text{mg/dL}$     | $\times 10$     |

| Marker                    | Units          | Factor |
|---------------------------|----------------|--------|
| total/HDL/LDL cholesterol | mmol/L ↔ mg/dL | ×38.67 |
| triglycerides             | mmol/L ↔ mg/dL | ×88.57 |
| urea/BUN                  | mmol/L ↔ mg/dL | ×2.801 |
| uric acid                 | μmol/L ↔ mg/dL | ×59.48 |
| bilirubin                 | μmol/L ↔ mg/dL | ×17.1  |

## 2.7 Error hierarchy

falconage/core/errors.py

```

FalconError
├─ ConfigurationError
│   └─ MissingUnitsError
│       └─ InvalidDeviceError
├─ DataError
│   ├── FeatureMismatchError      # 100% of a clock's features absent
│   ├── ExcessiveMissingnessError # above the configured threshold
│   ├── PlatformMismatchError     # EPIC v2 probes not aggregated, etc.
│   ├── MissingCovariateError     # clock needs age/sex and it is not there
│   └─ OutOfRangeError           # beta outside [0,1], negative counts
├─ RegistryError
│   ├── UnknownClockError
│   ├── WeightsUnavailableError
│   └─ SchemaValidationError
├─ DownloadError
│   ├── AccessionNotFoundError
│   ├── ChecksumMismatchError
│   └─ CredentialsRequiredError
└─ ComputeError
    ├── InsufficientMemoryError
    └─ NumericalInstabilityError

```

Every error carries `.remedy`, a one-sentence statement of what to do about it, which the R bridge surfaces as the condition message and the CLI prints after the traceback.

## 3. Clock registry

### 3.1 Why the registry is a separate artefact

The registry is versioned independently of the package, packaged into both language distributions, and validated in its own CI job. That separation exists because adding a clock and releasing software are different events with different risk profiles: a user pinning `registry_version: "1.0.0"` gets the same coefficients in a year regardless of package upgrades, and a registry fix does not require a PyPI release.

### 3.2 Schema

**v1.0 stores this differently.** All 161 entries live in one file, `python/src/falconage/registry/data/clocks.yaml`, loaded into frozen dataclasses by `registry/registry.py`. There is no `registry/` directory at the repository root, no file per clock, and no pydantic.

The per-clock tree below is the right shape for a registry that outside contributors edit concurrently, one file per pull request, no merge conflicts. At 161 entries maintained by one author it bought nothing and cost a directory walk on every import, plus 161 files to keep a schema version in step across. The single file is 161 entries the loader validates in one pass, and it travels inside the wheel so a clock catalogue is never a download. Pydantic went the same way: the fields are flat, the loader coerces and defaults them in about forty lines, and a runtime dependency that exists to check a file the package itself ships is a dependency earning nothing.

The field vocabulary below is otherwise accurate: it is what the loader reads.

```
# one entry in falconage/registry/data/clocks.yaml
id: grimage2
display_name: GrimAge2
schema_version: "1.0"
clock_version: "1.0.0"

provenance:
  year: 2022
  citation: "Lu AT, Binder AM, Zhang J, et al. DNA methylation GrimAge version 2. Aging.
2022;14(23):9484-9549."
  doi: "10.18632/aging.204434"
  last_author: "Steve Horvath"
  journal: "Aging"
  approved_by_author: false
  evidence: paper-confirmed          # author | code | paper | supplement-confirmed

license:
  spdx: null
  research_only: true
  redistributable: true
  notes: "Research use only; commercial use requires a licence from UCLA."

biology:
  data_type: dna_methylation
  species: homo_sapiens
  tissue: [whole_blood]
  platform: [illumina_450k]
  population: older_adults
  generation: second                # first|second|pace|causal|mitotic|system|foundation
  predicts: [mortality_risk]
  training_target: [mortality]
```

```

output:
  unit: years
  scale_type: age_years          # governs which downstream operations are legal
  plausible_range: [0, 150]      # a value outside this is flagged, not clipped

features:
  n_features: 1032
  id_type: cpg_illumina
  requires_covariates: [age, sex]
  reference_values: true

architecture:
  kind: multi_stage_cox
  sub_models:                    # order is load-bearing, see §4.3
    - {name: GDF15,      n_features: 138}
    - {name: B2M,        n_features: 92}
    - {name: CystatinC,  n_features: 88}
    - {name: TIMP1,      n_features: 43}
    - {name: ADM,        n_features: 187}
    - {name: PAI1,       n_features: 211}
    - {name: Leptin,     n_features: 187}
    - {name: PACKYRS,    n_features: 173}
    - {name: LogCRP,     n_features: 132}
    - {name: A1C,        n_features: 87}
  covariate_order: [age, female]
  requires_fp64: false

preprocess:
  - op: fill_nan_with_reference

postprocess:
  - op: cox_to_years
    params:
      cox_mean: 15.370829484122
      cox_std:  1.09534876966487
      age_mean: 66.0943807965085
      age_std:  9.05974444998421

weights:
  uri: "hf://bhagesh-h/falconage-registry/clocks/grimage2/weights.safetensors"
  sha256: "..."
  size_bytes: 8421376
  reference_uri: "hf://bhagesh-h/falconage-registry/clocks/grimage2/reference.
safetensors"
  reference_sha256: "..."

reliability:
  technical_icc: {value: 0.83, source: "Higgins-Chen 2022", n: null}

```

```

biological_icc: {value: 0.61, source: "TranslAGE 2025", n: null}

benchmark:
  computage:
    aa2_passed: null
    aa1_passed: null
    medae: null
    median_bias: null
    total_score: null

  gold_standard:
    input: "tests/vectors/grimage2/input.parquet"
    expected: "tests/vectors/grimage2/expected.parquet"
    tolerance: 1.0e-6
    source: "pyaging 0.3.0, FP64, CPU"

  known_discrepancies: []

```

### 3.3 The discrepancy field is load-bearing

The science, §15.2 documents the eleven clocks where the published paper and the coefficient set in circulation disagree. Every implementation in the field silently inherits one side of each disagreement. FALCONAge records both and warns at score time:

```

known_discrepancies:
  - field: n_features
    paper_says: 96
    implementation_has: 251
    source: "pyaging audit_report.md"
    resolution: implementation
    note: >
      The paper and supplement report a 96-CpG primary ultrasound model. The
      third-party coefficient file everyone uses carries 251 CpGs and cannot be
      mapped to any of the paper's six LASSO variants. No author-hosted 251-CpG
      model was located.

```

The nine to encode at minimum: `bohlin` (96 vs 251), `cvdwesterman` (1,305 vs 235), `zhangmortality` (integer quartile count vs continuous weighted sum), `yingcausage` (586 vs 585), `yingdamage` (1,090 vs 1,089), `yingadaptage` (1,000 vs 999), `synchronoage` (188 vs 187), `sencultureage` (141 vs 142), `senmortalityage` (89 vs 91), plus `bocklandt` (2 CpGs vs 1) and `downsyndrome` (EWAS estimates repurposed as a zero-intercept score).

Warning behaviour is a config knob: `discrepancy_warnings: once | always | never`, defaulting to `once` per clock per session.

### 3.4 Controlled vocabulary

`registry/vocabulary.yaml` fixes the allowed values for `data_type`, `species`, `tissue`, `platform`, `population`, `generation`, `predicts`, `training_target`, `unit`, `scale_type` and `architecture.kind`. This is the mechanism that kept `pyaging`'s 173 entries coherent through an audit that collapsed 94 free-text tissue strings to



30 controlled terms (`pyaging/clocks/metadata/audit_report.md`). `registry/validate.py` fails CI on any value outside the vocabulary, any missing required field, any weight URI without a checksum, and any clock without a gold-standard vector.

### 3.5 Registry API

`falconage/registry/registry.py`

```
class ClockRegistry:
    @classmethod
    def load(cls, version: str | None = None) -> ClockRegistry

    def get(self, clock_id: str) -> ClockMeta
    def list(self) -> list[str]
    def search(self, query: str) -> list[ClockMeta]           # free text over name,
citation, notes
    def filter(self, **criteria) -> list[ClockMeta]           # data_type=, species=,
generation=, ...
    def by_doi(self, doi: str) -> list[ClockMeta]
    def compatible_with(self, data: FalconData,
                        min_coverage: float = 0.8) -> list[ClockMeta]
    def cite(self, clock_id: str, style: str = "plain") -> str
```

`compatible_with` is the function that makes a 173-clock registry usable: hand it a loaded dataset and it returns the clocks whose species, platform and feature coverage actually match, sorted by coverage. This replaces the current field practice of running everything and reading the missingness column afterwards.

### 3.6 Coefficient provenance and the three availability tiers

A clock is two things: an **architecture**, which is a procedure described in a paper, and a **coefficient set**, which is data produced by fitting that procedure to a cohort. FALCONAge writes every architecture from scratch. It cannot write a coefficient set from scratch, because there is no procedure that regenerates Horvath’s 353 weights without Horvath’s training data.

So the question for each clock is only ever *where do its numbers come from and may we ship them*. Surveying `pyaging`’s 173 notebooks for the URL each one fetches gives the answer for the 159 methylation and clinical clocks in v1:

| Provenance                                           | n  | Traceable to the authors             |
|------------------------------------------------------|----|--------------------------------------|
| Publisher supplementary file                         | 31 | yes, the paper’s own Additional file |
| Author’s own code repository                         | 48 | yes                                  |
| Author-hosted file share (Box, Google Drive)         | 24 | yes, on fragile links                |
| Copied package-to-package, primary source unrecorded | 37 | not without tracing                  |
| No public primary source                             | 13 | no                                   |
| Embedded inline, small enough to read off the paper  | 6  | yes                                  |

The 37 in the fourth row are the reason the eleven discrepancies in §3.3 exist. `epiTOC2`’s coefficients in `pyaging` are a file lifted from `biolearn`’s repository; the McCartney scores and most of the deconvolution

panels are the same story. Copying a copy is how a transcription error becomes a field-wide convention, and tracing each back to its supplement is the only way to find out which reading is right.

### 3.7 The three tiers

Every registry entry carries `availability`, one of three values. This governs what ships, what is fetched, and what the user has to do.

| Tier                           | n  | Architecture | Coefficients                                                              | User action                  |
|--------------------------------|----|--------------|---------------------------------------------------------------------------|------------------------------|
| <b>A - open</b>                | 38 | ships        | ship, extracted from the primary source and redistributed                 | none                         |
| <b>B - fetch on first use</b>  | 95 | ships        | fetched from the author's URL at first use, cached, licence printed once  | none, but needs network once |
| <b>C - permission required</b> | 28 | ships        | never fetched; the user obtains them and points FALCONAge at a local file | obtain a licence or file     |

Tier A is where the coefficient set came from a publisher supplement or a permissively licensed author repository and may be redistributed. Includes PhenoAge, KDM and homeostatic dysregulation, which need no network at all.

Tier B is where the coefficients are public but FALCONAge has no clear right to re-host them, or where the primary source has not yet been traced. The scaffold ships; the numbers arrive on first use. Forty-eight of these are author repositories whose licences have not been individually reviewed and will move to A as each is checked.

Tier C is the subject of §4.7.

```
# in every registry entry
availability: B
coefficient_source:
  tier: T2
  url: "https://github.com/rsinghlab/AltumAge/raw/main/example_dependencies/..."
  format: csv
  sha256: "...
  extractor: extractors/altumage.py
  licence: "MIT (repository)"
  redistributable: true
  retrieved: "2026-08-07"
  primary_source_traced: true
```

`primary_source_traced: false` is a visible debt, counted in the registry validation report, and `falconage clocks list --untraced` prints them.

### 3.8 Coefficient extraction

`registry/extractors/`. One small script per clock, or per family where the format is shared. Each reads the primary source in whatever shape it was published, `xlsx`, `csv`, a zip of `csvs`, an `RData` object, a supplementary PDF table, and emits a normalised two-column file plus a provenance record.

```
registry/extractors/
├─ _base.py           # fetch, checksum, normalise, write, record
├─ horvath2013.py     # Springer MOESM3 csv
├─ hannum.py          # Elsevier mmc2.xlsx, sheet 2, cols B:C
├─ ying.py            # Nature Aging MOESM6 zip, three clocks in one archive
└─ ...
```

```
class Extractor(ABC):
    clock_id: str
    source_url: str
    expected_sha256: str
    expected_n_features: int

    @abstractmethod
    def parse(self, raw: bytes) -> pd.DataFrame:      # feature_id, coefficient
        ...

    def run(self) -> ExtractionResult:
        # fetch -> verify checksum -> parse -> assert n_features -> assert no
        # duplicate feature ids -> compare against any prior extraction -> write
```

Two assertions do most of the work. `expected_n_features` catches a supplement whose layout changed or a sheet index that was off by one. The comparison against the prior extraction turns an upstream file being silently revised into a failed CI run rather than a changed result.

**Diffing against the field.** For every clock where `pyaging`, `biolearn`, `computage` or `methylCIPHER` also carries coefficients, extraction runs a comparison and writes the result to `registry/diffs/<clock>.md`: features present in one and not the other, coefficients differing beyond  $1e-9$ , and a verdict. Disagreements are published rather than resolved silently. This is what turns the eleven known discrepancies from inherited folklore into a documented finding, and it is expected to surface more.

### 3.9 Weight loading

`falconage/registry/weights.py`

```
def resolve_weights(meta: ClockMeta, cache_dir: Path | None = None) -> Path
def load_weights(meta: ClockMeta, spec: DeviceSpec) -> dict[str, torch.Tensor]
def register_local_weights(clock_id: str, path: Path, sha256: str | None = None) -> None
```

Rules:

- **safetensors only.** `torch.load(weights_only=False)`, which `pyaging` uses (`pyaging/predict/_pred_utils.py:90`), executes arbitrary code from a downloaded file. Architecture lives in the registry YAML and in `models/`; tensors live in `safetensors`; nothing pickled is ever loaded.

- **Checksum verified** against the registry before use; a mismatch raises rather than warns.
- **Hugging Face cache** for tier A, the same cache for tier B, so weights are shared with other tools and survive a reinstall.
- **Offline mode:** `FALCONAGE_OFFLINE=1` uses only what is cached and raises `WeightsUnavailableError` naming the missing clock rather than attempting a network call.
- **Bundling:** nothing over 1 MB ships inside the PyPI or CRAN artefact. The clinical-chemistry sets are the exception. PhenoAge is ten numbers, so that path works with no network at all.
- **Licence surfacing:** tier B prints the source licence once per clock per session on first fetch. Tier C prints the obtain-it-yourself instructions.

## 4. Algorithms, the complete operation catalogue

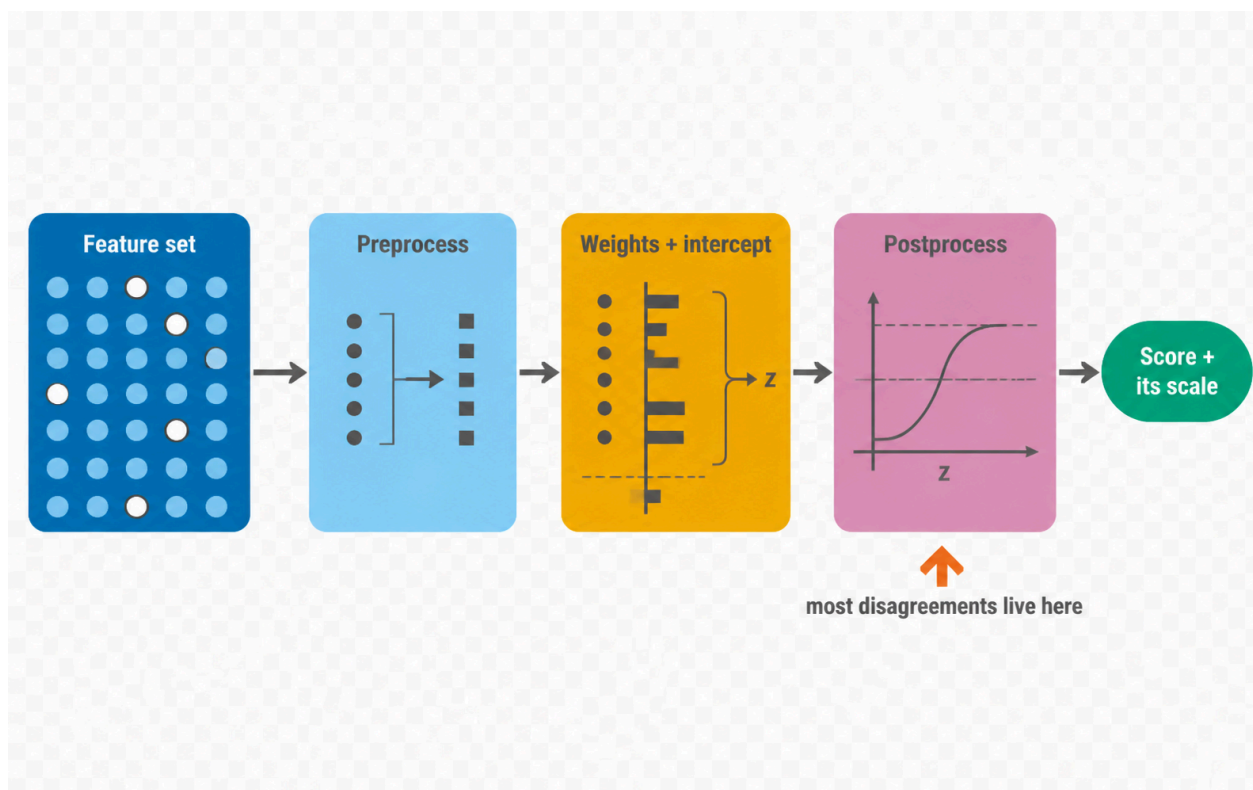


Figure 15: The four parts every clock has, and where implementations disagree

The shape is the field's, not this package's: the reviews that survey it describe the same four parts [59] [60].

Every clock in the registry is expressible as `select features → preprocess ops → architecture forward → postprocess ops`, with ten exceptions that need their own forward pass (§4.3). The ops are the vocabulary; implementing all of them is what makes adding a clock a YAML edit rather than a code change.

The weights are the part everybody agrees on, because they are published. The disagreements between implementations of the same clock are almost all in the last box: whether an anti-log is applied, which of two published transforms was meant, whether an intercept was already folded into the coefficients. That is why the registry stores the postprocess chain as declared data rather than burying it in each model class, and why the clocks whose packaged coefficients contradict their own paper carry the discrepancy as a warning at score time.

**v1.0 implements seven of the fifteen preprocess ops below and nine postprocess transforms**, the ones the 23 scoring clocks reach for. The rest are specified and unbuilt, because each is needed by exactly one clock family whose coefficients are not redistributable, and an op with no reachable test vector is an op that is wrong in a way nobody will notice. They are kept here because the vocabulary is the point: when a coefficient set does arrive, adding the clock stays a registry edit.

#### 4.1 Preprocess ops

`falconage/models/ops.py`, dispatched through `PREPROCESS` and applied by `apply_chain()`.

They are plain functions of `(x, xp, **params)`, not `nn.Module` subclasses as originally specified. `xp` is the array module the device resolver handed back, numpy or torch, so one implementation serves both backends and the CPU path never imports torch. An `nn.Module` would have forced torch on every install for a 500-CpG dot product that numpy finishes before torch has finished importing. All are batched and vectorised; none contains a Python loop over samples. Source for each is [Appendix E](#).

```
def apply_chain(x, chain: Sequence[dict], table: dict, *, xp) -> Any:
    """Run an ordered op list from a registry entry. Order is load-bearing:
    beta_to_m then scale is not scale then beta_to_m."""
```

Shipped: `beta_to_m`, `m_to_beta`, `scale`, `rank_normalize`, `quantile_normalize`, `binarize`, `simplex_projection`. The full specified catalogue follows; the shipped seven cover the first, third, fifth, sixth, ninth and tenth rows in substance under more general names.

| Op                                         | Computation                                                        | Stored params                          | Clocks                                            |
|--------------------------------------------|--------------------------------------------------------------------|----------------------------------------|---------------------------------------------------|
| <code>identity</code>                      | <code>x</code>                                                     | -                                      | default                                           |
| <code>fill_nan_with_reference</code>       | <code>where(isnan(x), ref, x)</code>                               | <code>ref[n_features]</code>           | ~90, the <code>LinearReferenceClock</code> family |
| <code>scale_median</code>                  | <code>(x - median) / std</code>                                    | <code>median[], std[]</code>           | AltumAge, CpGPTGrimAge3                           |
| <code>scale_mean</code>                    | <code>(x - mean) / std</code>                                      | <code>mean[], std[]</code>             | CpGPTPCGrimAge3                                   |
| <code>scale_row</code>                     | <code>(x - rowmean) / rowstd, guarding zero SD</code>              | -                                      | ZhangEN, ZhangBLUP                                |
| <code>quantile_normalize_gold</code>       | rank map onto sorted gold means, then column subset                | <code>gold[20000], idx[]</code>        | DunedinPACE                                       |
| <code>quantile_normalize_scale_gold</code> | as above, then <code>(x - gold_mean)/gold_std</code> , then subset | <code>gold[], gold_std[], idx[]</code> | Stubbs                                            |
| <code>tpm_loglp</code>                     | <code>loglp(1e6 · (1000 · x / len) / rowsum)</code> , then subset  | <code>lengths[], idx[]</code>          | OcampoATAC1/2                                     |
| <code>binarize_row_median</code>           | mask zeros, threshold at each row's non-zero median                | -                                      | BiTAge                                            |
| <code>median_fill_rank</code>              | fill NaN with global median, then per-row average ranks            | -                                      | Pasta, PastaMouse, REG                            |
| <code>mean_over_present</code>             | mean of entries <code>!= -1</code> per row                         | -                                      | epiTOC1, HypoClock, epiCMT×2                      |

| Op                                 | Computation                                                     | Stored params                  | Clocks                |
|------------------------------------|-----------------------------------------------------------------|--------------------------------|-----------------------|
| <code>quantile_over_present</code> | q-quantile of entries != -1 per row                             | <code>q=0.95</code>            | stemTOC, stemTOCvitro |
| <code>nan_to_zero</code>           | <code>nan_to_num(x, 0)</code>                                   | -                              | epiTOC2, epiTOC3      |
| <code>autosomal_zscore</code>      | z-score every probe against that sample's autosomal mean and SD | <code>autosome_idx[]</code>    | XChrom, YChrom        |
| <code>expand_with_reference</code> | scatter the provided features into a wider reference vector     | <code>full_ref[], idx[]</code> | PastaMouse            |

Three of these need care.

**quantile\_normalize\_gold.** The DunedinPACE gold standard is 20,000 probes, of which only 173 are scored; the other 19,827 exist to define the sample's rank distribution ([the science notes](#), §4.8). Dropping them produces a plausible wrong number, so the op raises `FeatureMismatchError` when background coverage falls below a configurable 90%. Vectorised form:

```
sorted_gold = torch.sort(gold)[0] # [20000]
q_idx = torch.linspace(0, len(sorted_gold) - 1, x.shape[1]).long()
order = torch.argsort(x, dim=1) # [B, F]
out = torch.empty_like(x)
out.scatter_(1, order, sorted_gold[q_idx].expand_as(x))
return out[:, self.scoring_idx]
```

One `argsort` and one `scatter_` per batch, replacing pyaging's per-row Python loop ([pyaging/models/\\_models.py:255](#)).

**median\_fill\_rank.** Average ranks with tie handling. pyaging's implementation is a nested `while` loop per row and is the slowest path in that package. The vectorised form uses the double-argsort identity for ordinal ranks and a segment-mean over `unique_consecutive` for ties:

```
order = torch.argsort(x, dim=1)
ordinal = torch.empty_like(order)
ordinal.scatter_(1, order, torch.arange(x.shape[1], device=x.device).expand_as(order))
# tie correction: for each run of equal sorted values, replace with the run's mean rank
```

**autosomal\_zscore.** Mean and SD over autosomal probes only, computed per sample with NaN masking, then applied to all probes including the sex chromosomes ([the science notes](#), §3.1).

## 4.2 Postprocess transforms

`falconage/models/ops/postprocess.py`. Scalar in, scalar out, applied elementwise over the batch. Constants come from [the science notes](#) §4.20 and §5.

| Op                           | Formula                                          | Params                    | Clocks                                                        |
|------------------------------|--------------------------------------------------|---------------------------|---------------------------------------------------------------|
| <code>identity</code>        | $y$                                              | -                         | most                                                          |
| <code>anti_log_linear</code> | $y < 0: (1+A)e^y - 1;$<br>$y \geq 0: (1+A)y + A$ | <code>adult_age=20</code> | Horvath2013, SkinAnd-Blood, PedBE, Han, PCHorvath2013, PCSki- |

| Op                     | Formula                                                                                                            | Params           | Clocks                                                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------|
|                        |                                                                                                                    |                  | nAndBlood, IntrinsicClock, CorticalClock, Pipek×3          |
| anti_log_linear_months | as above with A=48, then /12                                                                                       | adult_age=48     | Wu                                                         |
| anti_logp2             | $e^y - 2$                                                                                                          | -                | Mammalian1                                                 |
| anti_log               | $e^y$                                                                                                              | -                | MammalianLifespan                                          |
| anti_log_log           | $e^{(-e^{-y})}$                                                                                                    | -                | Mammalian2 relative-age step                               |
| mortality_to_phenoage  | $1 - \exp(-\exp(y) (e^{\{120\lambda\}} - 1) / \lambda)$ then $141.50225 + \ln(-0.00553 \cdot \ln(1-m)) / 0.090165$ | $\lambda=0.0192$ | PhenoAge                                                   |
| cox_to_years           | $((y - \mu_c) / \sigma_c) \cdot \sigma_a + \mu_a$                                                                  | four constants   | GrimAge, GrimAge2, CpG-PTGrimAge3, CpGTPC-GrimAge3         |
| petkovich_blood        | $((y+1.712)/0.1666)^{(1/0.4185)} / 30.5$                                                                           | -                | Petkovich                                                  |
| stubbs_multitissue     | $(\exp(0.1207y^2 + 1.2424y + 2.5440) - 3) \cdot 7/30.5$                                                            | -                | Stubbs                                                     |
| divide                 | $y / d$                                                                                                            | d                | Meer (30.5), Bohlin (7), EPICGA (7)                        |
| sigmoid                | $1/(1+e^{-y})$                                                                                                     | -                | McCartney ×6, MammalianFemale, CVDWesterman, ADBahadoSingh |
| one_minus              | $1 - y$                                                                                                            | -                | HypoClock                                                  |
| add_constant           | $y + c$                                                                                                            | c                | REG, Lin (+12.2169841), Ying ×3                            |
| scale_and_shift        | $y \cdot s + o \cdot s$                                                                                            | s, o             | Pasta, PastaMouse                                          |

mortality\_to\_phenoage clamps the mortality score to  $[1e-12, 1 - 1e-12]$  before the logarithm, following mcp-phenoage-clock ([mcp-phenoage-clock/src/phenoage.ts](#)), and records `clamped: true` in the result when it fires, an extreme biomarker combination that saturates the Gompertz CDF is information, not an implementation detail.

**Intercept convention.** ComputAge and biolearn carry the clock intercept inside the transform lambda; pyaging bakes it into the `nn.Linear` bias. FALCONAge normalises on import: the intercept always lives in the linear bias, and `add_constant` is reserved for genuine post-hoc offsets that the source publishes separately. The registry records `intercept_source: bias | transform` for each imported clock so the normalisation is auditable.

### 4.3 Model classes

`falconage/models/`

```
class FalconClock(nn.Module, ABC):
    meta: ClockMeta
```



```

features: list[str]
reference_values: Tensor | None

def preprocess(self, x: Tensor) -> Tensor      # composed from meta.preprocess
@abstractmethod
def core(self, x: Tensor, ctx: ClockContext) -> Tensor
def postprocess(self, y: Tensor) -> Tensor      # composed from meta.postprocess
def forward(self, x: Tensor, ctx: ClockContext) -> Tensor

```

`ClockContext` carries what a clock may need beyond its feature matrix, age, sex, species, and the outputs of other clocks it depends on:

```

@dataclass
class ClockContext:
    age: Tensor | None
    female: Tensor | None      # 1.0 / 0.0; unknown is an error, never 0.5
    species: Tensor | None     # one-hot for mammalian clocks
    upstream: dict[str, Tensor] # e.g. grimage for DNAmFitAge

```

Sex encoding deserves its own note. `ComputAge` maps unknown sex to 0.5 (`ComputAge/computage/models_library/model.py`), which silently produces a half-male half-female prediction. `FALCONAge` raises `MissingCovariateError` and offers `sex_unknown="error" | "impute_from_methylation" | "half"`, where the middle option runs `XChrom/YChrom` first and the last must be chosen deliberately.

| Class                           | Core computation                                                                       | Clocks                                                                   | v1.0           |
|---------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------|
| <code>LinearClock</code>        | $xW + b$                                                                               | ~120                                                                     | <b>shipped</b> |
| <code>ClinicalClock</code>      | PhenoAge / KDM / HD on tabular biomarkers                                              | 3                                                                        | <b>shipped</b> |
| <code>ScaffoldClock</code>      | refuses, and says where the coefficients come from                                     | the 28 tier C clocks                                                     | <b>shipped</b> |
| <code>PCLinearClock</code>      | $((x - \mu)R)W + b$                                                                    | PCHorvath2013, PCHannum, PCPhenoAge, PCSkinAndBlood, PCDNAmTL, PCGrimAge | specified      |
| <code>MultiStageCoxClock</code> | sub-linear-models $\rightarrow$ ordered concat with covariates $\rightarrow$ Cox layer | GrimAge, GrimAge2, CpG-PTGrimAge3, CpGPTPC-GrimAge3                      | specified      |
| <code>NeuralClock</code>        | arbitrary network from a registry-declared spec                                        | AltumAge                                                                 | specified      |
| <code>AggregationClock</code>   | mean / quantile / transmission-model reduction                                         | epiTOC1/2/3, stemTOC $\times$ 2, HypoClock, epiCMIT $\times$ 2           | specified      |
| <code>DeconvolutionClock</code> | pseudo-inverse $\rightarrow$ simplex projection $\rightarrow$ column select            | 6-cell and 12-cell EPIC                                                  | specified      |
| <code>CompositeClock</code>     | multiple sub-clocks combined by a declared rule                                        | SystemsAge $\times$ 12, DNAmFitAge                                       | specified      |
| <code>SpeciesClock</code>       | split CpG block from species one-hot, species-                                         | Mammalian $\times$ 9                                                     | specified      |



| Class                      | Core computation                                          | Clocks         | v1.0      |
|----------------------------|-----------------------------------------------------------|----------------|-----------|
|                            | dependent inverse transform                               |                |           |
| <code>SexScoreClock</code> | autosomal z-score $\rightarrow$ sex-chromosome projection | XChrom, YChrom | specified |

`ScaffoldClock` is the class that is not in the original design and turned out to matter most. Every “specified” row above is needed by a clock family whose coefficients cannot be redistributed, so writing the class first would produce code with no real answer to test against. Instead the registry entry, the feature list, the transform chain and the expected shapes all ship, `build()` returns a `ScaffoldClock`, and calling it raises `WeightsUnavailableError` naming the laboratory to ask, the licence that applies, and the open clocks that answer the same question. A silent skip would leave a results table with a column quietly missing.

#### 4.4 The ten special forward passes

Each gets an explicit implementation and its own gold-standard vector.

**1. GrimAge / GrimAge2** (`models/cox.py`). Ten (v2) or eight (v1) sub-linear-models on disjoint feature index sets, then a fixed concat order, then the Cox layer, then `cox_to_years`. The order is [GDF15, B2M, CystatinC, TIMP1, ADM, PAI1, Leptin, PACKYRS, (LogCRP, A1C), Age, Female] and is load-bearing, the Cox coefficients are positional (`pyaging/models/_models.py:1742`). The registry declares the order; a mismatch between declared and stored sub-model count fails validation.

**2. PCGrimAge.** Centre 78,464 CpGs, rotate to 1,933 PCs, append `Female` and `Age`, run eight PC-space sub-clocks, then the Cox layer. `requires_fp64: true`.

**3. CpGPTPCGrimAge3.** Scale  $\rightarrow$  split age from the 30 proxies  $\rightarrow$  rotate 30  $\rightarrow$  29 PCs  $\rightarrow$  concat  $\rightarrow$  rescale the PC block with a second mean/std pair  $\rightarrow$  linear  $\rightarrow$  `cox_to_years`. The two-stage scaling is not a mistake in the source; both scalers are stored.

**4. DNAmFitAge.** Split the batch by sex, run `GaitF/GaitM`, `GripF/GripM` and a shared `V02Max`, standardise each with the published sex-specific offset and slope, then a sex-specific Kleméra–Doubal layer, then recombine in the original row order. The eight constants ([the science notes](#), §4.17) live in the registry, not in code.

**5. Mammalian clocks** (`models/species.py`). Input is [CpGs | `species_one_hot`] with the one-hot width declared per clock (1,756 for clock-2 variants, 1,707 for clock-3). The species index selects gestation time, average maturity age and maximum lifespan from a packaged `AnAge` table, and the inverse transform is clock-specific:

```
clock 1: age = exp( $\hat{y}$ ) - 2
clock 2: age = e-(e- $\hat{y}$ )) · (max_age + gestation) - gestation
clock 3:  $\hat{m} = 5$  (gestation / maturity)0.38
           $\hat{y} \geq 0$  : age =  $\hat{m}$  (maturity + gestation)( $\hat{y} + 1$ ) - gestation
           $\hat{y} < 0$  : age =  $\hat{m}$  (maturity + gestation) e $\hat{y}$  - gestation
```

A species not in the `AnAge` table raises rather than falling back to a default.

**6. SystemsAge** (`models/composite.py`). Seven steps ([the science notes](#), §4.16): centre and rotate 125,175 CpGs  $\rightarrow$  project onto the system vector  $\rightarrow$  11 per-system linear modules  $\rightarrow$  a separate PC $\rightarrow$ age model with a quadratic correction then /12  $\rightarrow$  concat the 12-vector  $\rightarrow$  optional composite PCA layer  $\rightarrow$  affine

calibration  $((\text{raw} - c_0)/c_1) \cdot c_3 + c_2 \rightarrow /12$ . Twelve registry entries share one weight file and differ only in `prediction_index` (0–10, or –1 for the composite), so the weights are loaded once and cached across the twelve.

**7. Deconvolution** (`models/deconvolution.py`). Pseudo-inverse of the signature matrix, then exact Euclidean projection onto the probability simplex, then column selection. The projection is already batched in `pyaging` and that implementation is kept:

```
sorted_v, _ = torch.sort(p_raw, dim=1, descending=True)
cssv = torch.cumsum(sorted_v, dim=1)
inds = torch.arange(1, n + 1, device=..., dtype=...)
rho = (sorted_v - (cssv - 1) / inds > 0).sum(dim=1) - 1
theta = (cssv[torch.arange(B), rho] - 1) / (rho.to(dtype) + 1)
p = torch.clamp(p_raw - theta.unsqueeze(1), min=0)
```

The twelve 12-cell entries and six 6-cell entries share a weight file and differ by `cell_index`. A convenience wrapper `deconvolve(data, panel="epic_12cell")` returns all proportions as one frame, which is what users actually want.

**8. epiTOC2 / epiTOC3.**  $\text{tnsc} = (2/k) \sum_c (\beta_c - \beta_{0_c}) / (\delta_c(1 - \beta_{0_c}))$ , with per-CpG  $\delta$  and  $\beta_0$  vectors stored as weights and a guard against  $\delta(1-\beta_0) = 0$ .

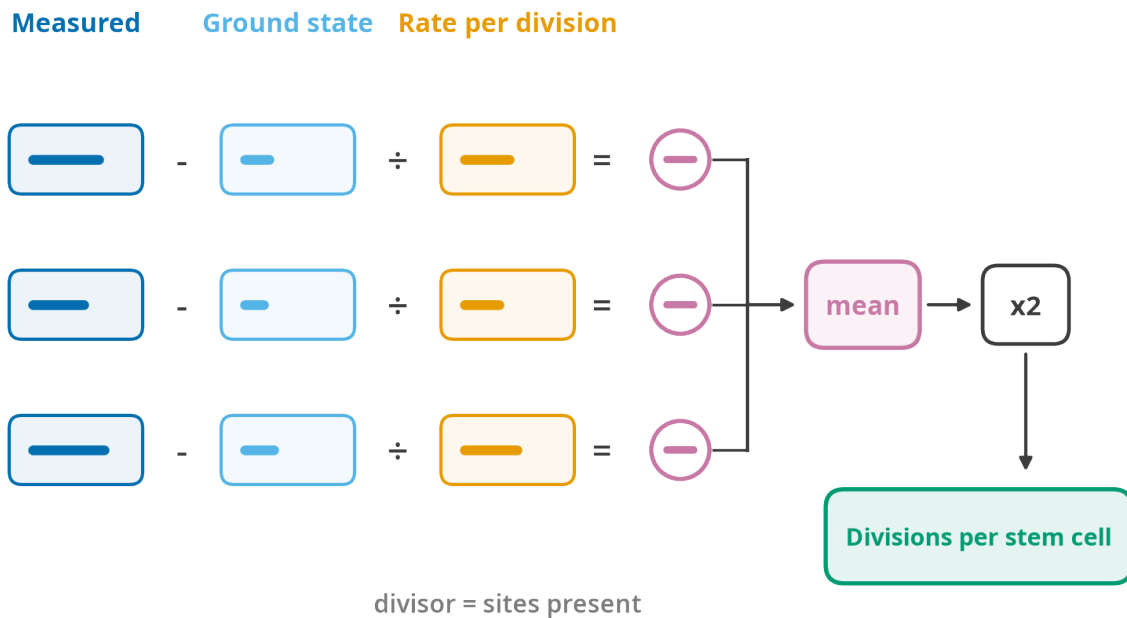


Figure 16: The division model inverted per site, then averaged

This is not a weighted sum with a different name. Each site is inverted on its own: the drift observed since the ground state is divided by that site’s drift per division, which turns a methylation fraction into a division count. Only then are the per-site estimates averaged, and  $k$  is the number of sites actually present

rather than the number the clock declares, so a missing site drops out of both the sum and the divisor instead of contributing a zero. The factor of two converts from the per-allele rate to divisions per stem cell.

**9. XChrom / YChrom.** Autosomal z-score, then  $\sum (z_{\text{sex}} - \text{mean}) \cdot \text{coef}$  over the sex-chromosome probe subset. Exposed both as clocks and through `estimate_sex(data) -> DataFrame[x_score, y_score, predicted_sex, confidence]`, since sex estimation is a preprocessing need as much as a clock.

**10. PastaMouse.** Expand the mouse feature vector into the full 8,113-gene human space using reference values for human-only genes, then run the shared Pasta forward.

#### 4.5 Clinical chemistry clocks

`falconage/models/clinical.py`. These are not methylation clocks and do not share the tensor path; they operate on a tabular biomarker frame and are implemented in a way that mirrors `BioAge/R/` so the numbers can be checked against it directly.

**PhenoAge.** Two variants, selected explicitly.

```
def phenoage(
    data: pd.DataFrame,
    units: Literal["si", "conventional"],          # no default
    variant: Literal["levine2018", "liu2019"] = "levine2018",
) -> pd.DataFrame                                # phenoage, mortality_score,
    phenoage_advance, clamped
```

The two coefficient sets and their Gompertz constants are in [the science notes §5.1](#) and are stored in the registry, not in code. `units` has no default because the SI and conventional parameterisations are not interconvertible by scaling the inputs, the coefficients differ too. The `0.090165` constant is the corrected one; `BioAge`'s April 2026 commit fixed a widespread `0.09165` and the test suite pins the corrected value.

**KDM.** Fit-then-project, with the fit reusable:

```
def kdm_fit(train: pd.DataFrame, biomarkers: list[str], age_col: str = "age",
            weights: str | None = None) -> KDMFit
def kdm_project(data: pd.DataFrame, fit: KDMFit, s_ba2: float | None = None) -> pd.DataFrame
def kdm(data, biomarkers, fit=None, by_sex=True) -> pd.DataFrame
```

The math is [the science notes §4.4](#), transcribed from `BioAge/R/kdm_calc.R` including the details that matter: `r_char` from the ratio of two weighted sums, `s_r` from the age range and biomarker count, the `s_BA2` correction, and the rule that a sample with more than two missing biomarkers returns NA rather than a partial score. Packaged NHANES III/IV fits ship as a `KDMFit` so the common case needs no training data.

**Homeostatic dysregulation.** Mahalanobis distance from a young reference cohort:

```
def hd_fit(reference: pd.DataFrame, biomarkers: list[str]) -> HDFit
def hd(data, biomarkers, fit=None, log=True, by_sex=True) -> pd.DataFrame
```

`hd_fit` standardises the reference by its own mean and SD, then stores the covariance. It checks the condition number and raises when the covariance is near-singular. `BioAge` only warns on a single-row reference, which is the degenerate case rather than the common one. The reference cohort must be young,

non-pregnant and within the clinical windows in [the science notes §3.5](#); the packaged NHANES III fit satisfies this and is the default.

#### 4.6 Op composition and validation

At load time the registry entry's `preprocess` and `postprocess` lists are compiled into an `nn.Sequential`, and the composition is validated:

- an op requiring stored params must find them in the weight file
- `quantile_normalize_gold` must be followed by a column subset
- a clock with `scale_type: age_years` may not carry `sigmoid` as its final transform
- the declared `n_features` must match the weight tensor's leading dimension

These checks run in `registry/validate.py` in CI and again at load time, so a malformed registry entry fails before it produces a number.

#### 4.7 Scaffold-only clocks

Twenty-eight clocks are tier C: their architecture is described in a published paper and is implemented here in full, but their coefficients are not ours to distribute. For these FALCONAge ships **the scaffold and nothing else** - the model class, the feature list, the preprocess and postprocess chain, the shape and dtype of every expected tensor, and a loader that accepts a coefficient file the user supplies.

Everything about these clocks except the numbers is written from scratch, and all of it is testable. What cannot be done is invent the numbers.

##### *Which they are, and why*

| Family                                                                   | n  | Why permission is needed                                                                                                                                 |
|--------------------------------------------------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| GrimAge, GrimAge2 and its 10 sub-clocks, CpGPTGrimAge3, CpGPTPC-GrimAge3 | 14 | Coefficients were never published in the papers. UCLA licenses the clock; what circulates in other packages has no traceable primary source.             |
| SystemsAge and its 11 system variants                                    | 12 | Flagged research-use-only by the authors. Coefficients are on a lab Google Drive, published in Nature Aging 2025 but not under a redistribution licence. |
| PCGrimAge                                                                | 1  | A principal-component reparameterisation of GrimAge; inherits its restriction.                                                                           |
| DunedinPACE                                                              | 1  | Research-use-only. Distributed through the authors' R package under terms the user accepts on install.                                                   |

The GrimAge family is the sharp case. GrimAge2 is the most-used mortality clock in the field, and its coefficient set has no public primary source at all, pyaging re-hosts a copy whose origin its own audit could not establish. FALCONAge will not launder that. The scaffold is complete and correct; the numbers must come from UCLA or from the user's own licensed copy.

##### *What ships*

```
# registry/clocks/grimage2.yaml (excerpt)
availability: C
coefficient_source:
```

```

tier: T5
url: null
primary_source_traced: false
licence: "Research use only. Commercial licensing via UCLA TDG."
obtain: >
  GrimAge2 coefficients are not publicly redistributable. Request access from
  the Horvath laboratory, or use the authors' online calculator at
  dname.clockfoundation.org. Once you hold a coefficient file, register it
  with falconage.registry.register_local_weights("grimage2", path).
contact: "https://dname.clockfoundation.org"
scaffold:
  complete: true
  expects:
    - {name: "sub_models", shape: "10 x (n_features_i)", dtype: float64}
    - {name: "cox", shape: "(12,)", dtype: float64}
  validated_against: "published worked example, Lu 2022 Table 2"

```

### Supplying coefficients

```

import falconage as fa

fa.registry.register_local_weights(
    "grimage2",
    "~/licensed/grimage2_coefficients.csv",
)
res = fa.score(data, clocks=["grimage2"])

```

```

register_local_weights("grimage2", "~/licensed/grimage2_coefficients.csv")
res <- score(data, clocks = "grimage2")

```

`register_local_weights` parses the file, checks it against the scaffold's declared shapes and feature count, warns on any feature the scaffold does not expect, converts to `safetensors`, and caches it. A file that does not match the published architecture is rejected with the mismatch named, which also makes this the mechanism for validating a coefficient set someone hands you.

The registration is recorded in the run manifest as `weights_source: user_supplied, sha256: ...`, so a result computed from a licensed copy is distinguishable from one computed from a redistributed set.

### Behaviour without coefficients

Scoring a tier C clock with nothing registered raises rather than silently skipping:

```
WeightsUnavailableError: grimage2 is a scaffold-only clock.
```

Its architecture is implemented and tested, but its coefficients are research-use-only and are not distributed with FALCONAge.

Obtain them from the Horvath laboratory ([dname.clockfoundation.org](https://dname.clockfoundation.org)), then: `falconage.registry.register_local_weights("grimage2", <path>)`

```
Open alternatives predicting mortality risk:
phenoage          clinical chemistry, 10 markers, tier A
dnamphenoaage     DNA methylation, 513 CpGs, tier A
zhangmortality    DNA methylation, 10 CpGs, tier A
```

Naming the open alternatives matters. A user who wanted a mortality clock and hit a wall should leave with one that works, not with an error.

`clocks="compatible"` excludes unregistered tier C clocks by default and reports the exclusion. `clocks="all"` includes them and fails loudly. `falconage clocks list --tier C` shows the whole set with its obtain instructions.

### *Testing a clock whose weights you do not have*

The scaffold is still fully tested, three ways:

1. **Synthetic weights.** Generate a coefficient set of the right shape; run it, and assert the forward pass produces the algebraically expected value. This tests the architecture, the sub-model ordering, the concat order, the Cox transform constants, without needing the real numbers. A GrimAge2 scaffold given a coefficient vector of all zeros except one must return the value that hand-computation says it should.
2. **Published worked examples.** Several papers report a reference sample's inputs and its resulting age. Where one exists it is the gold-standard vector, and it is a better oracle than another package's output because it comes from the authors.
3. **Shape and invariant contracts.** Feature count, dtype, sub-model count, concat order, output range and scale type are all asserted from the registry entry regardless of which weights are loaded.

A tier C clock without a published worked example carries `gold_standard.source: "synthetic weights, architecture only"` and the docs say so on its page. It is honest about being an untested-against-truth implementation of a correctly-described architecture.

### *Could they be reimplemented from scratch instead?*

For the architecture, yes, and that is what §4.4 already specifies, the GrimAge two-stage structure, the PC rotation and SystemsAge's seven-step pass are all described well enough in their papers to write independently, and are written here.

For the coefficients, no. They are the output of an elastic net fit to cohorts that are not public. Framingham, the Women's Health Initiative, the Dunedin Study. Refitting the same architecture on different data produces a different clock that must not carry the same name. That path is real and is scoped as a v1.5 milestone (§13), where a FALCONAge-native mortality clock trained on the open ComputAgeBench split would be a new contribution rather than a substitute.

## 5. Download module

### 5.1 Contract

One verb, one accession, files plus a normalised sample table.

```
def download(
    accession: str | Sequence[str],
```

```

dest: Path | None = None,          # None -> the managed cache
what: Literal["raw", "processed", "metadata", "auto"] = "auto",
credentials: Path | None = None,
dry_run: bool = False,
max_size_gb: float | None = None,
resume: bool = True,
n_jobs: int = 4,
) -> DownloadResult

```

```

@dataclass
class DownloadResult:
    accession: str
    source: str          # "geo" | "arrayexpress" | ...
    files: list[FileRecord] # path, role, sha256, size, url
    samples: pd.DataFrame  # the normalised sample table
    raw_metadata: dict     # whatever the source gave, unmodified
    manifest_path: Path

```

`what="auto"` prefers raw over processed when raw exists, because a beta matrix someone else normalised is not reproducible. `dry_run=True` returns the same object with `files` populated and sizes filled but nothing transferred, which is how a user finds out that a series is 12 GB before committing to it.

## 5.2 Source dispatch

`falconage/download/resolve.py` recognises the accession by pattern and returns the right `Downloader`. Unrecognised input raises with the list of recognised patterns rather than guessing.

| Pattern                                      | Source                    | Backend                                                                                    | Ships |
|----------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------|-------|
| GSE\d+                                       | GEO series                | GEOparse plus supplementary-file fetch; <code>methylnprep</code> when the series has IDATs | now   |
| GSM\d+                                       | GEO sample                | same, single sample                                                                        | now   |
| GPL\d+                                       | GEO platform              | manifest only                                                                              | now   |
| E-\w+-\d+                                    | ArrayExpress / BioStudies | BioStudies REST v1                                                                         | now   |
| SR[PRXS]\d+,<br>PRJNA\d+, ERP\d+,<br>DRP\d+  | SRA / ENA                 | <code>pysradb</code> for metadata, ENA FTP or <code>fasterq-dump</code> for reads          | now   |
| 10\.\d{4,}/.*                                | Zenodo, Figshare, Dryad   | DOI resolution then the host REST API                                                      | now   |
| numeric<br>with <code>source="zenodo"</code> | Zenodo record             | Zenodo REST                                                                                | now   |
| hf://owner/repo[/path]                       | Hugging Face              | <code>huggingface_hub</code>                                                               | now   |
| TCGA-\w+, GDC UUID                           | GDC                       | GDC files/data API                                                                         | now   |
| PXD\d+                                       | PRIDE                     | PRIDE REST                                                                                 | later |
| MTBLS\d+                                     | MetaboLights              | MetaboLights REST                                                                          | later |

| Pattern          | Source                 | Backend                                   | Ships |
|------------------|------------------------|-------------------------------------------|-------|
| ST\d{6}          | Metabolomics Workbench | MW REST                                   | later |
| syn\d+           | Synapse                | <code>synapseclient</code> , credentialed | later |
| EGAD\d+, EGAS\d+ | EGA                    | documented, credentialed                  | later |
| phs\d+           | dbGaP                  | documented, not automated                 | -     |

### 5.3 GEO, in detail

GEO is where most of the traffic goes, and it has three shapes that need different handling.

```
class GEODownloader(Downloader):
    def probe(self, gse: str) -> GEOSeriesInfo
    def fetch(self, gse: str, what: str, dest: Path) -> DownloadResult
```

`probe` reads the MINiML metadata and reports what is actually available before anything is transferred:

```
@dataclass
class GEOSeriesInfo:
    gse: str
    title: str
    n_samples: int
    platforms: list[str]          # GPL ids, mapped to falconage platform names
    has_idats: bool
    has_series_matrix: bool
    supplementary_types: list[str]
    total_size_bytes: int
    characteristics_fields: list[str] # the free-text keys that might hold age and sex
```

The three shapes:

1. **IDATs present** - roughly 40% of methylation series. Download the `_Grn.idat.gz` and `_Red.idat.gz` pairs plus the platform manifest, and hand off to the preprocess module. This is the only path that yields a reproducible beta matrix.
2. **Series matrix only**. Parse the `!Sample_` header block for metadata and the tab-delimited body for the value matrix. The values are whatever the submitter normalised, which is recorded in the manifest as `normalisation: submitter-reported` and carried into the result, so a downstream comparison across series knows the matrices are not commensurable.
3. **Supplementary processed files**. A CSV, TSV or RData of betas with no fixed convention. `read_betas` sniffs orientation, delimiter, and whether values are betas in `[0,1]` or M-values, and asks rather than guesses when it cannot tell.

### 5.4 Metadata normalisation

`falconage/download/metadata.py`. The hardest part of the download module is not transfer: it is that the GEO `characteristics_ch1` field is free text and every submitter writes age differently.



```
def normalise_samples(raw: pd.DataFrame, source: str,
                     mapping: dict[str, str] | None = None) -> pd.DataFrame
```

Target schema, which is what the rest of the package consumes:

| Column              | Type     | Notes                                                      |
|---------------------|----------|------------------------------------------------------------|
| sample_id           | str      | index; GSM id for GEO                                      |
| title               | str      | submitter sample title                                     |
| age                 | float    | years; months and weeks converted, source unit recorded    |
| sex                 | category | male, female, unknown                                      |
| tissue              | str      | controlled vocabulary where mappable, raw string otherwise |
| condition           | str      | HC for controls where identifiable                         |
| platform            | str      | falconage platform name                                    |
| subject_id          | str      | for repeated measures                                      |
| timepoint           | float    | for longitudinal series                                    |
| raw_characteristics | str      | the original text, always kept                             |

Extraction is rule-based and conservative: regex patterns for the common age spellings (`age: 54`, `age (years): 54`, `agey: 54`, `54 yr`), a sex vocabulary, and a tissue lexicon. Every extracted value keeps its source string in `raw_characteristics`, and the fraction of samples where extraction succeeded is reported. When it falls below a threshold the module says so loudly rather than returning a half-empty age column:

```
WARNING age extracted for 41/120 samples (34%) from field 'characteristics_ch1'
unparsed examples: "age at diagnosis: 54y", "AGE=54", "54 years old"
supply mapping={"age": "characteristics_ch1"} with a custom parser, or
edit the returned sample table directly
```

No LLM, no fuzzy matching that might silently mis-assign. A wrong age is worse than a missing one because age acceleration is computed against it.

## 5.5 Transfer, cache, credentials

`transfer.py` - HTTP range requests for resume, parallel connections capped at `n_jobs`, SHA-256 verification against the published checksum where one exists and against a recorded one on re-download, exponential backoff on 429 and 5xx, and a per-source rate limit (NCBI allows 3 requests per second without an API key, 10 with one).

`cache.py` - content-addressed under `FALCONAGE_CACHE`, default `~/.cache/falconage`, laid out as `<source>/<accession>/<sha256-prefix>/<filename>` with a `manifest.json` per accession.

```
falconage cache ls           # accession, source, size, last used
falconage cache size
falconage cache rm GSE40279
falconage cache verify      # re-check every stored checksum
falconage cache gc --older-than 90d --keep-under 20GB
```

`credentialled.py` covers the sources that need authorisation. Synapse and EGA get working implementations behind an explicit credentials file; dbGaP and UK Biobank get documentation and a clear error explaining that access is granted per project and cannot be automated:

```
raise CredentialsRequiredError(
    "dbGaP accession phs000424 requires an approved data access request. "
    "FALCONAge cannot download it for you.",
    remedy="Download via the dbGaP repository key and SRA Toolkit, then point "
    "falconage preprocess at the resulting files.",
)
```

Credentials come from a file or the environment, never from a function argument that would land in a shell history or a notebook.

## 5.6 CLI

```
falconage download GSE40279 --dest data/ --what raw
falconage download GSE40279 --dry-run           # sizes and file list only
falconage download GSE40279 GSE42861 --n-jobs 8
falconage download 10.5281/zenodo.18763485
```

As shipped, `--dry-run` is what answers “what is in this series?”: it prints every URL and byte count and transfers nothing. The specification also had a separate `probe` verb for that; it was not built, because it would have been the same code behind a second name.

## 6. Preprocess module

### 6.1 Contract

Raw or public data in, a `FalconData` the scoring module accepts out. One entry point per modality, all converging on samples by features with exact feature identifiers and declared units.

```
def preprocess_methylation(
    source: Path | pd.DataFrame | DownloadResult,
    platform: str | None = None,          # None -> auto-detect
    pipeline: str = "noob_bmiq",          # "noob_bmiq" | "sesame" | "noob" | "raw"
    detection_p: float = 0.01,
    mask_probes: bool = True,             # cross-reactive, SNP, sex-chromosome
    aggregate_epicv2: bool = True,
    imputation: str = "none",             # dataset-level; clock-level is separate
    sample_table: pd.DataFrame | None = None,
    device: str = "auto",
) -> FalconData
```

Analogous `preprocess_clinical`, plus `preprocess_expression` and `preprocess_proteomics`.

### 6.2 Methylation

**IDAT to beta.** The chain is [the science notes](#) section 3.1:

```

IDAT pair (Grn, Red)
-> parse binary IDAT, match to platform manifest
-> out-of-band background correction and dye bias (noob)
-> probe-type bias correction (BMIQ), or SeSAmE as a single-step alternative
-> detection p-value filter
-> cross-reactive / SNP / sex-chromosome probe masking
-> beta = M / (M + U + 100)
-> beta matrix, samples by probes

```

Implementation route: `methylnprep` for IDAT parsing and `noob`, a native BMIQ port (beta-mixture quantile normalisation is around 150 lines, and having it in-tree avoids an R dependency inside the Python core), and SeSAmE through `rrpy2` only when the user asks for it. The pipeline choice is recorded in the manifest because it changes the answer, the [science notes](#) section 3.1 notes that `noob + BMIQ` and SeSAmE both perform well but are not interchangeable, and `funnorm` is preferred when global methylation differs between groups.

### Platform detection.

```

def detect_platform(feature_ids: Sequence[str]) -> PlatformInfo

```

Decides from probe-ID pattern and count: EPIC v2 if any ID carries a `_TC21`-style suffix, then by overlap against packaged manifest probe sets for 27K, 450K, EPIC v1, EPIC v2, MSA, MammalMethylChip40 and MammalMethylChip320. Returns the best match with its overlap fraction, and raises `PlatformMismatchError` when the best overlap is under 50% rather than proceeding with the wrong manifest.

**EPIC v2 aggregation.** Split each column name on `_`, group by the CpG prefix, average replicates. Mandatory, the [science notes](#) section 16 records that skipping it means every clock sees zero overlapping features. When `detect_platform` says EPIC v2 and `aggregate_epicv2=False`, the module raises rather than producing a silently empty result.

**Cross-platform harmonisation.** A 450K clock applied to EPIC data uses the probes present on both. The module reports the intersection size per clock before scoring, and records `platform_native` alongside `platform_applied`, so a result computed on 340 of Horvath's 353 probes says so.

**Series matrix and beta CSV** paths run the same masking, aggregation and detection steps, minus the normalisation the submitter already applied, with `normalisation: submitter-reported` in the manifest.

## 6.3 Clinical chemistry

```

def preprocess_clinical(
    data: pd.DataFrame,
    units: dict[str, str],           # required, see core/units.py
    validate_ranges: bool = True,
    range_action: str = "warn",     # "warn" | "na" | "error"
) -> FalconData

```

Three steps: map column names to canonical marker names through a synonym table, so `ALB`, `albumin` and `Albumin (g/dL)` all resolve to `albumin`; convert every marker to the canonical unit the clocks expect;

validate against physiological ranges. The ranges are the BioAge clinically-healthy windows ([the science notes](#) section 3.5) for the reference-cohort case, and wider physiological bounds for the general case, a CRP of 40 mg/L is outside the healthy window but is a real measurement, so it warns rather than nulls by default.

## 6.4 Imputation, kept in three distinct places

[the science notes](#) section 6.3 separates three stages that are routinely conflated. Keeping them separate is what makes the missingness report meaningful.

| Stage          | Where                          | Options                                                                                                | Recorded as                     |
|----------------|--------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------|
| Dataset-level  | <code>preprocess.impute</code> | <code>none</code> , <code>mean</code> , <code>median</code> , <code>constant</code> , <code>knn</code> | <code>imputation.dataset</code> |
| Clock-level    | <code>score.features</code>    | reference values from the registry; zero only when no reference exists                                 | <code>imputation.clock</code>   |
| Model-internal | inside the clock               | the model's own stored statistic                                                                       | <code>imputation.model</code>   |

Dataset-level KNN is the expensive one and is GPU-accelerated when a device is available: `cuml.KNNImputer` when RAPIDS is installed, a chunked CuPy distance otherwise, sklearn on CPU.

Clock-level filling uses `reference_values` from the registry. Filling a missing beta with zero means “fully unmethylated”, which is biologically wrong for most sites, so a clock with no reference vector and missing features emits a warning naming the fraction affected. Above `max_missing`, default 0.30, scoring raises `ExcessiveMissingnessError` rather than returning a number nobody should use; `max_missing=1.0` restores the permissive behaviour deliberately.

## 6.5 QC report

```
@dataclass
class QCReport:
    n_samples: int
    n_features: int
    platform: PlatformInfo
    missingness_overall: float
    missingness_per_sample: pd.Series
    beta_distribution: dict          # quantiles, bimodality statistic
    failed_detection: pd.Series     # per sample
    predicted_sex: pd.DataFrame     # XChrom/YChrom, against declared sex
    sex_mismatches: list[str]
    age_available: float            # fraction of samples with age
    clock_coverage: pd.DataFrame   # clock, n_present, n_total, pct
    warnings: list[str]
```

`clock_coverage` prints before scoring, not after. A user who can see that GrimAge2 will run on 61% of its features gets to decide whether to proceed; a user who discovers it afterwards has already written the number down.

## 6.6 CLI

```
falconage preprocess methylation data/GSE40279/ --out prepared.h5ad --pipeline noob_bmiq
falconage preprocess methylation betas.csv --platform illumina_450k --out prepared.h5ad
falconage preprocess clinical labs.csv --units units.yaml --out prepared.h5ad
```

The specification had a separate `qc` verb writing its own HTML. Not built: `preprocess` already prints the QC summary, and `report` embeds the full set of QC figures, so a third path to the same numbers was a third thing to keep in step.

## 7. Scoring, analysis, plots, report

### 7.1 `score()`

```
def score(
    data: FalconData,
    clocks: str | Sequence[str] | Literal["all", "compatible"] = "compatible",
    device: str = "auto",
    dtype: str = "float64",
    batch_size: int | None = None,
    max_missing: float = 0.30,
    imputation: str = "reference",
    sex_unknown: str = "error",
    keep_features: bool = False,
    progress: bool = True,
) -> FalconResult
```

`clocks="compatible"` is the default and calls `registry.compatible_with(data)`, which is the behaviour a first-time user wants. `"all"` runs everything and warns about the ones that will be mostly imputed.

Loop, per clock:

1. Resolve metadata; warn on `research_only` and on any `known_discrepancies`.
2. Check required covariates; raise `MissingCovariateError` naming the clock and the covariate.
3. Align features into `obsm[f"X_{clock}"]`, filling absences from `reference_values`. Record `percent_na` and the missing-feature list. Raise at 100% missing, or above `max_missing`.
4. Promote dtype if `requires_fp64`.
5. Batch through the model under `torch.inference_mode()`.
6. Write predictions to `obs[clock]` and metadata to `uns[f"{clock}_metadata"]`.
7. Range-check against `plausible_range`; flag, never clip.
8. Free the `obsm` slice unless `keep_features`, then `gc.collect()` and `torch.cuda.empty_cache()`.

### 7.2 `FalconResult`

```
class FalconResult:
    data: FalconData
    manifest: RunManifest

    def table(self, long: bool = True) -> pd.DataFrame
```

```

def wide(self) -> pd.DataFrame
def acceleration(self, method: str = "both",
                  within: str | None = None) -> pd.DataFrame
def qc(self) -> pd.DataFrame
def to_csv(self, path) -> None
def to_h5ad(self, path) -> None

```

The long table carries every provenance field from [the science notes](#) section 8.1:

```

sample_id, clock, clock_version, registry_version, value, unit, scale_type,
chronological_age, aa_absolute, aa_relative, percent_features_missing,
n_features_missing, imputation, preprocess_ops, postprocess_ops,
species, tissue, platform_native, platform_applied, research_only,
flags, citation, doi

```

flags is a comma-separated list drawn from a fixed vocabulary: out\_of\_range, high\_missingness, clamped, dtype\_promoted, platform\_mismatch, known\_discrepancy, extrapolated\_age\_range, sex\_imputed.

### 7.3 Age acceleration

```

def acceleration(result, method="both", within=None, covariates=None) -> pd.DataFrame

```

Three conventions ([the science notes](#) section 1.1), all guarded by scale\_type:

- **absolute** - value - chronological\_age. Legal only for scale\_type in {age\_years, age\_months, age\_days, age\_hours, gestational\_weeks}. Attempting it on rate, log\_hazard, probability or proportion raises IncompatibleScaleError: DunedinPACE is not an age, and subtracting CA from it is a category error.
- **relative** - residual of `lm(value ~ age)`, optionally with covariates. Legal for any continuous scale. This is the convention that removes the regression-dilution bias every penalised clock carries.
- **within** - residual against a group mean, for the single-cell and spatial case where the reference is the animal rather than the calendar, following [SpatialAgingClock](#).

method="both" returns absolute and relative side by side, and omits absolute for scales where it is illegal rather than emitting NaN.

### 7.4 Survival, association, reliability, benchmark

```

# survival.py
def cox(result, time, status, clocks=None, covariates=("age", "sex"),
          standardise_within=("sex",)) -> pd.DataFrame      # HR, CI, p, n, events
def c_index(result, time, status, clocks=None) -> pd.DataFrame
def time_dependent_auc(result, time, status, horizons=(5, 10, 15)) -> pd.DataFrame

# association.py
def associate(result, outcome, family="gaussian", covariates=("age", "sex")) ->
pd.DataFrame
def interactions(result, outcome,
                  modifiers=("bmi", "sex", "smoking", "education")) -> pd.DataFrame

```

```
# reliability.py
def icc(result, subject_col, replicate_col, kind="ICC2_1") -> pd.DataFrame
def pool_icc(iccs: Sequence[pd.DataFrame]) -> pd.DataFrame          # Fisher-Z

# benchmark.py
def aa2_test(result, condition_col, control="HC", assumption="normal") -> pd.DataFrame
def aa1_test(result, condition_col, control="HC", assumption="normal") -> pd.DataFrame
def ca_accuracy(result, control="HC") -> pd.DataFrame          # MedAE, MAE, RMSE, r, rho,
slope, MedE
def total_score(aa2, aa1, accuracy) -> pd.DataFrame
def run_benchmark(result, condition_col, ...) -> BenchmarkResult
```

cox standardises the acceleration within sex before fitting, so the hazard ratio is per sex-specific SD - the BioAge convention ([BioAge table\\_surv.R](#)), which is what makes effect sizes comparable across clocks with different variances.

The benchmark functions implement the ComputAgeBench methodology ([the science notes](#) section 11.1):

```
AA2:   one-sided Welch t-test (normal) or Mann-Whitney U (none), AAC against HC
AA1:   one-sided one-sample t-test or Wilcoxon signed-rank, AAC against 0
FDR:   Benjamini-Hochberg per model across datasets, threshold 0.05
total = aa2_passed + aa1_passed * (1 - max(0, MedE) / MedAE)
```

The bias penalty in the total is what stops a clock that simply over-predicts everyone from sweeping AA1. `total_score` returns the components alongside the total, and the report never prints the total on its own.

`run_benchmark(..., dataset="computage")` fetches the 65-study benchmark set from Hugging Face and runs the whole thing, so a user can place their own clock on the published leaderboard.

## 7.5 Agreement

```
def clock_correlation(result, method="spearman", cluster=True) -> pd.DataFrame
def clock_embedding(result, method="umap", n_neighbors=15) -> pd.DataFrame
def concordance(result, reference_clock: str) -> pd.DataFrame
```

`clock_correlation` with `cluster=True` returns the matrix plus a dendrogram order, which is the pyaging Figure 1a analysis: telomere predictors cluster together, and chronological predictors separate from mortality predictors.

## 7.6 Plots

Every plotting function returns the figure and has a parallel `*_data()` function returning the frame it plotted, so the R side renders the same numbers with ggplot2 rather than passing matplotlib figures across the bridge. The fourteen from [the science notes](#) section 10:

| Function              | Figure                                                                       |
|-----------------------|------------------------------------------------------------------------------|
| <code>ba_vs_ca</code> | BA against CA, faceted by clock, coloured by sex, identity line, r annotated |

| Function                          | Figure                                                                       |
|-----------------------------------|------------------------------------------------------------------------------|
| <code>clock_corr</code>           | correlation matrix: scatter below the diagonal, r above, labels on it        |
| <code>clock_dendrogram</code>     | hierarchical clustering of clocks                                            |
| <code>clock_umap</code>           | samples embedded in clock space                                              |
| <code>trajectory</code>           | prediction across a timepoint variable, one line per clock                   |
| <code>class_bench</code>          | boolean clock by dataset:condition:test grid with per-class column groups    |
| <code>medae_bar</code>            | median absolute error per clock, healthy controls only                       |
| <code>bias_bar</code>             | median error per clock, symmetric x-limits                                   |
| <code>forest</code>               | hazard ratios per clock per outcome                                          |
| <code>acceleration_density</code> | violin or density of age acceleration by group                               |
| <code>system_profile</code>       | radar or heatmap over SystemsAge systems and organAge-ing organs             |
| <code>qc_density</code>           | beta distribution per sample at each preprocessing stage                     |
| <code>calibration</code>          | predicted against observed, for probability outputs                          |
| <code>missingness</code>          | percent of features absent per clock, drawn before the results               |
| <code>kaplan_meier</code>         | survival of the fastest-ageing tail against the slowest, with the log-rank p |
| <code>volcano</code>              | effect size against evidence, thresholded at the BH cut rather than a raw p  |

The last two close the gap against the figure conventions in current papers. Both estimators are written out rather than imported: the product-limit form and the two-sample log-rank statistic are about forty lines together, and `survival` and `lifelines` are heavier dependencies than that deserves. The R side calls the Python core for the log-rank p and redraws the curve in `ggplot2`, so the statistic cannot differ between the languages while the figure still carries the R theme.

`volcano` draws the Benjamini-Hochberg threshold, taken from the `q` column that `associate()` already returns rather than recomputed. Drawing a raw p-value cut is the usual error and, across thousands of tests, calls noise significant.

Conventions enforced in `theme.py`: the identity line is always drawn on BA-against-CA; rate clocks never share an axis with age clocks; facets are ordered by clock generation rather than alphabetically; every panel is annotated with n, r and missingness; colour encodes group and shape encodes platform, with one palette across a whole report.

## 7.7 Report

```
def report(result, path: Path, sections="all", embed_data=True) -> Path
```

One self-contained HTML file, inlined CSS, base64 figures, no external requests, carrying the run manifest, the QC block, the results table, the standard figures, and the caveats that apply to the clocks that actually ran. `embed_data=True` attaches the results CSV as a data URI so the report is a complete record.



## 7.8 CLI

**The command lines in this section, in 5.5, 5.6 and 6.6, are the specified interface and several of them do not parse against the shipped one.** The verbs are right; the argument shapes are not. `score`, `bench` and `preprocess` take `--input` rather than a positional, `preprocess` has no `methylation` or `clinical` sub-verb, `clocks` requires an action (`list`, `info`, `cite`) before any filter, `power` takes `--clock`, and `cache` has `ls` and `rm` but not `size`, `verify` or `gc`.

Left as written because this is the design record and the spec is what it records. The shipped interface is in [Getting started](#) and in `--help`, which is generated from the parser and cannot drift. `docs/check_api_docs.py` parses every command in every other document against that parser and skips this page for exactly this reason.

```
falconage score prepared.h5ad --clocks horvath2013,grimage2,dunedinpace --out results/
falconage score prepared.h5ad --clocks compatible --device cuda --dtype float32
falconage bench results/ --condition-col condition --control HC
falconage report results/ --out report.html
falconage clocks --tier A --search mortality
falconage power horvath2013 --effect 0.5 --sd 6
falconage consensus results/ --group condition
```

Two departures from the specification. `analyse --cox` was not built as a verb: survival and association are two functions with several arguments each, and a command line is a poor place to specify a model formula, so they stay in `score()`'s Python and R surfaces where the arguments can be named. And `clocks` took `list/info/cite` sub-verbs in the specification; as shipped it is one verb with flags, because the three did the same lookup and differed only in what they printed. `power` and `consensus` are new since the specification was written.

## 8. R package

### 8.1 What “identical” means here

FALCONAge is not a reimplementation. Every number it returns is computed by the Python core through `reticulate`, so the R and Python answers are the same bytes because they are the same computation. What the R package adds is an interface an R user would have written: S3 classes with `print`, `summary`, `plot` and `as.data.frame`; `data.frame` in and `data.frame` out; roxygen2 documentation; a pkgdown site; and errors that arrive as R conditions rather than Python tracebacks.

This is the design tAge uses (tAge/R/predict.R), with one difference: tAge asks the user to set `RETICULATE_PYTHON` by hand, and FALCONAge manages the environment itself.

### 8.2 Environment management

R/install.R

```
falconage_install(method = c("auto", "uv", "virtualenv", "conda"),
                  envname = "r-falconage",
                  version = NULL,           # NULL = the version matching this R package
                  gpu = FALSE,
```

```

        force = FALSE)

falconage_config()      # what Python, what version, what device, what registry
falconage_available()  # logical, does a working core exist
falconage_update()

```

`method = "auto"` prefers `uv` when it is on the `PATH`, because it resolves in seconds rather than minutes, and falls back to `virtualenv`. `gpu = TRUE` installs the CUDA torch build.

`version = NULL` pins the Python core to the exact version that matches the R package, read from `DESCRIPTION`'s `Config/falconage/python-version` field. This is the mechanism that stops an R package from silently driving a Python core it was never tested against.

`falconage_config()` prints the same block the CLI prints, so a bug report from an R user and one from a Python user carry comparable information:

```

FALCONAge 1.0.0 (R)
python core  1.0.0  /home/x/.virtualenvs/r-falconage/bin/python
torch        2.6.0  CUDA 12.4
device       cuda:0  NVIDIA RTX A4000, 16 GB
dtype        float64
registry     1.1.0  (161 clocks)
cache        /home/x/.cache/falconage  (2.1 GB)

```

### 8.3 The bridge

R/zzz.R and R/bridge.R

```

.falconage <- new.env(parent = emptyenv())

.onLoad <- function(libname, pkgname) {
  .falconage$py <- reticulate::import("falconage", delay_load = list(
    environment = "r-falconage",
    on_load = function() check_version_compatibility(),
    on_error = function(e) NULL          # defer the error to first use
  ))
  set_default_options()
}

```

`delay_load` matters: without it, `library(FALCONAge)` fails on a machine with no Python environment yet, which breaks R CMD check on CRAN's build farm. With it, the package loads, `falconage_available()` returns `FALSE`, and every exported function that needs the core calls `require_core()`, which raises a condition telling the user to run `falconage_install()`.

**Type marshalling.** Explicit, never implicit.

| R                       | Python                        | Direction            |
|-------------------------|-------------------------------|----------------------|
| <code>data.frame</code> | <code>pandas.DataFrame</code> | both, via reticulate |

| R                          | Python                                                                | Direction                                                                |
|----------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>falcon_data</code>   | <code>.h5ad</code> path                                               | both: the object is passed as a file path, never marshalled cell by cell |
| <code>falcon_result</code> | <code>.h5ad</code> path plus a results <code>data.frame</code>        | both                                                                     |
| <code>character(1)</code>  | <code>str</code>                                                      | both                                                                     |
| <code>NULL</code>          | <code>None</code>                                                     | both                                                                     |
| <code>NA</code>            | <code>float('nan')</code> for numerics, <code>None</code> for strings | R to Python                                                              |
| <code>matrix</code>        | <code>numpy.ndarray</code>                                            | both                                                                     |

The `falcon_data` rule is the important one. Passing an 850,000-column matrix through reticulate cell by cell is slow and copies twice. Instead the R object holds a path to an `.h5ad` file plus a cached summary (dimensions, platform, sample table), and both languages read the same file. This is also what makes a mixed-language workflow possible: preprocess in R, score in Python, plot back in R, one artefact throughout.

**Error translation.** `bridge.R` wraps every call:

```
py_call <- function(fn, ...) {
  tryCatch(fn(...), error = function(e) {
    info <- parse_python_error(e)      # class, message, .remedy
    rlang::abort(
      message = info$message,
      class   = paste0("falconage_", info$class),
      remedy  = info$remedy,
      parent  = e
    )
  })
}
```

A `MissingUnitsError` from Python becomes an R condition of class `falconage_missing_units_error`, so R users can `tryCatch` on it, and the message is the one-sentence remedy rather than a 40-line traceback. The Python traceback stays available as `parent`.

## 8.4 Public API - one-to-one with Python

Every Python verb has an R counterpart with the same name in snake\_case and the same argument names. This table is the contract the conformance tests enforce (§10.5).

| Python                                 | R                                     | Returns                      |
|----------------------------------------|---------------------------------------|------------------------------|
| <code>fa.download()</code>             | <code>download()</code>               | <code>falcon_download</code> |
| <code>fa.download(dry_run=True)</code> | <code>download(dry_run = TRUE)</code> | <code>data.frame</code>      |
| <code>fa.prepare()</code>              | <code>prepare()</code>                | <code>falcon_data</code>     |
| <code>fa.prepare_clinical()</code>     | <code>preprocess_clinical()</code>    | <code>falcon_data</code>     |
| <code>fa.qc()</code>                   | <code>qc()</code>                     | <code>falcon_qc</code>       |
| <code>fa.score()</code>                | <code>score()</code>                  | <code>falcon_result</code>   |

| Python                                     | R                                | Returns                       |
|--------------------------------------------|----------------------------------|-------------------------------|
| <code>fa.acceleration()</code>             | <code>acceleration()</code>      | <code>data.frame</code>       |
| <code>fa.cox_hazard()</code>               | <code>cox()</code>               | <code>data.frame</code>       |
| <code>fa.icc()</code>                      | <code>icc()</code>               | <code>data.frame</code>       |
| <code>fa.run_benchmark()</code>            | <code>run_benchmark()</code>     | <code>falcon_benchmark</code> |
| <code>fa.registry.list()</code>            | <code>list_clocks()</code>       | <code>data.frame</code>       |
| <code>fa.registry.get()</code>             | <code>clock_info()</code>        | <code>list</code>             |
| <code>fa.registry.cite()</code>            | <code>cite_clock()</code>        | <code>character</code>        |
| <code>fa.registry.compatible_with()</code> | <code>compatible_clocks()</code> | <code>data.frame</code>       |
| <code>fa.plot.ba_vs_ca()</code>            | <code>plot_ba_vs_ca()</code>     | <code>ggplot</code>           |
| <code>fa.report()</code>                   | <code>report()</code>            | <code>path, invisibly</code>  |

Plotting is the one place the two diverge deliberately. Python returns matplotlib figures; R calls the Python `*_data()` function to get the frame and renders it with ggplot2, so an R user gets a `ggplot` object they can modify. The data is identical; the rendering is native to each language.

### 8.5 S3 classes

R/classes.R

```
# falcon_data
print.falcon_data      # dimensions, modality, platform, species, age availability
summary.falcon_data    # plus per-clock coverage
dim.falcon_data
as.data.frame.falcon_data

# falcon_result
print.falcon_result    # clocks run, samples, warnings
summary.falcon_result  # per-clock mean, SD, missingness, flags
as.data.frame.falcon_result # the long table
plot.falcon_result     # dispatches to plot_ba_vs_ca()
`[,.falcon_result`     # subset by clock or sample

# falcon_qc
print.falcon_qc
plot.falcon_qc
```

`print.falcon_result` shows what a user needs before trusting the numbers:

```
<falcon_result> 120 samples x 5 clocks

  clock      unit  mean   sd  missing  flags
horvath2013 years  54.2   9.1    0.8%   -
hannum      years  51.7   8.4    0.0%   -
phenoage    years  56.9  11.2    0.0%   -
grimage2    years  61.4   8.8   12.4%   -
```

```
dunedinspace    ratio    1.03    0.11    2.1%    -
```

```
2 warnings. See warnings(x).
grimage2 is research-use-only.
```

## 8.6 DESCRIPTION

The file as it ships, not as it was specified: the differences are noted below it:

```
Package: FALCONAge
Title: Multiomic Biological Age and Aging Clock Scoring
Version: 1.0.0
Authors@R:
  person("Bhagesh", "Hunakunti", , "bhunakun@uni-bonn.de", role = c("aut", "cre"),
    comment = c(ORCID = "0000-0002-5957-8005"))
Description: Scores DNA methylation and clinical chemistry data against a
  catalogue of 161 published aging clocks, and provides the downstream
  statistics the field expects: age acceleration in its three conventions,
  survival and association models, technical reliability, and the AA1/AA2
  benchmark. Numerical work is delegated to the 'falconage' Python package
  through 'reticulate', so results from R and from Python are the same bits
  rather than two implementations that agree approximately. Twenty-three
  clocks carry coefficients inside the package and run offline; twenty-eight
  are implemented as scaffolds whose research-use-only coefficients the user
  supplies.
License: GPL (>= 3)
URL: https://github.com/bhagesh-h/FALCONAge,
  https://bhagesh-h.github.io/FALCONAge/,
  https://bhagesh-h.github.io/FALCONAge/r
BugReports: https://github.com/bhagesh-h/FALCONAge/issues
Depends:
  R (>= 4.1)
Imports:
  grDevices,
  reticulate (>= 1.34),
  stats,
  utils,
  yaml
Suggests:
  ggplot2,
  knitr,
  optparse,
  patchwork,
  rmarkdown,
  testthat (>= 3.0.0)
VignetteBuilder: knitr
biocViews: Epigenetics, DNAMethylation, DifferentialMethylation,
  MethylationArray, Software
```

```
Config/testthat/edition: 3  
Encoding: UTF-8  
Language: en-GB  
Roxygen: list(markdown = TRUE)  
RoxygenNote: 7.3.2
```

Four departures from the specification, each deliberate:

| Specified                                                       | Shipped                                                  | Why                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ggplot2, cli, rlang, jsonlite</code> in Imports           | <code>ggplot2</code> in Suggests; the other three absent | Plotting returns a data frame and draws with base graphics, so <code>grDevices</code> is the hard dependency and <code>ggplot2</code> is optional. Four fewer required packages is four fewer ways for a CRAN check to fail on a machine we do not control.             |
| Methods implemented include ...<br><doi:...> in the description | Absent                                                   | CRAN wants DOIs for methods the package <i>implements in R</i> . Every method here is implemented in the Python core; the per-clock references live in the registry and come back from <code>cite()</code> , which is a truer place for them than a fixed list of five. |
| SystemRequirements: Python (>= 3.10) ...                        | Absent                                                   | On reticulate $\geq 1.41$ the environment is provisioned automatically on first use, so Python is not a system requirement the user must satisfy. Still to be added before a CRAN submission, since the field is also documentation.                                    |
| Config/falconage/python-version                                 | Absent                                                   | The R package pins the Python core by git tag inside <code>falconage_install()</code> , so the field would be a second copy of a number that already exists in one place.                                                                                               |

Every example and every test that needs the core is wrapped in `skip_if_not(falconage_available())`, so R CMD check passes on a machine with no Python.

## 8.7 Roxygen conventions

Following the cyRAVEN register, see `R/thresholds.R` in that package for the model. Every exported function gets:

- an opening paragraph saying **what** it does, then **why** it does it that way when the choice is not obvious
- `@param` on every argument, naming the unit or the allowed values
- `@return` on every export, describing the object rather than naming its class
- `@examples`, runnable, wrapped in `\donttest{}` when they need the core or the network
- `@seealso` linking the Python equivalent by name
- `@references` with the DOI for anything implementing a published method

Worked example:

```
#' Score biological age with one or more aging clocks
#
# WHAT: applies published aging-clock definitions to a prepared dataset and
# returns one value per sample per clock, with the provenance needed to
# interpret it.
#
# WHY THE DEFAULT IS `"compatible"` RATHER THAN `"all"`: the registry holds
# clocks for several species, platforms and molecular layers, and most of them
# do not apply to any given dataset. Running everything produces a table whose
# majority of rows were computed almost entirely from imputed reference values,
# which looks like a result and is not one. `"compatible"` selects the clocks
# whose species, platform and feature coverage actually match the data, and
# reports what it excluded.
#
# @param data A `falcon_data` object from [preprocess_methylation()] or
# [preprocess_clinical()].
# @param clocks Clock identifiers, or `"compatible"` (default) to select
# automatically, or `"all"`. See [list_clocks()].
# @param device `"auto"`, `"cpu"`, `"cuda"`, or a specific device such as
# `"cuda:1"`. `"auto"` uses CUDA when it is available.
# @param dtype `"float64"` (default) or `"float32"`. Clocks whose registry
# entry requires double precision are promoted regardless, with a warning.
# @param max_missing Largest fraction of a clock's features that may be absent
# before scoring that clock is an error rather than a warning. Default 0.30.
# @param sex_unknown What to do when a clock needs sex and it is not recorded:
# `"error"` (default), `"impute_from_methylation"`, or `"half"`.
#
# @return A `falcon_result`. [as.data.frame()] gives one row per sample per
# clock with the value, its unit, its scale type, the fraction of features
# that had to be imputed, and any flags raised. The underlying dataset,
# including the run manifest, is reachable through the object.
#
# @examples
```



```

#' \donttest{
#' if (falconage_available()) {
#'   data <- preprocess_methylation(
#'     system.file("extdata", "example_betas.csv", package = "FALCONAge"),
#'     platform = "illumina_450k"
#'   )
#'   res <- score(data, clocks = c("horvath2013", "phenoage"))
#'   head(as.data.frame(res))
#' }
#' }
#'
#' @seealso [acceleration()] for age acceleration, [list_clocks()] to browse the
#'   registry, and the Python equivalent `falconage.score()`.
#' @references
#' Horvath S (2013). DNA methylation age of human tissues and cell types.
#' Genome Biology 14:R115. \doi{10.1186/gb-2013-14-10-r115}
#' @export
score <- function(data, clocks = "compatible", device = "auto",
  dtype = "float64", max_missing = 0.30,
  sex_unknown = "error", ...) { }

```

## 8.8 Command-line front end

inst/scripts/falconage.R, invoked the cyRAVEN way:

```

Rscript "$(Rscript -e 'cat(system.file("scripts", "falconage.R", package="FALCONAge"))')" \
  score --input prepared.h5ad --clocks horvath2013,grimage2 --outdir results/

```

Options are built in R/cli-options.R with `optparse`, and the flag names match the Python CLI exactly, so a script written against one runs against the other with only the interpreter changed.

## 8.9 pkgdown reference groups

r/\_pkgdown.yml. Group titles are identical to the quartodoc sections in §11, which is what makes the two sites navigable as one.

```

url: https://bhagesh-h.github.io/FALCONAge

template:
  bootstrap: 5
  light-switch: true

reference:
- title: Setup
  desc: >
    FALCONAge computes through a Python core. These functions create and inspect
    that environment; everything else assumes it exists.
  contents: [falconage_install, falconage_config, falconage_available, falconage_update]

```

```

- title: Download
  desc: >
    Public data by accession. One verb covers GEO, ArrayExpress, SRA, Zenodo,
    Figshare, Hugging Face and GDC, and returns files alongside a sample table
    with age and sex extracted from whatever free-text field held them.
  contents: [download, probe, cache_ls, cache_rm, cache_verify]

- title: Preprocess
  desc: >
    Raw intensities or a public matrix into a scoreable one. Normalisation is
    selectable and recorded, because it changes the answer.
  contents: [preprocess_methylation, preprocess_clinical, detect_platform,
    aggregate_epicv2, estimate_sex, qc]

- title: Score
  desc: One call runs any number of clocks. Every value carries its provenance.
  contents: [score, acceleration]

- title: The clock registry
  desc: >
    What is available, what it predicts, in what unit, on what platform, and
    where the published definition and the circulating coefficients disagree.
  contents: [list_clocks, clock_info, compatible_clocks, cite_clock,
    registry_version]

- title: Analysis
  desc: >
    Tests on per-sample values. Age acceleration is standardised within sex
    before any hazard ratio, so effect sizes compare across clocks.
  contents: [cox, c_index, associate, interactions, icc, run_benchmark]

- title: Figures
  contents: [starts_with("plot_"), theme_falconage]

- title: Reporting
  contents: [report, run_manifest]

articles:
- title: Get started
  navbar: ~
  contents: [FALCONAge, methylation]
- title: Working with data
  navbar: Working with data
  contents: [download, clinical, clocks]
- title: Going further
  navbar: Going further
  contents: [gpu, benchmarking]

```

## 9. Docker

Two images from one Dockerfile family, following the cyRAVEN pattern in its `inst/scripts/Dockerfile`: pinned base, dependency layer copied before source, dependencies read from the declaration rather than a second list, and a build-time smoke test so a broken image fails at build instead of an hour into an analysis.

### 9.1 Images

| Tag                               | Base                                                | Contains                               |
|-----------------------------------|-----------------------------------------------------|----------------------------------------|
| <code>falconage:1.0.0-cpu</code>  | <code>python:3.12-slim-bookworm</code>              | Python core, R, <b>FALCONAge</b> , CLI |
| <code>falconage:1.0.0-cuda</code> | <code>nvidia/cuda:12.4.1-runtime-ubuntu22.04</code> | the same plus the CUDA torch build     |
| <code>falconage:1.0.0-slim</code> | <code>python:3.12-slim-bookworm</code>              | Python core and CLI only, no R         |

### 9.2 Dockerfile.cpu, annotated

```
# =====
# FALCONAge - reproducible CPU environment
# =====
#
# WHY A PINNED BASE. Clock coefficients are fixed, but the arithmetic around
# them is not: a different BLAS changes the last bits of a 78,464-dimensional
# PC rotation, and a different scikit-learn changes what SimpleImputer stores.
# Neither moves a published age by a year, but both make two runs of the same
# data disagree in the seventh decimal, which is exactly what the gold-standard
# tests assert on. `latest` would silently re-roll that on a rebuild.
#
# Versions this image reproduces:
#   Python 3.12.8 . torch 2.6.0 (CPU) . numpy 2.2.1 . pandas 2.2.3
#   anndata 0.11.3 . scikit-learn 1.6.1 . R 4.4.3 . reticulate 1.40.0
#
# BUILD, from the repository root:
#   docker build -f docker/Dockerfile.cpu -t falconage:1.0.0-cpu .
#
# RUN. Nothing is baked in: no data, no cohort, no credentials.
#   docker run --rm \
#     -v "$PWD/data:/data:ro" -v "$PWD/results:/results" \
#     falconage:1.0.0-cpu \
#     score --input /data/prepared.h5ad --clocks compatible --outdir /results
#
#   On Windows PowerShell replace "$PWD" with "${PWD}".
#
# ITERATING WITHOUT REBUILDING. Mount a checkout and set FALCONAGE_SOURCE; the
# entrypoint runs that tree instead of the installed copy. A rebuild is still
# needed when the source starts requiring a package the image does not have.
#   -v "$PWD:/src:ro" -e FALCONAGE_SOURCE=/src
# =====
FROM python:3.12-slim-bookworm AS base
```

```

LABEL org.opencontainers.image.title="FALCONAge" \
        org.opencontainers.image.description="Multiomic biological age and aging clock
scoring" \
        org.opencontainers.image.source="https://github.com/bhagesh-h/FALCONAge" \
        org.opencontainers.image.licenses="GPL-3.0-or-later"

ENV PYTHONUNBUFFERED=1 \
    PIP_NO_CACHE_DIR=1 \
    # torch and numpy are OpenMP-threaded. Unset they take every core on the
    # host, which is antisocial on a shared machine and causes thermal
    # throttling on a laptop. Override with -e OMP_NUM_THREADS=8.
    OMP_NUM_THREADS=4 \
    # Weights land here. Mount a host directory over it to keep them between
    # runs instead of re-downloading a gigabyte each time.
    HF_HOME=/opt/falconage/cache \
    FALCONAGE_CACHE=/opt/falconage/cache \
    LANG=C.UTF-8

# Each library is here because a specific wheel needs it, not as a blanket
# build-essential block.
# libhdf5      -> h5py, which anndata writes .h5ad through
# libcurl/ssl  -> the download module
# libxml2      -> lxml, used by the GEO MINiML parser
# libgomp1     -> OpenMP runtime for torch and numpy
RUN apt-get update && apt-get install -y --no-install-recommends \
    libhdf5-dev libcurl4-openssl-dev libssl-dev libxml2-dev \
    libgomp1 zlib1g-dev libbz2-dev liblzma-dev ca-certificates curl \
    && rm -rf /var/lib/apt/lists/*

WORKDIR /opt/falconage

# ---- dependency layer -----
# Only the declaration, before any source. This layer costs several minutes and
# rebuilding it on every edit to src/ is the single biggest waste in a naive
# Dockerfile.
COPY python/pyproject.toml python/uv.lock /opt/falconage/python/
RUN pip install --no-cache-dir uv \
    && cd python && uv export --frozen --no-dev --format requirements-txt > /tmp/req.txt \
    && pip install --no-cache-dir -r /tmp/req.txt

# ---- R layer -----
FROM base AS with-r
RUN apt-get update && apt-get install -y --no-install-recommends \
    r-base r-base-dev libfontconfig1-dev libfreetype6-dev libpng-dev \
    && rm -rf /var/lib/apt/lists/*

# install_deps.R parses r/DESCRIPTION instead of repeating the package list, so
# the image can never install a different set than the package declares.

```

```

COPY r/DESCRIPTION docker/install_deps.R /opt/falconage/r/
RUN Rscript /opt/falconage/r/install_deps.R /opt/falconage/r/DESCRIPTION

# ---- source -----
FROM with-r AS final
COPY . /opt/falconage/src
RUN pip install --no-cache-dir --no-deps /opt/falconage/src/python \
  && R CMD INSTALL --no-multiarch --with-keep.source /opt/falconage/src/r \
  && cp /opt/falconage/src/docker/entrypoint.sh /opt/falconage/entrypoint.sh \
  && chmod +x /opt/falconage/entrypoint.sh

# reticulate must find the interpreter that already has the core installed,
# rather than trying to build an environment inside a read-only container.
ENV RETICULATE_PYTHON=/usr/local/bin/python

# ---- build-time verification -----
# A broken image fails HERE, not an hour into someone's analysis. This exercises
# the CLI parser, the registry load, the R bridge and every import both sides
# need.
RUN falconage --version \
  && falconage clocks list --limit 5 \
  && falconage config \
    && Rscript -e "library(FALCONAge); stopifnot(falconage_available());
print(falconage_config())" \
    && python -c "import falconage; print(falconage.__version__,
len(falconage.registry.load().list()))"

VOLUME ["/results", "/opt/falconage/cache"]
WORKDIR /work
ENTRYPOINT ["/opt/falconage/entrypoint.sh"]
CMD ["--help"]

```

Dockerfile.cuda differs only in the base image, the torch index URL, and a build-time torch.cuda.is\_available() check that is allowed to report `False` at build (there is no GPU on the builder) but prints the compiled CUDA version so a mismatch is visible.

### 9.3 entrypoint.sh

Same indirection as cyRAVEN's: run the installed copy by default, or a mounted source tree when FALCONAGE\_SOURCE is set, so routine edits need no rebuild.

```

#!/bin/sh
set -e

if [ -n "${FALCONAGE_SOURCE}" ]; then
  # -e is deliberately editable-install-free: prepending to PYTHONPATH is
  # instant and leaves the image's site-packages untouched, which matters
  # because most run configurations mount the filesystem read-only.

```

```

PYTHONPATH="${FALCONAGE_SOURCE}/python/src:${PYTHONPATH}"
export PYTHONPATH
fi

# `falconage` with no recognised verb is a help request, not an error.
case "$1" in
  ""|-h|--help) exec falconage --help ;;
esac

exec falconage "$@"

```

## 9.4 docker-compose.yml

```

services:
  falconage:
    image: falconage:1.0.0-cpu
    volumes:
      - ./data:/data:ro
      - ./results:/results
      - falconage-cache:/opt/falconage/cache
    environment:
      OMP_NUM_THREADS: 8

  falconage-gpu:
    image: falconage:1.0.0-cuda
    deploy:
      resources:
        reservations:
          devices: [{driver: nvidia, count: all, capabilities: [gpu]}]
    volumes:
      - ./data:/data:ro
      - ./results:/results
      - falconage-cache:/opt/falconage/cache

volumes:
  falconage-cache:

```

The named cache volume is not cosmetic: the PC clocks are several hundred megabytes each and re-downloading them on every container start is the difference between a two-minute run and a twenty-minute one.

## 10. Testing

### 10.1 Gold-standard vectors

The load-bearing test. For every clock: an input matrix and the output the reference implementation produces, stored under `tests/vectors/<clock>/`, asserted to `1e-6` in FP64, and consumed by both languages.

```

tests/vectors/
├─ manifest.yaml          # clock -> {input, expected, tolerance, source, generated}
├─ horvath2013/
│   ├─ input.parquet      # 20 synthetic samples x 353 CpGs, seeded
│   ├─ expected.parquet   # sample_id, value
│   └─ provenance.json    # pyaging 0.3.0, FP64, CPU, commit hash, date
└─ ...

```

Inputs are synthetic and seeded rather than real data: no consent question, no file-size problem, and a seeded generator gives coverage of the edge cases real data rarely contains, all-zero rows, a constant feature, values at exactly 0 and 1, and a controlled missingness pattern.

Generation is a one-off script per source implementation (`scripts/generate_vectors_pyaging.py`, `scripts/generate_vectors_bioblearn.py`, `scripts/generate_vectors_bioage.R`), run in a pinned environment and committed with its provenance. Regenerating is a deliberate act recorded in the changelog, never a fix for a failing test.

`registry/validate.py` fails when a clock has no vector, so the registry cannot grow past the test suite.

### *Which oracle*

Validating against pyaging makes pyaging the arbiter of correctness, and its bugs become ours. Oracles are therefore ranked, and each vector records which tier it used:

| Rank | Oracle                                                                                          | Available for                                                        |
|------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1    | A worked example published by the clock's own authors, reference inputs and the age they report | a minority of clocks, but the only truth that is not somebody's code |
| 2    | The authors' own reference implementation, run in a pinned environment                          | most clocks with an author repository                                |
| 3    | Hand computation from the published equation, for clocks small enough                           | PhenoAge, Weidner, Vidal-Bralo, Bocklandt, Garagnani, ZhangMortality |
| 4    | An independent third-party package, with at least two agreeing                                  | the remainder                                                        |
| 5    | Synthetic weights, architecture only                                                            | tier C clocks with no published worked example (§4.7)                |

A rank-4 vector where pyaging and bioblearn disagree is not a vector; it is an open question, and it goes to `registry/diffs/` rather than to `tests/vectors/`.

## 10.2 Cross-implementation comparison

A separate, non-blocking job compares FALCONAge against pyaging, bioblearn, computage, methylCIPHER and BioAge on shared clocks, and writes a table of agreements and disagreements to `docs/comparison.md`. Disagreements are published rather than hidden, [the science notes §15.2](#) shows the field already carries several, and a user comparing a FALCONAge number to a pyaging number deserves to know which ones differ and why.

## 10.3 Property tests

Hypothesis in Python, `testthat` with generated inputs in R.

| Property                                                         | Holds for                                                                                         |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| deconvolution proportions sum to 1 and are non-negative          | all deconvolution clocks                                                                          |
| probability outputs lie in [0, 1]                                | <code>scale_type: probability</code>                                                              |
| permuting feature column order does not change the result        | every clock                                                                                       |
| duplicating a sample does not change either copy's score         | every clock except per-sample-normalised ones                                                     |
| per-sample-normalised clocks DO change under sample duplication  | ZhangEN, ZhangBLUP, DunedinPACE, Pasta, REG - this is a property, not a bug, and the test pins it |
| a synthetic monotone age gradient produces a monotone prediction | chronological clocks                                                                              |
| <code>score(x)[i] == score(x[i])</code> for a single sample      | every clock without per-sample normalisation                                                      |
| FP32 and FP64 agree within the registry's per-clock tolerance    | every clock                                                                                       |
| scale-type guards reject illegal operations                      | <code>acceleration(absolute)</code> on a rate clock                                               |

## 10.4 Unit and integration

Unit tests per op with hand-computed expected values, the 15 preprocess ops and 15 postprocess transforms each get one, because these are where an off-by-one in a quantile index or a wrong constant hides. Integration tests run the full path (download a small cached fixture, preprocess, score, analyse, report) on both a methylation and a clinical fixture.

## 10.5 R/Python conformance

A CI gate, not an aspiration.

```
# tests/conformance/test_r_python_parity.py
@pytest.mark.parametrize("clock", REGISTRY.list())
def test_identical_scores(clock, fixture_h5ad, tmp_path):
    py = falconage.score(fa.FalconData.read_h5ad(fixture_h5ad), clocks=[clock])
    r = run_r(f'''
        library(FALCONAge)
        d <- read_falcon_data("{fixture_h5ad}")
        write.csv(as.data.frame(score(d, clocks = "{clock}")), "{tmp_path}/r.csv")
    ''')
    assert_frame_equal(py.table(), pd.read_csv(tmp_path / "r.csv"), atol=0, rtol=0)
```

`atol=0, rtol=0` - exact equality, not approximate. The two languages run the same computation, so anything other than bit equality means the bridge is doing something it should not, such as round-tripping a float through a string.

A second conformance test asserts API symmetry: every function in the Python public namespace has an R export with the matching name and argument set, checked against the table in §8.4.

## 10.6 Performance regression

Benchmarks on a fixed synthetic dataset, run nightly, with results committed so a slowdown is visible in the history rather than discovered later. Tracked: per-clock scoring throughput at 1,024 / 8,192 / 32,768 samples on CPU and CUDA; KNN imputation on a 500 by 400,000 matrix; IDAT parsing throughput; peak memory for the largest PC clock.



## 10.7 Coverage targets

90% line coverage on `models/`, `score/` and `registry/`, which are where a wrong number comes from; 75% elsewhere. Coverage on `download/` is deliberately lower because most of it is network error handling exercised by mocks rather than by real transfers.

## 11. Documentation

### 11.1 One site, two languages

A single Quarto site at `docs/`, with the Python API rendered by `quartodoc` and the R API by `pkgdown`, sharing a theme and navigation. Section titles are identical on both sides (§8.9), so a reader moving between them sees the same structure.

```
docs/
├─ _quarto.yml
├─ index.qmd                # what it is, in four paragraphs
├─ installation.qmd         # pip, R, Docker, GPU
├─ quickstart.qmd          # the same analysis in both languages, side by side
├─ guide/
│   ├─ concepts.qmd        # BA, AA, generations, units - from [the science notes]
│   (science.qmd) 1 and 8
│   ├─ download.qmd
│   ├─ preprocess.qmd
│   ├─ scoring.qmd
│   ├─ choosing-a-clock.qmd
│   ├─ analysis.qmd
│   ├─ benchmarking.qmd
│   ├─ gpu.qmd
│   ├─ reproducibility.qmd
│   └─ caveats.qmd        # [the science notes](science.qmd) 16, compressed
├─ clocks/
│   ├─ index.qmd           # searchable registry table, generated from YAML
│   └─ <clock>.qmd         # one page per clock, generated
├─ reference/              # quartodoc, Python
├─ r/                      # pkgdown output, mounted into the site
└─ _extensions/
```

### 11.2 \_quarto.yml

```
project:
  type: website
  output-dir: _site
  pre-render:
    - python scripts/build_clock_pages.py    # registry YAML -> clocks/*.qmd
    - Rscript scripts/build_pkgdown.R        # pkgdown -> docs/r/

website:
  title: "FALCONAge"
```

```

favicon: assets/logo.png
navbar:
  left:
    - text: Get started
      href: quickstart.qmd
    - text: Guide
      menu: [guide/concepts.qmd, guide/download.qmd, guide/preprocess.qmd,
            guide/scoring.qmd, guide/choosing-a-clock.qmd, guide/analysis.qmd]
    - text: Clocks
      href: clocks/index.qmd
    - text: Reference
      menu:
        - {text: "Python API", href: reference/index.qmd}
        - {text: "R API", href: r/reference/index.html}
  right:
    - icon: github
      href: https://github.com/bhagesh-h/FALCONAge

format:
  html:
    theme: {light: cosmo, dark: darkly}
    toc: true
    code-copy: true
    code-annotations: hover

quartodoc:
  package: falconage
  dir: reference
  style: pkgdown # section-grouped, matching _pkgdown.yml
  sections:
    - title: Download
      desc: >
        Public data by accession. One verb covers GEO, ArrayExpress, SRA,
        Zenodo, Figshare, Hugging Face and GDC, and returns files alongside a
        sample table with age and sex extracted from whatever free-text field
        held them.
      contents: [download, probe, cache.ls, cache.rm, cache.verify]
    # ... the remaining sections carry the same titles and desc text as
    # r/_pkgdown.yml, which is what makes the two sites read as one.

```

quartodoc's style: `pkgdown` produces a grouped reference index with the same shape `pkgdown` produces, which is why the two sides can share section titles and `desc` paragraphs verbatim. The Quarto version is pinned in CI and in the Docker image, because Quarto 2 is a full rewrite due late 2026 and a documentation build should not break on a toolchain upgrade nobody asked for.

### 11.3 Bilingual examples

Every guide page shows the same operation in both languages with Quarto tabsets:

```

::: {.panel-tabset}

## Python

```python
import falconage as fa

data = fa.prepare(fa.read_betas("betas.csv"))
res = fa.score(data, clocks=["horvath2013", "phenoage"])
res.table().head()
```

## R

```r
library(FALCONAge)

data <- preprocess_methylation("betas.csv", platform = "illumina_450k")
res <- score(data, clocks = c("horvath2013", "phenoage"))
head(as.data.frame(res))
```

:::

```

A CI job extracts both blocks from every page and runs them, so an example that has drifted from the API fails the docs build rather than misleading a reader.

### 11.4 Generated clock pages

`scripts/build_clock_pages.py` turns each registry YAML into a page carrying: what it predicts and in what unit, the platform and species it applies to, the feature count, the architecture, the preprocess and postprocess chain, the reliability ICCs, the benchmark scores, the citation and DOI, any known discrepancies, and a runnable example. The index is a searchable, filterable table over all of them. This is generated, never hand-written, so the docs cannot drift from the registry.

### 11.5 Docstring conventions

numpydoc in Python, roxygen2 in R, deliberately parallel:

| Concept         | numpydoc      | roxygen2                       |
|-----------------|---------------|--------------------------------|
| Summary         | first line    | first line                     |
| Rationale       | Notes section | body paragraph after the title |
| Argument        | Parameters    | @param                         |
| Return          | Returns       | @return                        |
| Example         | Examples      | @examples                      |
| Cross-reference | See Also      | @seealso                       |
| Citation        | References    | @references                    |

Both carry the same rationale text where a function exists in both languages, so a reader gets the same explanation whichever site they land on.

## 12. CI/CD and publishing

### 12.1 Workflows

**FALCONAge runs four workflows, not nine, and the difference is a decision rather than a backlog.** The table below is the specification. What `.github/workflows/` actually holds:

| Shipped                       | Covers                                                                                           | Specified rows it replaces                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>python-test.yaml</code> | tests, lint, nine guards, the unpinned-resolution job                                            | <code>python-test.yaml</code> , <code>registry-validate.yaml</code> |
| <code>R-CMD-check.yaml</code> | <code>R CMD check --as-cran</code> with the Python core, plus a no-Python job                    | <code>r-cmd-check.yaml</code> , <code>conformance.yaml</code>       |
| <code>docs.yaml</code>        | Quarto site with the embedded pkg-down tree, deployed to gh-pages                                | <code>docs.yaml</code>                                              |
| <code>release.yaml</code>     | one workflow, a <code>python</code> job and an <code>r</code> job, both behind the version guard | <code>release-pypi.yaml</code> , <code>release-r.yaml</code>        |

Registry validation and R/Python conformance are **jobs inside the two check workflows**, not workflows of their own: they share a checkout and an install with the suite they belong to, and splitting them would pay for that twice to gain a second badge saying the same thing.

`docker.yaml` and `benchmark.yaml` do not exist. Nothing publishes an image to a registry; `README.md` tells a user to build `docker/Dockerfile.cpu` locally, and the `falconage:1.0.0-cpu` tag it names is the one that build produces. The nightly benchmark would commit timings from a shared runner whose neighbours decide the numbers, which is a regression detector that cries wolf; the measurements that exist were taken on known hardware and are in **GPU support**.

The `python-test.yaml` row below is the one to disregard outright. There is no matrix and no Codecov upload, and the workflow's own header explains at length why: the lock file pins one resolution, `uv export --frozen` reproduces it, and on any other interpreter the pinned wheels either do not exist or do not match, so a matrix job fails before reaching a test. Three red badges where one would do teaches people to ignore the badge. Coverage was dropped after printing sixty lines of site-packages paths that nobody read.

| Workflow                      | Trigger  | Does                                                                                                                                                           |
|-------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>python-test.yaml</code> | push, PR | pytest on 3.10–3.13 × {ubuntu, macOS, windows}; gold-standard vectors; property tests; coverage to Codecov                                                     |
| <code>r-cmd-check.yaml</code> | push, PR | <code>R CMD check --as-cran</code> on ubuntu-release with the Python core installed; a second job with no Python, asserting the package still loads and checks |

| Workflow                            | Trigger                          | Does                                                                   |
|-------------------------------------|----------------------------------|------------------------------------------------------------------------|
| <code>conformance.yaml</code>       | push, PR                         | R/Python parity across every clock; API symmetry                       |
| <code>registry-validate.yaml</code> | push to <code>registry/**</code> | schema, vocabulary, checksums, gold-standard presence                  |
| <code>docs.yaml</code>              | push to main                     | Quarto site including pkgdown, examples executed, deployed to gh-pages |
| <code>docker.yaml</code>            | push to main, tags               | build and push cpu/cuda/slim to GHCR, multi-arch on cpu                |
| <code>release-pypi.yaml</code>      | tag <code>v*</code>              | version guard, build sdist and wheel, publish via trusted publishing   |
| <code>release-r.yaml</code>         | tag <code>v*</code>              | version guard, R CMD build, --as-cran, attach tarball to the release   |
| <code>benchmark.yaml</code>         | nightly                          | performance regression, results committed                              |

## 12.2 The version guard

Borrowed directly from cyRAVEN's `release.yaml`, extended to four files. A tag and a version that disagree cannot be repaired afterwards, because the tag is what a citation points at.

```
- name: Versions must agree
  run: |
    TAG="${GITHUB_REF#refs/tags/v}"
    PY=$(grep -m1 '^version' python/pyproject.toml | sed 's/.*"\(.*\)"/\1/')
    R=$(grep -m1 '^Version:' r/DESCRIPTION | awk '{print $2}')
    PYREQ=$(grep -m1 'Config/falconage/python-version:' r/DESCRIPTION | awk '{print $2}')
    for pair in "python:$PY" "R:$R" "R-python-pin:$PYREQ"; do
      name="${pair%*:}"; val="${pair#*:}"
      [ "$val" = "$TAG" ] || { echo "::error::$name is $val, tag is $TAG"; exit 1; }
    done
    echo "all versions agree at $TAG"
```

## 12.3 R CMD check without Python

The second job in `r-cmd-check.yaml` is the one that matters for CRAN. It runs on a machine with no Python core at all and asserts that `library(FALCONAge)` succeeds, `falconage_available()` returns `FALSE`, every example is skipped rather than erroring, and the check reports no `ERROR` or `WARNING`. The `delay_load` in `.onLoad` (§8.3) is what makes this possible, and this job is what stops it from silently regressing.

## 12.4 Publishing

`PUBLISHING.md` carries the same honest gap table cyRAVEN uses, per target.

**PyPI** - trusted publishing, no long-lived token. `sdist` plus a pure-Python wheel; no compiled extensions, so no `cibuildwheel`. Test on TestPyPI first for the initial release.

**r-universe** - the first R route, no review. Add the repository to a `packages.json` in `bhagesh-h.r-universe.dev`, and binaries for Windows, macOS and Linux appear within minutes.

**CRAN** - needs, beyond the above: `cran-comments.md` naming platforms checked and explaining every NOTE; `\value` on every exported function; runnable examples; total runtime under 10 minutes; no writing outside `tempdir()`; win-builder and macOS results. The reticulate dependency is the reviewable point, and the answer is the no-Python check job in §12.3 plus `SystemRequirements`.

**Bioconductor** - the better long-term home given the `biocViews` and the audience. Needs version `0.99.x` at submission, `BiocCheck` clean, a `BiocStyle` vignette, package under 5 MB (satisfied, since no weights ship), no `sapply` or `1:n` in package code, and a support-site subscription.

**Docker** - GHCR on every main push and tag; Docker Hub mirror for the release tags.

**Zenodo** - DOI minted from each GitHub release, which is what a methods section cites.

## 13. Build order

Fourteen milestones. Each has an exit criterion that is testable, not a feeling.

| #  | Milestone                                                                                                       | Exit criterion                                                                                                                                                                                                  |
|----|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Repo scaffold: monorepo layout, <code>pyproject.toml</code> , <code>DESCRIPTION</code> , licences, CI skeletons | <code>pip install -e python/</code> and <code>R CMD INSTALL r/</code> both succeed; empty test suites pass on CI                                                                                                |
| 2  | Core engine: device, dtype, <code>FalconData</code> , manifest, logging, config, units, errors                  | Round-trip a <code>FalconData</code> through <code>.h5ad</code> unchanged; unit conversion table tested both directions; <code>resolve_device</code> correct on CPU and CUDA                                    |
| 3  | Registry: schema, validator, vocabulary, weight resolution, 5 hand-written clock YAMLs                          | <code>registry-validate</code> green; <code>resolve_weights</code> fetches, verifies and caches; offline mode raises correctly                                                                                  |
| 4  | Op catalogue: all 15 preprocess ops, all 15 postprocess transforms, vectorised                                  | Each op has a hand-computed unit test; no Python loop over samples anywhere in <code>ops/</code>                                                                                                                |
| 4b | Coefficient extraction: one extractor per primary source, tier assignment, diffs against pyaging/biolearn       | Every tier A and B clock extracted from its own source with a checksum; <code>registry/diffs/</code> written; untraced count reported                                                                           |
| 5  | Linear and PC clocks + <code>score()</code> : Horvath, Hannum, Lin, PCHorvath2013, PCPhenoAge                   | Gold-standard vectors pass at 1e-6 FP64; missingness accounting correct                                                                                                                                         |
| 6  | Clinical clocks: PhenoAge both variants, KDM with fit/project, HD                                               | Matches BioAge on NHANES to 1e-6; <code>units</code> omission raises; packaged NHANES fits load                                                                                                                 |
| 7  | Remaining architectures: GrimAge family, AltumAge, mitotic, deconvolution, SystemsAge, mammalian, sex           | Every tier A and B clock has a passing gold-standard vector; every tier C clock has a passing synthetic-weight architecture test and a working <code>register_local_weights</code> ; registry covers 161 clocks |
| 8  | Preprocess: IDAT chain, platform detection, EPIC v2 aggregation, clinical, QC                                   | Beta matrix from a known GEO series matches the published one within noise; EPIC v2 without aggregation raises                                                                                                  |
| 9  | Download: GEO, ArrayExpress, SRA, Zenodo, Figshare, HF, GDC; cache; metadata normalisation                      | <code>download("GSE40279", dry_run=True)</code> reports correct sizes; age extracted for a known series matches its published table                                                                             |
| 10 | Analysis and plots: acceleration, Cox, ICC, AA1/AA2, the 14 figures                                             | Cox HRs reproduce the BioAge NHANES IV table; AA1/AA2 reproduce a published ComputAgeBench row                                                                                                                  |
| 11 | R package: bridge, S3 classes, all exports, roxygen, vignettes                                                  | Conformance suite green at <code>atol=0</code> ; <code>R CMD check --as-cran</code> clean with and without Python                                                                                               |

| #  | Milestone                                                                                                  | Exit criterion                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | Docker: cpu, cuda, slim; entrypoint; compose                                                               | Build-time smoke test passes; a GPU run on a CUDA host produces the same numbers as CPU within FP tolerance                                 |
| 13 | Documentation: Quarto site, quartodoc, pkgdown, generated clock pages, bilingual examples                  | Docs build green with every example executed; both reference trees render with matching sections                                            |
| 14 | Release 1.0.0                                                                                              | Version guard passes on all four files; PyPI, r-universe, GHCR and Zenodo all carry 1.0.0                                                   |
| 15 | (v1.5) Native clock training: elastic net, Cox elastic net and KDM fitted on the open ComputAgeBench split | A FALCONAge-native chronological and mortality clock, benchmarked against the registry, distributed under GPL with no permission constraint |



Milestones 1–7 are the critical path; 4b gates 5 and 7, because a clock cannot be tested before its coefficients have a traced source. 8–10 can proceed in parallel once 2 and 3 land; 11 needs 7; 12 and 13 need 11. Milestone 15 is post-1.0 and is the answer to “can the permission-gated clocks be replaced rather than licensed”, not with the same clocks, but with open ones.

## 14. Open questions

**Weight redistribution, resolved into policy, but the letters still need writing.** §3.7 sorts every clock into tier A, B or C, and §4.7 specifies what a tier C clock ships. That closes the design question. What remains is correspondence: the maintainers of the GrimAge, SystemsAge and DunedinPACE coefficient sets should be asked directly whether FALCONAge may redistribute them, and each answer recorded in the registry entry. A yes moves a clock from C to A and is worth having; a no costs nothing, because the scaffold already works with a user-supplied file.

The same question arrives again for the transcriptomic and proteomic layers: tAge models are MGB Open Access non-commercial, and organAging is an Academic Non-Commercial License. Both are likely tier B or C.

**Whether the 37 package-to-package clocks can all be traced.** §3.8 asserts that every tier B clock will get a primary source. Some may not have one, a coefficient file that entered circulation through a personal communication and was never published leaves no trail. Those end up as tier C by default, and the count in §3.7 could grow. The honest position is that the number is not yet known, and `primary_source_traced: false` makes the debt visible rather than hidden.

**CRAN name.** `available::available("FALCONAge")` has not been run. PyPI is clear for both `falconage` and `FALCONAge`. If CRAN objects, the Python name stays and only the R name moves.

**How much benchmark data to vendor.** `ComputAgeBench` is 10,404 samples across 65 studies, several gigabytes. Fetching on demand from Hugging Face keeps the package small but makes the benchmark untestable offline and in CI. Likely answer: vendor a small deterministic subset (3 studies, around 200 samples) for CI and tests, fetch the full set on demand.

**IDAT parsing dependency.** `methylnprep` is the only maintained pure-Python IDAT reader, and it is a heavy dependency with an opinionated pipeline of its own. The alternatives are vendoring the IDAT binary parser (roughly 300 lines, well-specified format) or requiring R and `minfi` through `rp2`. Leaning toward vendoring the parser and implementing `noob` and `BMIQ` natively, with `methylnprep` as an optional accelerated path.

**Which normalisation is the default.** `noob + BMIQ` and `SeSAMe` both perform well and are not interchangeable. Defaulting to either makes results from the two paths incomparable while looking identical. Current plan is to require the choice explicitly on the first call and record it everywhere, with `noob_bmiq` as the documented default only because it is what most published clock coefficients were derived under.

**Mammalian species table.** The `AnAge` lookup that the mammalian clocks need covers 1,756 species, and its licence and redistribution terms need checking before it is packaged.

## 15. The layer that says what a score is worth

Sections 1–14 describe the architecture that computes a score. This one describes the layer that says how much that score is worth. It is deliberately a *layer*: nothing below changed, no score moved, and every new artefact is reported rather than applied.

## 15.1 New modules

| Module                                       | What it holds                                                                                                                                                           |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>falconage.uncertainty</code>           | <code>technical_se</code> , <code>interval</code> , <code>conformal_interval</code> , <code>icc_from_replicates</code> , and the loaders for the two calibration tables |
| <code>falconage.core.tissue</code>           | the specimen alias table, families, and the published discordance between them                                                                                          |
| <code>falconage.core.prealanalytical</code>  | the recognised <code>obs</code> fields, their thresholds, and the manifest block                                                                                        |
| <code>falconage.preprocess.batch</code>      | <code>fit_batch_reference</code> , <code>apply_batch_reference</code> , <code>BatchReference</code>                                                                     |
| <code>falconage.models.ops.DERIVATIVE</code> | an analytic derivative per transform, and an explicit refusal for the four that destroy the information                                                                 |
| <code>python/tools/</code>                   | the three derivation scripts for the shipped data artefacts                                                                                                             |

## 15.2 New registry artefacts

Each is derived by a script in `python/tools/`, digest-checked, and carries its provenance into the run manifest. None can be regenerated in CI, two need the 586 MB corpus, one needs a 72 MB published spreadsheet, so CI checks that the shipped files parse and cover what they claim instead, and the local `--check` catches staleness before a commit.

| File                           | Rows    | Source                              | Built by                            |
|--------------------------------|---------|-------------------------------------|-------------------------------------|
| <code>probe_icc.csv.gz</code>  | 283,860 | ADNI duplicates, EPIC v1, CC BY 4.0 | <code>build_probe_icc.py</code>     |
| <code>platform_bias.csv</code> | 30      | this corpus, masked and rescored    | <code>build_platform_bias.py</code> |
| <code>conformal.csv</code>     | 48      | healthy controls in this corpus     | <code>build_conformal.py</code>     |
| <code>evidence.yaml</code>     | 18      | multi-clock comparison studies      | hand-curated, DOI enforced at load  |

## 15.3 Where the checks fire

The registry already declared three properties that scoring enforced. Two more were added late, and they use the same shape, a declared property of the clock, checked in `score()`, refusing when the substitution is a category error and warning when it is an offset.

| Declared                                         | Refuses when                                                                                |
|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| <code>scale_type</code> → <code>LEGAL_OPS</code> | an operation is undefined for the output's scale                                            |
| coverage / coefficient mass                      | too much of the model is absent                                                             |
| <code>availability</code> tier                   | the coefficients are not distributable                                                      |
| <code>requires_cohort</code>                     | a cohort-centred clock is given one sample                                                  |
| <code>tissue_policy</code>                       | a clock is scored off a tissue with no counterpart, or on cell-free DNA                     |
| <code>**age_years_relative</code>                | the absolute acceleration convention is asked of a clock whose origin moves between cohorts |

The last one also fixed a latent gap rather than only adding a rule. `LEGAL_OPS` has listed `acceleration` and `residual` as separate entries since v1.0, but `acceleration()` checked "acceleration" whatever convention was asked for, so the distinction in the data model was never read. It is now: `absolute` and `both` need `acceleration`, `residual` and `within_group` need `residual`.

#### 15.4 Figure inventory, updated

Twenty-six figure types, four of them new:

| Figure                          | Answers                                                      |
|---------------------------------|--------------------------------------------------------------|
| <code>reliability_forest</code> | which of these clocks can be trusted to move                 |
| <code>score_interval</code>     | which two samples are actually distinguishable               |
| <code>platform_bias</code>      | what this array costs each clock, in years                   |
| <code>consensus_plot</code>     | did the difference hold up across clocks, or is it one clock |

`save_all()` takes `se=`, `conformal=` and `consensus=` and emits them when given; `test/run_all.py` computes all three per dataset, so the gallery figures come from real public data rather than a synthetic demonstration.

#### 15.5 What did not change

No score. The earlier numbers are bit-identical for any dataset whose clocks were legal to score in the first place, the two integration tests that changed did so because they were scoring placenta clocks on cord blood and buccal clocks on adult blood, and those results were wrong before the check existed rather than different after it.

## Chapter 8. References

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Citations across this site are numbers in square brackets and every one of them points here. The numbering is global rather than per page, so a source keeps its number wherever it is cited, and the same list is bound into the PDF as its own chapter.

Where a source is available at a stable identifier the entry links to it. Some of these are preprints and are cited as preprints on purpose: the field moves faster than journal review, and a claim resting on one is worth knowing about.

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